MODULE 2

8.1 INTRODUCTION

8.2 CIRCUIT-SWITCHED NETWORK

8.3 PACKET-SWITCHING

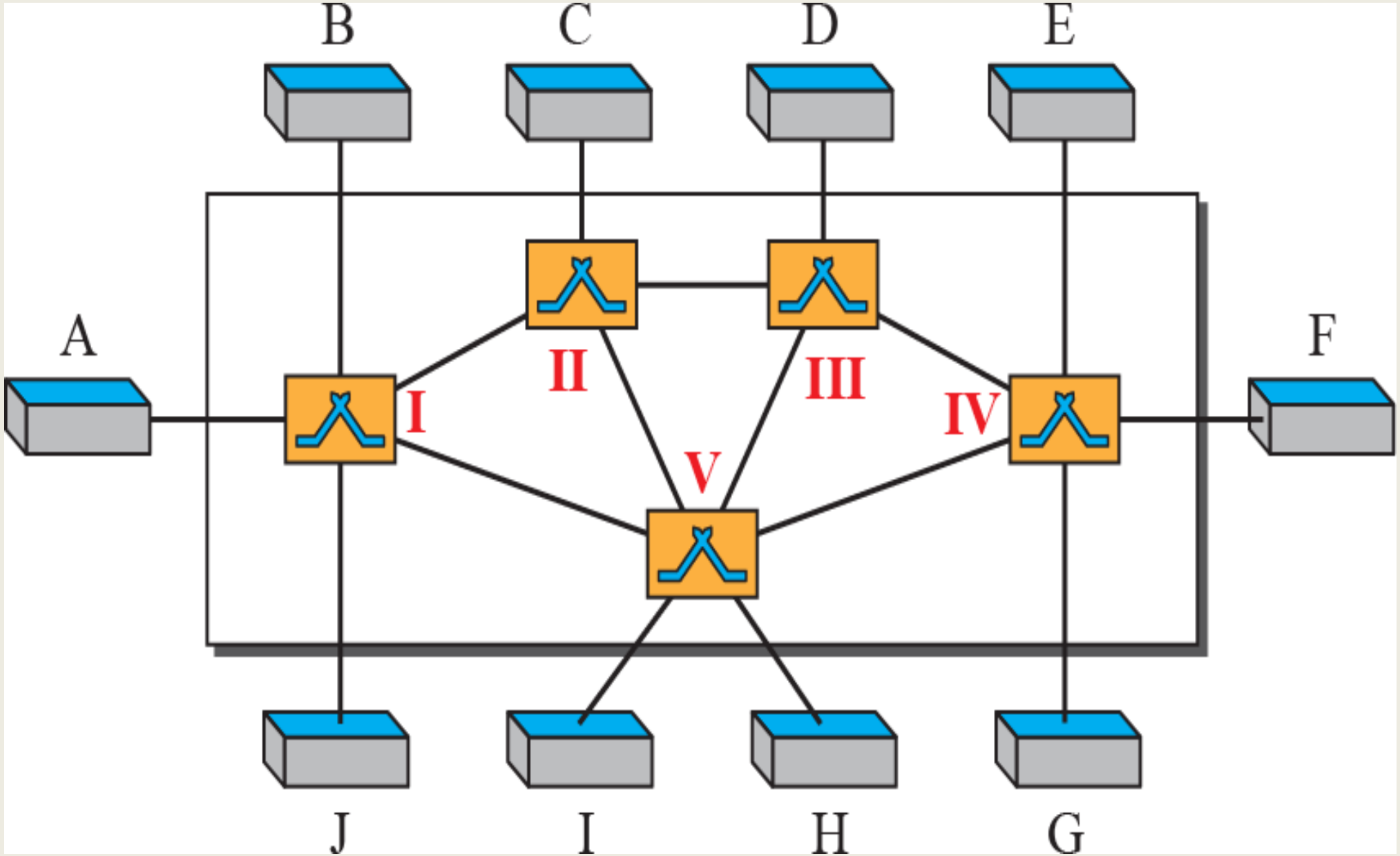
Objective

- It mentions three methods of switching: circuit switching, packet switching, and message switching. Also defines the switching methods that can occur in some layers of the Internet model.
- The second section discusses circuit-switched networks. It first defines three phases in these types of networks and describes the efficiency and delay of these networks.
- The third section briefly discusses packet-switched networks. It describes datagram networks and virtual circuit networks

INTRODUCTION

- A network is a set of connected devices. Whenever we have multiple devices, the problem is how to connect them to make one-to-one communication possible.
- The solution is switching. A switched network consists of a series of interlinked nodes, called switches.
- Switches are devices capable of creating temporary connections between the devices connected to it.
- Some of these node are connected to the end systems(Comp. or Telephones)

Switched network



➢Three Methods of Switching

- Traditionally, three methods of switching have been discussed:

  i) Circuit switching,

  ii) Packet switching

  iii) Message switching.

- The first two are commonly used today. The third has been phased out in general communications but still has applications.

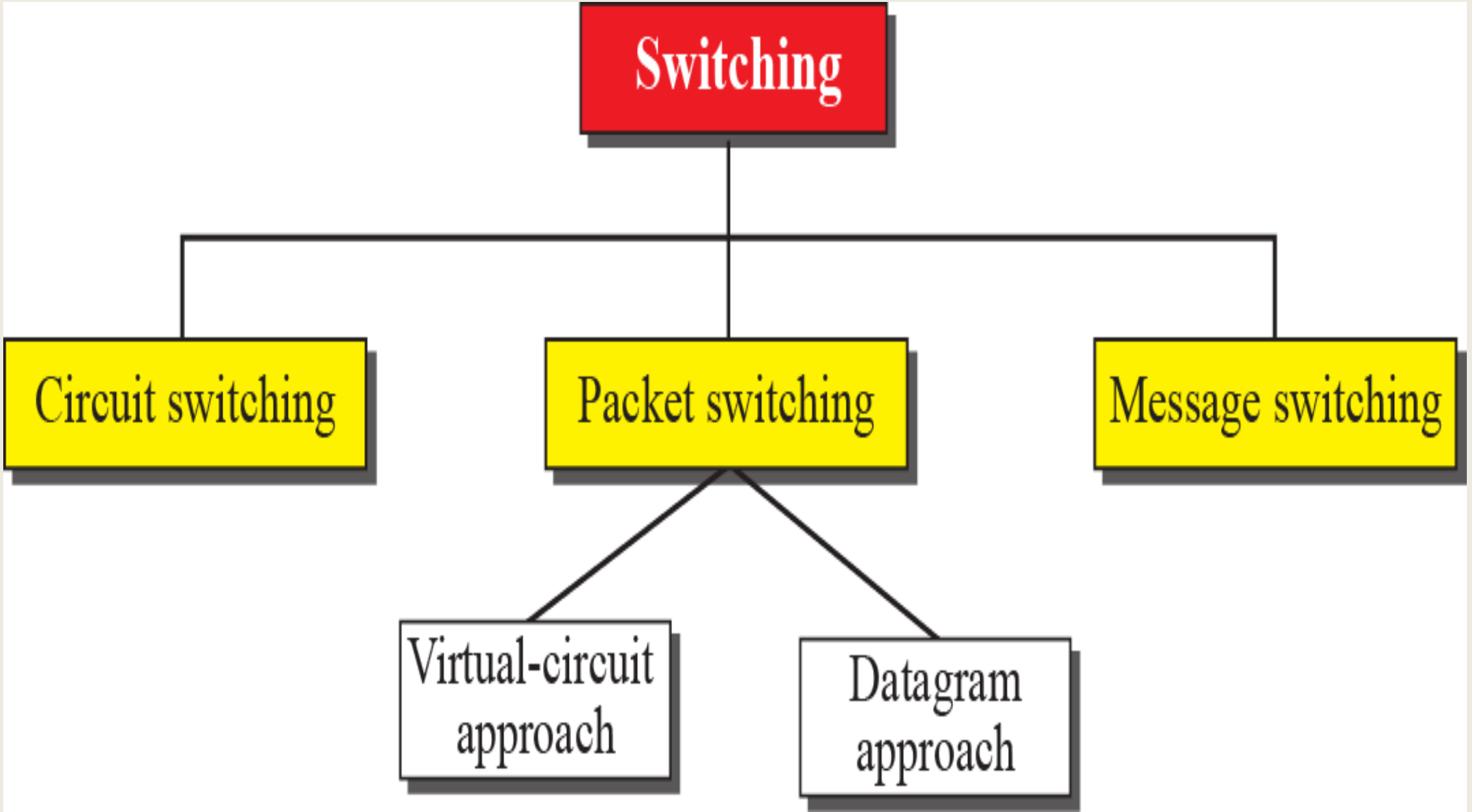
  ➢Packet switching can further be divided into two subcategories

  ➢ virtual-circuit approach and datagram approach.

Switching and TCP/IP Layers

- *Switching can happen at several layers of the TCP/IP protocol suite:*
  - At physical layer: Only circuit switching used. No packets exchanged. Switches allow signals to travel in one path to another.
  - At data-link layer: Packet switching is used. Packet here means frames or cells. Virtual circuit approach is used.
  - At network layer: Packet switching using virtual or datagram approach can be used.
  - At Application Layer: Only Message switching is used. Communication occurs by exchanging messages

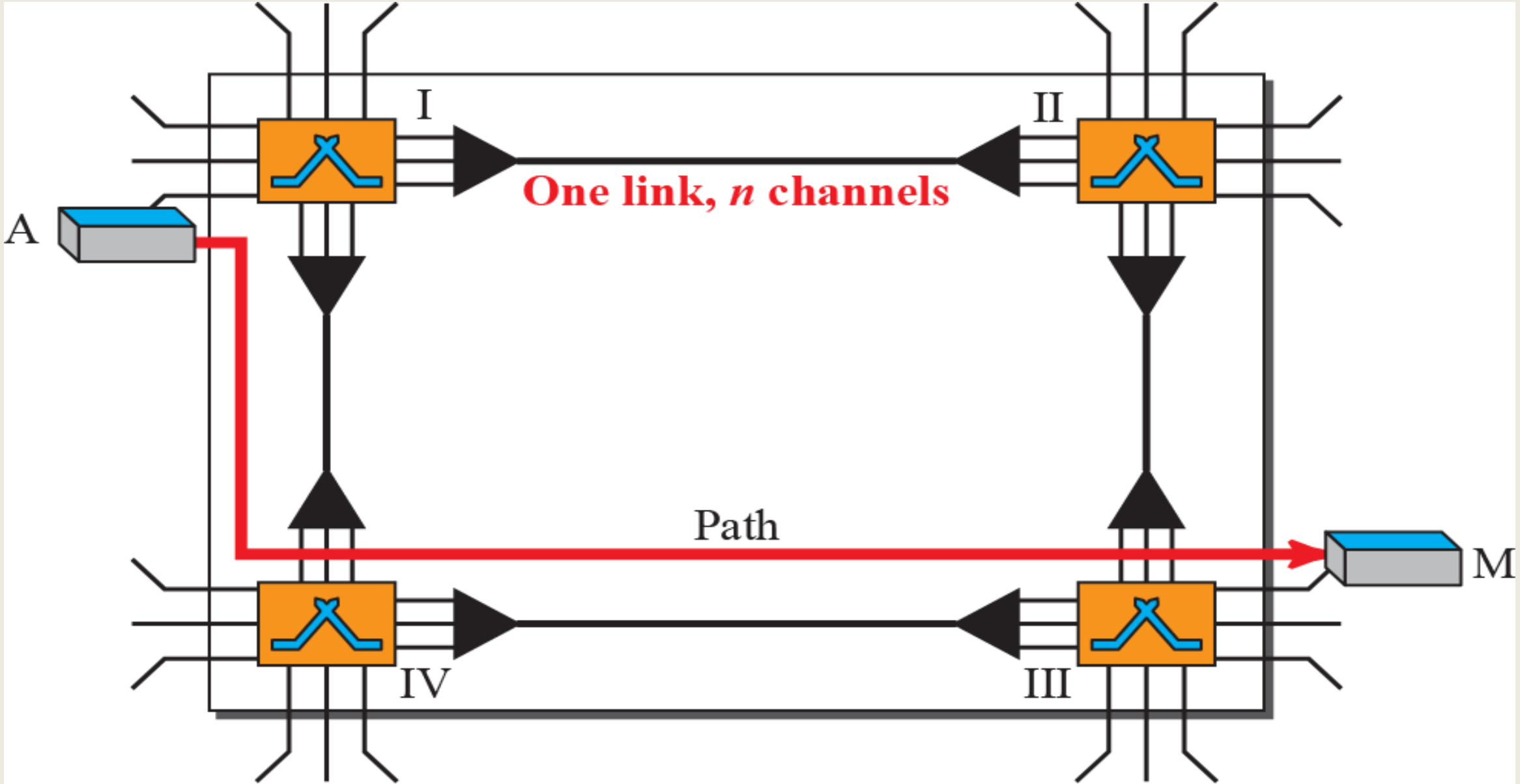
Taxonomy of switched networks



8.2 CIRCUIT-SWITCHED NETWORKS

- A circuit-switched network consists of a set of switches connected by physical links.
- Connection between two stations has a dedicated path made of one or more links.
- However, each connection uses only one dedicated channel on each link.
- Each link is normally divided into n channels by using FDM or TDM.

➢A trivial circuit-switched network



➢Circuit switched network

- When end system A need to communicate with end system M, Connection request must be accepted by all switches and M itself.(Setup phase)
- A circuit is reserved on each link. A combination of circuits or channels defines dedicated paths .data transfer . Finally tearing down the connection.
- Circuit switching takes place at Physical Layer.
  - Before starting communication, the stations must make a reservation for the resources to be used during communication.
  - These resources are band width in FDM, time slots in TDM, switch buffers, processing time, switch i/o ports, must remain dedicate for the entire duration of data transfer until tear down phase.No addressing involved in

Example-1

➢As a trivial example, letus use a circuit-switched network to connect eight telephones in a small area.

•Communication is through 4-kHz voice channels.

•Assume that each link uses FDM to connect a maximum of two voice channels.

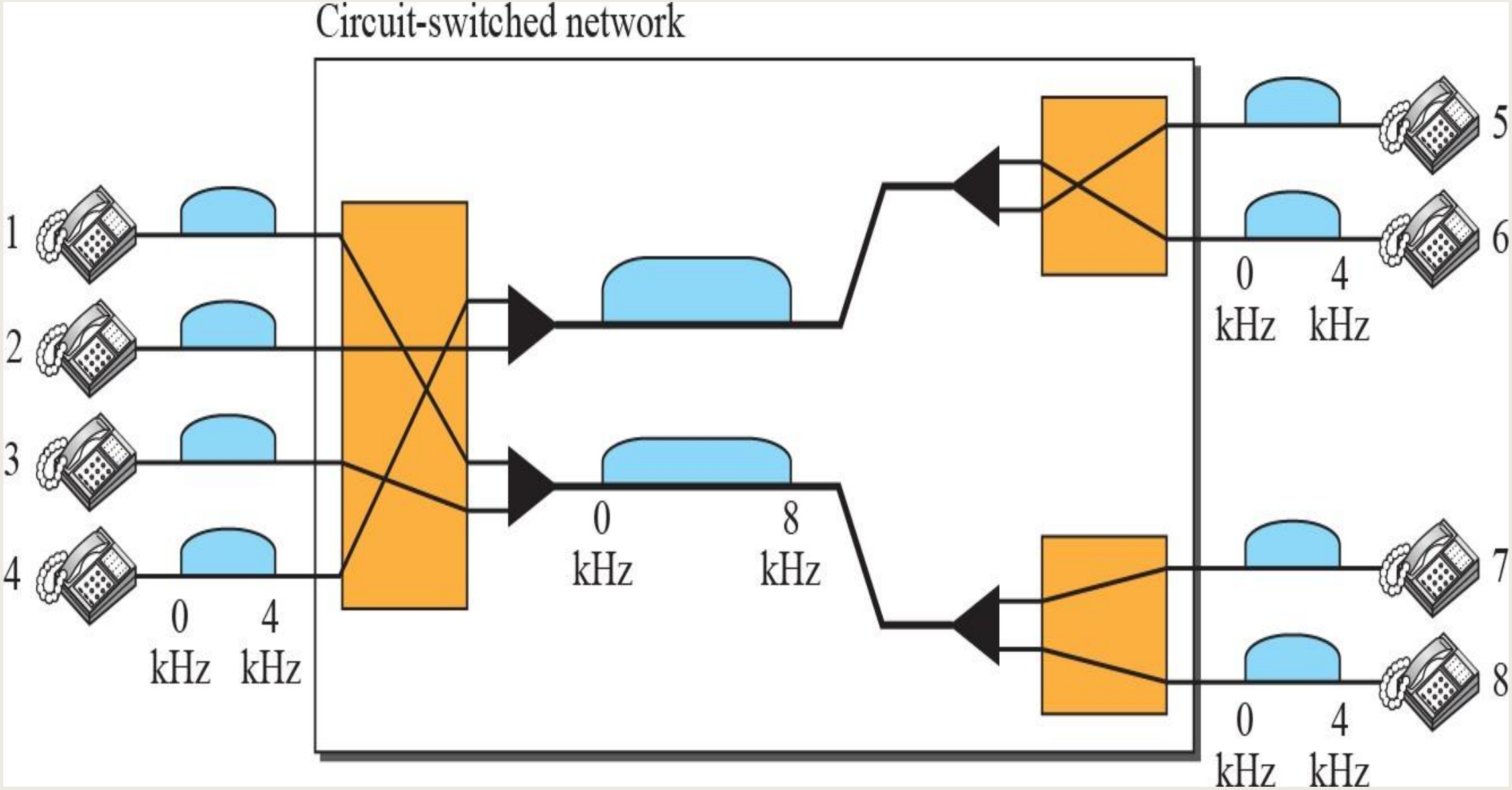
•The bandwidth of each link is then 8 kHz.

•Figure below shows the situation.

•Telephone 1 is connected to telephone 7; 2 to 5; 3 to 8; and 4 to 6.

• The switch controls the connections.

➢Circuit-switched network



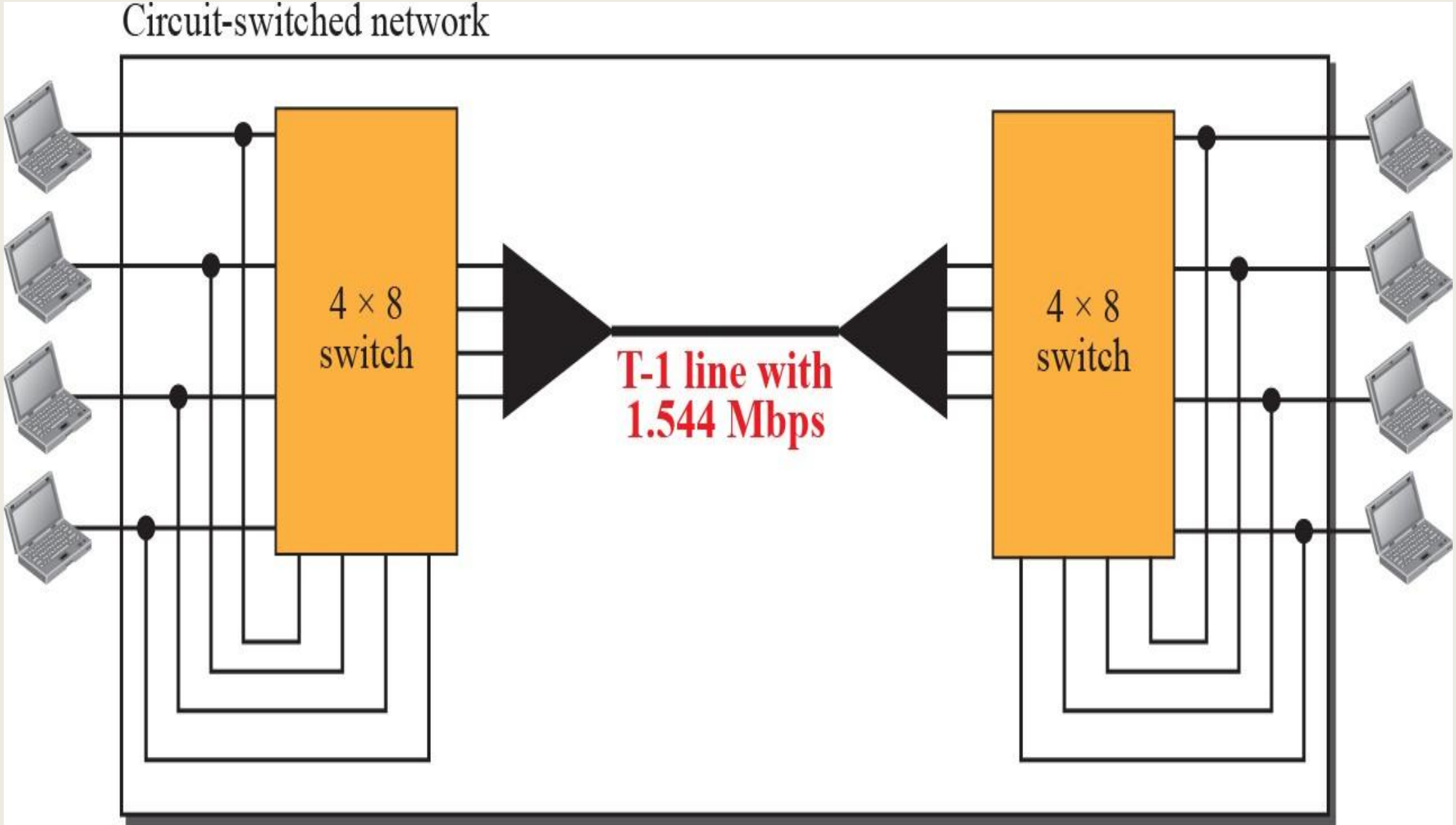
Example-2

Consider a circuit-switched network that connects computers in two remote offices of a private company.

- The offices are connected using a T-1 line leased from a communication service provider.
- There are two 4 × 8 (4 inputs and 8 outputs) switches in this network.
- For each switch, four output ports are folded into the input ports to allow communication between computers in the same office.
- Four other output ports allow communication between the two offices.

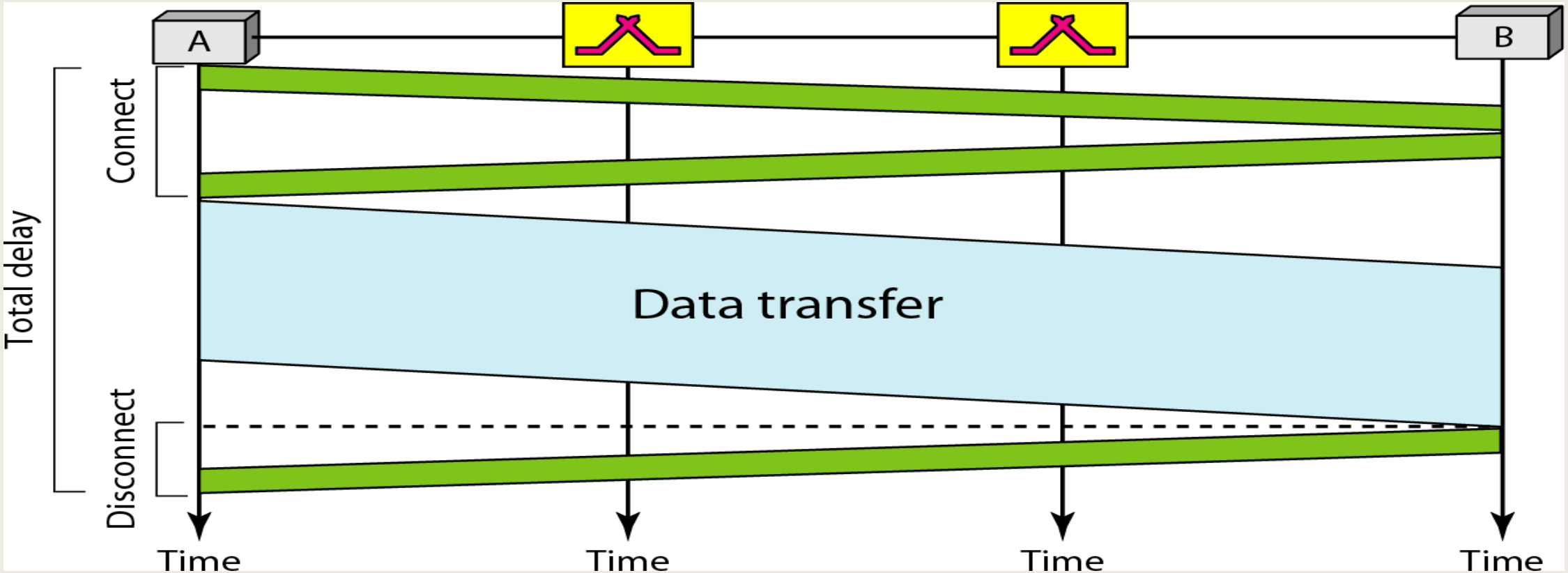
Figure below shows the situation

➢Circuit-switched network



➢Connection phases in circuit switched networks

1. Connection Setup phase
2. Data transfer phase
3. Tear down phase



Three phases(Continued..)

- Before two parties can communicate, a dedicated channel needs to be established.
- Connection set up means creating dedicated channel. Eg: System A would like to communicate to M.

➢ The setup request must pass through all the switches between A and M.

➢ Switches have to find a dedicated channel.

➢ Destination should send back the Ack. to source in opposite direction to system A.Only after System A receives the ACK. , connection is established.

➢ End-to-end addressing is required to set up a connection

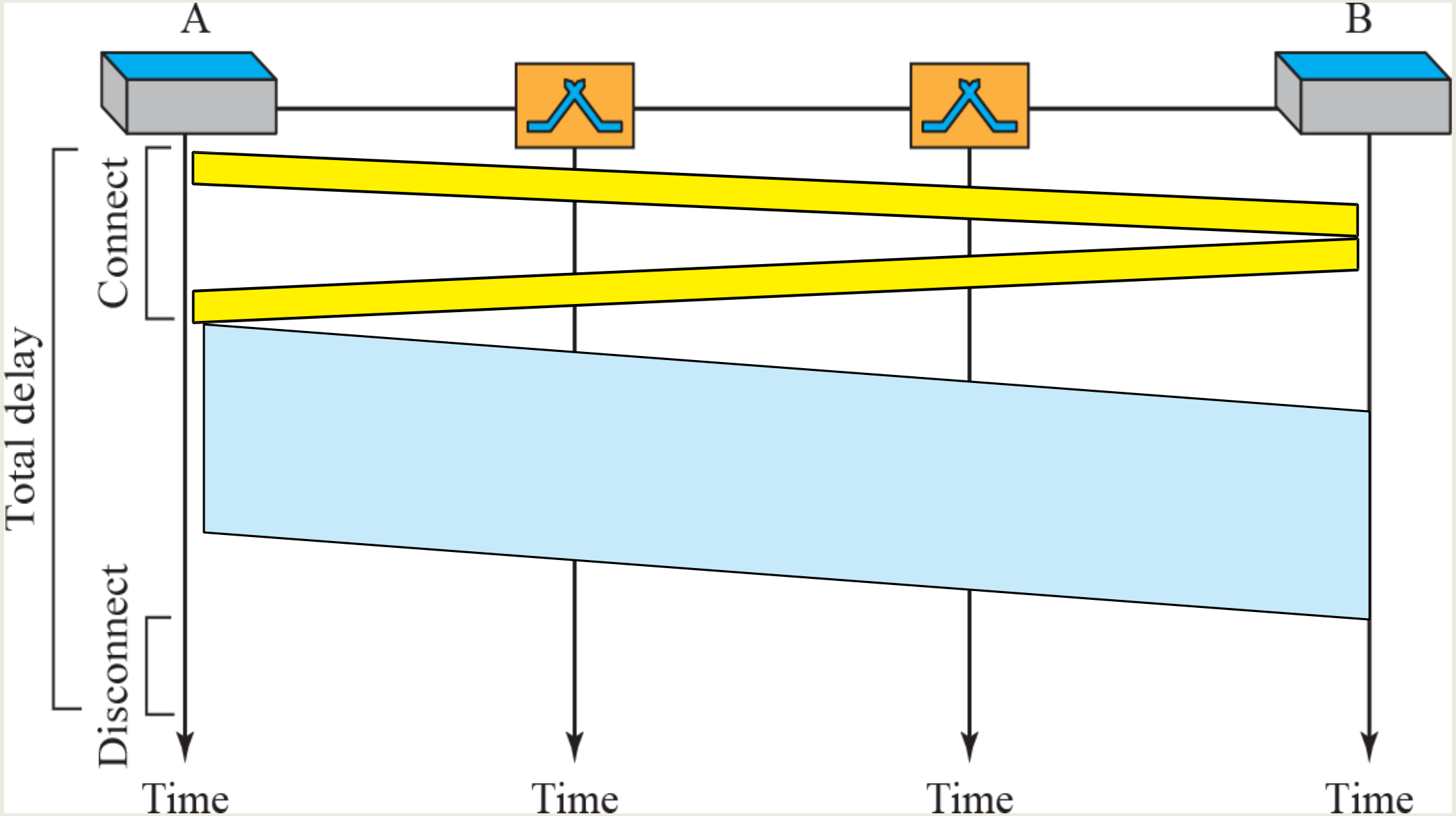
Contd..

- Data transfer phase: After dedicated channel is established, two parties can transfer the data.
- Tear down phase: When no data to transfer, a signal is sent to all switches in between source and destination to release the resources.

Efficiency: As resources are allocated before data transfer for the entire duration of connection,

- There is an argument that the CSN is not efficient. because the resources are unavailable for other connections.
- In telephone N/W, the moment conversation is over, connection is released,
- Whereas in Computer N/Ws, a computer can be connected to other even if no activity for a long time.

➢Delay in a circuit-switched network



➢Delay

- Circuit Switched Network has low efficiency, delay is minimal.
- During data transfer, data are not delayed at switches.
- The total delay will be due to time needed to setup connection, data transfer and tear down of connection.
- Delay caused by set up is sum of four parts

➢ Propagation time + Request signal transfer time+ Propagation time of ack. from the destination +Signal transfer time of ACK.

- The delay due to data transfer is sum of two parts

delay= Propagation time+ Data transfer time.

- Finally time required to tear down the connection.

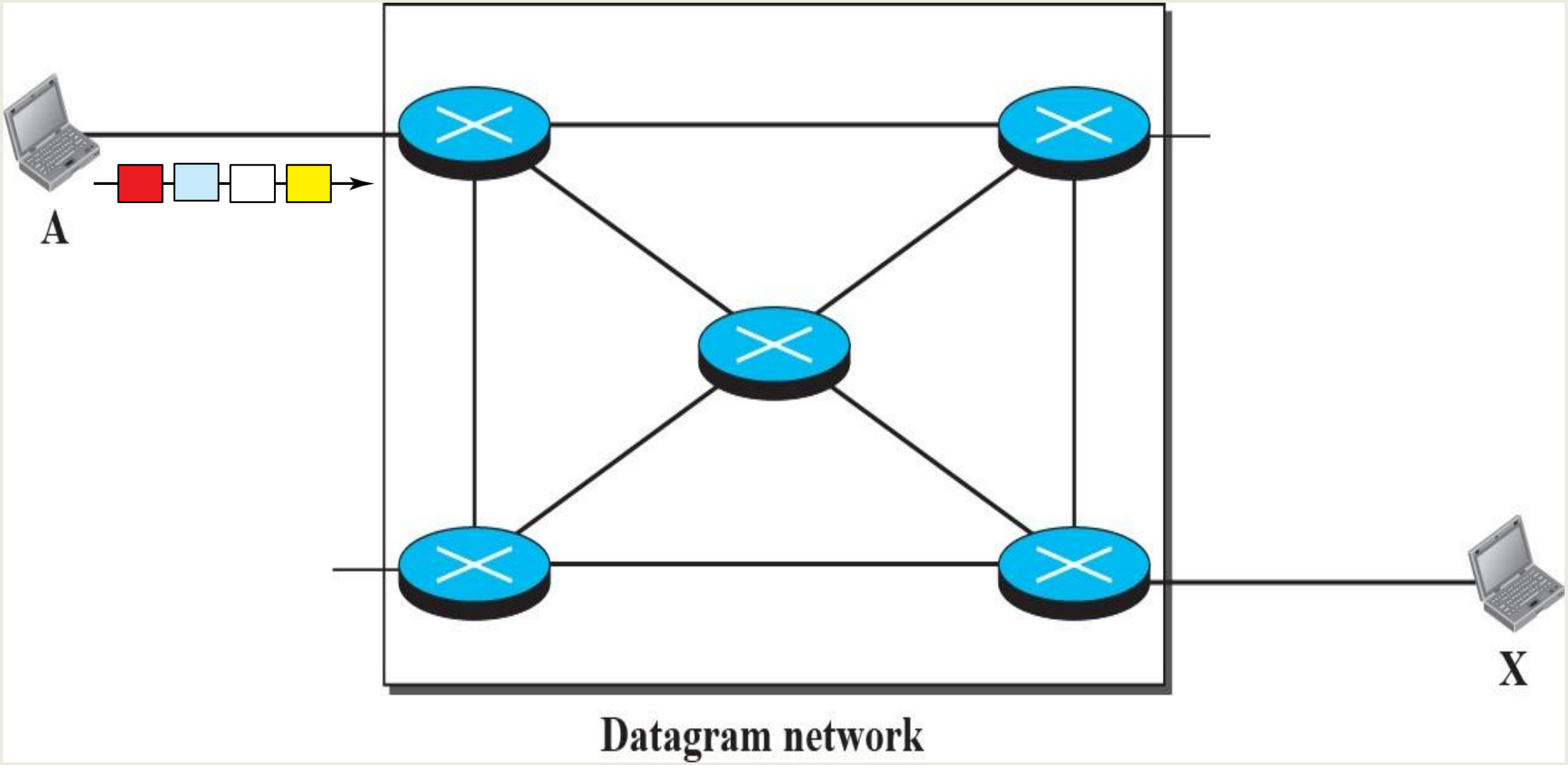
8.3 PACKET SWITCHING

- In data communications, user's need to send messages from one end system to another.
- If the message is going to pass through a packet-switched network, it needs to be divided into packets of fixed or variable size.
- The size of the packet is determined by the network and the governing protocol. No resource allocation for a packet i.e no reserved bandwidth on the links, no scheduled processing time for each packet.
- Resources are allocated on demand using first-come, first-served basis.
- When switch receives a packet, it has to wait if switch is processing other packet.Lack of reservation creates delay.

➢Datagram Networks

- In a datagram network, each packet is treated independently of all others.
- Even if a packet is part of a multi packet transmission, the network treats it as though it existed alone.
- Packets in this approach are referred to as data grams. Data gram switching is done at the Net work Layer.
- Data gram N/w's are sometimes referred to as connectionless N/W's means that the switch does not keep information about the connection state.
- No set up and tear down phase.
- Each packet is treated same by a switch regardless of its source or destination.

A Datagram network with four switches (routers)



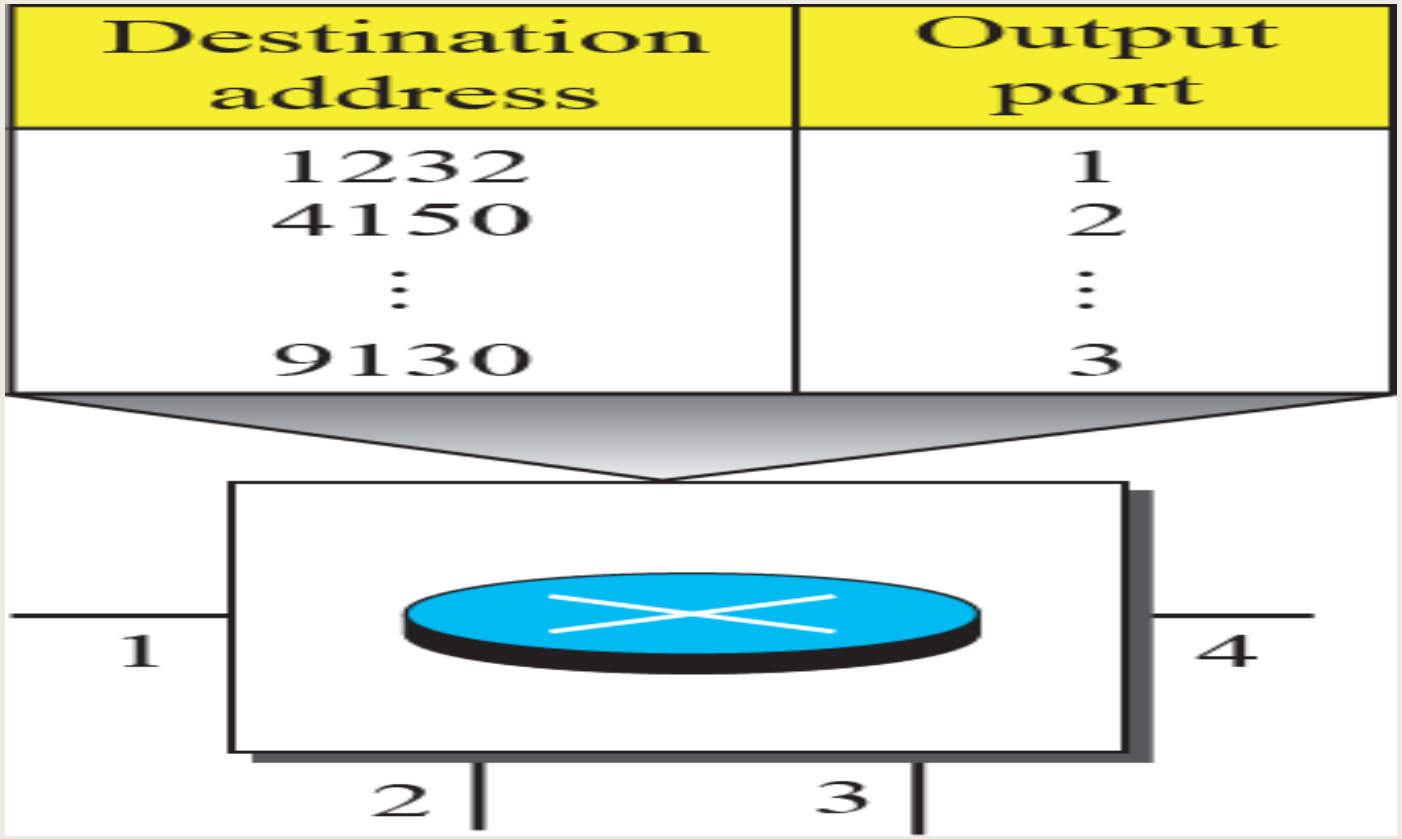
➢Routing table

- How packets are routed in datagram network?????
- Each switch has a routing table which is based on destination address. Routing tables are dynamic and updated periodically.
- The destination addresses and the corresponding forwarding o/p ports are recorded in the routing table.
- Every packet in the datagram N/W, carries the destination address along with the other header information.
- When switch receives a packet
  i)It examines the destination address.
  ii) Checks routing table to find the corresponding port through which packet should be forwarded.

Contd….

- Efficiency of datagram N/W:

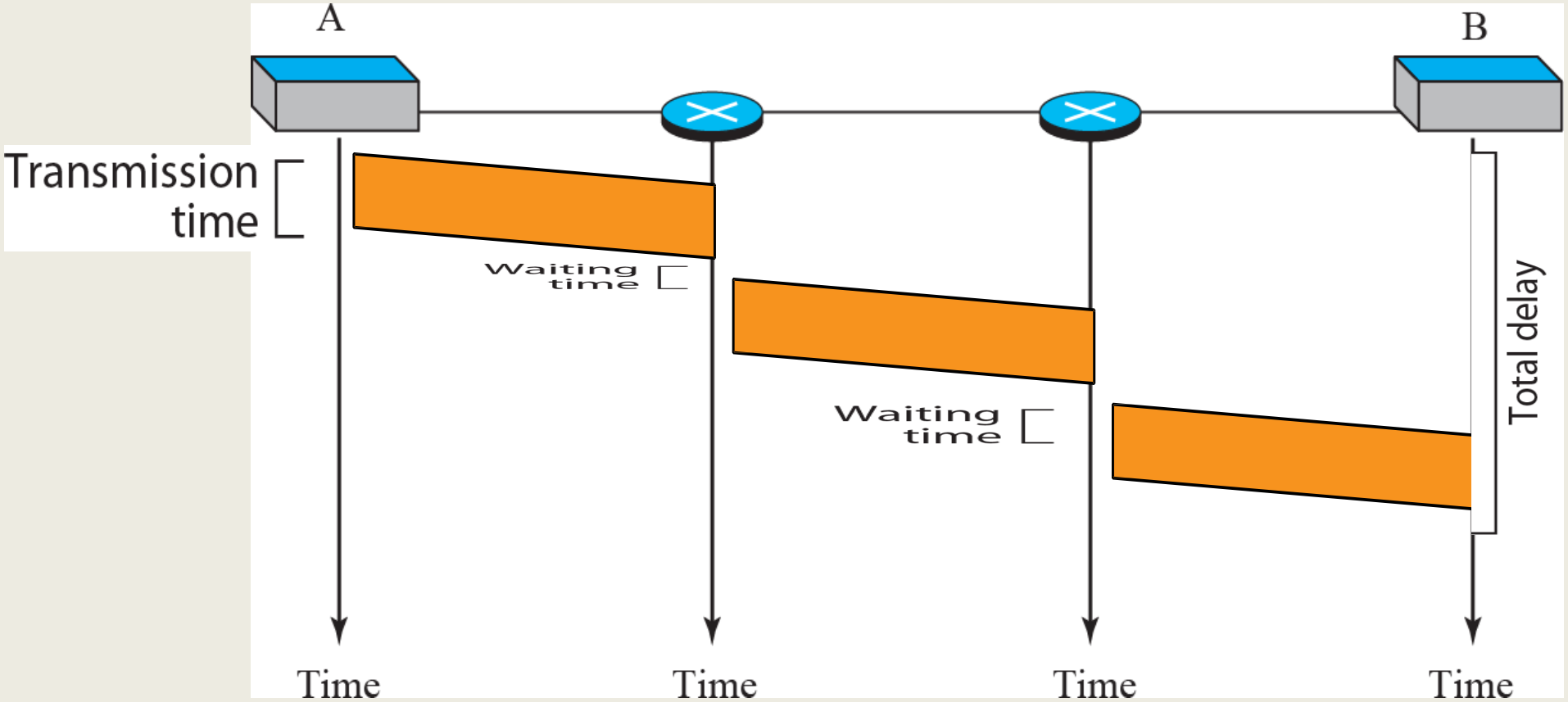
i)Resources are allocated when the packet is to be transferred.



➢Efficiency

- Datagram approach is better as resources are allocated only when there are packets to be transferred.
- If source sends packet and if there is a delay of few minutes before another packet sent, resources are reallocated from other sources.

➢Delays in a datagram network



Contd..

- Greater delay and it is not uniform for every packet in the message.
- As packet passes through 2 switches,

  i)There are three transmission times(3T)

  ii)Three propagation delays (3τ of the lines)

  iii)Two waiting times(w1+w2)

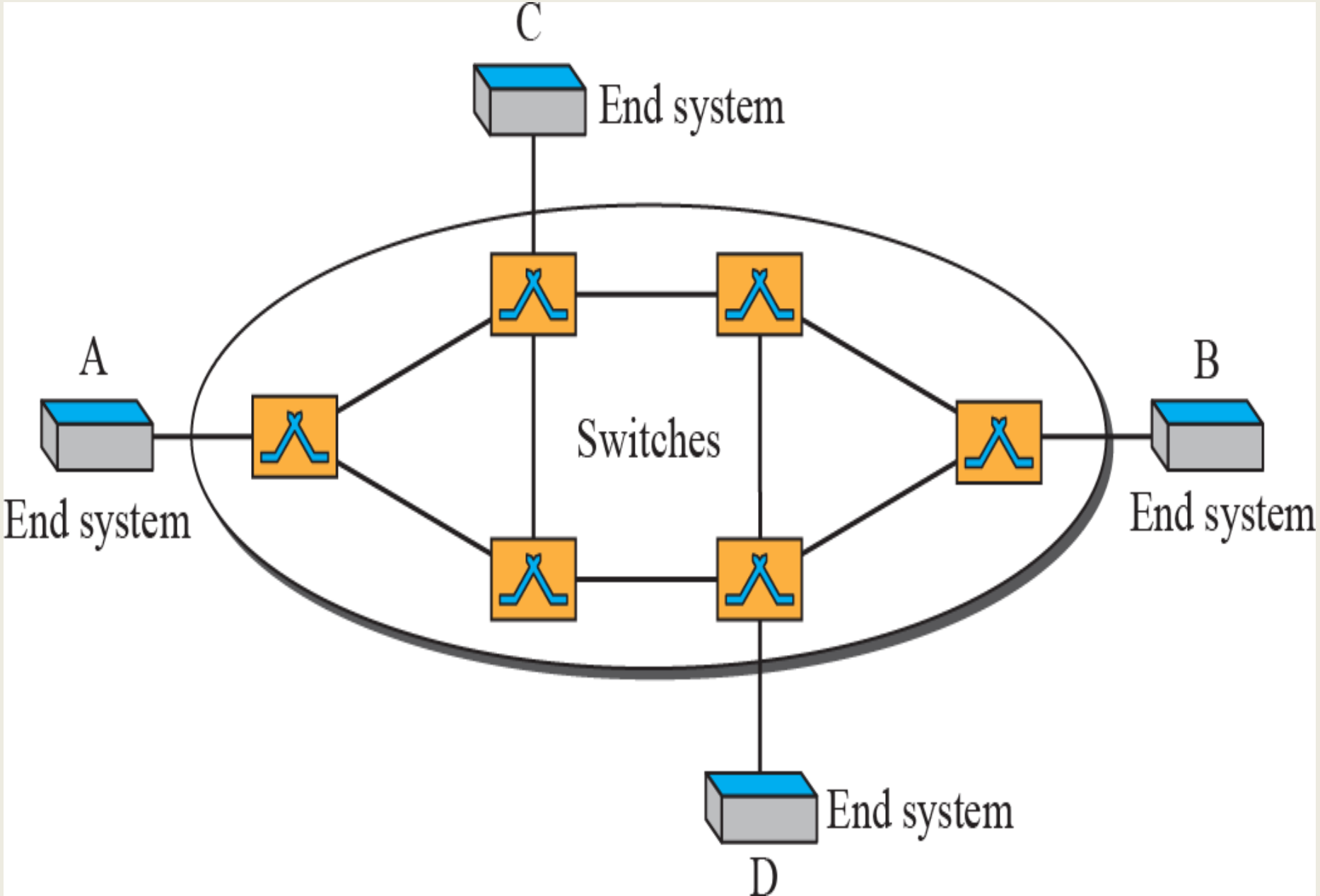
  Total delay = 3T+ 3τ+w1+w2.
- Internet has chosen Datagram approach to switch packets at the Network Layer.
- It uses universal addresses defined in the network layer to route packet from source to destination.

➢Virtual –Circuit Networks

- A virtual-circuit network is a cross between a circuit-switched network and a datagram network. It has some characteristics of both.

1. There is setup and tear down phase in addition to the data transfer phase.
2. Resources are allocated during setup phase.
3. Data are packetized and each packet carries an address in the header.
   - Address in the packet has local jurisdiction which defines which is the next switch and the channel to forward the packet.
4. All packets follow the same path established during the connection setup phase.
5. VCN is implemented in DLL, Circuit switched network in Physical layer and Datagram N/W in Network Layer

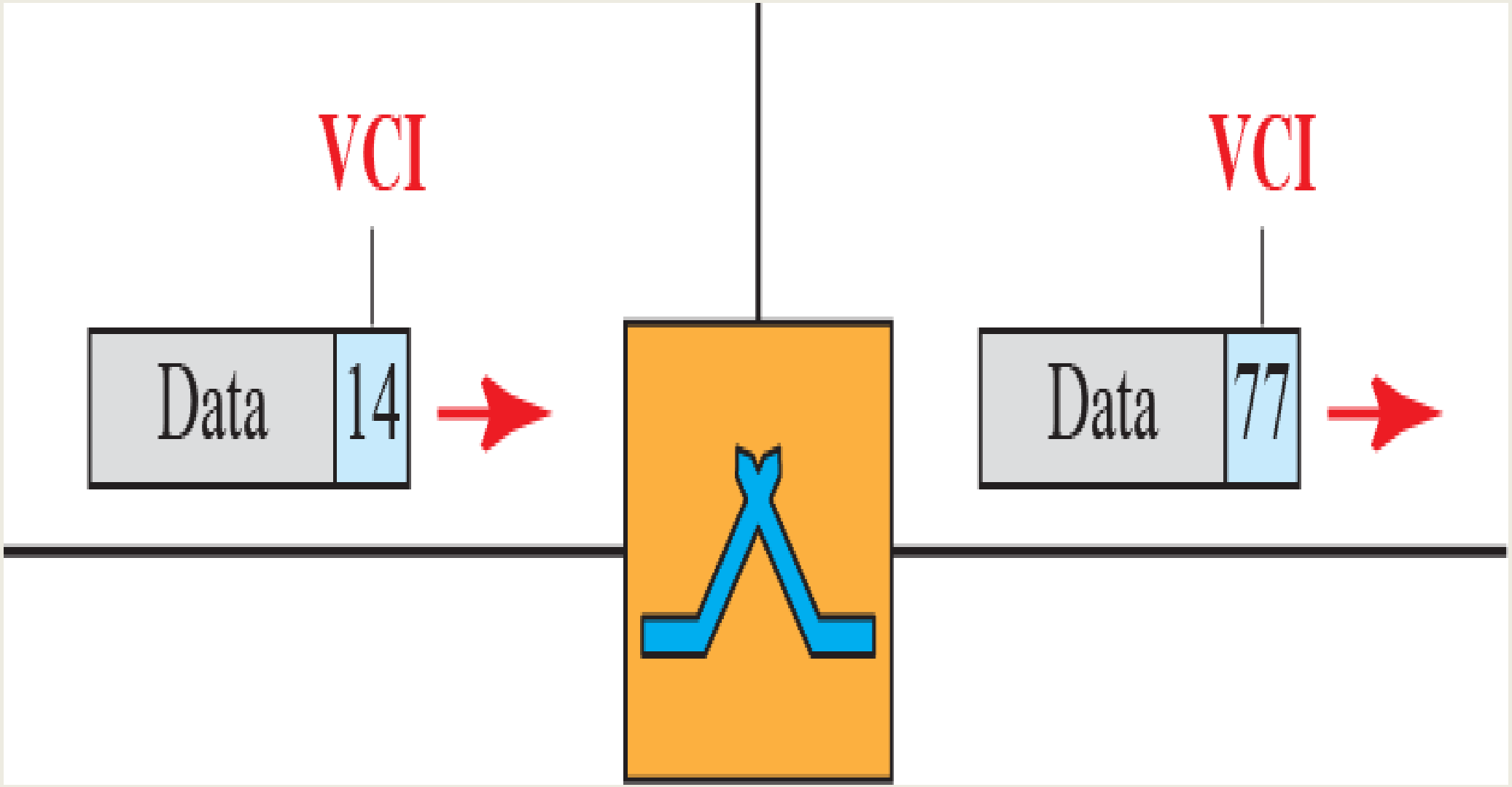
➢Virtual-circuit network



➢Addressing in Virtual circuit switching

- Two types of addresses are used: Global and Local
- Source and destination need to have global addresses i.e address which is unique through out the Internet.
- Global addresses in VC are used only to create a virtual circuit identifier.
- The identifier which is used for data transfer is called VCI.
- VCI is a small number that has only switch scope.
- It is used by a frame between two switches and changes as it moves from switch to switch.

➢Virtual-circuit identifier(VCI)



➢ Phases in VCI

There are 3 Phases

i) Setup phase:

- In setup phase, source and destination uses their global addresses to help switch to make table entries for the connection.

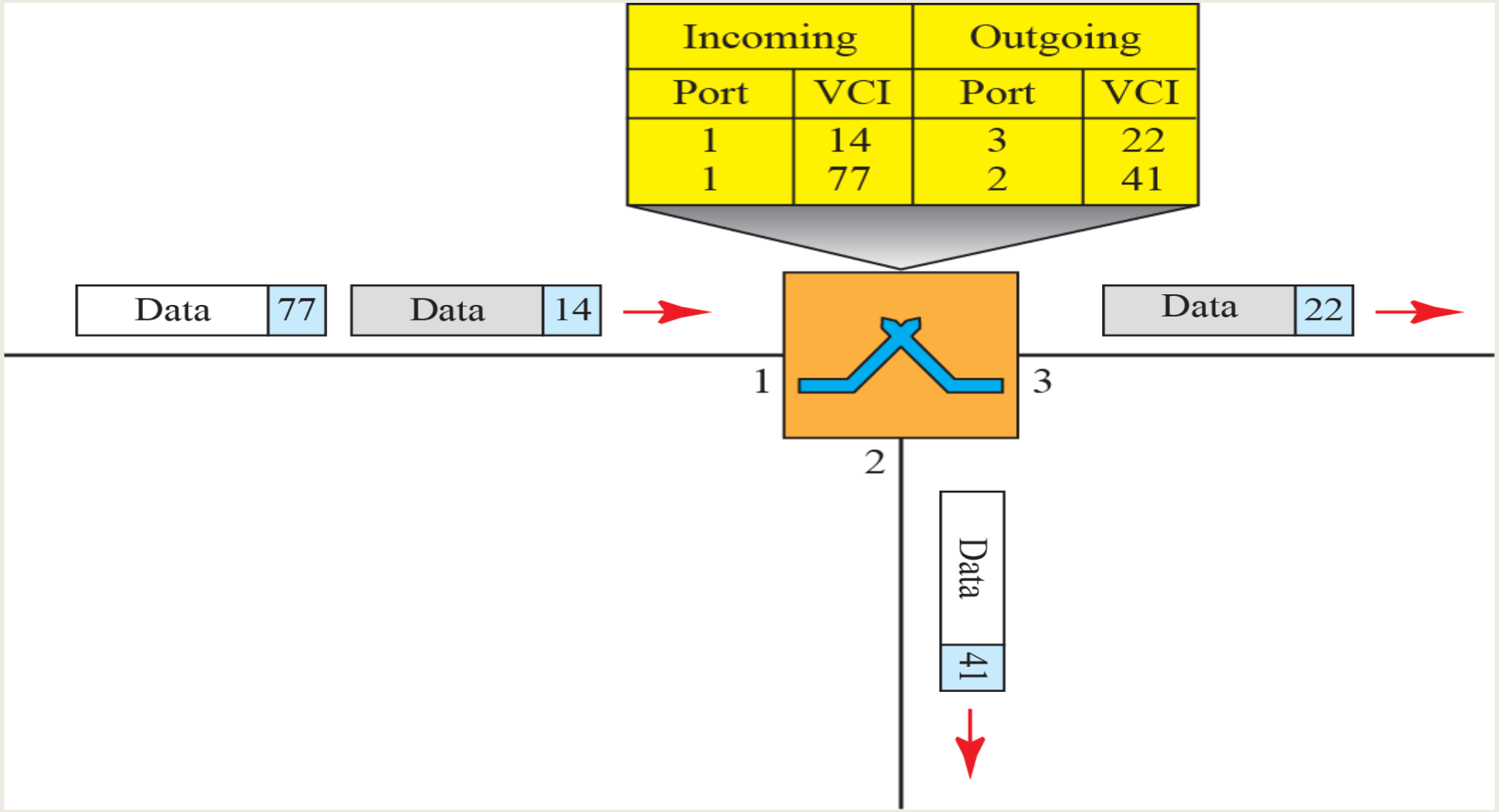
ii. Data transfer :

- In Data transfer phase, data transfer occurs

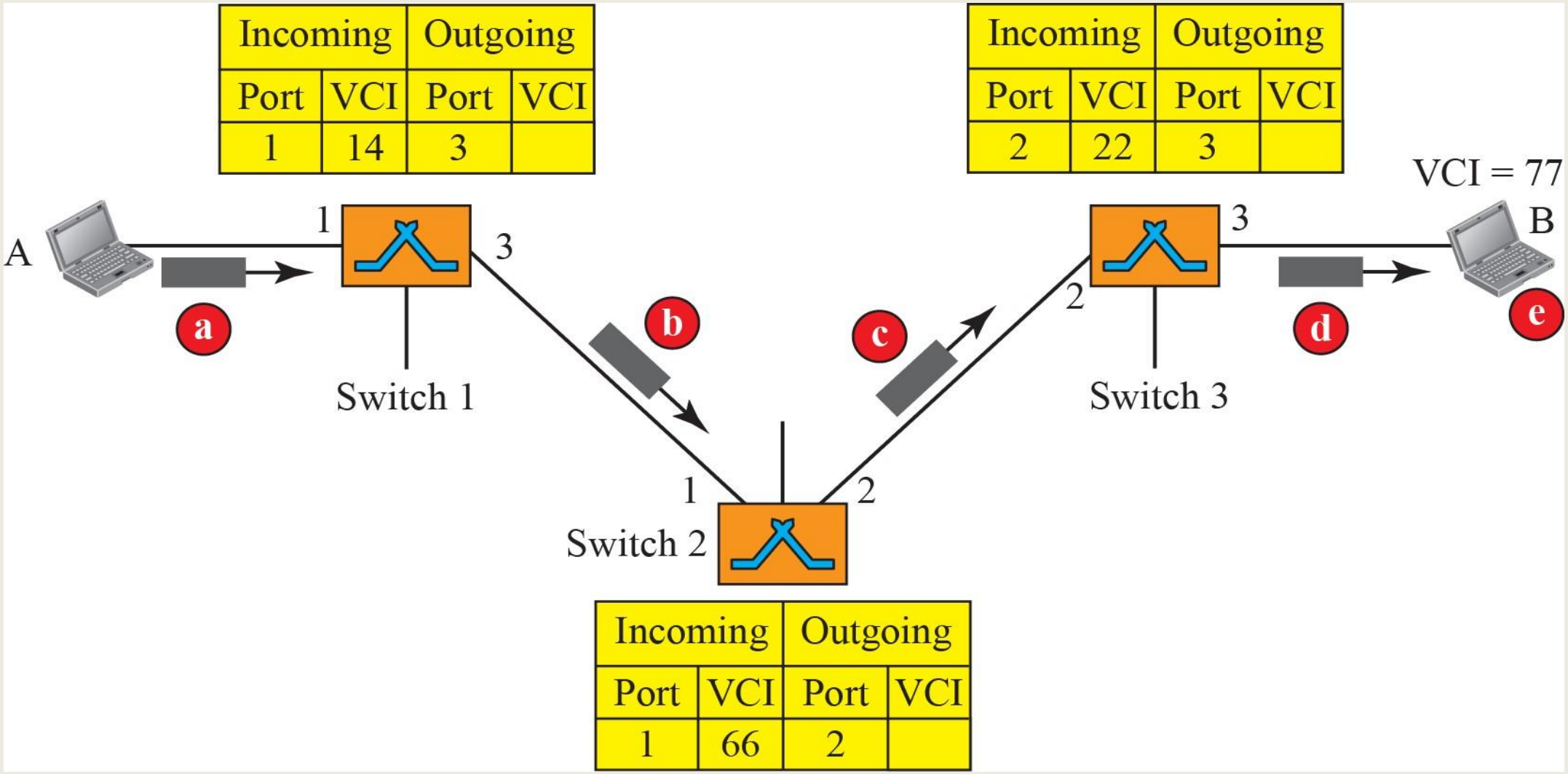
iii. Tear down:

- In tear down phase, source and destination informs switch to remove the corresponding entry from the table.

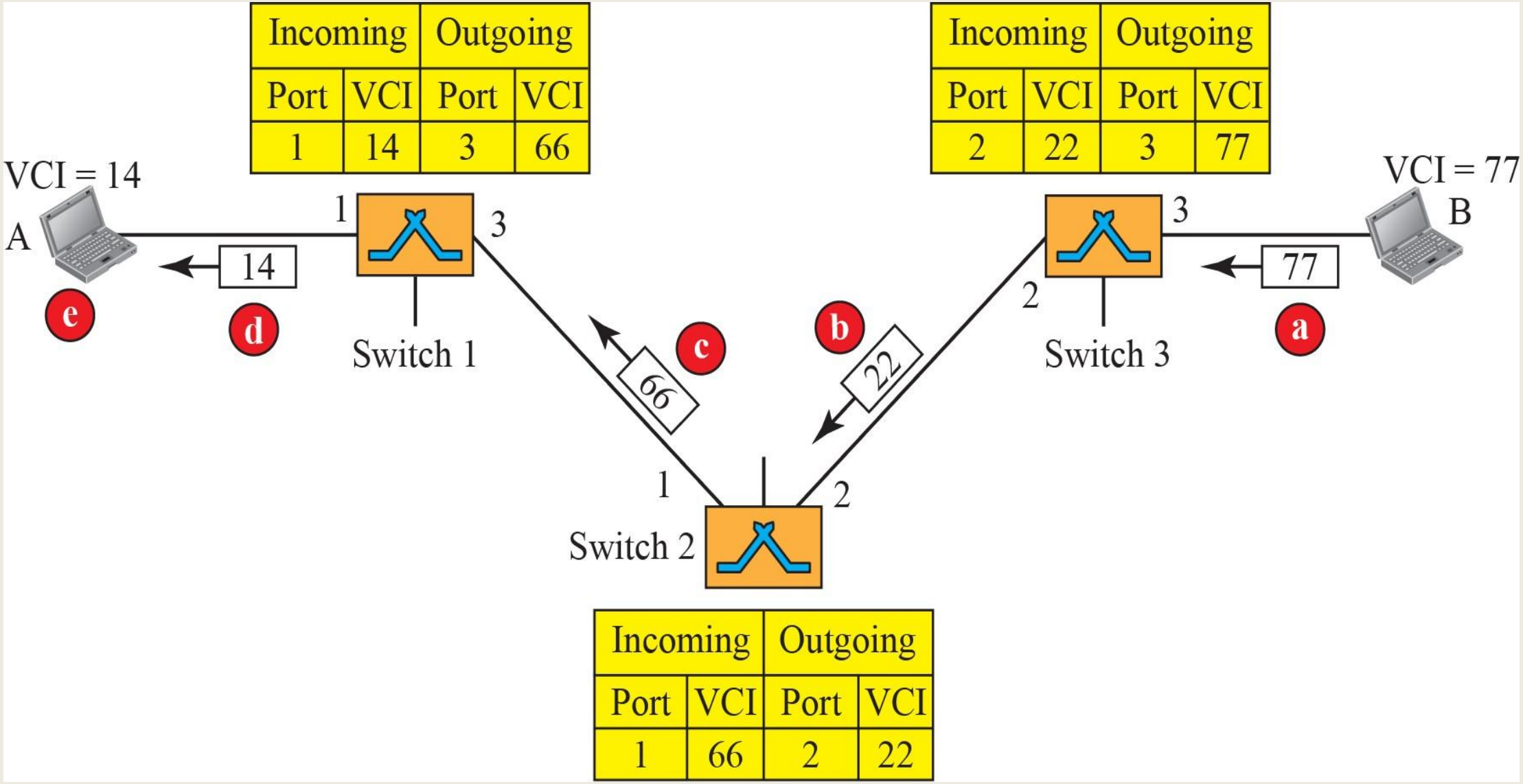
➢Switching table for a virtual-circuit network



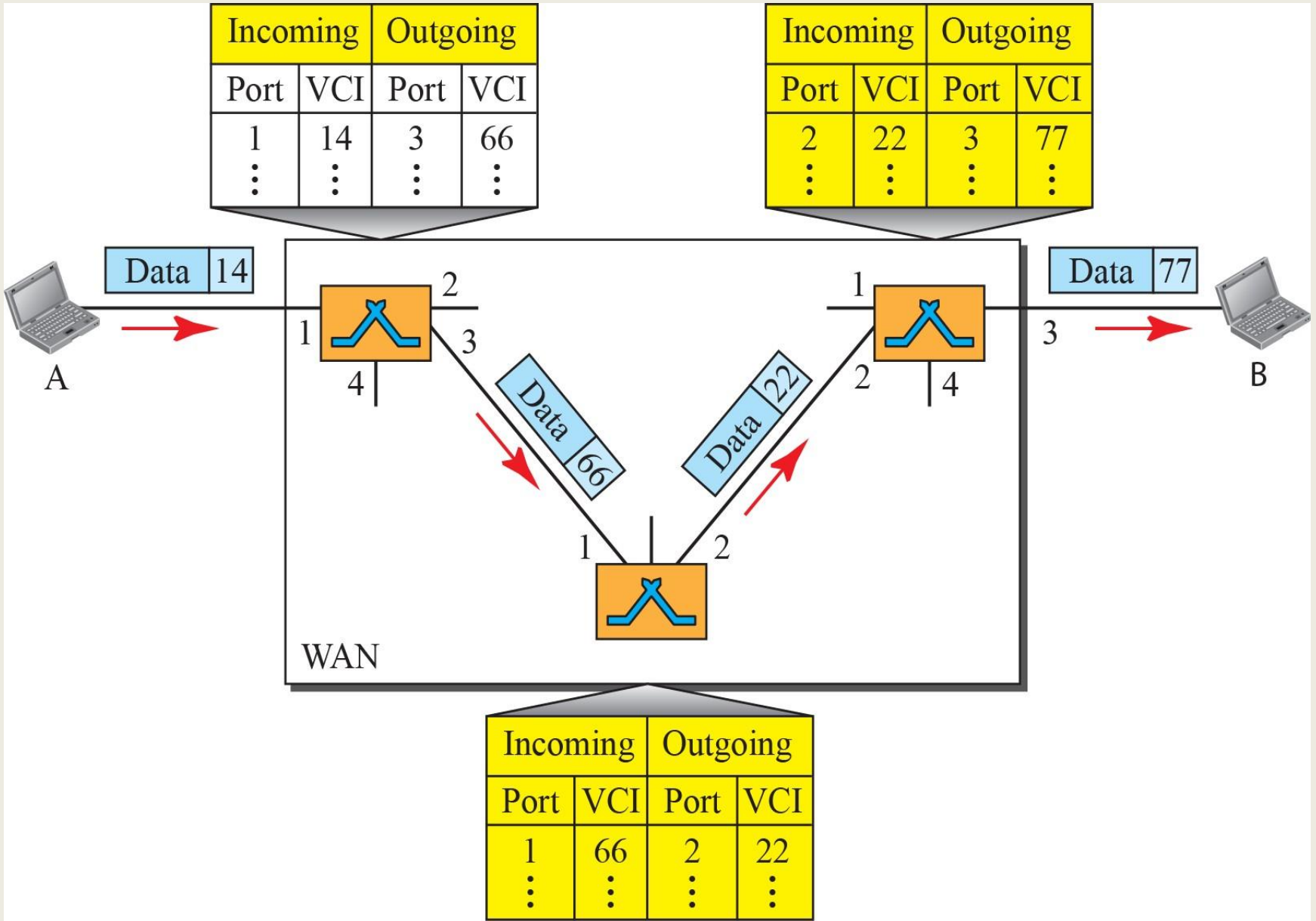
➢Setup request in a virtual-circuit network



➢Setup acknowledgment in a virtual-circuit network



➢Data transfer in a circuit-switch network Source-to-destination



➢Efficiency and Delay in VCI

- As resources are reserved in VCN during connection setup or can be on demand during data transfer phase.

i) If connection set up, delay for each packet is same.

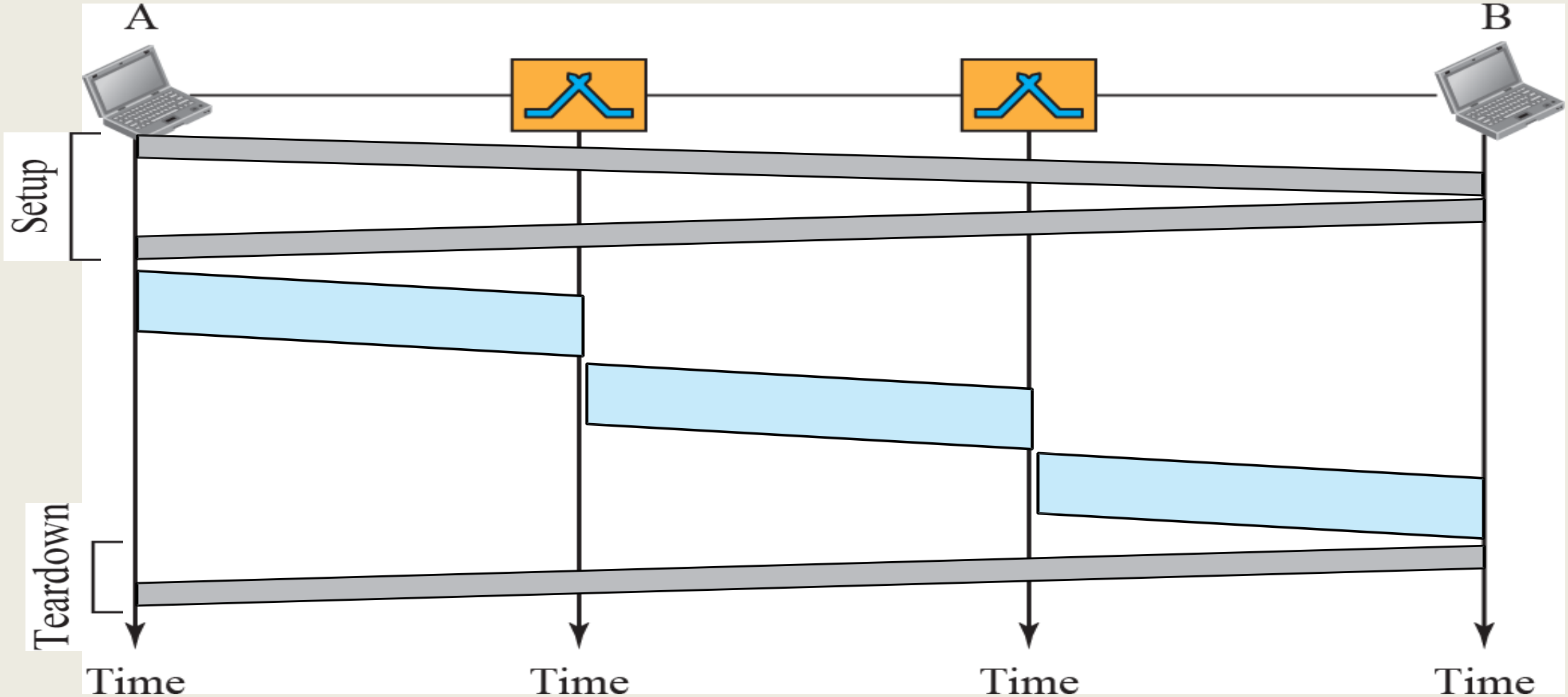
ii)During data transfer phase,each packet may encounter different delays.

➢ Delay

One time delay only for connection set up and one time delay for tear down.

- If resources allocated during set up phase, no wait time for individual packets.
- Total delay = 3T+ 3τ+setup delay and tear down delay.
- VCN's are used in switched WAN's such as frame relay and ATM N/W.

➢Delay in a virtual-circuit network



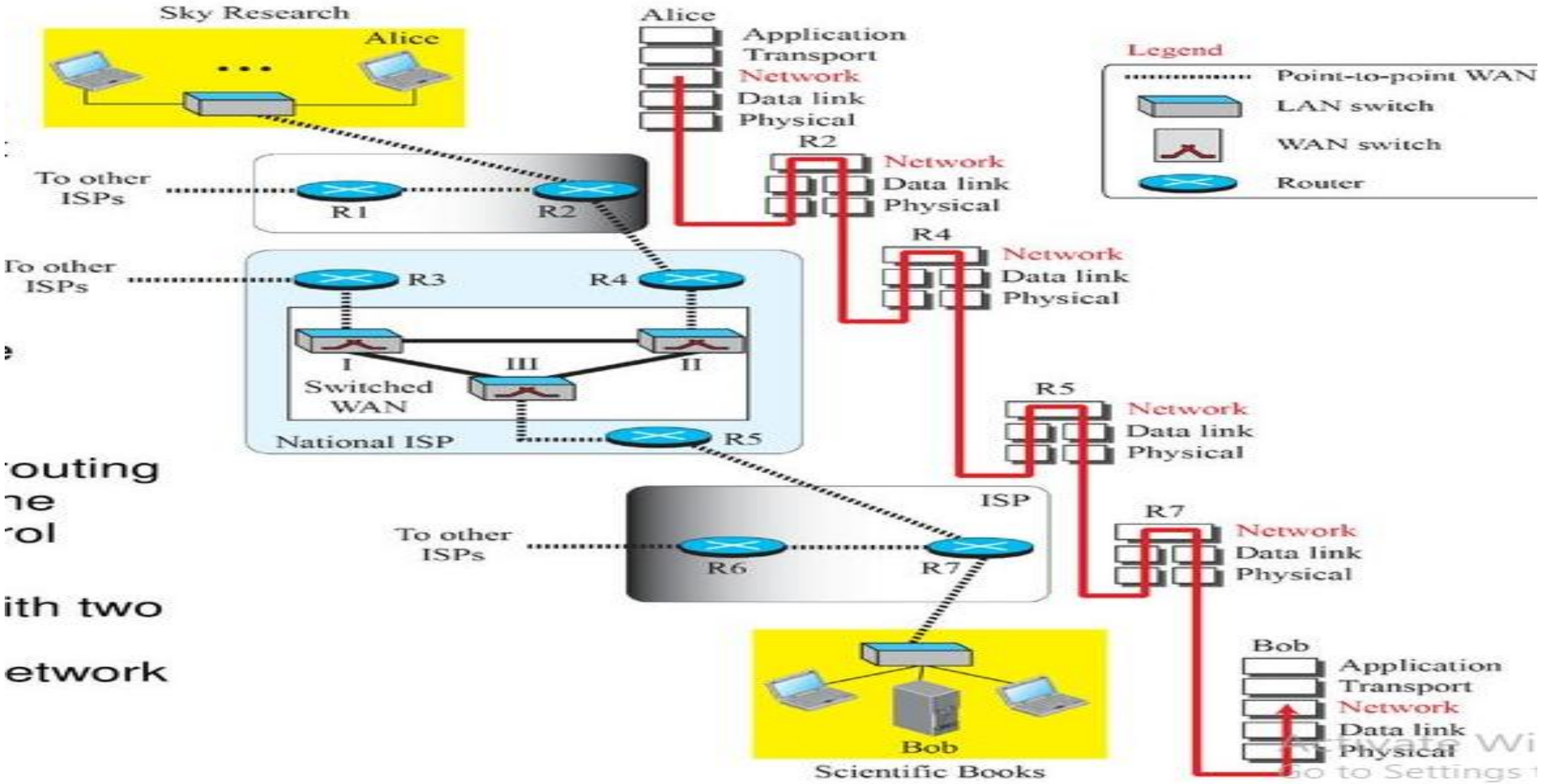
Module 2 Network Layer

Contents

1. ***Network layer Services***
2. ***Packet Switching***
3. ***IPv4 Address, IPv4 Datagram***
4. ***IPv6 Datagram***
5. ***Introduction to Routing Algorithms***
6. ***Unicast Routing Protocols: DVR, LSR, PVR,***
7. ***Unicast Routing protocols: RIP, OSPF, BGP,***

Multicasting Routing-MOSPF

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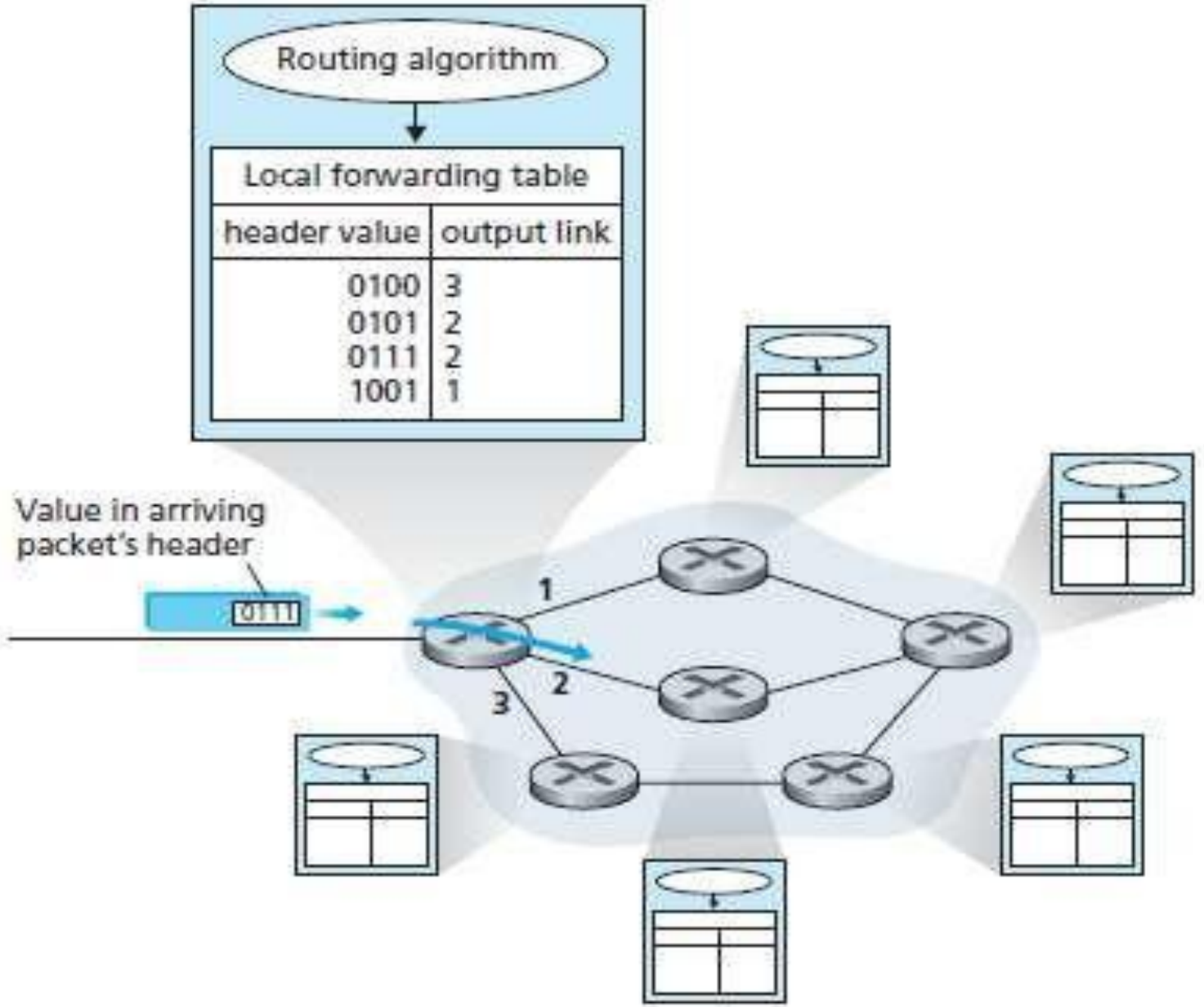
- the network layer is involved at the source host, destination host, and all routers in the path (R2, R4, R5, and R7).
- At the source host (Alice), the network layer accepts a packet from a transport layer, encapsulates the packet in a datagram, and delivers the packet to the data-link layer.
- At the destination host (Bob), the datagram is decapsulated, and the packet is extracted and delivered to the corresponding transport layer.
- The routers use three layers if they are routing packets only; however, they may need the transport and application layers for control purposes.
- A router in the path is normally shown with two data-link layers and two physical layers, because it receives a packet from one network and delivers it to another network.

Packetizing

- The first duty of the network layer is definitely packetizing:
- encapsulating the payload in a network-layer packet at the source and decapsulating the payload from the network-layer packet at the destination.
- The network layer is doing the service of a carrier such as the postal office, which is responsible for delivery of packages from a sender to a receiver without changing or using the contents

Routing and Forwarding

- **Routing** :This means that there is more than one route from the source to the destination, the network layer is responsible for finding the best one. This is done by running some routing protocols to help the routers coordinate their knowledge about the neighborhood and to come up with consistent tables to be used when a packet arrives.
- **Forwarding**:The action applied by each router when a packet arrives at one of its interfaces. A router normally use a decision-making table for applying this action. It is called (the forwarding table) or (the routing table).

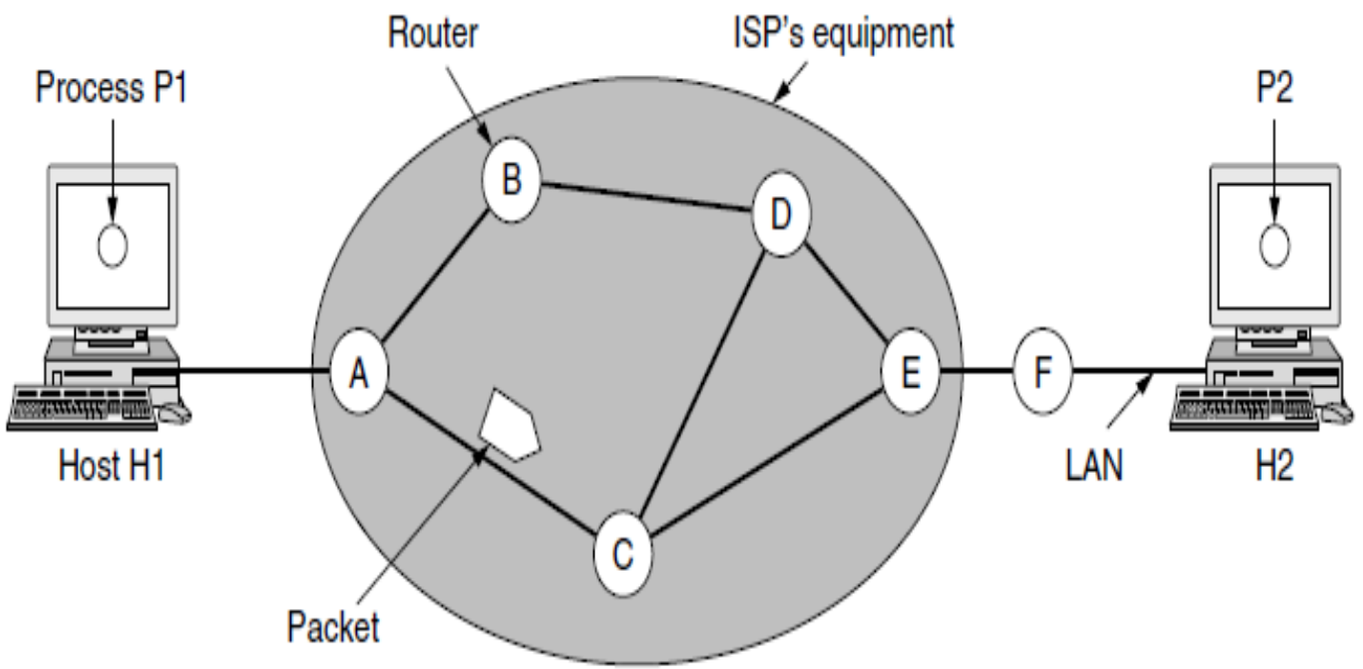

Routing algorithm
Local forwarding table
header value
output link
0100 3
0101 2
0111 2
1001 1
Value in arriving packet's header
0111
1
2
3

Other Services

- **Error Control**
- **Flow Control**
- **Congestion Control**:
- **Quality of Service**:

18.2 Packet Switching

1.Store-and-Forward Packet Switching



2.Services Provided to the Transport Layer

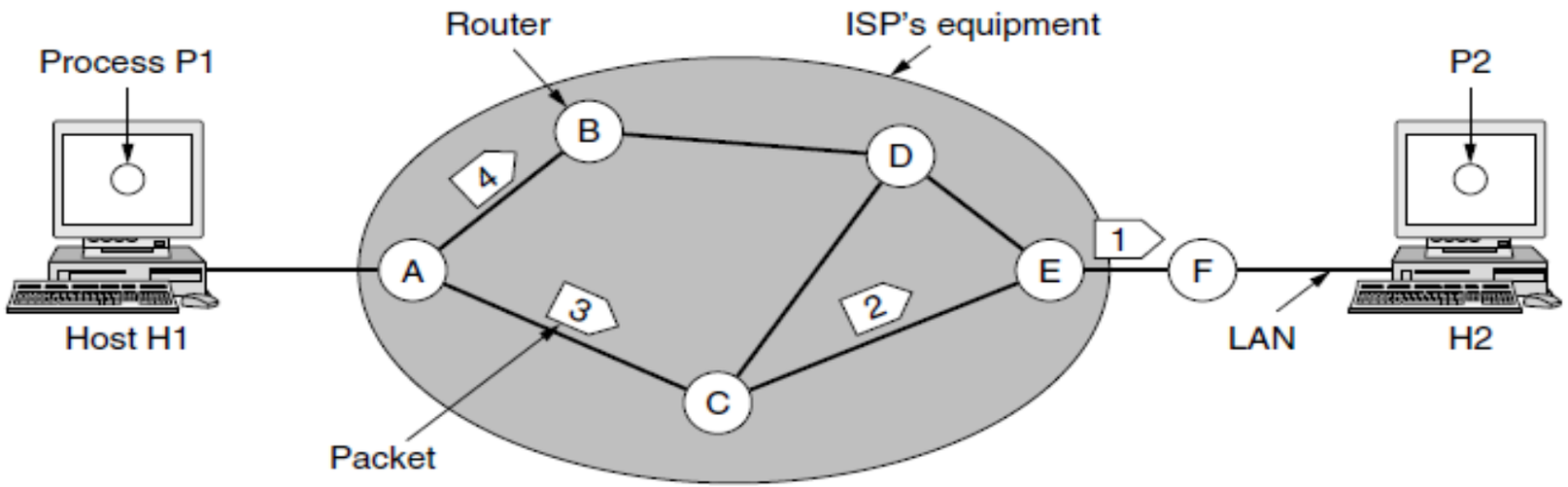
what kind of services the network layer provides to the transport layer:

1. The services should be independent of the router technology.

2. The transport layer should be shielded from the number, type, and topology of the routers present.

3. The network addresses made available to the transport layer should use a uniform numbering plan, even across LANs and WANs.

3.Implementation of Connectionless Service



A's table (initially)

Dest.	Line
A	–
B	B
C	C
D	B
E	C
F	C

A's table (later)

A	–
B	B
C	C
D	B
E	B
F	B

C's table

A	A
B	A
C	–
D	E
E	E
F	E

E's table

A	C
B	D
C	C
D	D
E	–
F	F

Fig:5.2 Routing within a datagram network.

- The transport layer code runs on H1, typically within the operating system.
- It prepends a transport header to the front of the message and hands the result to the network layer, probably just another procedure within the operating system
- Let us assume for this example that the message is four times longer than the maximum packet size, so the network layer has to break it into four packets, 1, 2, 3, and 4, and send each of them in turn to router A using some point-to-point protocol,

- Routing algorithm and IP

4.Implementation of Connection-Oriented Service

- For connection-oriented service, we need a virtual-circuit network.
- The idea behind virtual circuits is to avoid having to choose a new route for every packet sent,
- as in Fig. 5-2. Instead, when a connection is established,a route from the source machine to the destination machine is chosen as part of the connection setup and stored in tables inside the routers.
- That route is used for all traffic flowing over the connection, exactly the same way that thetelephone system works. When the connection is released, the virtual circuit is also terminated.

4.Implementation of Connection-Oriented Service

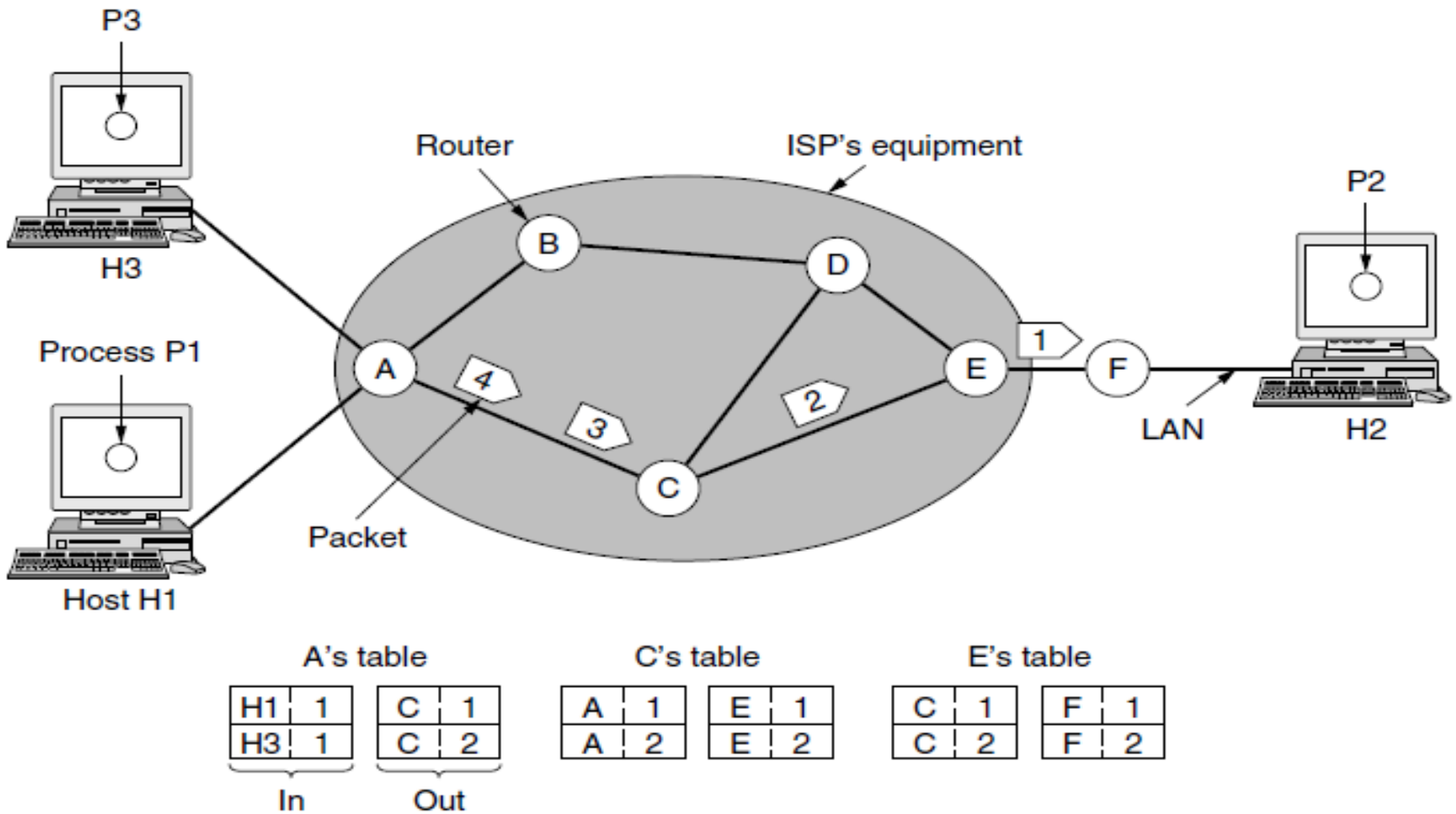


Figure 5-3. Routing within a virtual-circuit network.

- host H1 has established connection 1 with host H2. This connection is remembered as the first entry in each of the routing tables.
- The first line of A's table says that if a packet bearing connection identifier 1 comes in from H1, it is to be sent to router C and given connection identifier 1.
- Similarly, the first entry at C routes the packet to E,also with connection identifier 1.
- Now let us consider what happens if H3 also wants to establish a connectionto H2.
- It chooses connection identifier 1and tells the network to establish the virtual circuit
- This leads to the second row in the tables. Note that we have a conflict here becauseconnection 1 packets from H1 from onnection 1 packets from H3, C cannot do this.

➢For this reason, A assigns a different connection identifier to the outgoing traffic for the second connection.

➢Avoiding conflicts of this kind is why routers need the ability to replace connection identifiers in outgoing packets. this process is called **label switching**

An example of a connection-oriented network service is MPLS .

➢It is used within ISP networks in the Internet, with IP packets wrapped in an MPLS header having a 20-bit connection identifier or label

➢MPLS is often hidden from customers,

5.Comparison of Virtual-Circuit and Datagram Networks

Issue	Datagram network	Virtual-circuit network
Circuit setup	Not needed	Required
Addressing	Each packet contains the full source and destination address	Each packet contains a short VC number
State information	Routers do not hold state information about connections	Each VC requires router table space per connection
Routing	Each packet is routed independently	Route chosen when VC is set up; all packets follow it
Effect of router failures	None, except for packets lost during the crash	All VCs that passed through the failed router are terminated
Quality of service	Difficult	Easy if enough resources can be allocated in advance for each VC
Congestion control	Difficult	Easy if enough resources can be allocated in advance for each VC

Contd..

A related issue is that the destination addresses used in datagram networks are longer than circuit numbers used in virtual-circuit networks because they have a global meaning.

- If the packets tend to be fairly short, including a full destination address in every packet may represent a significant amount of overhead, and hence a waste of bandwidth
- another issue is the amount of table space required in router memory.
- A datagram network needs to have an entry for every possible destination, whereas a virtual-circuit network just needs an entry for each virtual circuit.
- However, thisa dvantage is somewhat illusory since connection setup packets

Module-2

The Network Layer

1.Network Layer Design Issues

2.Routing Algorithms

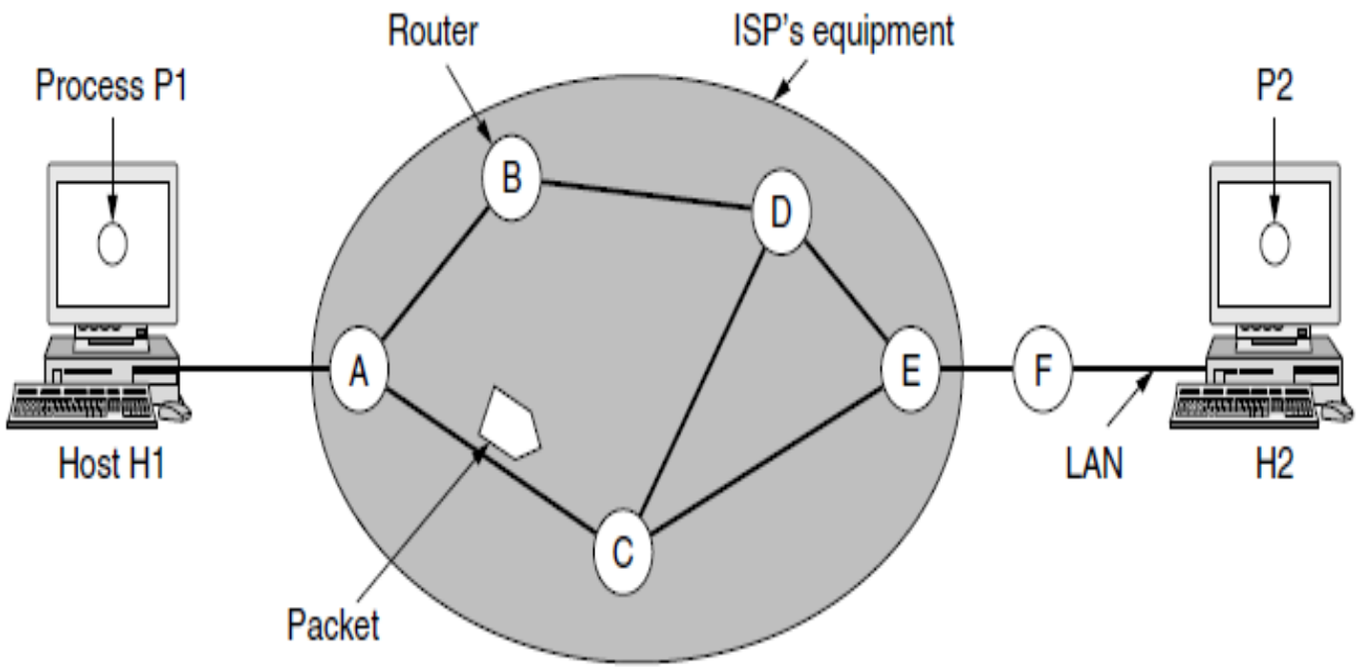
3.Congestion Control Algorithms

4.QoS.

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5.1 NETWORK LAYER DESIGN ISSUES

1.Store-and-Forward Packet Switching



2.Services Provided to the Transport Layer

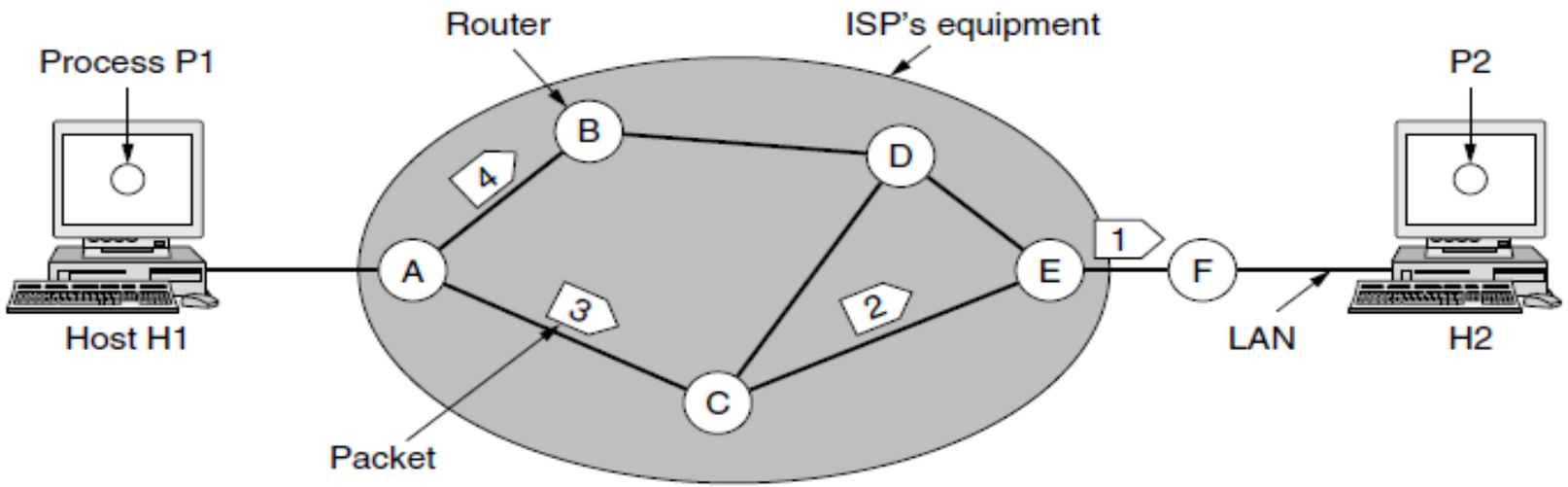
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D	B
E	C
F	C

A's table (later)

A	–
B	B
C	C
D	B
E	B
F	B

C's table

A	A
B	A
C	–
D	E
E	E
F	E

E's table

A	C
B	D
C	C
D	D
E	–
F	F

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- The transport layer code runs on H1, typically within the operating system.
- It prepends a transport header to the front of the message and hands the result to the network layer, probably just another procedure within the operating system
- Let us assume for this example that the message is four times longer than the maximum packet size, so the network layer has to break it into four packets, 1, 2, 3, and 4, and send each of them in turn to router A using some point-to-point protocol,

- Routing algorithm and IP

4.Implementation of Connection-Oriented Service

- For connection-oriented service, we need a virtual-circuit network.
- The idea behind virtual circuits is to avoid having to choose a new route for every packet sent,
- as in Fig. 5-2. Instead, when a connection is established,a route from the source machine to the destination machine is chosen as part of the connection setup and stored in tables inside the routers.
- That route is used for all traffic flowing over the connection, exactly the same way that thetelephone system works. When the connection is released, the virtual circuit is also terminated.

4.Implementation of Connection-Oriented Service

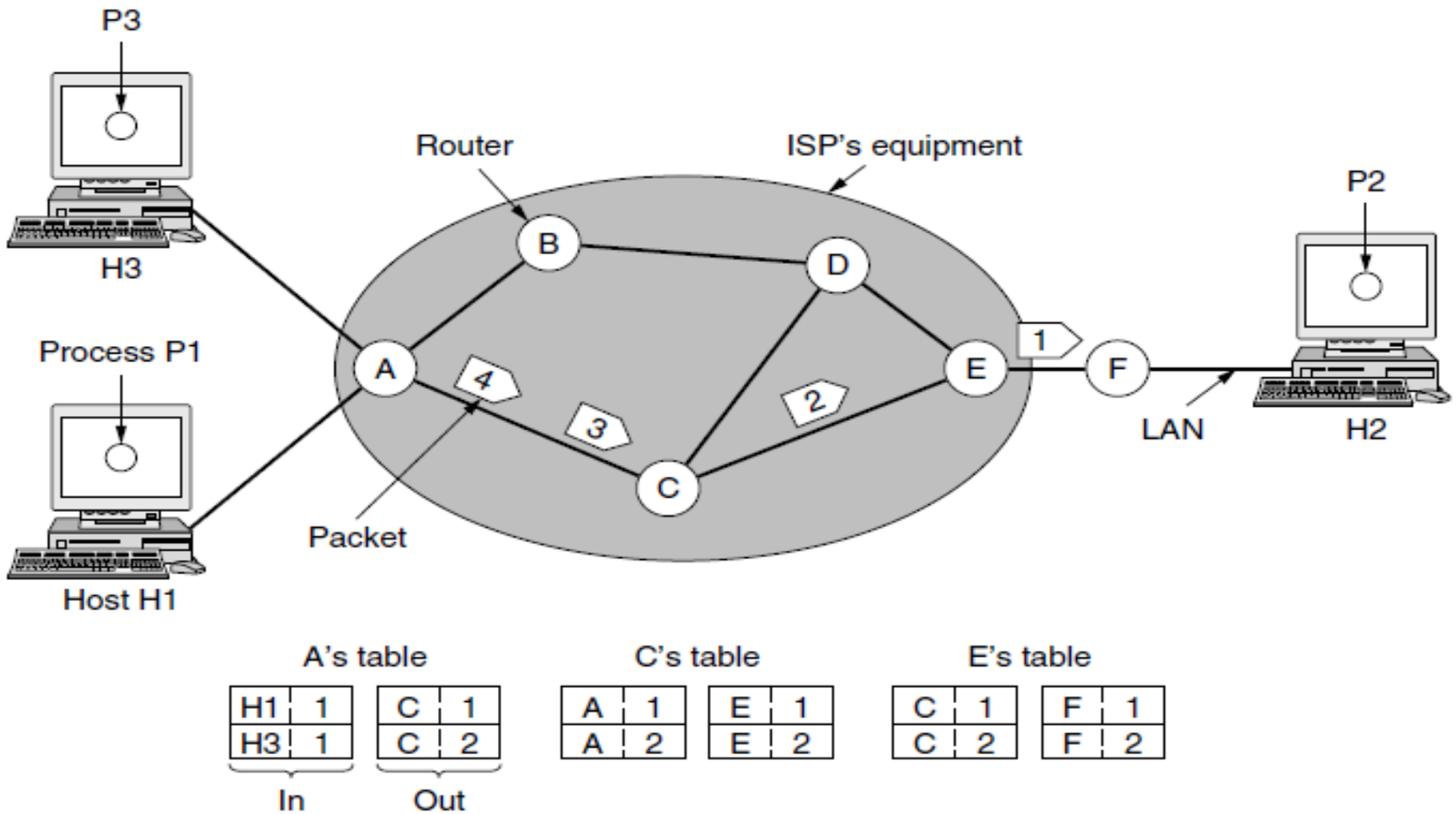


Figure 5-3. Routing within a virtual-circuit network.

- host H1 has established connection 1 with host H2. This connection is remembered as the first entry in each of the routing tables.
- The first line of A's table says that if a packet bearing connection identifier 1 comes in from H1, it is to be sent to router C and given connection identifier 1.
- Similarly, the first entry at C routes the packet to E,also with connection identifier 1.
- Now let us consider what happens if H3 also wants to establish a connectionto H2.
- It chooses connection identifier 1and tells the network to establish the virtual circuit
- This leads to the second row in the tables. Note that we have a conflict here becauseconnection 1 packets from H1 from onnection 1 packets from H3, C cannot do this.

➢For this reason, A assigns a different connection identifier to the outgoing traffic for the second connection.

➢Avoiding conflicts of this kind is why routers need the ability to replace connection identifiers in outgoing packets. this process is called **label switching**

An example of a connection-oriented network service is MPLS .

➢It is used within ISP networks in the Internet, with IP packets wrapped in an MPLS header having a 20-bit connection identifier or label

➢MPLS is often hidden from customers,

5.Comparison of Virtual-Circuit and Datagram Networks

Issue	Datagram network	Virtual-circuit network
Circuit setup	Not needed	Required
Addressing	Each packet contains the full source and destination address	Each packet contains a short VC number
State information	Routers do not hold state information about connections	Each VC requires router table space per connection
Routing	Each packet is routed independently	Route chosen when VC is set up; all packets follow it
Effect of router failures	None, except for packets lost during the crash	All VCs that passed through the failed router are terminated
Quality of service	Difficult	Easy if enough resources can be allocated in advance for each VC
Congestion control	Difficult	Easy if enough resources can be allocated in advance for each VC

Contd..

A related issue is that the destination addresses used in datagram networks are longer than circuit numbers used in virtual-circuit networks because they have a global meaning.

- If the packets tend to be fairly short, including a full destination address in every packet may represent a significant amount of overhead, and hence a waste of bandwidth
- another issue is the amount of table space required in router memory.
- A datagram network needs to have an entry for every possible destination, whereas a virtual-circuit network just needs an entry for each virtual circuit.
- However, thisa dvantage is somewhat illusory since connection setup packets

- Virtual circuits have some advantages in guaranteeing quality of service and avoiding congestion within the network because resources can be reserved in advance, when the connection is established.
- With a datagram network, congestion avoidance is more difficult.

5.2 ROUTING ALGORITHMS

- The only notable exception is for broadcast networks, but even here routing is an issue if the source and destination are not on the same network segment.
- The algorithms that choose the routes and the data structures that they use are a major area of network layer design.
- The routing algorithm is that part of the network layer software responsible for deciding which output line an incoming packet should be transmitted on.
- If the network uses virtual circuits internally, routing decisions are made only when a new virtual circuit is being set up.

Obejectives of Routing Algoithm

The main ingredients of a good routing algorithm depend on the objective function that one is trying to optimize:

1. Rapid and accurate delivery of packets.
2. Adaptability to changes in network topology resulting from node or link failures.
3. Adaptability to varying source-destination traffic loads.
4. Ability to route packets away from temporarily congested links
5. Ability to determine the connectivity of the network
6. Ability to avoid routing loops.
7. Low overhead.

Routing Algorithm Classification

1.static

2.dynamic (or adaptive).

- In **static routing** paths are precomputed based on the network topology, link capacities, and other information When the computation is completed,the paths are loaded to the routing table and remain fixed for a relatively long period of time.
- Static *routing may suffice if the network size is small, the traffic load doesnot change appreciably, or the network topology is relatively fixed*

Static routing may become cumbersome as the network size increases.

- The biggest disadvantage of static routing is its inability to react rapidly to network failures

- In **dynamic (adaptive)** routing each node continuously learns the state of the network by communicating with its neighbors.
- Thus a change in a network topology is eventually propagated to all nodes.
- Based on the information collected, each node can compute the best paths to desired destinations
- One **disadvantage** of dynamic routing is the added complexity in the node.

network control center computes all paths and then uploads this information to the nodes in the network.

- In distributed routing nodes cooperate by means of message exchanges and perform their own routing computations
- Distributed routing algorithms generally scale better than centralized algorithms but are more likely to produce inconsistent results.
- If the paths calculated by different nodes are consistent, loops can develop.

➢The latter case is sometimes called **session routing** because a route remains in force for an entire session.

➢Stability is also an important goal for the routing algorithm. There exist routing algorithms that **never converge to a fixed set of paths**, no matter how long they run.

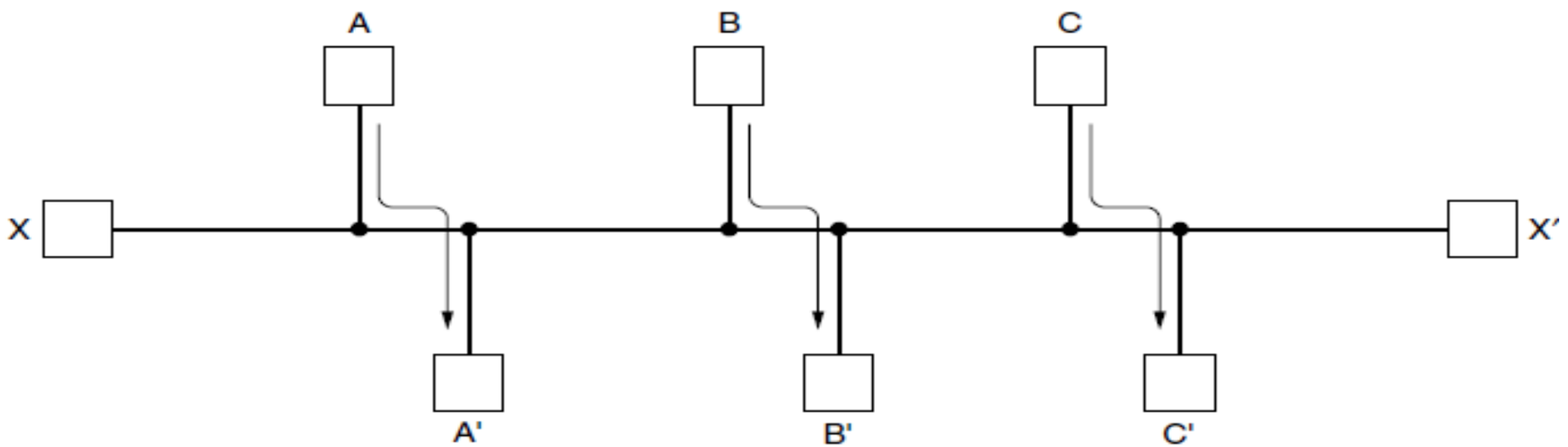


Fig:Network with a conflict between fairness and efficiency

The Optimality Principle

- It states that if router *J is on the optimal path from router I to router K,*then the optimal path from *J to K also falls along the same route.*
- *To see this, all* the part of the route from *I to J* ***r1*** *and the rest of the route* ***r2***
- If a route better than ***r2*** *existed from J to K, it could be concatenated with* ***r1*** *to improve the route from I to K, contradicting our statement that* ***r1 r2*** *is optimal.*

SHORTEST-PATH ROUTING

To better understand the purpose of these algorithms, consider a communication network as a graph consisting of a set of *nodes (or vertices) and a set of links (or edges), where each node represents a* packet switch and each link represents a transmission line between two packet switches.

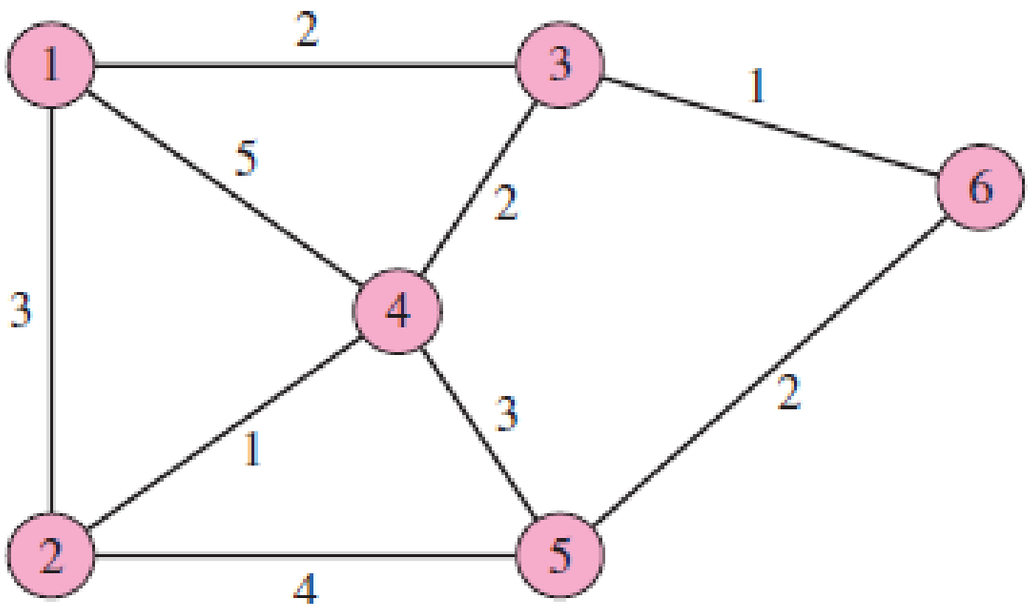


Fig:A network with associated link costs.packet

For example, ***the shortest path from node 2 to node 6 is 2-4-3-6, and the path cost is 4.***

Many metrics can be used to assign a cost to each link, depending on the objective function that is to be optimized.

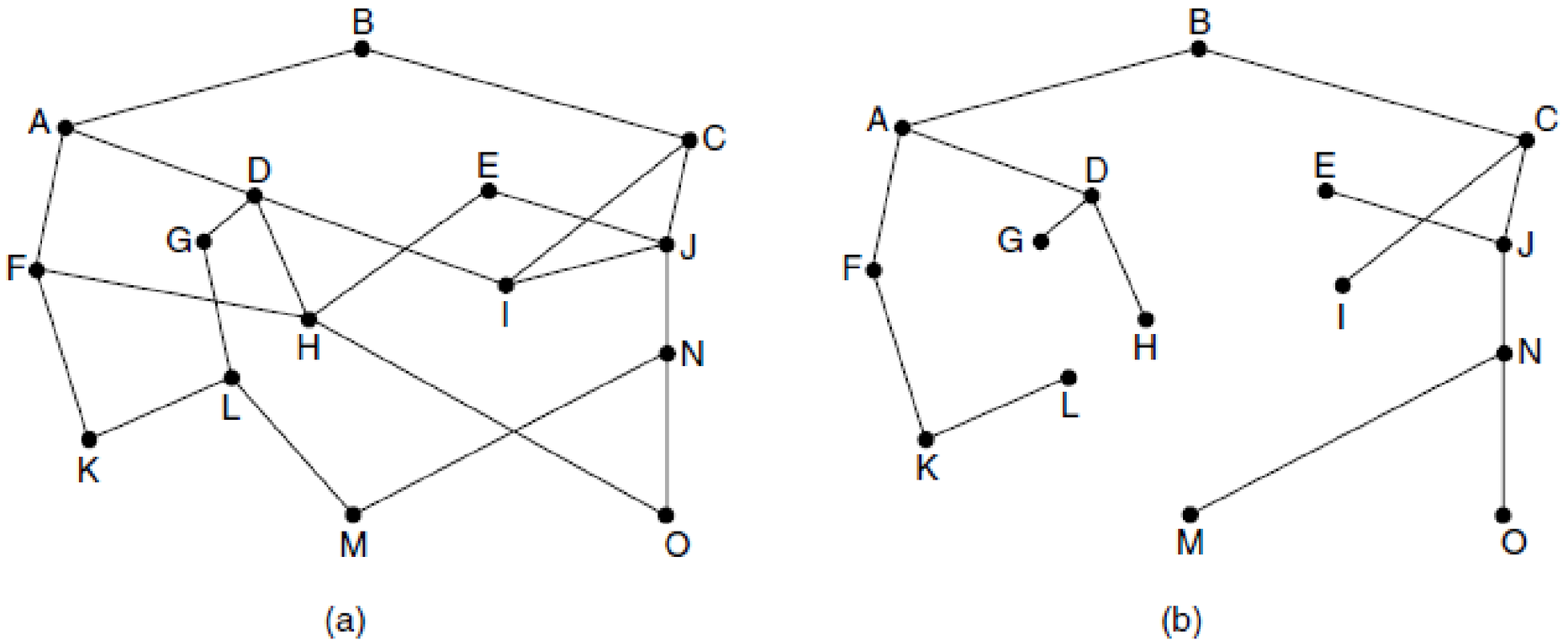
1. *Cost ~ 1/capacity.*
2. *Cost ~ packet delay.*
3. *Cost ~ congestion.*

Cost ~ 1/capacity The cost is inversely proportional to the link capacity. Here one assigns higher costs to lower-capacity links. The objective is to send a packet through a path with the highest capacity. If each link has equal capacity, then the shortest path is the path with the minimum number of hops

Cost ~ packet delay. The cost is proportional to an average packet delay, which includes queuing delay in the switch buffer and propagation delay in the link. The shortest path represents the fastest path to reach the destination.

Cost ~ congestion. The cost is proportional to some congestion measure, for example,traffic loading. Thus the shortest path tries to avoid congested links.

we can see that the set of optimal routes from all sources to a given destination form a tree rooted at the destination. Such a tree is called a **sink tree** and is illustrated in fig (b) A sink tree for router B



If we allow all of the possible paths to be chosen, the tree becomes a more general structure called a DAG (Directed Acyclic Graph).

Distance Vector Routing

- Computer networks generally use dynamic routing algorithms that are more complex than flooding, but more efficient because they find shortest paths for the current topology.
- Two dynamic algorithms in particular, **distance vector routing** and **link state routing**, are the most popular.
- **distance vector routing algorithm operates by having each router maintain** a table (i.e., a vector) giving the best known distance to each destination and which link to use to get there.
- These tables are updated by exchanging information with the neighbors.

The Bellman-Ford Algorithm

1. Initialization (destination node *d is distance 0 from itself)*

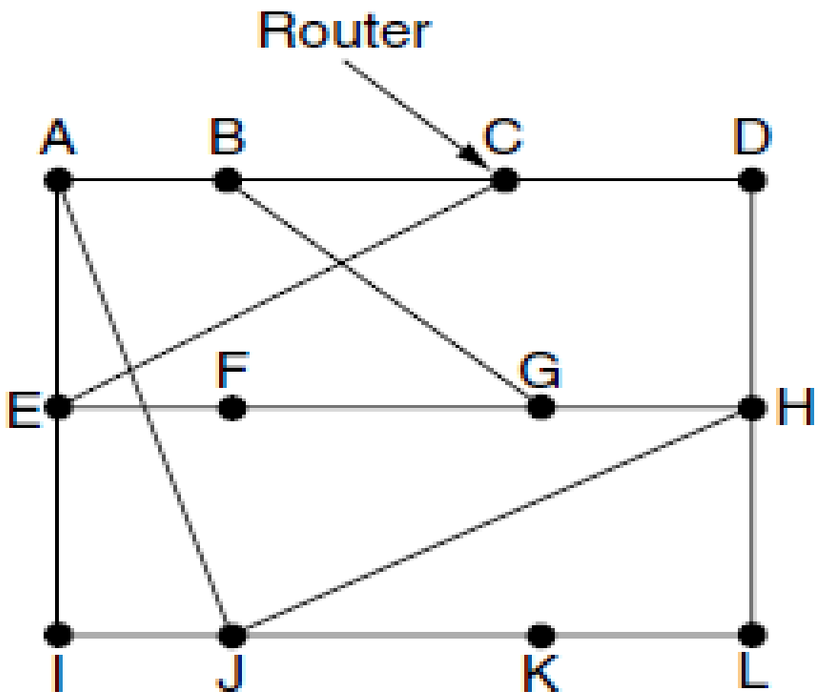
$$D_i = \infty, \quad \text{for all } i \neq d$$
$$D_d = 0$$

2. Updating (find minimum distance to destination through neighbors): For each *i = d,*

$$D_i = \min_j \{C_{ij} + D_j\}, \quad \text{for all } j \neq i$$

Repeat step 2 until no more changes occur in the iteration.

Example

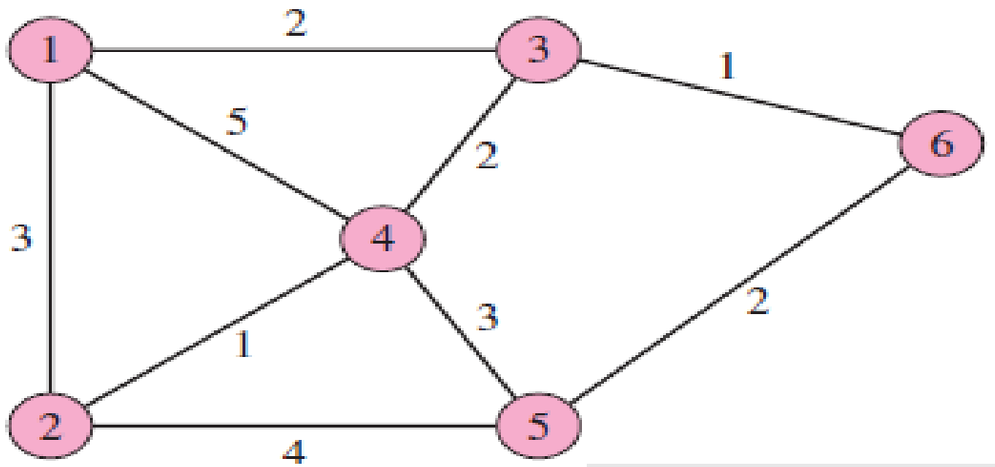


To	A	I	H	K	New estimated delay from J	Line
A	0	24	20	21	8	A
B	12	36	31	28	20	A
C	25	18	19	36	28	I
D	40	27	8	24	20	H
E	14	7	30	22	17	I
F	23	20	19	40	30	I
G	18	31	6	31	18	H
H	17	20	0	19	12	H
I	21	0	14	22	10	I
J	9	11	7	10	0	—
K	24	22	22	0	6	K
L	29	33	9	9	15	K
	JA delay is 8	JI delay is 10	JH delay is 12	JK delay is 6	New routing table for J	

Vectors received from J's four neighbors

(b)

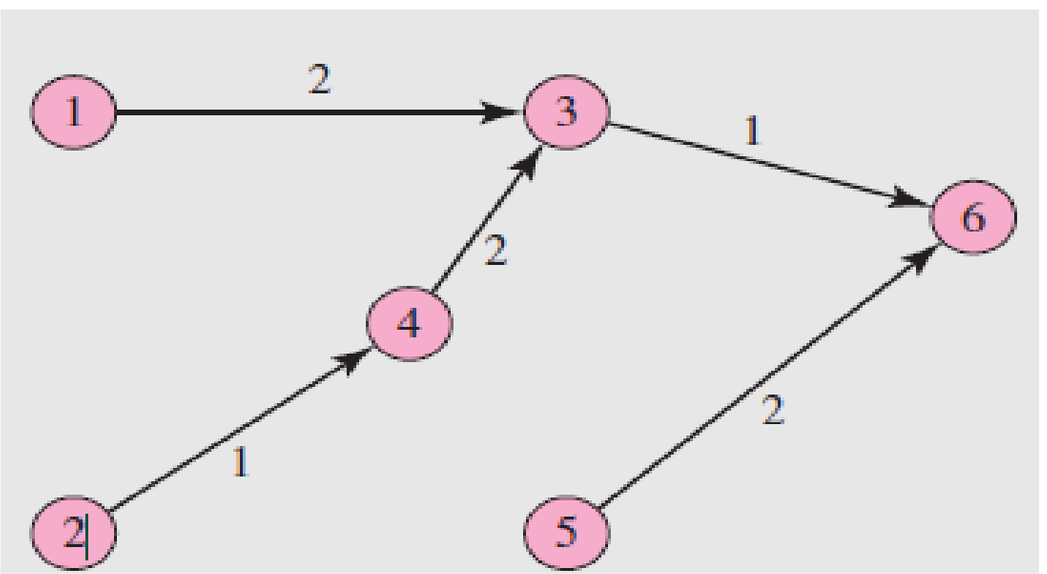
Ex:Apply Bellmanford algorithm to find the shortest path from node 2 to node 6 in given Network



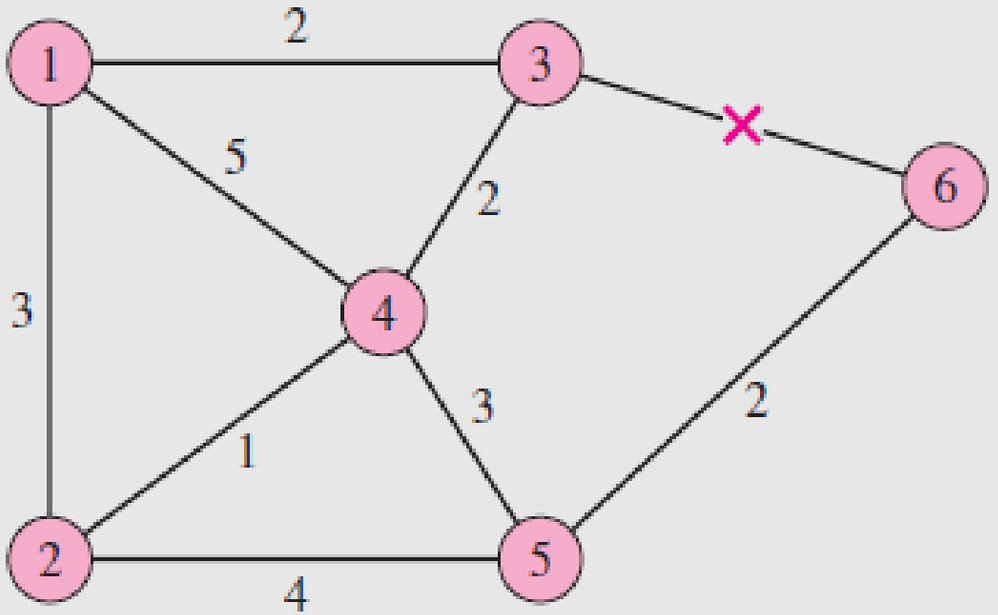
Iteration	Node 1	Node 2	Node 3	Node 4	Node 5
Initial	$(-1, \infty)$	$(-1, \infty)$	$(-1, \infty)$	$(-1, \infty)$	$(-1, \infty)$
1	$(-1, \infty)$	$(-1, \infty)$	(6, 1)	$(-1, \infty)$	(6, 2)
2	(3, 3)	(5, 6)	(6, 1)	(3, 3)	(6, 2)
3	(3, 3)	(4, 4)	(6, 1)	(3, 3)	(6, 2)
4	(3, 3)	(4, 4)	(6, 1)	(3, 3)	(6, 2)

Shortest-Path Tree

- From the preceding example, draw the shortest path from each node to the destination node. from the last row of Table, we see the next node of node 1 is node 3,
- the next node of node 2 is node 4, the next node of node 3 is node 6, and so forth. shows the shortest-path tree rooted to node 6.



- Suppose that after the distributed algorithm stabilizes for the network shown in Figure, the link connecting node 3 and node 6 breaks.
- Compute the minimum cost from each node to the destination node (node 6), assuming that each node immediately recompute its cost after detecting changes and broadcasts its routing updates to its neighbors. The new network topology is shown in Figure.

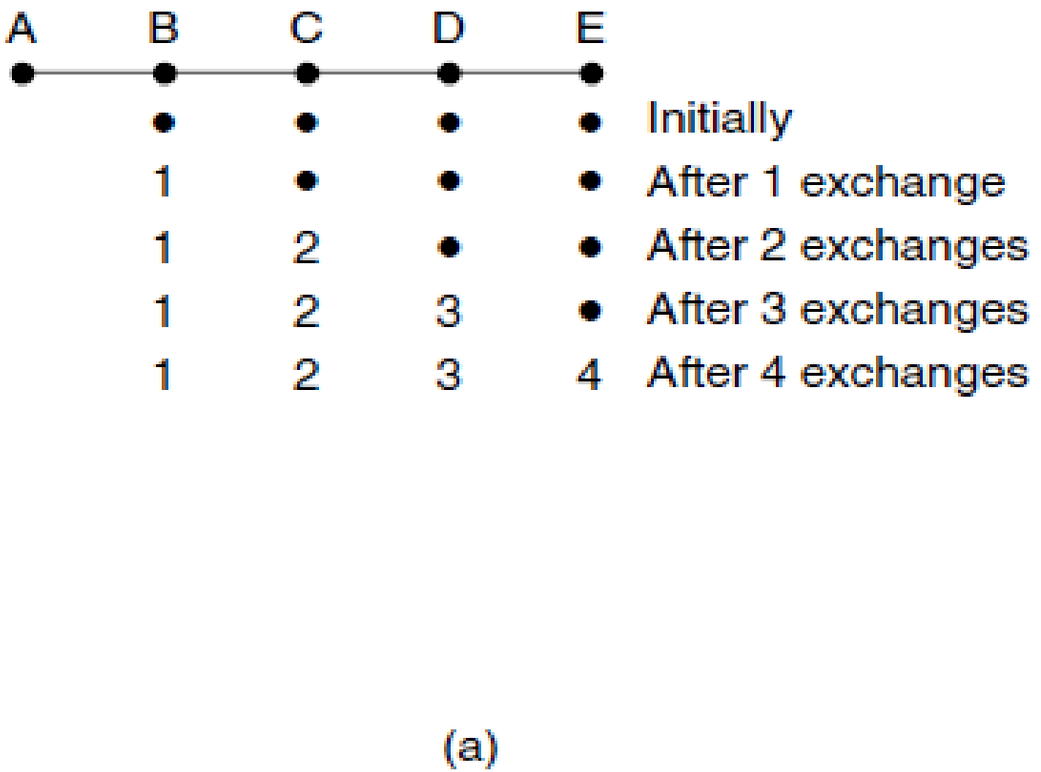


Update	Node 1	Node 2	Node 3	Node 4	Node 5
Before break	(3, 3)	(4, 4)	(6, 1)	(3, 3)	(6, 2)
1	(3, 3)	(4, 4)	(4, 5)	(3, 3)	(6, 2)
2	(3, 7)	(4, 4)	(4, 5)	(2, 5)	(6, 2)
3	(3, 7)	(4, 6)	(4, 7)	(2, 5)	(6, 2)
4	(2, 9)	(4, 6)	(4, 7)	(5, 5)	(6, 2)
5	(2, 9)	(4, 6)	(4, 7)	(5, 5)	(6, 2)

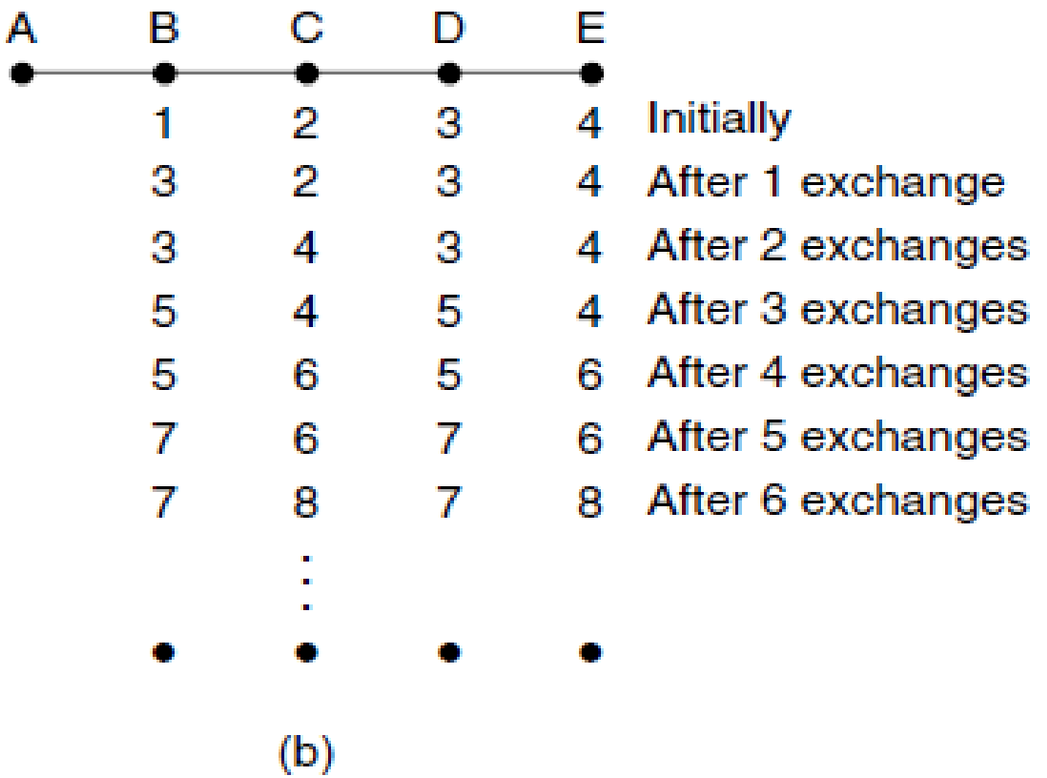
The Count-to-Infinity Problem

- The settling of routes to best paths across the network is called **convergence** Distance vector routing is useful as a simple technique by which routers can collectively compute shortest paths, but it has a serious drawback in practice
- In particular, it reacts rapidly to good news, but leisurely to bad news. Consider a router whose best route to destination X is long. If, on the next exchange,
- neighbor A suddenly reports a short delay to X, the router just switches over to using the line to A to send traffic to X. In one vector exchange, the good news is processed.

Example :count to infinity



(a)



(b)

- When *A comes up, the other routers learn about it via the vector exchanges.*

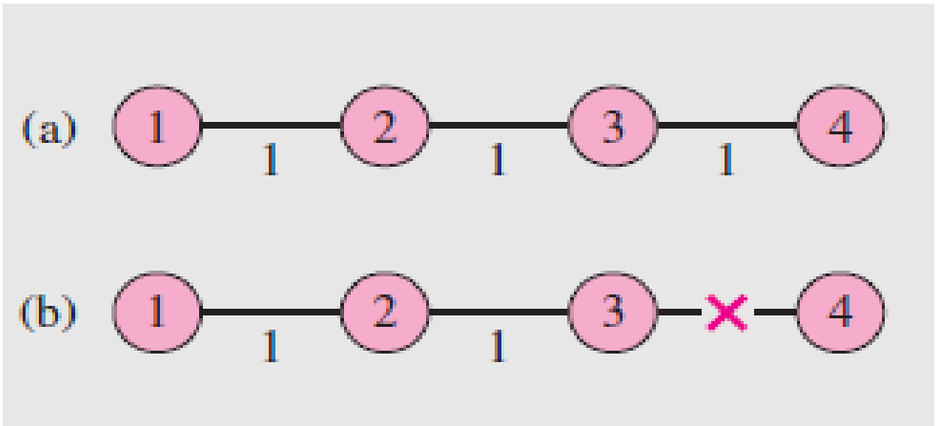
For simplicity, we will assume that there is a gigantic gong somewhere that is struck periodically to initiate a vector exchange at all routers simultaneously.

- At the time of the first exchange, *B learns that its left-hand neighbor has zero delay* to *A. B now makes an entry in its routing table indicating that A is one hop away* to the left.
- All the other routers still think that *A is down.* At this point, the routing table entries for *A are as shown in the second row of Fig. 5-10 a.*
- Now let us consider the situation of Fig. 5-10(b), in which all the links and routers are initially up. Routers *B, C, D, and E have distances to A of 1, 2, 3, and* 4 hops, respectively. Suddenly, either *A goes down or the link between A and B is* cut (which is effectively the same thing from *B's point of view).*

The **Reaction to Link Failure**

- This example shows that the distributed Bellman-Ford algorithm may react very slowly to a link failure. To see this, consider the topology shown in Figure a, with node 4 as the destination.
- Suppose that after the algorithm stabilizes, link (3,4) breaks, as shown in Figure b. Recomputed the minimum cost from each node to the destination node(node 4).

Example Count to infinity



Update	Node 1	Node 2	Node 3
Before break	(2, 3)	(3, 2)	(4, 1)
After break	(2, 3)	(3, 2)	(2, 3)
1	(2, 3)	(3, 4)	(2, 3)
2	(2, 5)	(3, 4)	(2, 5)
3	(2, 5)	(3, 6)	(2, 5)
4	(2, 7)	(3, 6)	(2, 7)
5	(2, 7)	(3, 8)	(2, 7)
. . .	. . .	. . .	. . .

Note: Dots in the last row indicate that the table continues to infinity

➢One particular method that is widely implemented is called the **split horizon**

For example, if node X thinks that the best route to node Y is via node Z, then node X should not send the corresponding minimum cost to node Z.

➢ Another variation called **split horizon with poisoned reverse allows a node** to

For example, if node X thinks that the best route to node Y is via node Z, then node X should set the corresponding minimum cost to infinity before sending it to node Z.

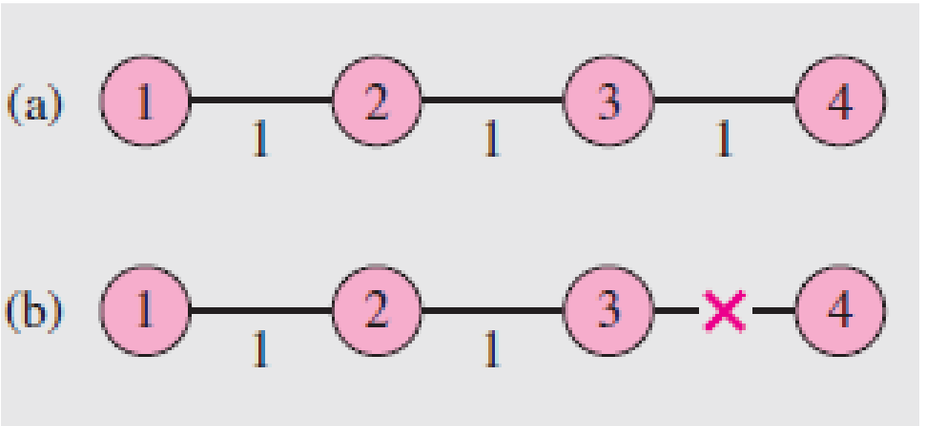


TABLE 7.5 Minimum costs by using split horizon with poisoned reverse.

Update	Node 1	Node 2	Node 3
Before break	(2, 3)	(3, 2)	(4, 1)
After break	(2, 3)	(3, 2)	$(-1, \infty)$
1	(2, 3)	$(-1, \infty)$	$(-1, \infty)$
2	$(-1, \infty)$	$(-1, \infty)$	$(-1, \infty)$

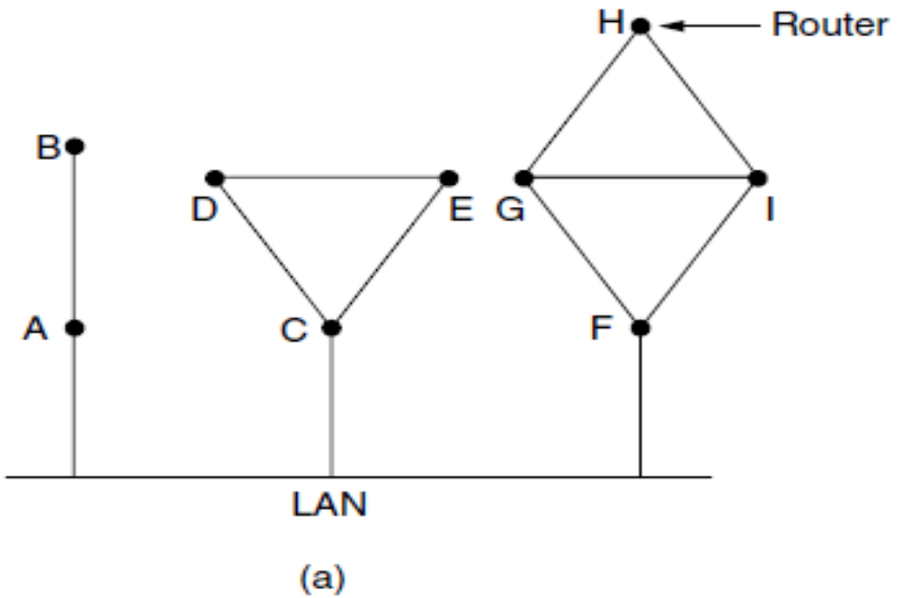
5.2.5 Link State Routing Distance

- The primary problem that caused its demise was that the algorithm often took too long to converge after the network topology changed (due to the count-to-infinity problem).
- Consequently, it was replaced by an entirely new algorithm, now called **link state routing.**
- **Dijkstra's algorithm is an alternative algorithm for finding *the shortest paths from*** *a source node to all other nodes in a network. It is generally more efficient than* *the DVR*
- Variants of link state routing called IS-IS and OSPF are the routing algorithms that are most widely used inside large networks and the Internet today.

The idea behind link state routing is fairly simple and can be stated as five parts. Each router must do the following things to make it work:

1. Discover its neighbors and learn their network addresses.
2. Set the distance or cost metric to each of its neighbors.
3. Construct a packet telling all it has just learned.
4. Send this packet to and receive packets from all other routers.
5. Compute the shortest path to every other router.

1.Learning about the Neighbors



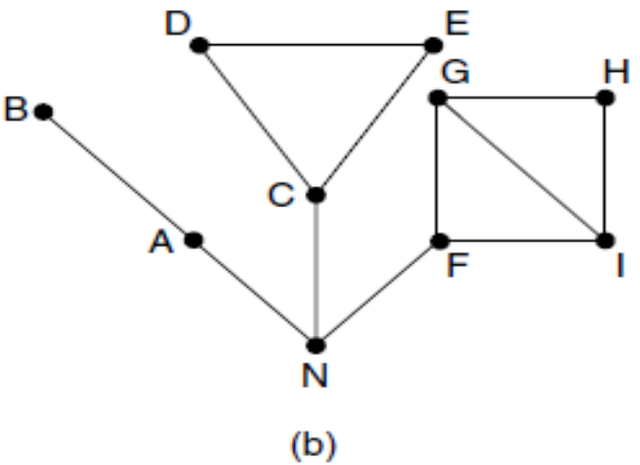


Figure : (a) Nine routers and a broadcast LAN. (b) A graph model of (a).

- The broadcast LAN provides connectivity between each pair of attached routers.
- However, modeling the LAN as many point-to-point links increases the sizeof the topology and leads to wasteful messages. A better way to model the LAN
- is to consider it as a node itself, as shown in Fig. (b). Here, we have introduced a new, artificial node, *N, to which A, C, and F are connected. One* ***designated*** **router on the LAN is selected to play the role of *N in the routing protocol***

Setting Link Costs

- The link state routing algorithm requires each link to have a distance or cost metric for finding shortest paths. The cost to reach neighbors can be set automatically, or configured by the network operator.
- A common choice is to make the cost inversely proportional to the bandwidth of the link. For example, 1-Gbps Ethernet may have a cost of 1 and 100-Mbps Ethernet a cost of 10. This makes higher-capacity paths better choices.

- If the network is geographically spread out, the delay of the links may be factored into the cost so that paths over shorter links are better choices.
- The most direct way to determine this delay is to send over the line a special ECHO packet that the other side is required to send back immediately. By measuring the round-trip time and dividing it by two, the sending router can get a reasonable estimate of the delay.

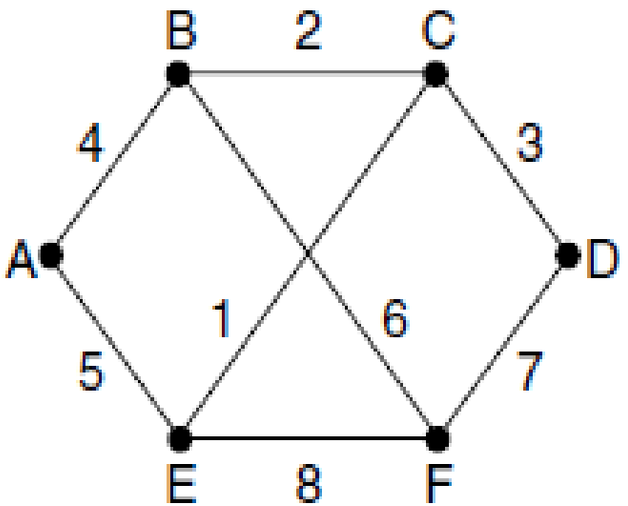
3.Building Link State Packets

- Once the information needed for the exchange has been collected,

the next step is for each router to build a packet containing all the data.

- The packet starts with the identity of the sender, followed by a sequence number and age (to be described later) and a list of neighbors.
-

example



(a)

Link State Packets

A		B		C		D		E		F	
Seq.		Seq.		Seq.		Seq.		Seq.		Seq.	
Age		Age		Age		Age		Age		Age	
B	4	A	4	B	2	C	3	A	5	B	6
E	5	C	2	D	3	F	7	C	1	D	7
		F	6	E	1			F	8	E	8

(b)

Figure . (a) A network. (b) The link state packets for this network.

4.Distributing the Link State Packets

- The fundamental idea is to use flooding to distribute the link state packets to all routers. To keep the flood in check, each packet contains a sequence number that is incremented for each new packet sent.
- Routers keep track of all the (source router, sequence) pairs they see
- When a new link state packet comes in, it is checked against the list of packets already seen. If it is new,
- it is forwarded on all lines except the one it arrived on. If it is a duplicate, it is discarded.
- If a packet with a sequence number lower than the highest one seen so far ever arrives, it is rejected as being obsolete as the router has more recent data

Drawbacks of LSA

- First, if the sequence numbers wrap around, confusion will reign. The solution here is to use a 32-bit sequence number. With one link state packet per second,
- it would take 137 years to wrap around, so this possibility can be ignored.
- Second, if a router ever crashes, it will lose track of its sequence number. If it starts again at 0, the next packet it sends will be rejected as a duplicate.
- Third, if a sequence number is ever corrupted and 65,540 is received instead of 4 (a 1-bit error), packets 5 through 65,540 will be rejected as obsolete, since the
- current sequence number will be thought to be 65,540.

Dijkstra's Algorithm

let us define *Di to be the current minimum cost from the source node (labeled s) to node i .*

1. Initialization:

$$N = \{s\}$$
$$D_j = C_{sj}, \quad \text{for all } j \neq s$$
$$D_s = 0$$

2. Finding the next closest node: Find node i / ∈ N such that

$$D_i = \min_{j \notin N} D_j$$

Add i to N.

If N contains all the nodes, stop.

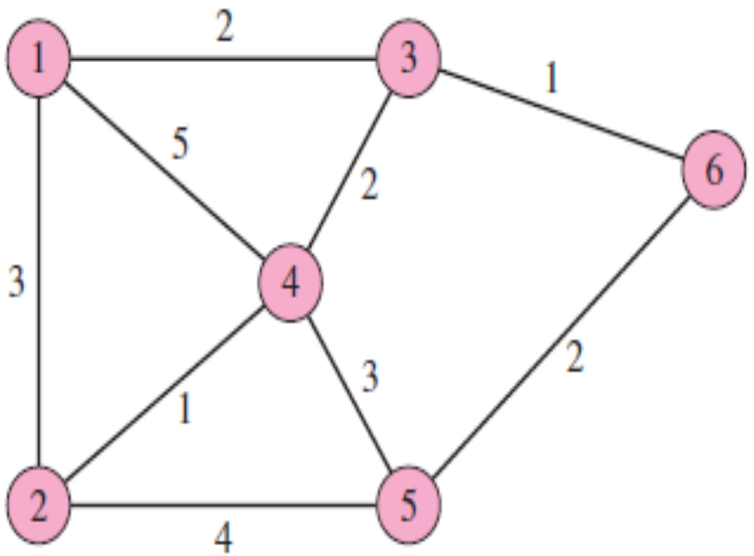
3. Updating minimum costs after node i added to N: For each node $j \notin N$

$$D_j = \min\{D_j, D_i + C_{ij}\}$$

Go to step 2**.**

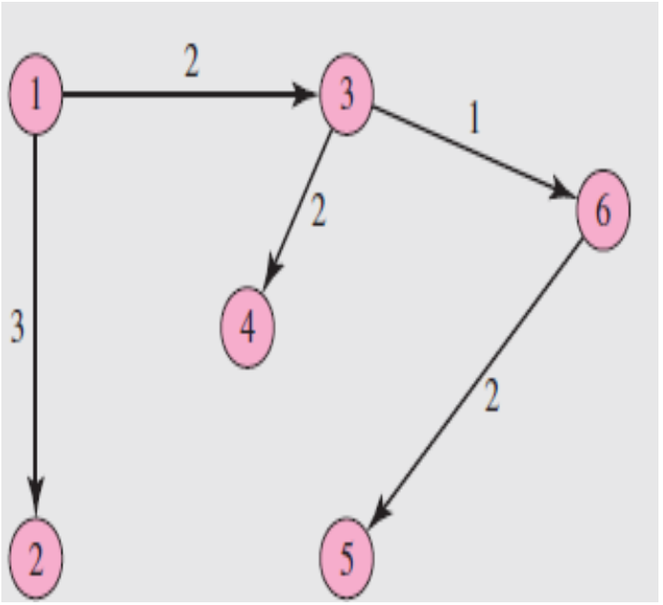
- The algorithm can be implemented by maintaining a set *N of permanently labeled nodes, which consists of those nodes whose shortest paths have been determined.*
- At each iteration the next-closest node is added to the set *N and the distance to the* remaining nodes via the new node is evaluated.

Apply Dijkstra's algorithm to find the shortest paths from the source node (assumed to be node 1) to all the other nodes



Iteration	N	D_2	D_3	D_4	D_5	D_6
Initial	{1}	3	2	5	∞	∞
1	{1,3}	3	2	4	∞	3
2	{1,2,3}	3	2	4	7	3
3	{1,2,3,6}	3	2	4	5	3
4	{1,2,3,4,6}	3	2	4	5	3
5	{1,2,3,4,5,6}	3	2	4	5	3

The Shortest-path tree from node 1 to other nodes.



Destination	Next node	Cost
2	2	3
3	3	2
4	3	4
5	3	5
6	3	3

Routing table at node 1 for Figure

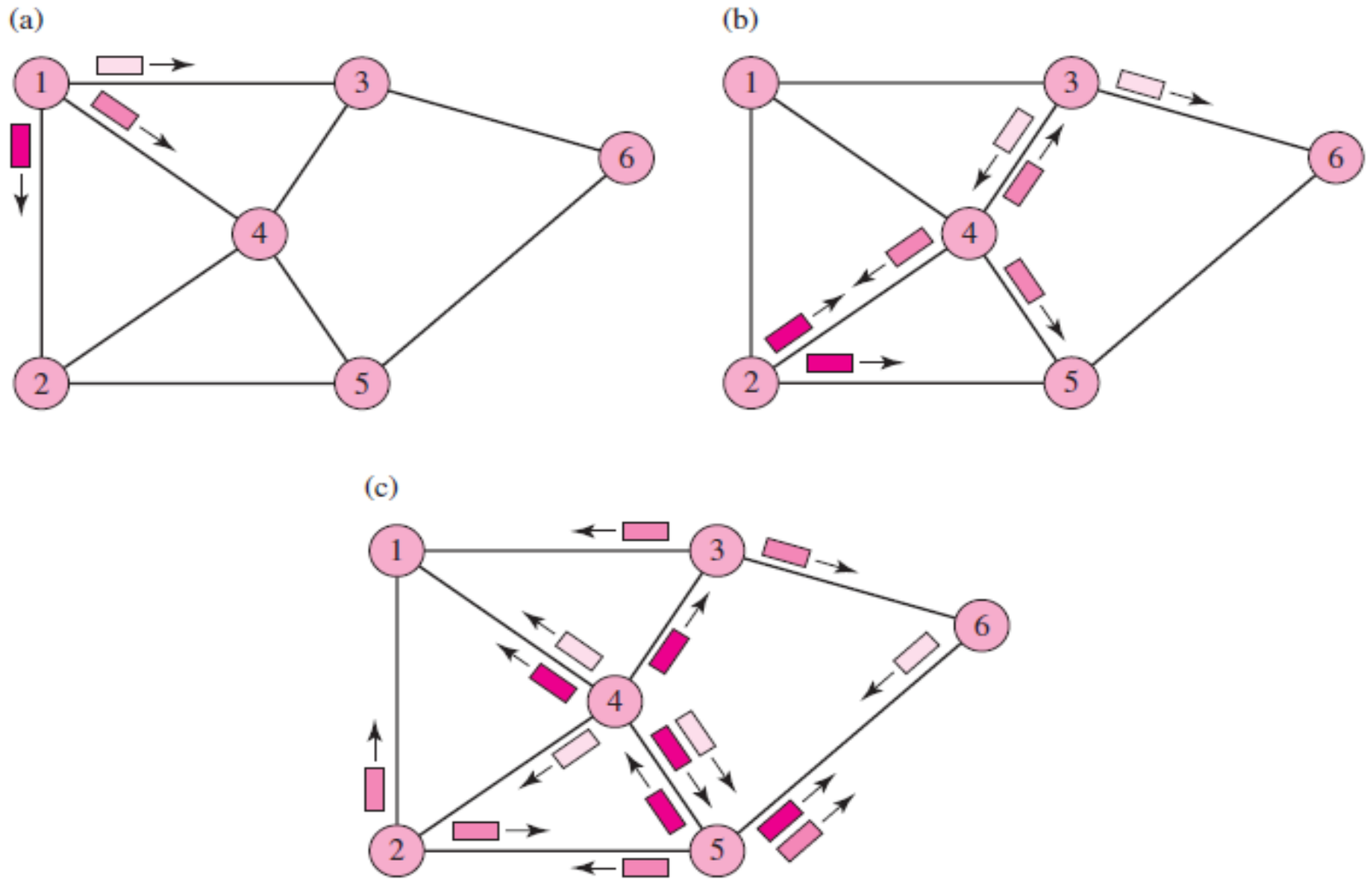
Specialized Routing :Flooding

- **flooding, in which every incoming packet is sent out on every** outgoing line except the one it arrived on.
- Flooding obviously generates vast numbers of duplicate packets, in fact, an infinite number unless some measures are taken to damp the proces
- One such measure is to have a hop counter contained in the header of each packet that is decremented at each hop, with the packet being discarded when the counter reaches zero.
- the hop counter should be initialized to the length of the path from source to

- it can initialize the counter to the worst case, namely, the full diameter of the network.
- A better technique for damming the flood is to have routers keep track of which packets have been flooded, to avoid sending them out a second time.
- One way to achieve this goal is to have the source router put a sequence number in each packet it receives from its hosts
- Each router then needs a list per source router telling which sequence numbers originating at that source have already been seen. If an incoming packet is on the list, it is not flooded.

Advantages:

- Flooding is an effective routing approach when the information in the routing tables is not available, such as during system startup, or when survivability is required, such as in military networks.
- Flooding is also effective when the source needs to send a packet to all nodes connected to the network (i.e., broadcast delivery).
- In wireless networks, all messages transmitted by a station can be received by all other stations within its radio range.
- Flooding may easily swamp the network as one packet creates multiple packets that in turn create multiples of multiple packets, generating an exponential growth rate as illustrated i



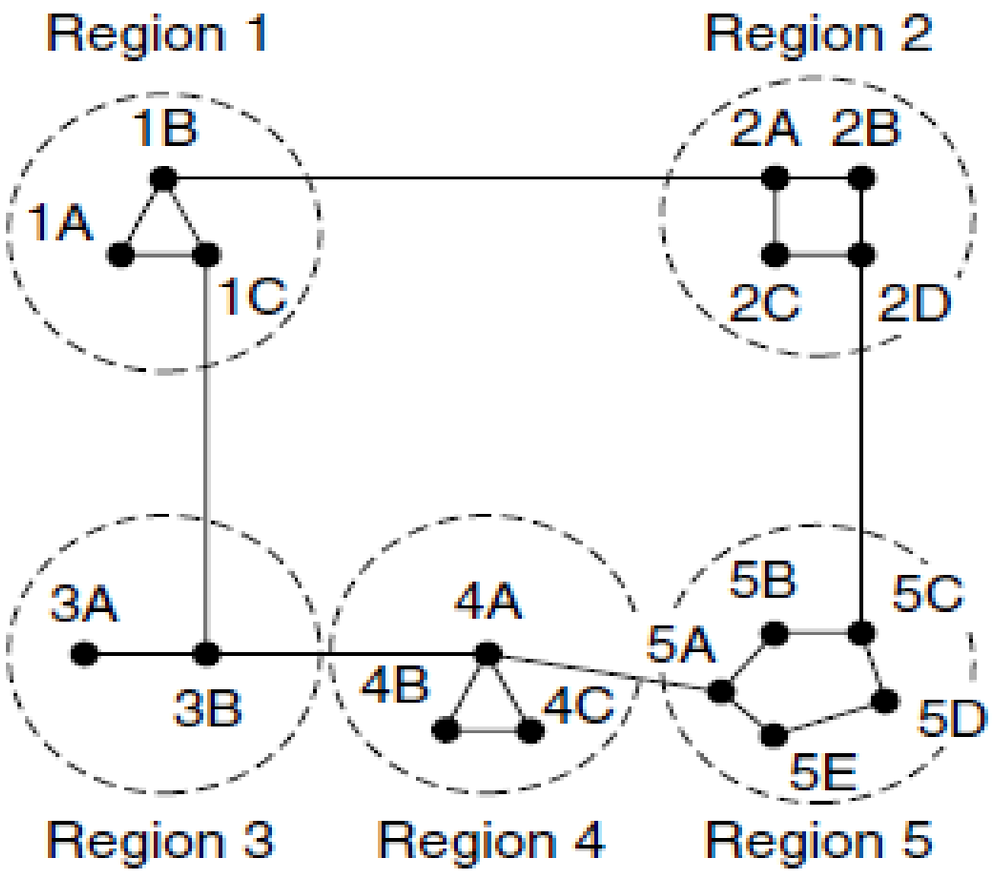
Flooding is initiated from node 1: (a) hop-1 transmissions, (b) hop-2 transmissions, and (c) hop-3 transmissions

- flooding needs to be controlled so that packets are not generated excessively. To reduce resource consumption in the network, one can implement a number of mechanisms
- One simple method is to use a time-to-live (TTL) field in each packet. When the source sends a packet, the TTL is initially set to some number.
- Each node decrements the TTL by one before flooding the packet. If the value reaches zero, the node discards the packet
- In the second method, each node adds its identifier to the header of the packet before it floods the packet. When a node receives a packet that contains the identifier of the node, it discards the packet since it knows that the packet already visited the node before. sequence number of the packet.

- When a node receives a packet, the node records the source address and the sequence number of the packet.
- If the node discovers that the packet has already visited the node, based on the stored source address and sequence number, it will discard the packet.

Hierarchical Routing

- When hierarchical routing is used, the routers are divided into what we will call **regions. Each router knows all the details about how to route packets to destinations** within its own region but knows nothing about the internal structure of other regions.
- When different networks are interconnected, it is natural to regard each one as a separate region to free the routers in one network from having to know the topological structure of the other ones
- it may be necessary to group the regions into clusters, the clusters into zones, the zones into groups, and so on, until we run out of names for aggregations



Full table for 1A

Dest.	Line	Hops
1A	–	–
1B	1B	1
1C	1C	1
2A	1B	2
2B	1B	3
2C	1B	3
2D	1B	4
3A	1C	3
3B	1C	2
4A	1C	3
4B	1C	4
4C	1C	4
5A	1C	4
5B	1C	5
5C	1B	5
5D	1C	6
5E	1C	5

(b)

Hierarchical table for 1A

Dest.	Line	Hops
1A	–	–
1B	1B	1
1C	1C	1
2	1B	2
3	1C	2
4	1C	3
5	1C	4

- Figure gives a quantitative example of routing in a two-level hierarchy with five regions. The full routing table for router *1A has 17 entries, as shown in* Fig. (b).
- When routing is done hierarchically, as in Fig. (c), there are entries for all the local routers, as before, but all other regions are condensed into a single router, so all traffic for region 2 goes via the *1B-2A line,*
- *but the rest of the* remote traffic goes via the *1C-3B line. Hierarchical routing has reduced the table* from 17 to 7 entries.
- As the ratio of the number of regions to the number of routers per region grows, the savings in table space increase

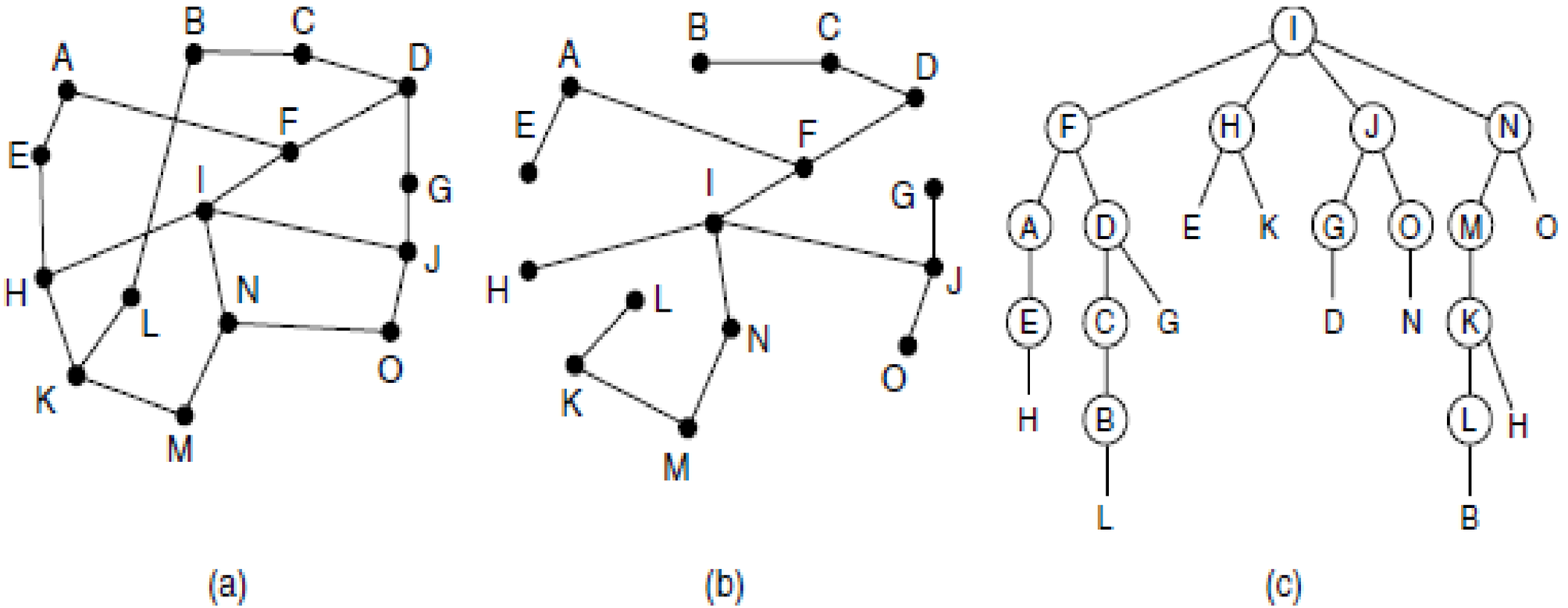
Broadcast Routing

In some applications, hosts need to send messages to many or all other hosts.Sending a packet to all destinations simultaneously is called **broadcasting.**

- For example, a service distributing weather reports, stock market updates, or live radio programs might work best by sending to all machines and letting those that are interested read the data

- An improvement is **multidestination routing, in which each packet contains** either a list of destinations or a bit map indicating the desired destinations.
- When a packet arrives at a router, the router checks all the destinations to determine the set of output lines that will be needed The router generates a new copy of the packet for each output line to be used and includes in each packet only those destinations that are to use the line.

Broadcast routing



Reverse path forwarding. (a) A network. (b) A sink tree. (c) The tree built by reverse path forwarding.

➢Fig c:shows how the reverse path algorithm works. On the first hop, I sends packets to F, H, J, and N, as indicated by the second row of the tree. Each of these packets arrives on the preferred path to I (assuming that the preferred path falls along the sink tree) and is so indicated by a circle around the letter.

➢On the second hop *eight packets are generated, two by each of the routers that received a packet on the first hop. As it turns out, all eight of these arrive at previously unvisited routers, and five of these arrive along the preferred line of the six packets generated.*

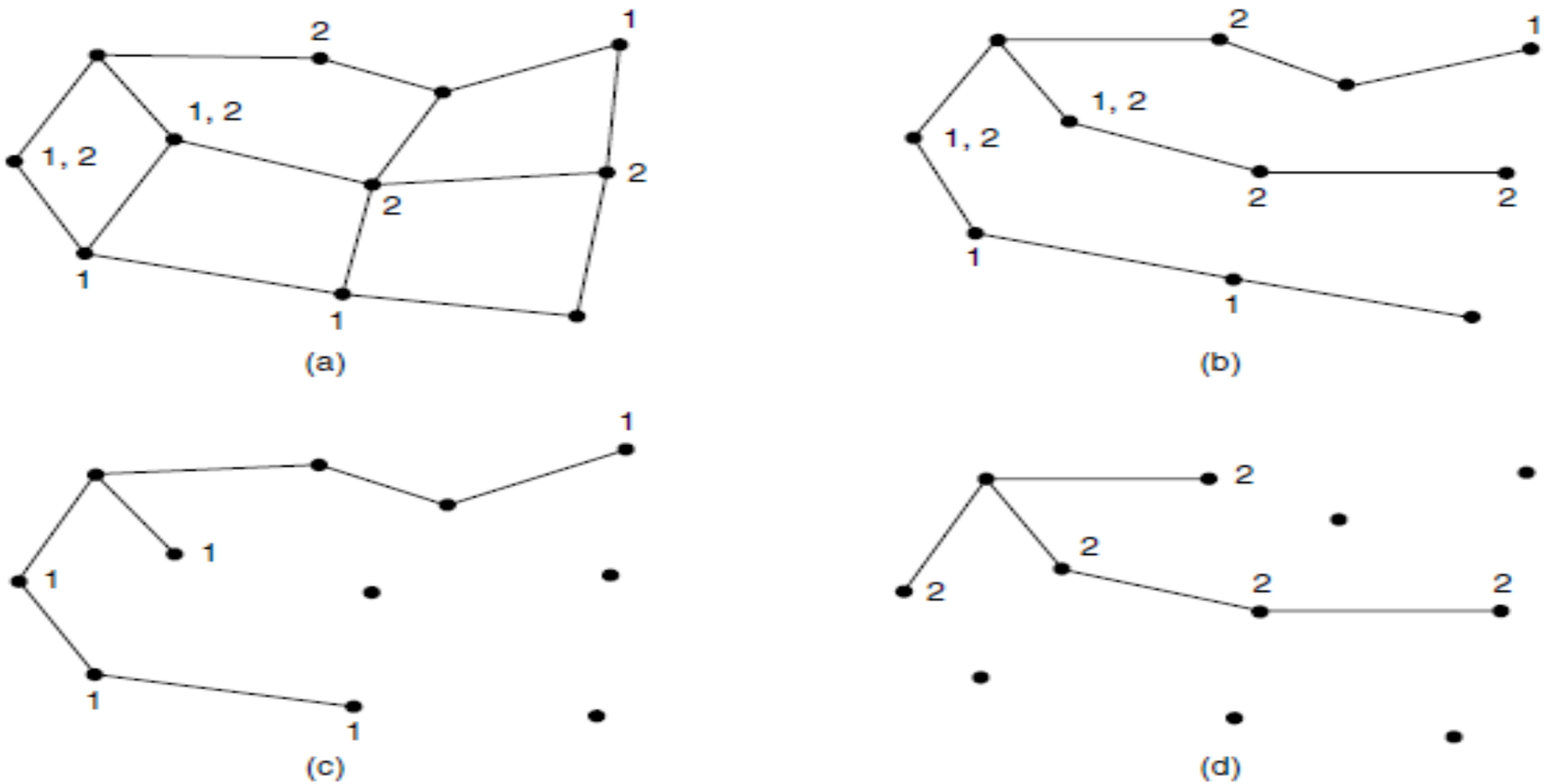
➢*on the third hop, only three arrive on the preferred path (at C, E, and K); the othersare duplicates. After five hops and 24 packets, the broadcasting terminates,*

➢*compared with four hops and 14 packets had the sink tree been followed exactly.*

- broadcast algorithm improves on the behavior of reverse path forwarding.It makes explicit use of the sink tree.
- spanning tree is a subset of the network that includes all the routers but contains no loops. Sink trees are spanning trees.
- If each router knows which of its lines belong to the spanning tree, it can copy an incoming broadcast packet onto all the spanning tree lines except the one it arrived on.
- This method makes excellent use of bandwidth, generating the absolute minimum number of packets necessary to do the job

Multicast Routing

- Some applications, such as a multiplayer game or live video of a sports event streamed to many viewing locations, send packets to multiple receivers. Unless the group is very small, sending a distinct packet to each receiver is expensive.
- *Thus, we need a way to send messages to well-defined groups that are numerically large in size but small compared to the network as a whole.*
- *Sending a message to such a group is called multicasting, and the routing algorithm used is called multicast routing.*
- *Multicast routing schemes build on the broadcast routing schemes we have already studied, sending packets along spanning trees to deliver the packets to the members of the group while making efficient use of bandwidth*



a) A network. (b) A spanning tree for the leftmost router. (c) A multicast tree for group 1. (d) A multicast tree for group 2.

➢A spanning tree for the leftmost router is shown in Fig. 5-16(b). This tree can be used for broadcast but is overkill for multicast,as can be seen from the two pruned versions that are shown next.

➢In Fig. 5-16(c), all the links that do not lead to hosts that are members of group 1 have been removed.

➢The result is the multicast spanning tree for the leftmost router to send to group 1. Packets are forwarded only along this spanning tree,which is more efficient than the broadcast tree because there are 7 links instead of 10.

➢ Fig. 5-16(d) shows the multicast spanning tree after pruning for group 2. It is efficient too, with only five links this time. It also shows that different multicast groups have different spanning trees.

Various ways of pruning the spanning tree

- The simplest one can be used if link state routing is used and each router is aware of the complete topology, including which hosts belong to which groups.
- Each router can then construct its own pruned spanning tree for each sender to the group in question by constructing a sink tree for the sender as usual and then removing all links that donot connect group members to the sink node.
- **MOSPF (Multicast OSPF) is an** example of a link state protocol that works in this way

2.Using distance vector routing

- A different pruning strategy can be followed.The basic algorithm is reverse path forwarding.
- However, whenever a router with no hosts interested in a particular group and no connections to other routers receives a multicast message for that group, it responds with a PRUNE message, telling the neighbor that sent the message not to send it any more multicasts from the sender for that group.

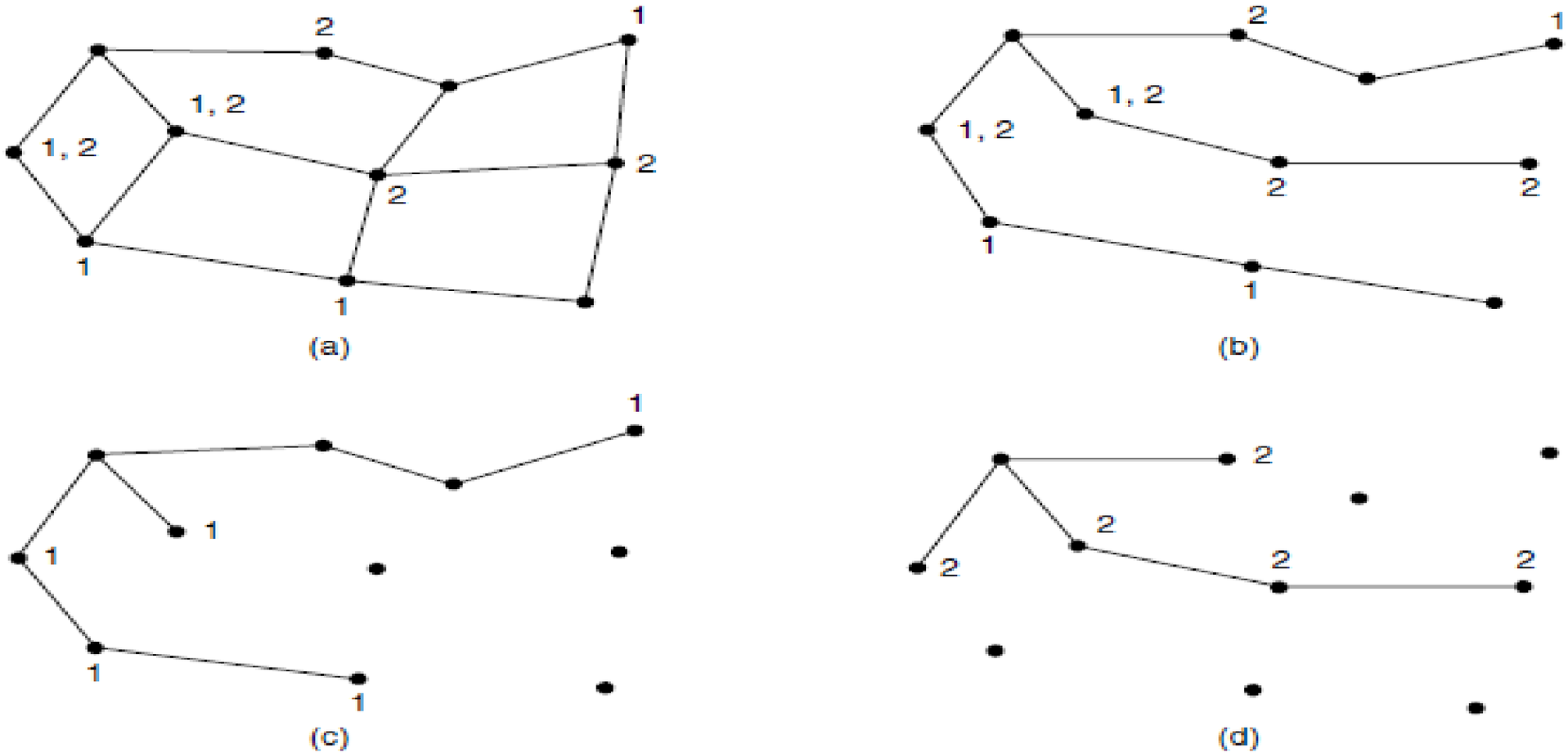
- When a router with no group members among its own hosts has received such messages on all the lines to which it sends the multicast, it,. too, can respond with a PRUNE message.

example DVMRP (Distance Vector Multicast Routing Protocol)

of a multicast routing protocol that works

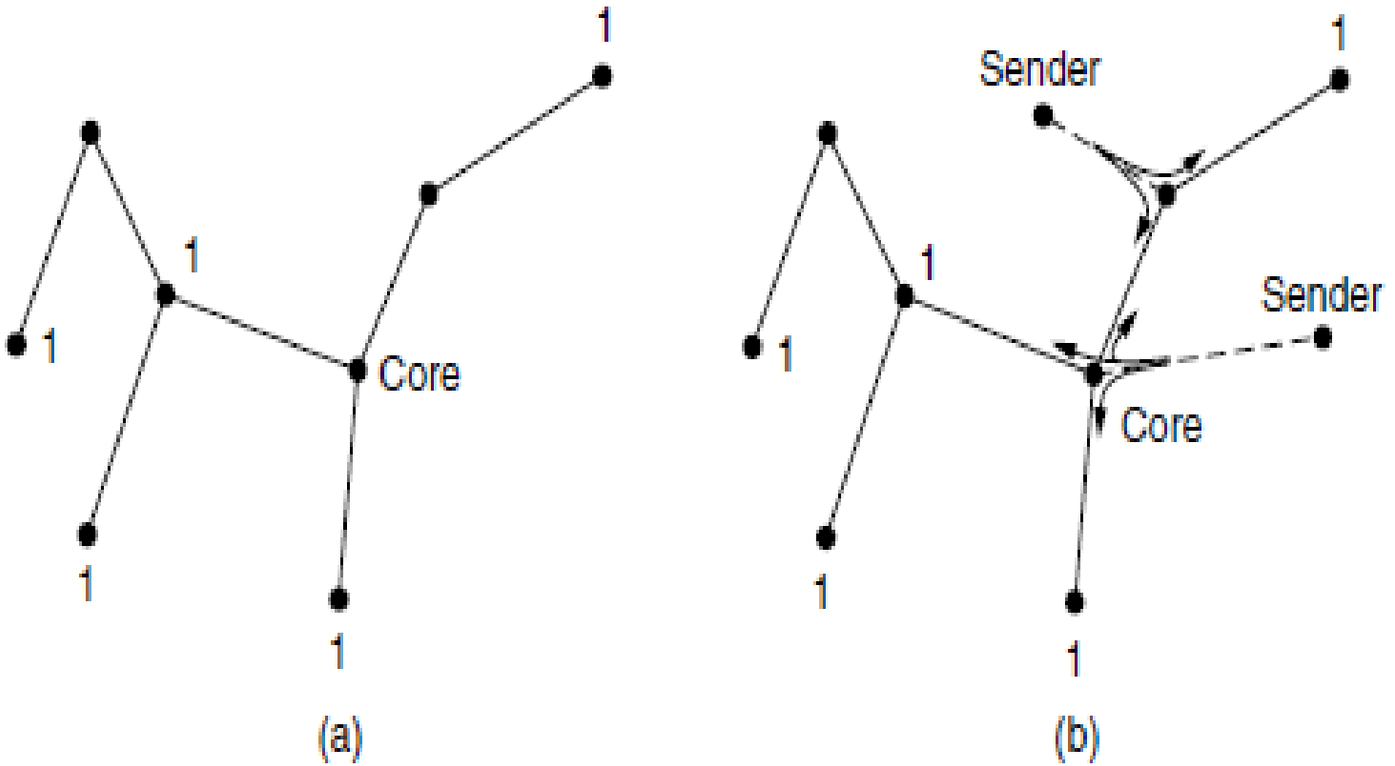
Drawback:

- Lots of work for routers, especially for large networks. Suppose that a network has n groups, each with an average of m nodes. At each router and for each group, m pruned spanning trees must be stored, for a total of mn trees.



(a) A network. (b) A spanning tree for the leftmost router. (c) Amulticast tree for group 1. (d) A multicast tree for group 2.

- An alternative design uses **core-based trees to compute a single spanning tree** for the group
- All of the routers agree on a root called the **core and build the tree by sending a packet from each** member to the root.
- The tree is the union of the paths traced by these packets.
- Fig. (a) shows a core-based tree for group 1.
- To send to this group, a sender sends a packet to the core. When the packet reaches the core, it is forwarded down the tree.



(a) Core-based tree for group 1. (b) Sending to group 1.

Fig. (b) for the sender on the right hand side of the network. As a performance optimization, packets destined for the group do not need to reach the core before they are multicast.

- As soon as a packet reaches the tree, it can be forwarded up toward the root, as well as down all the other branches.
- Also of note is that shared trees can be a major savings in storage costs, messages sent, and computation. Each router has to keep only one tree per group, instead of *m trees*.
- *Routers that are not part of the tree do no work at all to* support the group. For this reason, shared tree approaches like core-based trees are used for multicasting to sparse groups in the Internet as part of popular protocols such as **PIM (Protocol Independent Multicast)**

Applications of Multicast routing

- **Streaming:** This is one of the most widely used multicasting applications, allowing video or other media to be delivered to users across a corporate campus, school, or hotel property. Live streaming of events and concerts is also sometimes done via multicast.
- **Videoconferencing:** From Zoom calls to mass video training sessions, multicasting allows many entrants to participate in the same event without overwhelming the server.

Applications of Multicast routing

- **File distribution:** Over-the-air patch management, operating system imaging, and even financial stock tickers can all be multicast over corporate networks.
- **IoT devices:** The sensors embedded into IoT devices translate and send information to databases, AI, and countless other sources when it collects info. These systems may rely on multicasting to keep tabs on cybersecurity and ensure the correct recipients.

Anycast Routing

➢Another delivery model, called **anycast is** sometimes also useful. In anycast, a packet is delivered to the nearest member of a group Schemes that find these paths are called **anycast routing** .

For example, anycast is used in the Internet as part of DNS, anycast uses regular distance vector and link state routing can produce anycast routes. Suppose

- we want to anycast to the members of group 1. They will all be given the address“1,” instead of different addresses.
- Distance vector routing will distribute vectors as usual, and nodes will choose the shortest path to destination 1. This will result in nodes sending to the nearest instance of destination 1.
- The routes are shown in Fig. (a). This procedure works because the routing protocol does not realize that there are multiple instances of destination 1.
- That is, it believes that all the instances of node 1 are the same node, as in the topology shown in Fig. (b).

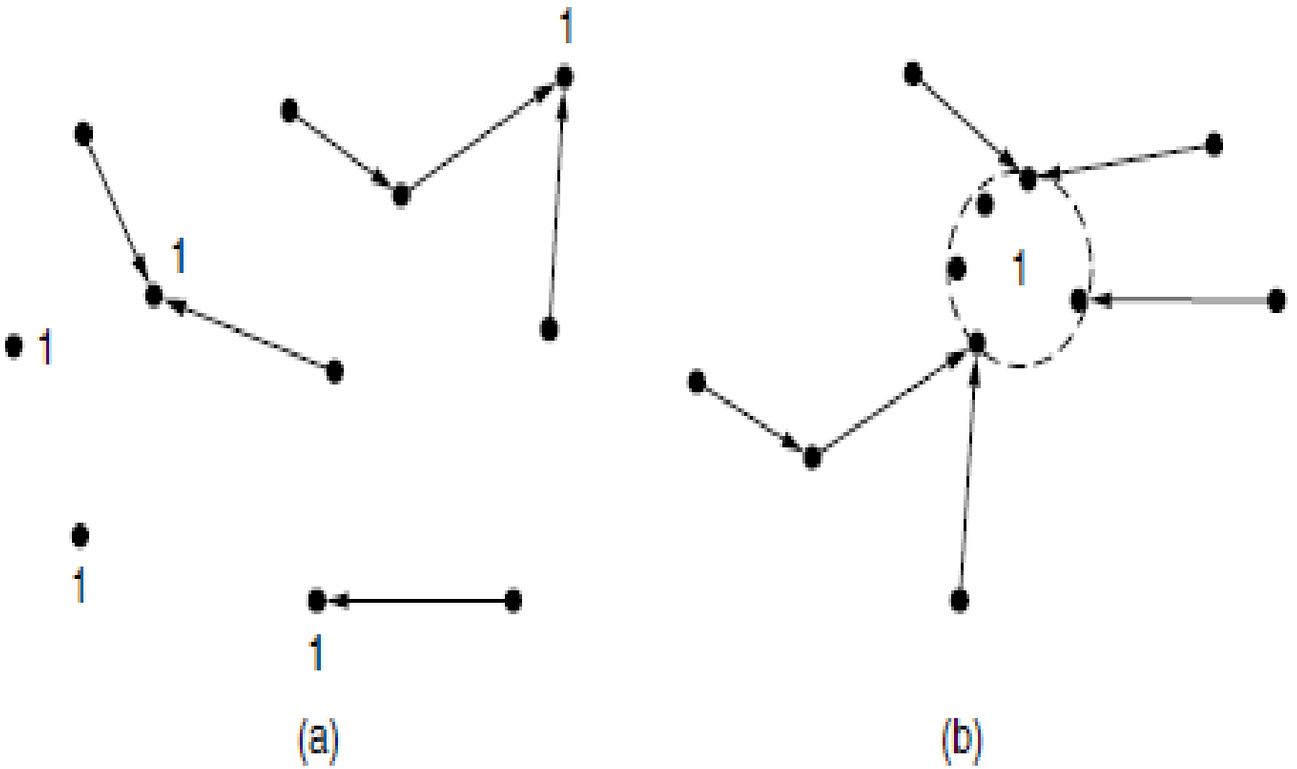


Figure 5-18. (a) Anycast routes to group 1. (b) Topology seen by the routing protocol.

Routing for Mobile Hosts

•The model of the world that we will consider is one in which all hosts are assumed to have a permanent **home location** The routing goal in systems with**that never changes.**

- mobile hosts is to make it possible to send packets to mobile hosts using their
- fixed home addresses and have the packets efficiently reach them wherever they
- may be. The trick, of course, is to find them

•The basic idea used for mobile routing in the Internet and cellular networks is for the mobile host to tell a host at the home location where it is now. This host, which acts on behalf of the mobile host, is called the **home agent.**

•Once it knows where the mobile host is currently located, it can forward packets so that they are delivered.

•A sender in the northwest city of Seattle wants to send a packet to a host normally located across the United States in New York.

•The case of interest to us is when the mobile host is not at home. Instead, it is temporarily in San Diego. The mobile host in San Diego must acquire a local network address before it can use the network

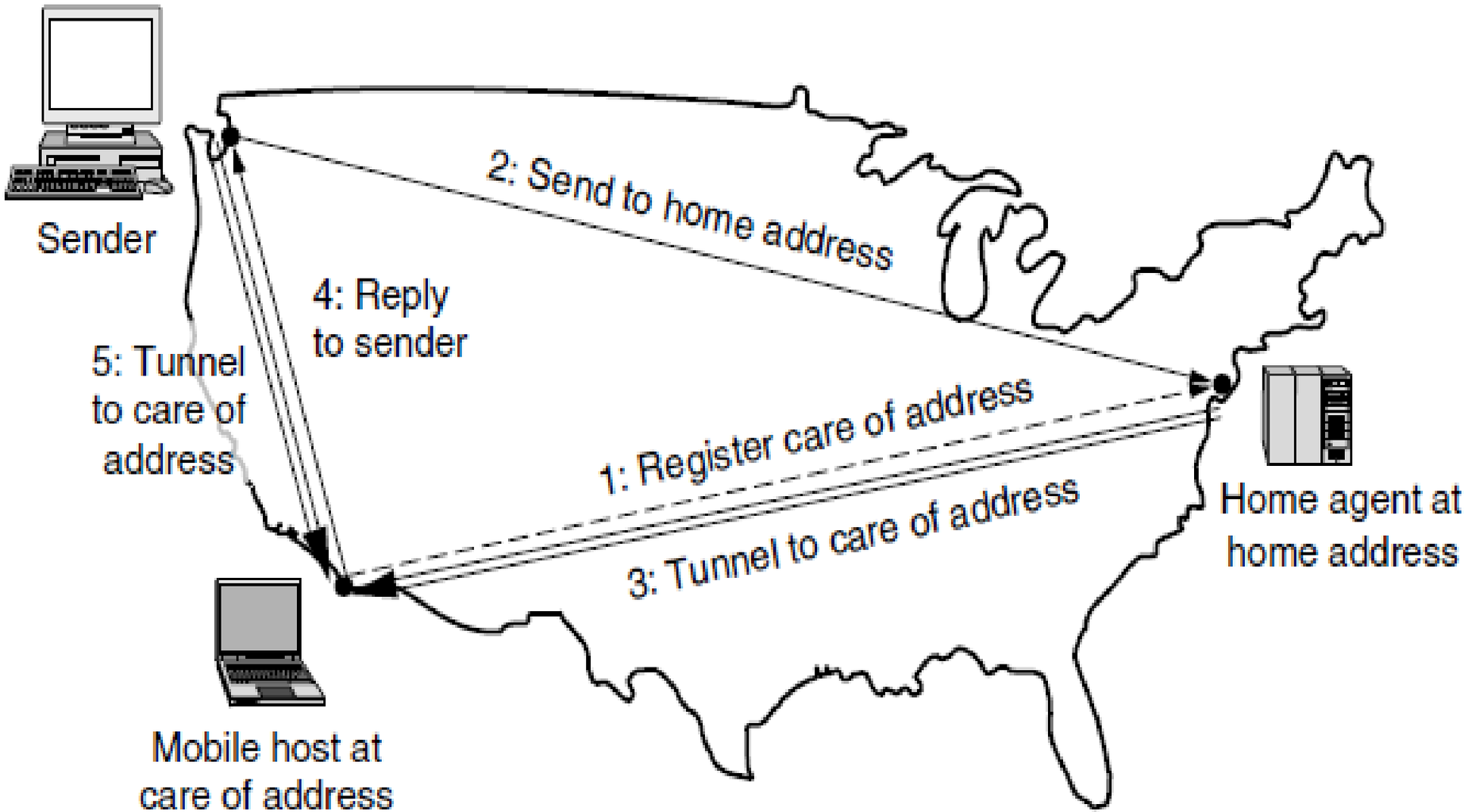


Fig:Packet routing for mobile hosts.

step 1: The local address is called a **care of address. Once the mobile host has this address,**it can tell its home agent where it is now. It does this by sending a registration message to the home agent with the care of address

step 2: the sender sends a data packet to the mobile host using its permanent address This packet is routed by the network to the host's home location because that is where the home address belongs.

step 3: In New York, the home agent intercepts this packet because the mobile host is away from home. It then wraps or encapsulates the packet with a new header and sends this bundle to the care of address. This mechanism is called **tunneling**

- step 4: When the encapsulated packet arrives at the care of address, the mobile host unwraps it and retrieves the packet from the sender. The mobile host then sends its reply packet directly to the sender.

The overall route is called **triangle routing because it may be circuitous if the remote location is far from the home** location.

step 5: the sender may learn the current care of address. Subsequent packets can be routed directly to the mobile host by tunneling them to the care of address bypassing the home location entirely.

- If connectivity is lost for any reason as the mobile moves, the home address can always be used to reach the mobile.

Routing in Ad Hoc Networks

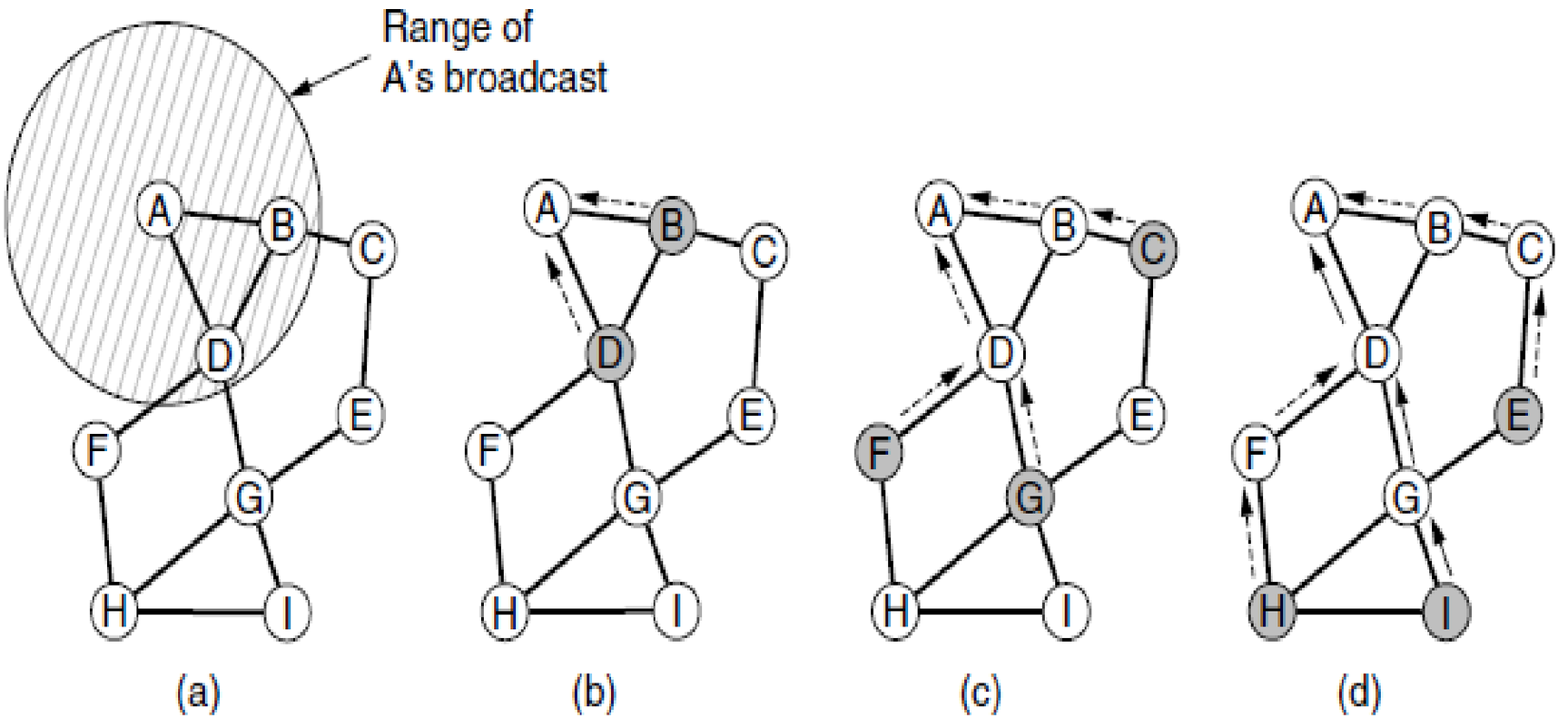
➢Among the possibilities are emergency workers at an earthquake site, military vehicles on a battlefield, a fleet of ships at sea, or a gathering of people with laptop computers in an area lacking 802.11.

➢In all these cases, and others, each node communicates wirelessly and acts as both a host and a router.

➢Networks of nodes that just happen to be near each other are called **ad hoc networks or MANETs (Mobile Ad hoc NETworks).**

- the most popular routing algorithms, **AODV**

Route Discovery



(a)Range of A's broadcast. (b) After B and D receive it. (c) After C, F, and G receive it. (d) After E, H, and I receive it. The shaded nodes are new recipients. The dashed lines show possible reverse routes. The solid lines show the discovered route.

- the topology of an ad hoc network can be described by agraph o connected nodes. Two nodes are connected if they can communicate directly using their radios.
- A basic but adequate model that is sufficient for our purposes is that each node can communicate with all other nodes that lie within its coverage circle.
- To locate I, A constructs a ROUTE REQUEST packet and broadcasts it using flooding,

- The transmission from *A reaches B and D, as* illustrated in Fig.(a). Each node rebroadcasts the request, which continues to reach nodes *F, G, and C in Fig.(c) and nodes H, E, and I in Fig. (d).*
- *A* sequence number set at the source is used to weed out duplicates during the flood.

For example, *D discards the transmission from B in Fig. (c) because it has already* forwarded the request.

- Eventually, the request reaches node *I, which constructs a ROUTE REPLY* packet. This packet is unicast to the sender along the reverse of the path followed by the request.
- For this to work, each intermediate node must remember the node that sent it

Route Maintenance

- Because nodes can move or be switched off, the topology can change spontaneously.

For example, in Fig. if *G is switched off, A will not realize that* the route it was using to *I (ADGI) is no longer valid. The algorithm needs to be* able to deal with this. Periodically, each node broadcasts a *Hello message.*

- *Each*of its neighbors is expected to respond to it. If no response is forthcoming, the broadcaster knows that that neighbor has moved out of range or failed and is no longer connected to it.
- Similarly, if it tries to send a packet to a neighbor that does not respond, it learns that the neighbor is no longer available.

5.3 Congestion Control Algorithm

- Too many packets present in (a part of) the network causes packet delay and loss that degrades performance.
- This situation is called congestion. The network and transport layers share the responsibility for handling congestion.

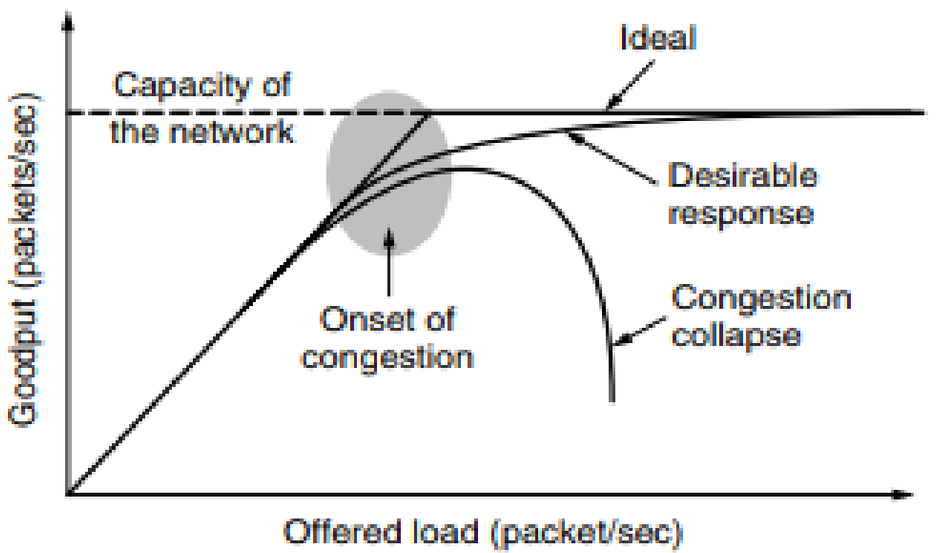


Fig: With too much traffic, performance drops sharply

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- Fig depicts the onset of congestion. When the number of packets hosts send into the network is well within its carrying capacity, the number delivered is proportional to the number sent.
- If twice as many are sent, twice as many are delivered.
- However, as the offered load approaches the carrying capacity, bursts of traffic occasionally fill up the buffers inside routers and some packets are lost.
- These lost packets consume some of the capacity, so the number of delivered packets falls below the idea

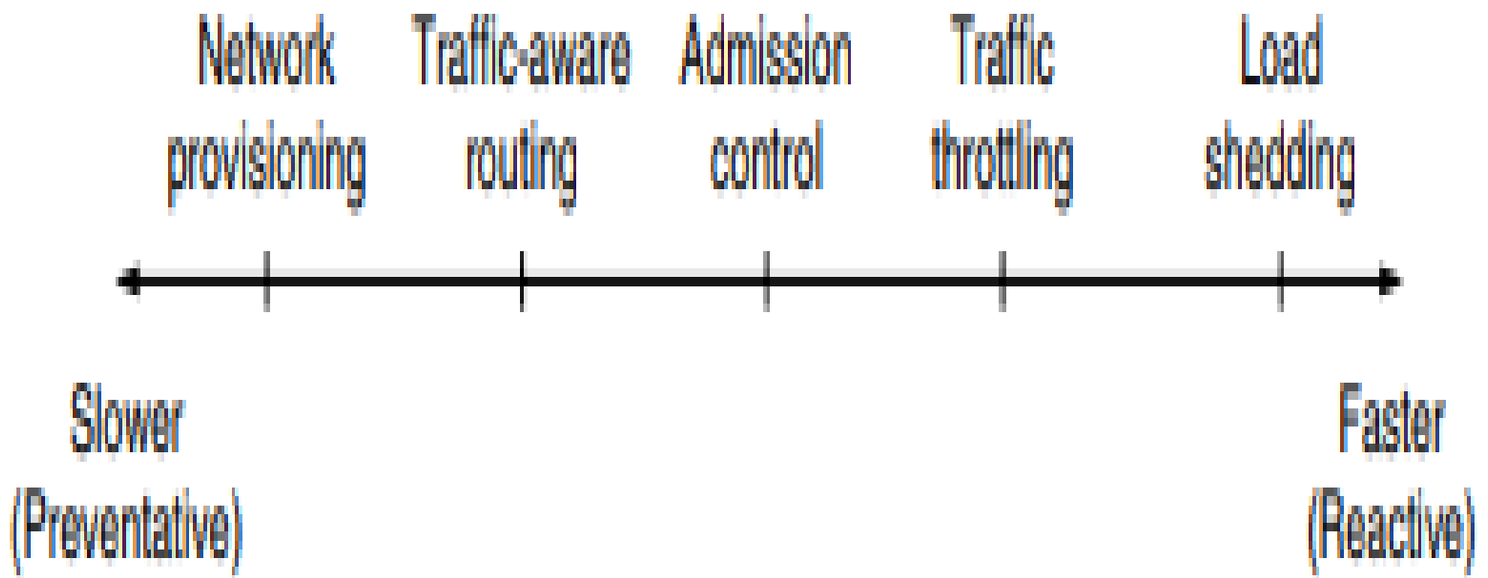
- difference between congestion control and flow control, as the relationship is a very subtle one. Congestion control has to do with
- making sure the network is able to carry the offered traffic. It is a global issue, involving the behavior of all the hosts and routers. Flow control, in contrast, relates to the traffic between a particular sender and a particular receiver. Its job is to make sure that a fast sender cannot continually transmit data faster than the re ceiver is able to absorb it

Example :

- consider a network made up of 100-Gbps fiber optic links on which a supercomputer is trying to force feed a large file to a personal computer that is capable of handling only 1 Gbps. Although there is no congestion.
- flow control is needed to force the supercomputer to stop frequently to give the personal computer a chance to breathe.

5.3.1 Approaches to Congestion Control

The presence of congestion means that the load is (temporarily) greater than the resources (in a part of the network) can handle.
Two solutions : increase the resources or decrease the load.



- The most basic way to avoid congestion is to build a network that is well matched to the traffic that it carries. If there is a low-bandwidth link on the path along which most traffic is directed, congestion is likely.
- Sometimes resource can be added dynamically when there is serious congestion, for example, turning on spare routers or enabling lines that are normally used only as backups (to make the system fault tolerant) or purchasing bandwidth on the open market.
- More often, links and routers that are regularly heavily utilized are upgraded at the earliest opportunity. This is called **provisioning** and happens on a time scale of months, driven by long-term traffic trends.

- To make the most of the existing network capacity, routes can be tailored to traffic patterns that change during the day as network users wake and sleep in dif ferent time zones.

For example, routes may be changed to shift traffic away from heavily used paths by changing the shortest path weights.

Some local radio stations have helicopters flying around their cities to report on road congestion to make it possible for their mobile listeners to route their packets (cars) around hotspots. This is called **traffic-aware routing**. Splitting traffic across multiple paths is also helpful.

sometimes it is not possible to increase capacity. The only way then to beat back the congestion is to decrease the load. In a virtual-circuit network, new connections can be refused if they would cause the network to become congested. This is called admission control.

when congestion is imminent the network can deliver feedback to the sources whose traffic flows are responsible for the problem. The network can request these sources to throttle their traffic, or it can slow down the traffic itself.

Two difficulties

1.How to identify the onset of congestion

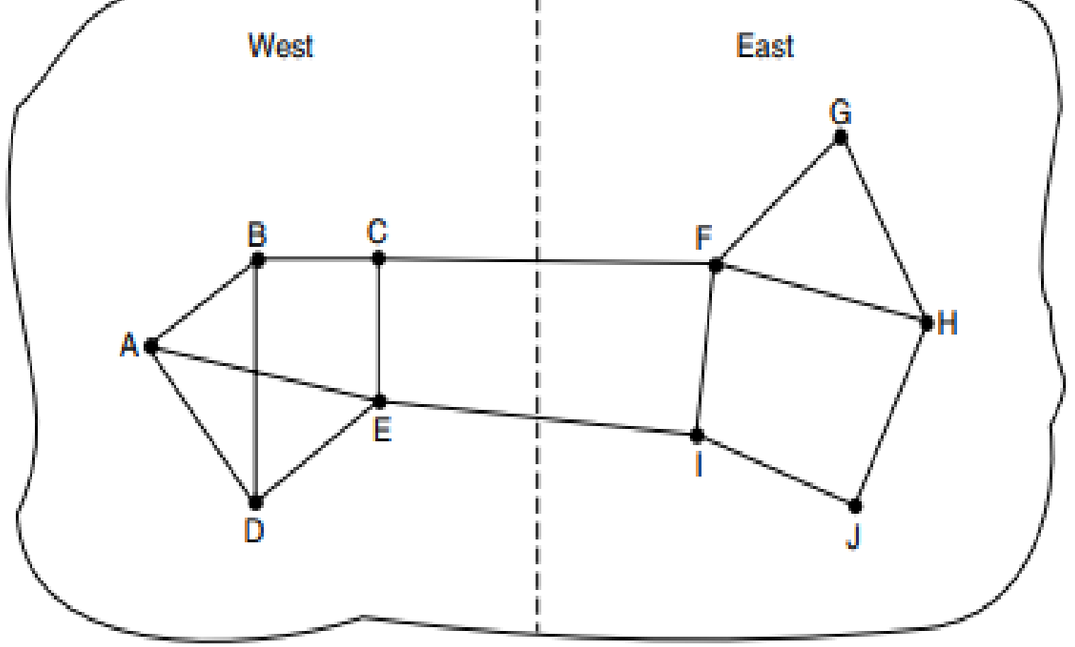
2.How to inform the source that needs to slow down

- To tackle the **first issue**, routers can monitor the average load, queueing delay, or packet loss. In all cases, rising numbers indicate growing congestion.
- To tackle the **second issue**, routers must participate in a feedback loop with the sources. For a scheme to work correctly, the time scale must be adjusted carefully.
- If every time two packets arrive in a row, a router yells **STOP** and every time a router is idle for 20 μsec, it yells **GO**, the system will oscillate wildly and never converge.

- Finally, when all else fails, the network is forced to discard packets that it cannot deliver.
- The general name for this is **load shedding**. A good policy for choosing which packets to discard can help to prevent congestion collapse.

5.3.2 Traffic aware Routing

- The routing schemes used fixed link weights. These schemes adapted to changes in topology, but not to changes in load.
- The goal in taking load into account when computing routes is to shift traffic away from hotspots that will be the first places in the network to experience congestion.
- The most direct way to do this is to set the link weight to be a function of the (fixed) link bandwidth and propagation delay plus the (variable) measured load or average queuing delay.
- Least-weight paths will then favor paths that are more lightly loaded, all else being equal. Traffic-aware routing was used



- Consider the network of Fig. which is divided into two parts, East and West, connected by two links, CF and EI.
- Suppose that most of the traffic between East and West is using link CF, and, as a result, this link is heavily loaded with long delays.
- Including queueing delay in the weight used for the shortest path calculation will make EI more attractive.
- After the new routing tables have been installed, most of the East-West traffic will now go over EI, loading this link.
- Consequently, in the next update, CF will appear to be the shortest path. As a result, the routing tables may oscil late wildly, leading to erratic routing and many potential problems.

If load is ignored and only bandwidth and propagation delay are considered, this problem does not occur.

Attempts to include load but change weights within a narrow range only slow down routing oscillations. Two techniques can contribute to a successful solution.

The first is multipath routing, in which there can be multiple paths from a source to a destination.

example this means that the traffic can be spread across both of the East to West links.

The second one is for the routing scheme to shift traffic across routes slowly enough that it is able to converge,

Given these difficulties, in the Internet routing protocols do not generally adjust their routes depending on the load. Instead, adjustments are made outside the routing protocol by slowly changing its inputs. This is called traffic engineering

5.3.3 Admission Control

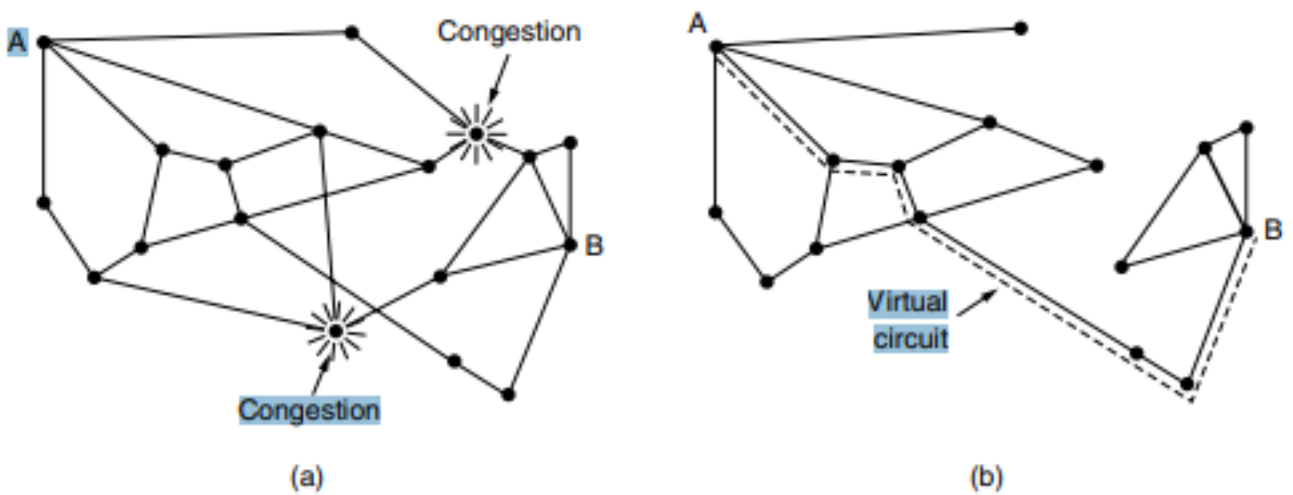
One technique that is widely used in virtual-circuit networks to keep congestion at bay is **admission control**.

The idea is simple, do not set up a new virtual circuit unless the network can carry the added traffic without becoming congested.

analogy, in the telephone system, when a switch gets overloaded it practices admission control by not giving dial tones.

For example, traffic that varies while browsing the Web is more difficult to handle than a streaming movie with the same long-term throughput because the bursts of Web traffic are more likely to congest routers in the network. A commonly used descriptor that captures this effect is the **leaky bucket or token bucket**

- Admission control can also be combined with traffic-aware routing by considering routes around traffic hotspots as part of the setup procedure.
- For example consider the network illustrated in Fig. (a), in which two routers are congested, as indicated. A Congestion Virtual circuit



- Suppose that a host attached to router A wants to set up a connection to a host attached to router B.
- Normally, this connection would pass through one of the congested routers. To avoid this situation, we can redraw the network as shown in Fig. (b), omitting the congested routers and all of their lines.
- The dashed line shows a possible route for the virtual circuit that avoids the congested routers. give a design for this kind of load-sensitive routing.

Open-Loop Control

- Open-loop control does not rely on feedback information to react to congestion. Instead, open-loop control is based on a principle that network performance is guaranteed to all traffic flows that have been admitted into the network. To guarantee network performance during the lifetime of admitted flows

Three mechanisms are:

1. Admission control
2. Policing
3. Traffic shaping

1.ADMISSION CONTROL

4.Quality Of Service

Quality of service mechanisms let a network with less capacity meet application requirements just as well at a lower cost. **overprovisioning** is based on expected traffic.

All bets are off if the traffic pattern changes too much. With quality of service mechanisms, the network can honor the performance guarantees that it makes even when traffic spikes, at the cost of turning down some requests.

Four issues must be addressed to ensure quality of service:

1.What applications need from the network.

2. How to regulate the traffic that enters the network.

3. How to reserve resources at routers to guarantee performance.

4. Whether the network can safely accept more traffic

Contd…

Traffic management is concerned with delivery of QoS to the end user and with efficient use of network resources. Based on traffic granularity, we can classify traffic management into three levels.

1. packet level
2. flow level
3. flow-aggregated level.

1.Application Requirements

A stream of packets from a source to a destination is called a **flow** .

A flow might be all the packets of a connection in a connection-oriented network, or all the packets sent from one process to another process in a connectionless network.

The needs of each flow can be characterized by four primary parameters:

-bandwidth,

-delay,

-Jitter

-loss.

All Together, these determine the QoS (Quality of Service) the flow requires

common applications and the stringency of their network requirements are listed in Fig.

Application	Bandwidth	Delay	Jitter	Loss
Email	Low	Low	Low	Medium
File sharing	High	Low	Low	Medium
Web access	Medium	Medium	Low	Medium
Remote login	Low	Medium	Medium	Medium
Audio on demand	Low	Low	High	Low
Video on demand	High	Low	High	Low
Telephony	Low	High	High	Low
Videoconferencing	High	High	High	Low

To accommodate a variety of applications, networks may support different categories of QoS.

An influential example comes from ATM networks, which were once part of a grand vision for networking but have since become a niche technology.

They support:

1. Constant bit rate (e.g., telephony).
2. Real-time variable bit rate (e.g., compressed videoconferencing).
3. Non-real-time variable bit rate (e.g., watching a movie on demand).
4. Available bit rate (e.g., file transfer)

2.Traffic Shaping

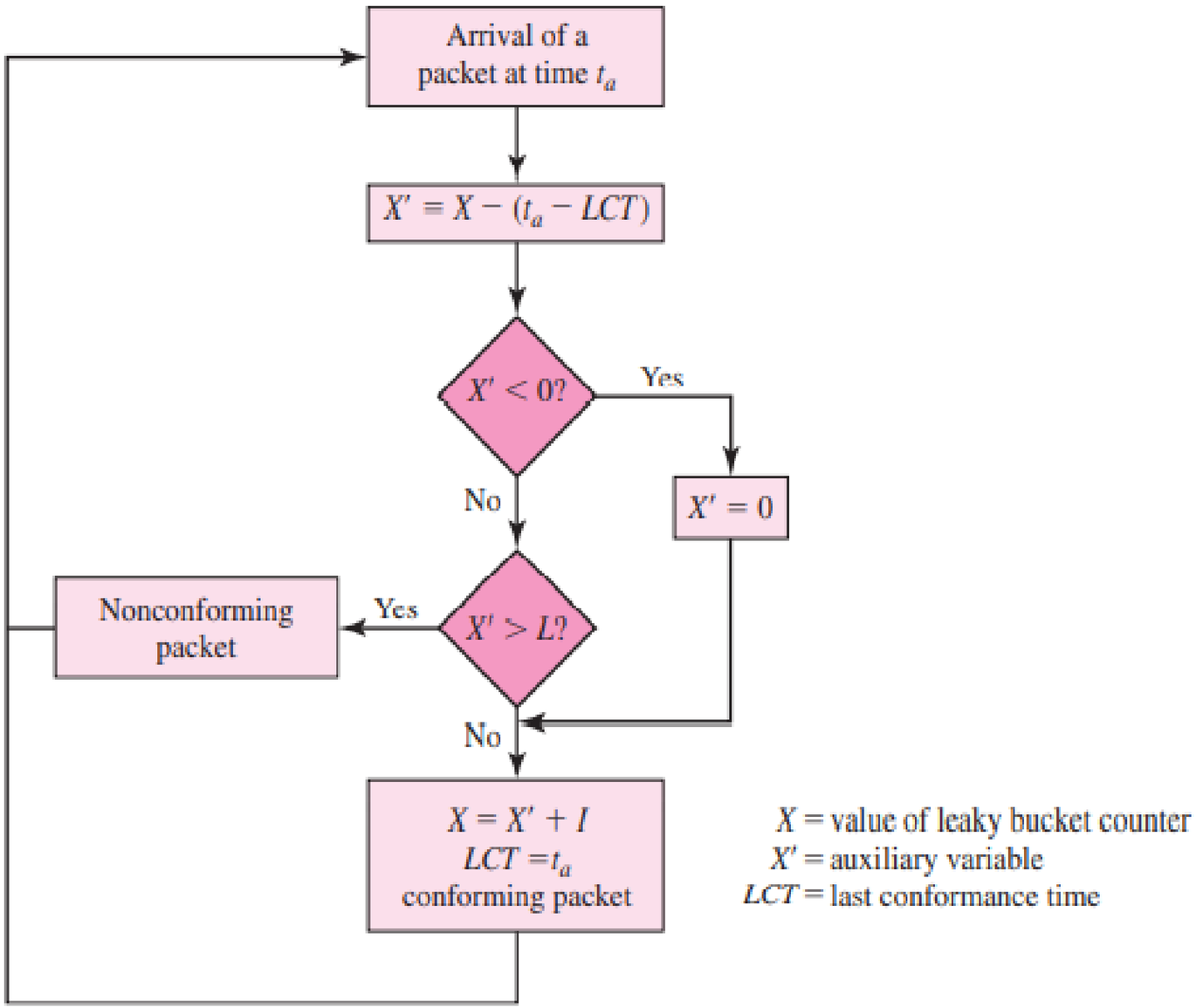
- Before the network can make QoS guarantees, it must know what traffic is being guaranteed.
- Bursts of traffic are more difficult to handle than constant-rate traffic because they can fill buffers and cause packets to be lost.
- Traffic shaping is a technique for regulating the average rate and burstiness of a flow of data that enters the network
- The goal is to allow applications to transmit a wide variety of traffic that suits their needs, including some bursts, yet have a simple and

- **Traffic shaping** reduces congestion and thus helps the network live up to its promise.
- The process of monitoring and enforcing the traffic flow is called **policing.**
- Shaping and policing are not so important for peer-to-peer and other transfers that will consume any and all available bandwidth,
- but they are of great importance for real-time data, such as audio and video connections, which have stringent quality-of-service requirements

Leaky and Token Buckets

- Most implementations of a policing device are based on the concept of a **leaky bucket**. Imagine the traffic flow to a policing device as water being poured into a bucket that has a hole at the bottom
- A new portion of water (that is, packet) is said to be **conforming** if the bucket does not overflow when the water is poured in the bucket
- When the bucket is full, the new portion of water is said to be **nonconforming** and the water can be discarded

Figure, shows the leaky bucket algorithm that can be used to police



- At the arrival of the first packet, the content of the bucket X is set to zero and the last conforming time (LCT) is set to the arrival time of the first packet.
- The depth of the bucket is L + I, where L typically depends on the traffic burstiness. At the arrival of the kth packet, the auxiliary variable X records the difference between the bucket.
- content at the arrival of the last conforming packet and the interarrival time between the last conforming packet and the kth packet.
- The auxiliary variable is constrained to be nonnegative. If the auxiliary variable is greater than L, the packet is considered nonconforming.
- Otherwise, the packet is conforming. The bucket content and the arrival time of the packet are then updated

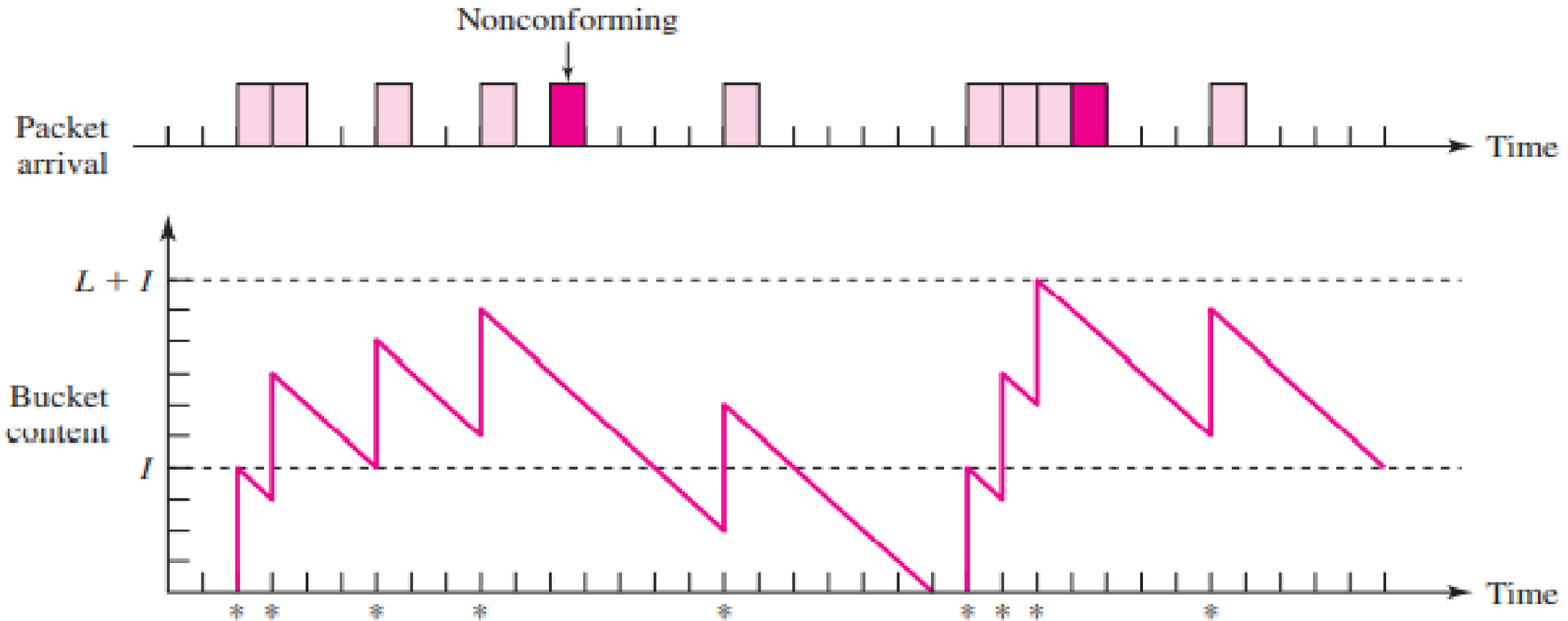
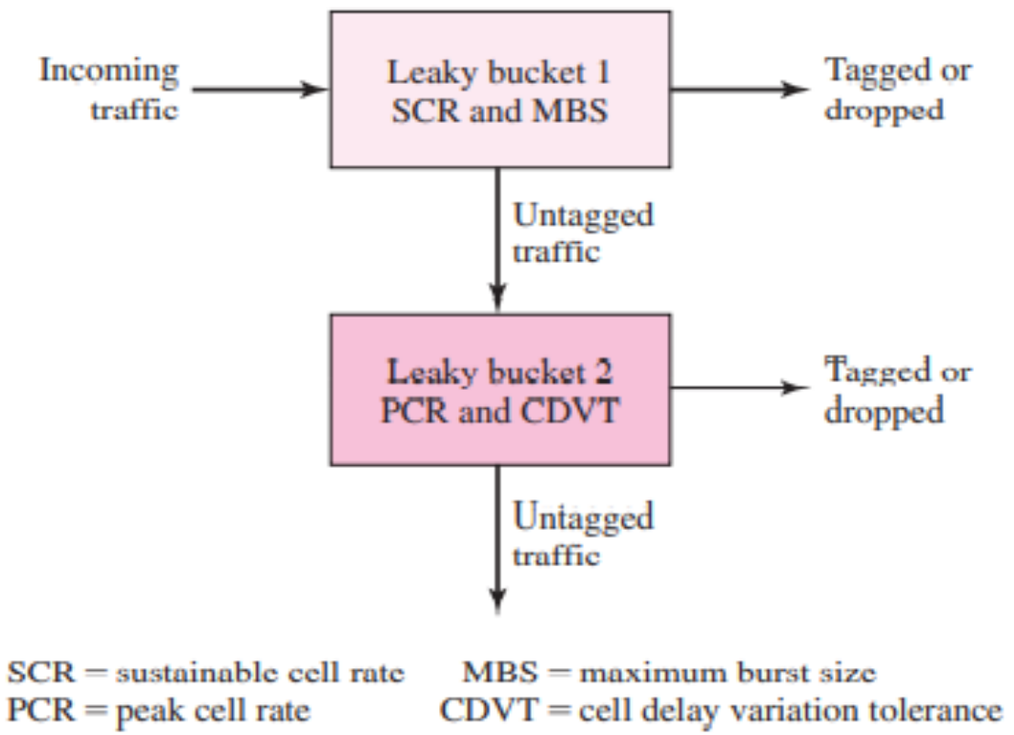


Fig:Behaviour of leaky bucket

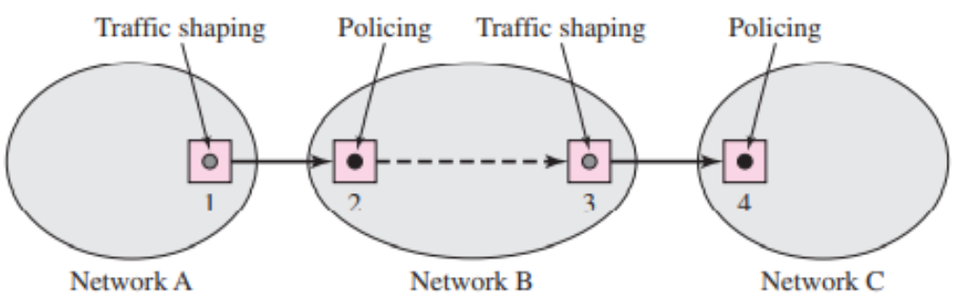
- The value of I is four packet times, and the value of L is six packet times. The arrival of the first packet increases the bucket content by four (packet times).
- At the second arrival the content has decreased to three, but four more are added to the bucket resulting in a total of seven.
- The fifth packet is declared as nonconforming since it would increase the content to 11, which would exceed L + I (10). Packets 7, 8, 9, and 10 arrive back to back after the bucket becomes empty. Packets 7, 8, and 9 are conforming, and the last one is nonconforming.
- If the peak rate is one packet/packet time, then the maximum number of packets that can be transmitted at the peak rate, called the maximum burst

A combination of leaky buckets can be used to police multiple traffic parameters (for example, the peak rate and the sustainable rate). In this situation dual leaky buckets such as the one shown in Figure 7.56 can be used.

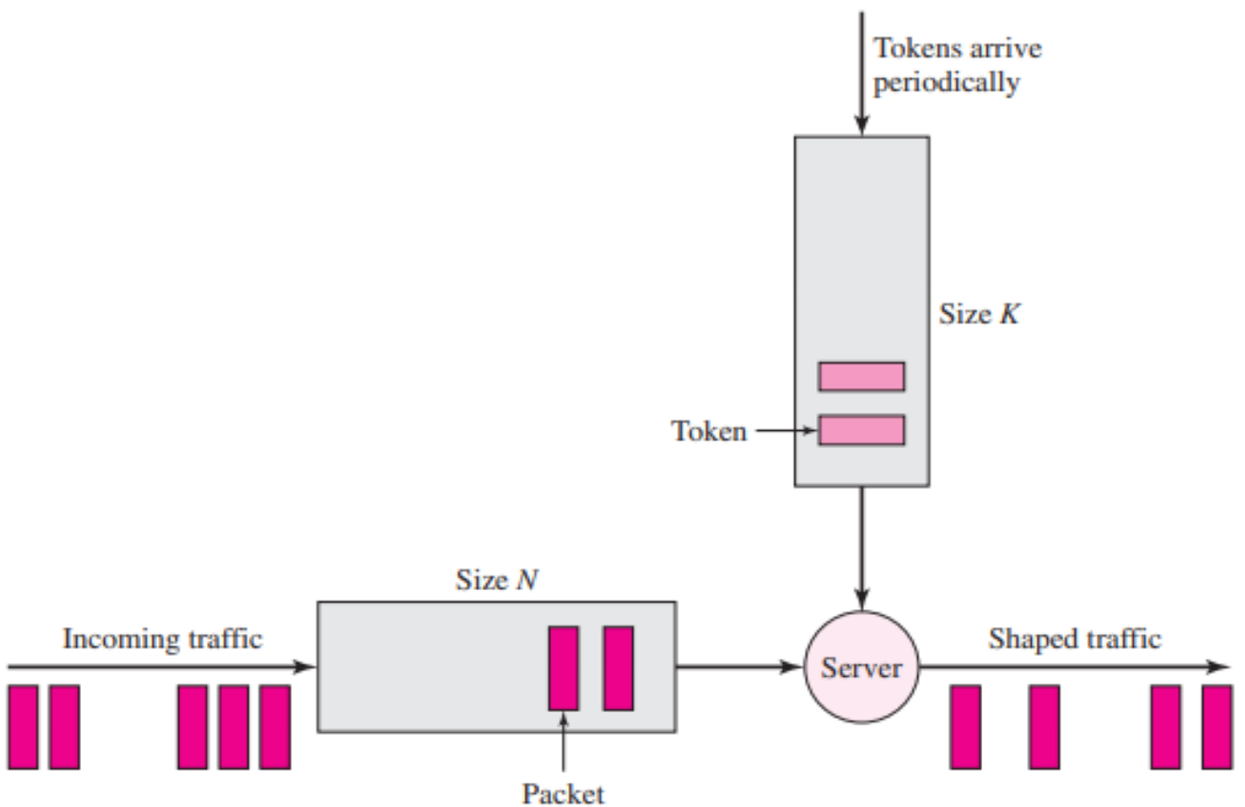


Traffic shaping

- Traffic shaping refers to the process of altering a traffic flow to ensure conformance
- As shown in Figure , typically the traffic shaping device is located at the node just before the traffic flow leaves a network (egress node) while the policing device is located at the node that receives the traffic flow from another network (ingress node).



- A more realistic shaper, called the **token bucket traffic shaper**, regulates only the packets that are not conforming. Packets that are deemed conforming are passed through without further delay.



1. Packet level

- The packet level, which is mainly concerned with packet queueing and packet scheduling at switches, routers, and multiplexers to provide differentiated treatment for packets belonging to different QoS classes.
- Packet-level traffic management operates in the smallest time scale on the order of packet transmission time.
- A packet switch also contains buffers to ensure that coincident packet arrivals are not lost. Thus the path traversed by a packet through a network can be modeled as a sequence of queueing systems as shown

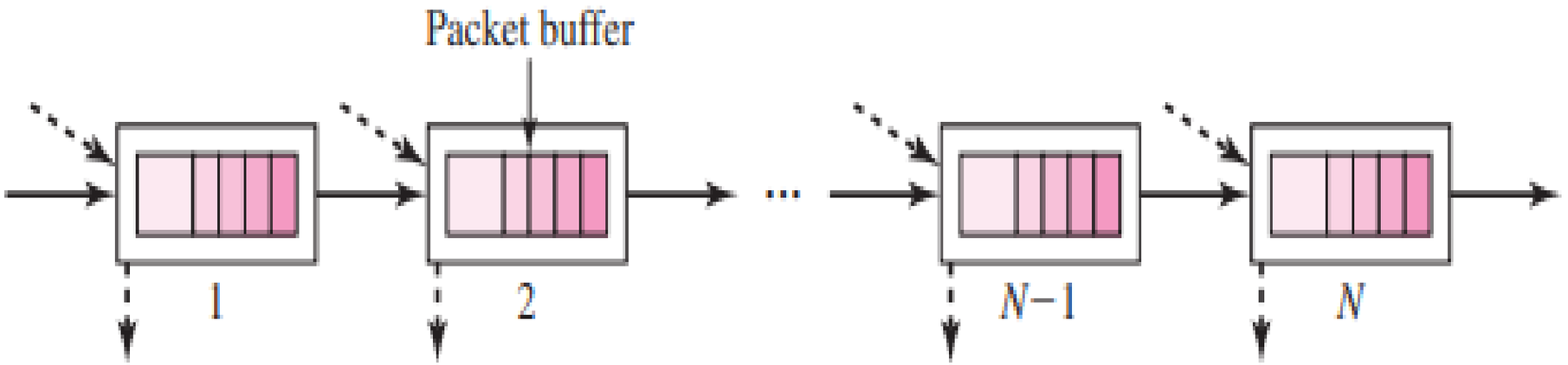


Fig:The end-to-end QoS of a packet along a path traversing N queueing systems

- The dashed arrows show packets from other flows that "interfere" with the packet of interest in the sense of contending for buffers and transmission along the path. We also note that these interfering flows may enter at one node and depart at some later node, since in general they belong to different source-destination pairs and follow different paths through the network.
- The performance experienced by a packet along the path is the accumulation of the performance experienced at the N queueing systems.

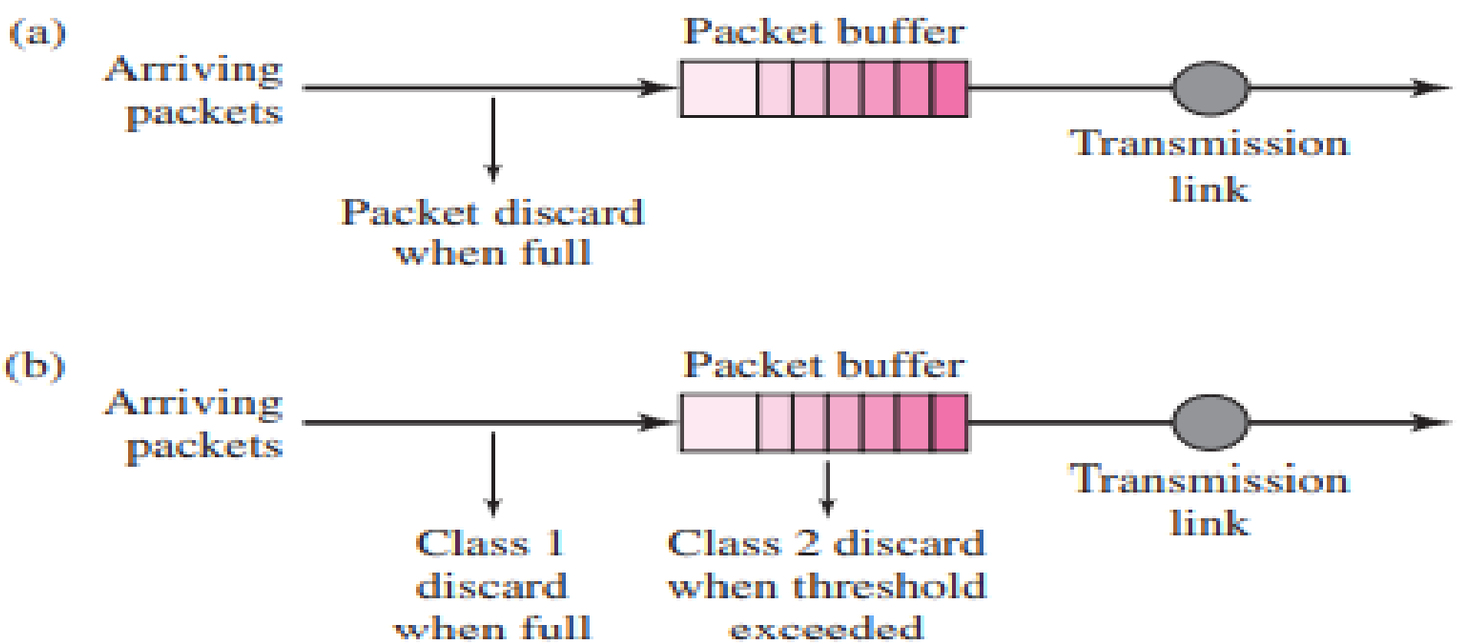
For example, the total end to-end delay is the sum of the individual delays experienced at each system. the average end-to-end delay is the sum of the individual average delays.

- The jitter experienced by packets is also of interest. The jitter measures the variability in the packet delays and is typically measured in terms of the difference of the minimum delay and some maximum value of delay.
- Packet loss performance is also of interest. Packet loss occurs when a packet arrives at a queueing system that has no more buffers available.
- Causes of packet loss include surges in packet arrivals to a buffer and increased transmission time due to long packets or congestion downstream.
- The end-to-end probability of packet loss is the probability of packet loss somewhere along the path and is bounded above by the sum of the packet loss probabilities at each system.

- Packet-switching networks are called upon to support a wide range of services with diverse QoS requirements.
- To meet the QoS requirements of multiple services, a queueing system must implement strategies for controlling the transmission bit rates that are provided to the various information flows **called queue scheduling**
- For managing how packets are placed in the queueing system called **queue management**.

1.FIFO and Priority Queues

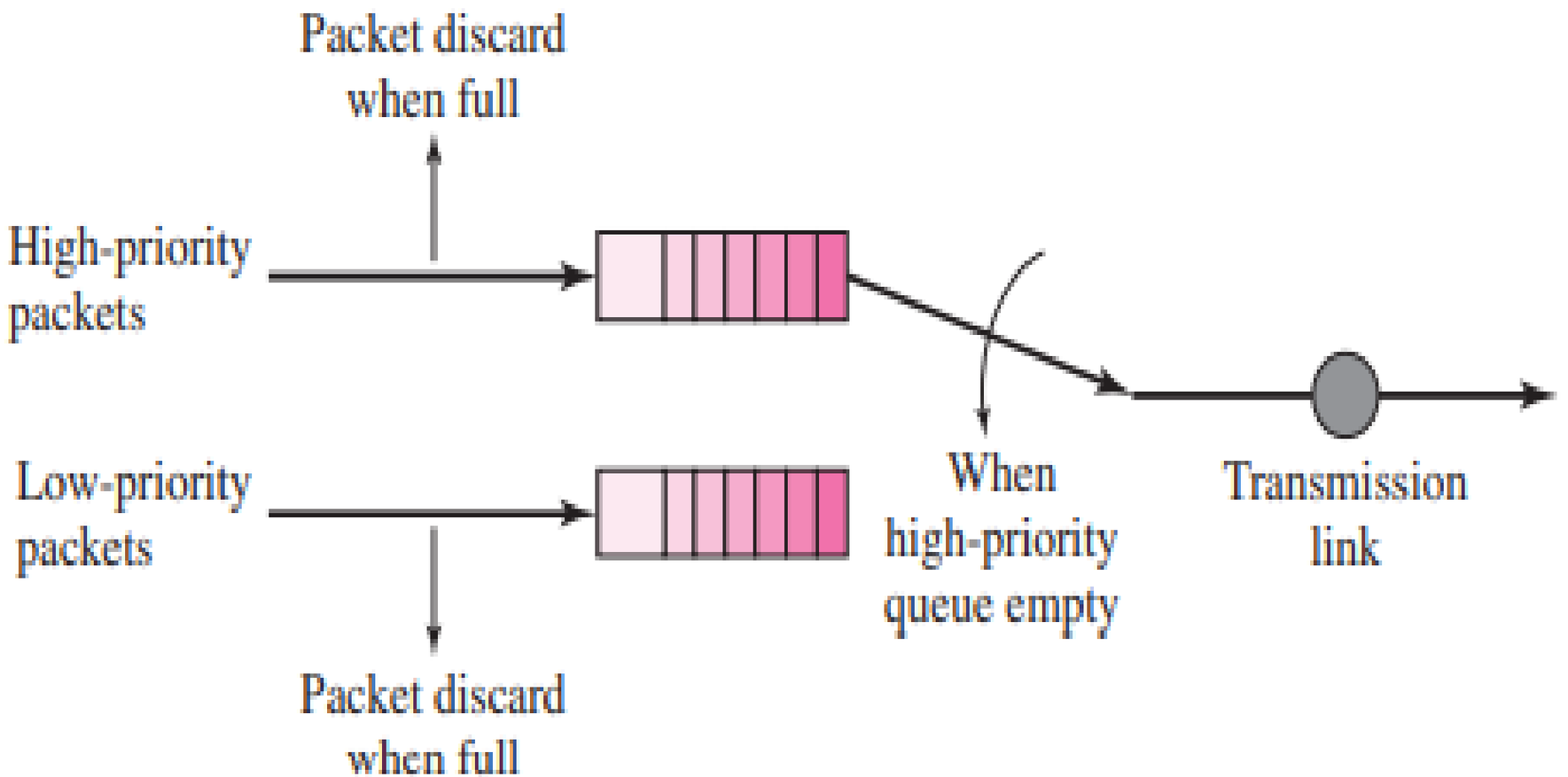
The simplest approach to queue scheduling involves **first-in, first-out** (FIFO) queuing where packets are transmitted in order of their arrival



- Packets are discarded when they arrive at a full buffer. The delay and loss experienced by packets in a FIFO queueing system depend on the packet interarrival times and on the packet lengths.
- As interarrivals become more bursty or packet lengths more variable, packets will tend to bunch up and queues will then build up, causing performance to deteriorate.
- Because FIFO queueing treats all packets in the same manner, it is not possible to provide different information flows with different qualities of service.
- FIFO queueing systems are also subject to hogging, which occurs when a user sends packets at a high rate and fills the buffers in the system, thus depriving other users of access to the buffer.

- The FIFO queue management can be modified to provide different characteristics of **packet-loss performance** to different classes of traffic.
- In Fig (b)with two classes of traffic. When the number of packets in a buffer reaches a certain threshold, arrivals of lower access priority (Class 2) are not allowed into the system.
- Arrivals of higher access priority (Class 1) are allowed as long as the buffer is not full. As a result, packets of lower access priority will experience a higher packet-loss probability

Head-of-line (HOL) priority queuing is a second queue scheduling approach that involves defining a number of priority classes.



For example, packets requiring **low delay** may be assigned a **high priority**, whereas packets that are not urgent may be assigned a **lower priority.**

- The size of the buffers for the different priority classes can be selected to meet different loss probability requirements.
- While priority queuing does provide different levels of service to the different classes, it still has shortcomings.

For example, it does not allow for providing some degree of guaranteed access to transmission bandwidth to the lower priority classes.

- Another problem is that it does not discriminate among users of the same priority. Fairness problems can arise here when a certain user hogs the bandwidth by sending an excessive number of packets.

2.Queue scheduling

A third approach is queue scheduling involves sorting packets in the buffer according to a **priority tag** that reflects the urgency with which each packet needs to be transmitted. This system is very flexible because the method for Defining priority is open and can even be defined dynamically

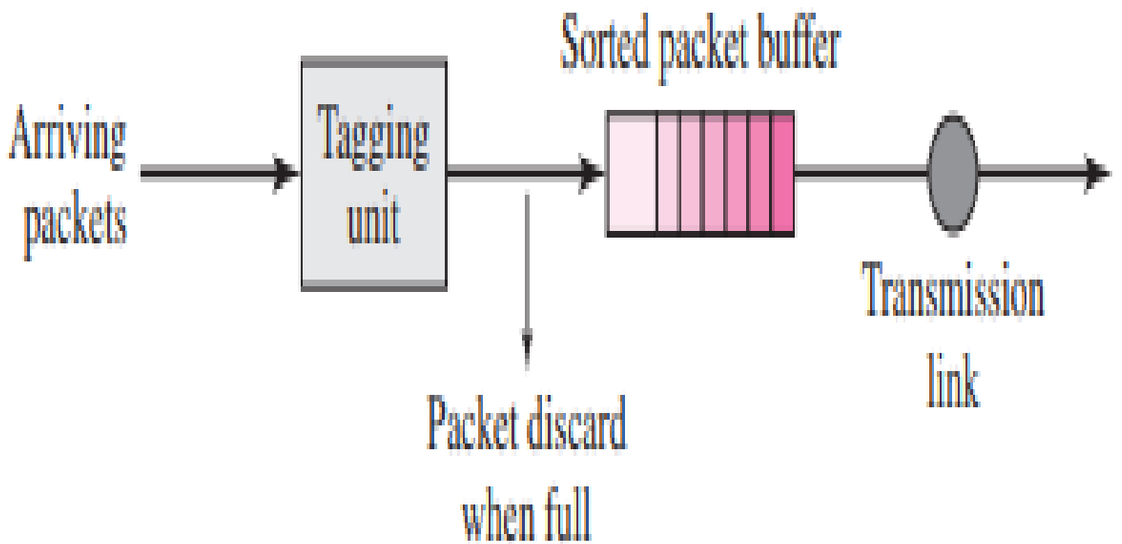


Fig:Sorting packets according to priority tag.

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For example, the priority tag could consist of a priority class followed by the arrival time of a packet.

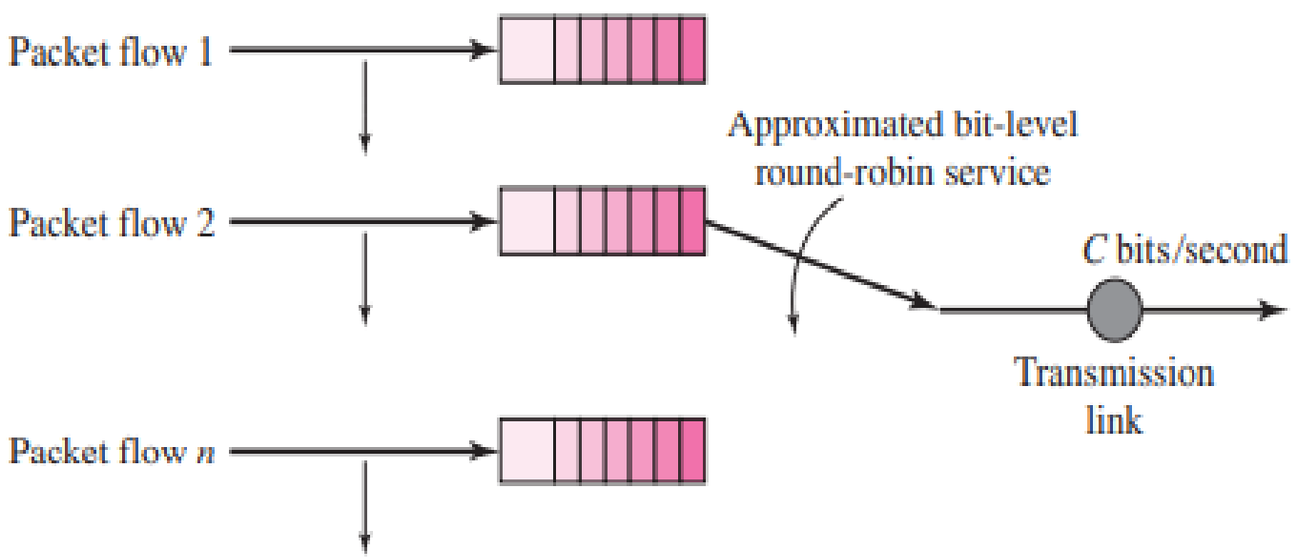
- The priority tag corresponds to a due date.
- Packets requiring smaller delays are assigned earlier due dates and are transmitted sooner.
- Packets without a delay requirement get indefinite or very long due dates, and are transmitted after all time-critical packets have been transmitted.

A third important example that can be implemented by the approach is fair queueing and weighted fair queueing,

3.Fair Queueing

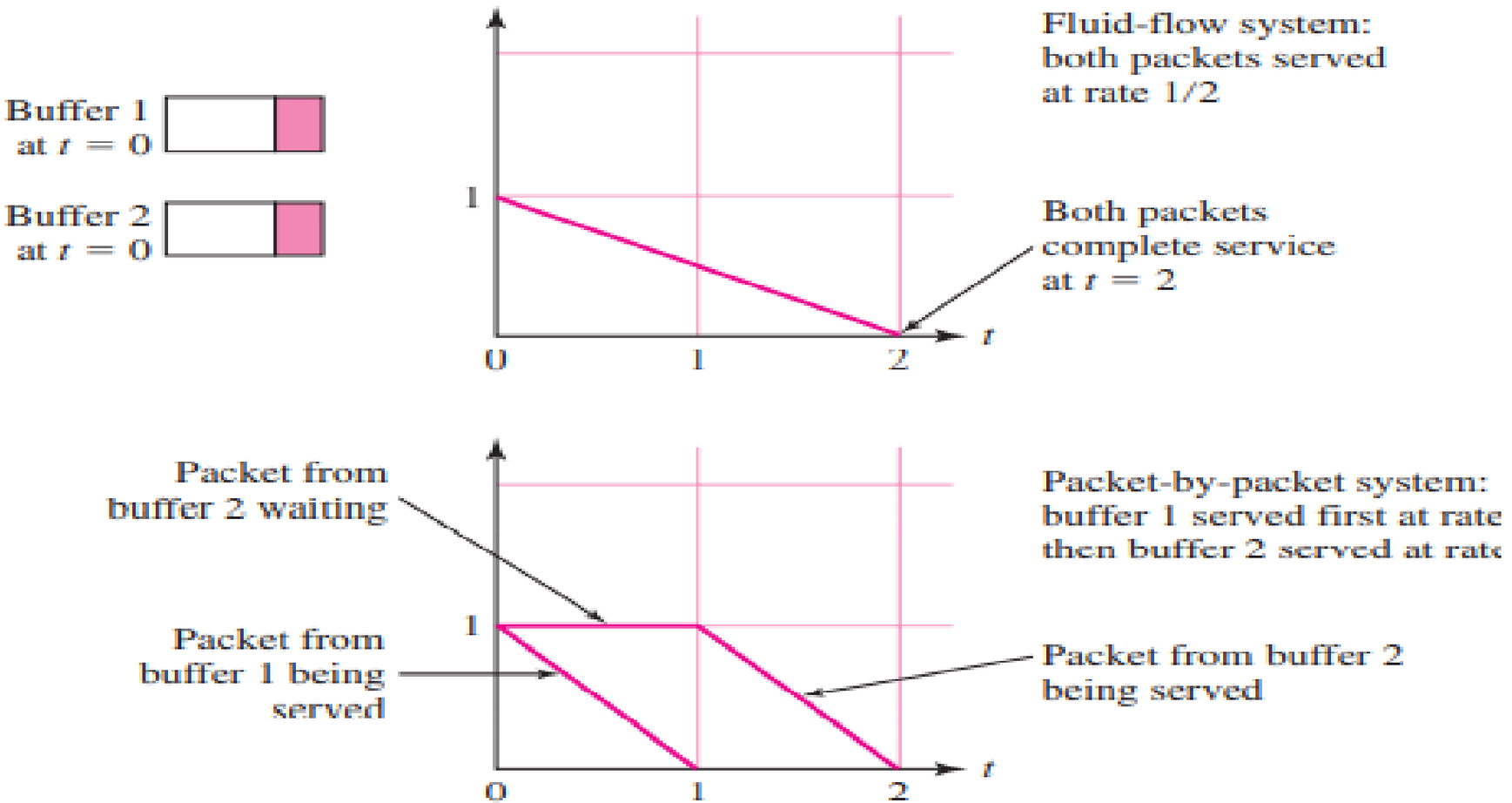
- Fair queueing attempts to provide **equitable access** to transmission bandwidth. Each user flow has its own logical buffer.
- In an ideal system the transmission bandwidth, say, C bits/second, is divided equally among the buffers that have packets to transmit.
- The contents of each buffer can then be viewed as a fluid that is drained continuously.
- The size of the buffer for each user flow can be selected to meet specific loss probability requirements so that the cells or packets of a given user will be discarded when that buffer is full.

- In the ideal fluid flow situation, the transmission bandwidth is divided equally among all nonempty buffers.
- Thus if the total number of flows in the system is **n** and the transmission capacity is **C**, then each flow is guaranteed at least **C/n bits/second**.
- In general, the actual transmission rate experienced may be higher because buffers will be empty from time to time, so a share larger than **C/n bits/second** is received at those times.



- In practice, dividing the transmission capacity exactly equally is not possible. As shown in Figure one approach could be to service each nonempty buffer one bit at a time in **round-robin fashion**.

Fluid-flow and packet-by-packet fair queueing (two packets of equal length)



- Figure illustrates the differences between ideal or “fluid flow” and packet-by packet fair queueing.

- The figure assumes that buffer 1 and buffer 2 each has a single L-bit packet to transmit at t = 0 and that no subsequent packets arrive.

- Assuming a capacity of C = L bits/second = 1 packet/second, the fluid-flow system transmits each packet at a rate of 1/2 and therefore completes the transmission of both packets exactly at time t = 2 seconds.

- The bit-by-bit system (not shown in the figure) would begin by transmitting one bit from buffer 1, followed by one bit from buffer 2, and so on.

- After the first bit each subsequent bit from buffer 1 would require 2/L seconds to transmit.

- Therefore, the transmission of the packet from buffer 1 would be completed after $1 + 2(L-1) = 2L-1$ bit-transmission times, which equals $2-1/L$ seconds.

The packet from buffer 2 is completed at time 2 seconds.

- The transmission of both packets in the fluid-flow system would be completed at time $2L/L = 2$ seconds.

- On the other hand, the **packet-by-packet fair-queueing** system transmits the packet from buffer 1 first and then transmits the packet from buffer 2,

- so the packet completion times are 1 and 2 seconds. In this case the first packet

For example, if the packets of **one flow are twice the size of packets** in another flow, then in the long run the first flow will obtain twice the bandwidth of the second flow.

- A better approach is to transmit packets from the user buffers so that the packet completion times approximate those of a fluid-flow fair-queueing system.
- Each time a packet arrives at a user buffer, the completion time of the packet is derived from a fluid-flow fair-queueing system.
- This number is used as a **finish tag** for the packet. Each time the transmission of a packet is completed, the next packet to be transmitted is the one with the smallest finish tag among all of the user buffers. this system as a **packet-by-packet fair-queueing** system.

- Consider how a finish tag may be computed. Assume that there are n flows, each with its own buffer. Suppose for now that each buffer is served one bit at a time.
- Let a round consist of a cycle in which all n buffers are offered service as shown in Figure .
- The number of rounds in a time period is the number of opportunities that each buffer has had to transmit a bit

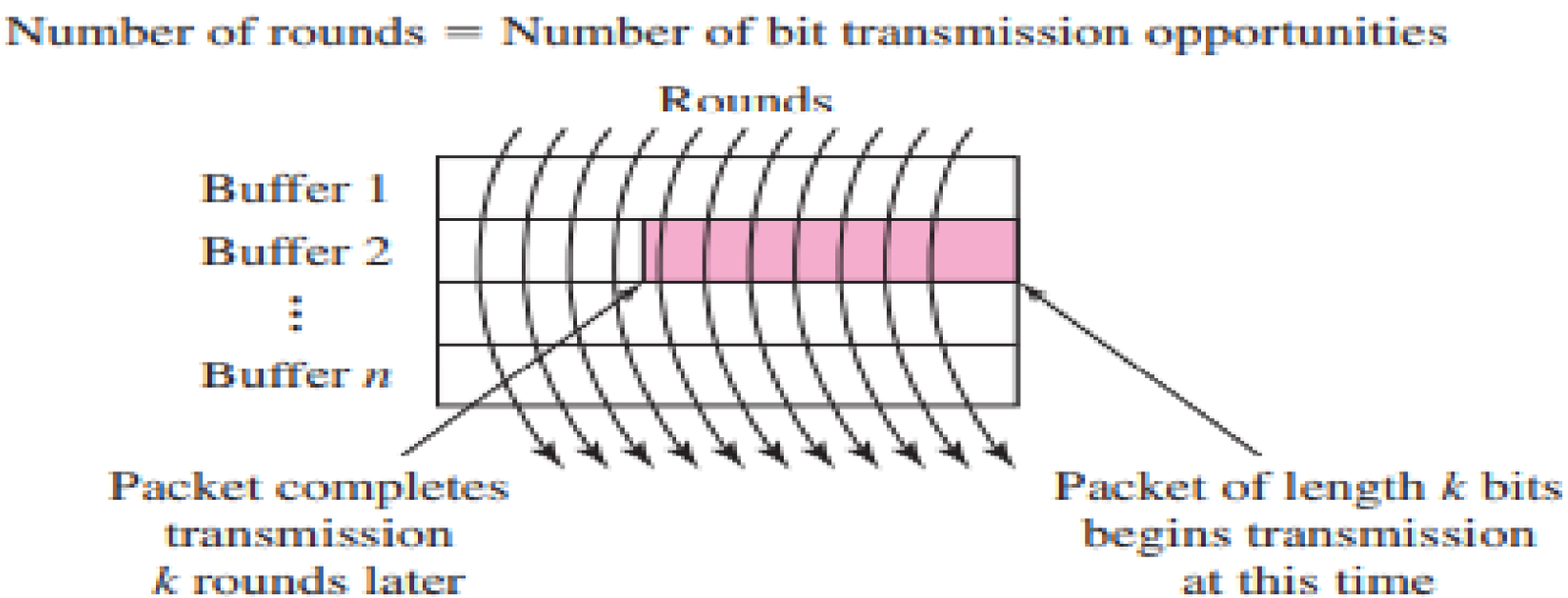


Fig:Computing the finishing time in packet-by-packet fair queuein

The actual duration of a given round is proportional to the actual number of buffers nactive(t) that have information to transmit

suppose that the buffers are served as in a fluid-flow system. Also suppose that the system is started at t = 0.

Let R(t) be the number of the rounds at time t, that is, the number of cycles of service to all n buffers.

Each time R(t) reaches a new integer value marks an instant at which all the buffers have been given an equal number of opportunities to transmit a bit.

Because of the fluid-flow assumption,

R(t) is a continuous function that increases at a rate that is inversely proportional to the number of active buffers; that is, d

R(t)/dt = C/n active(t)----------(1)

- Let us see how we can calculate the finish tags to approximate fluid-flow fair queueing.
- Suppose that the kth packet from flow i arrives at an empty buffer at time and suppose that the packet has t_k^i
- length P(i, k). This packet will complete its transmission when P(i, k) rounds have elapsed, one round for each bit in the packet.
- Therefore, the packet completion time will be the value of time t∗ when the R(t∗) reaches the value

$$F(i,k) = R(t_k^i) + P(i,k)$$

The two preceding equations can be combined into the following compact equation The two preceding equations can be combined into the following compact equation

$$F(i,k) = F(i,k-1) + P(i,k)$$

The two preceding equations can be combined into the following compact equation

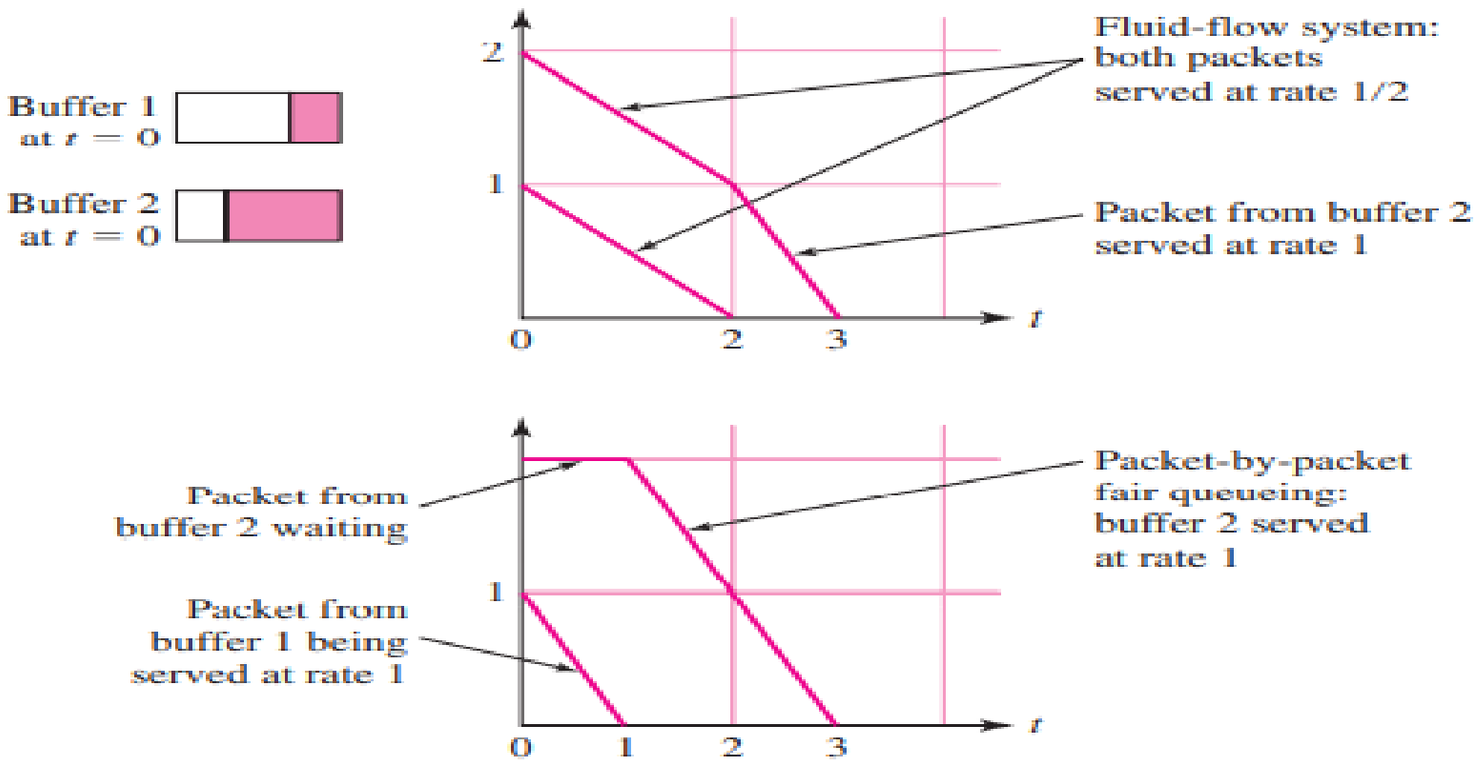
$$F(i,k) = \max\{F(i,k-1), R(t_k^i)\} + P(i,k) \quad \text{for fair queueing.}$$

Example, suppose that at time t = 0 buffer 1 has one packet of length one unit and buffer 2 has one packet of length two units. A fluid-flow system services each buffer at rate 1/2 as long as both buffers remain nonempty.

-finish tag of the packet of buffer 1 is F(1, 1) = R(0) + 1 = 1.

-The finish tag of the packet from buffer 2 is F(2, 1) = R(0) + 2 = 2.

-the system will service buffer 1 first. Thus the packet of buffer 1 completes its transmissions at time t = 1 and the packet of buffer 2 completes its transmissions at t =3



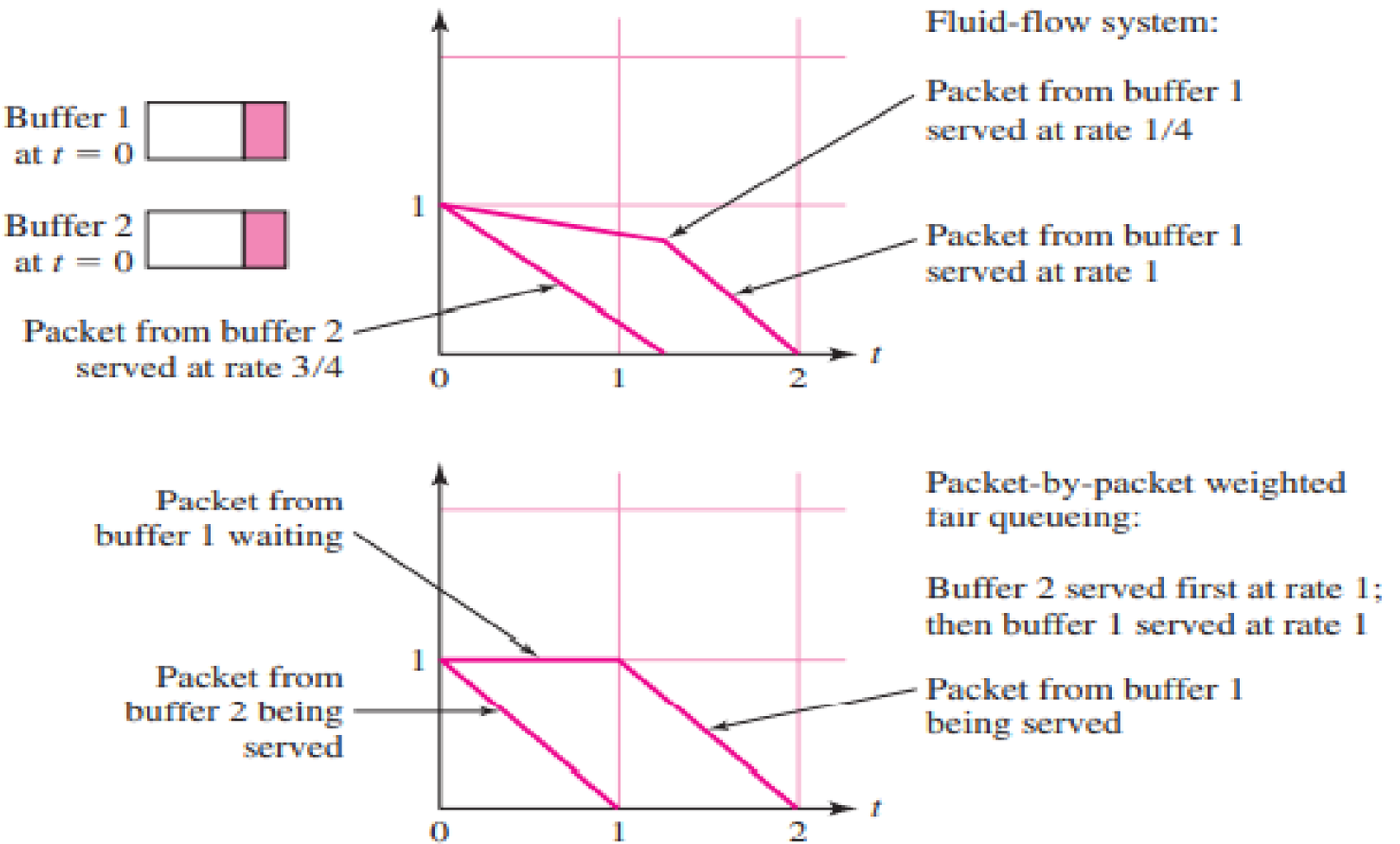
Weighted Fair Queueing

Weighted fair queueing addresses the situation in which different users have different requirements.

As before, each user flow has its own buffer, but each user flow also has a weight that determines its relative share of the bandwidth.

Thus if buffer 1 has weight 1 and buffer 2 has weight 3, then when both buffers are nonempty, buffer 1 will receive $1/(1 + 3) = 1/4$ of the bandwidth and buffer 2 will receive 3/4 of the bandwidth.

Fluid-flow and packet-by-packet weighted fair queueing



- Fig shows the completion times for the fluid-flow case where both buffers have a one-unit length packet at time t = 0.
- For the fluid-flow system, the transmission of the packet from buffer 2 is completed at time t = 4/3, and the packet from buffer 1 is completed at t = 2.
- The bit-by-bit approximation to weighted fair queueing would operate by allotting each buffer a different number of bits/round. In the preceding example, buffer 1 would receive 1 bit/round and buffer 2 would receive 3 bits/round.

Weighted fair-queueing systems are a means for providing QoS guarantees. Suppose that a given user flow has weight wi and suppose that the sum of the weights of all the user flows is W. In the worst case when all the user buffers are nonempty, the given user flow will receive a fraction wi/W of the bandwidth C.

$$F(i,k) = \max\{F(i,k-1), R(t_k^i)\} + P(i,k)/w_i \quad \text{for weighted fair queueing.}$$

When other user buffers are empty, the given user flow will receive a greater share. Thus the user is guaranteed a minimum long-term bandwidth of at least (wi/W)C bits/second.

Module-2: Outline

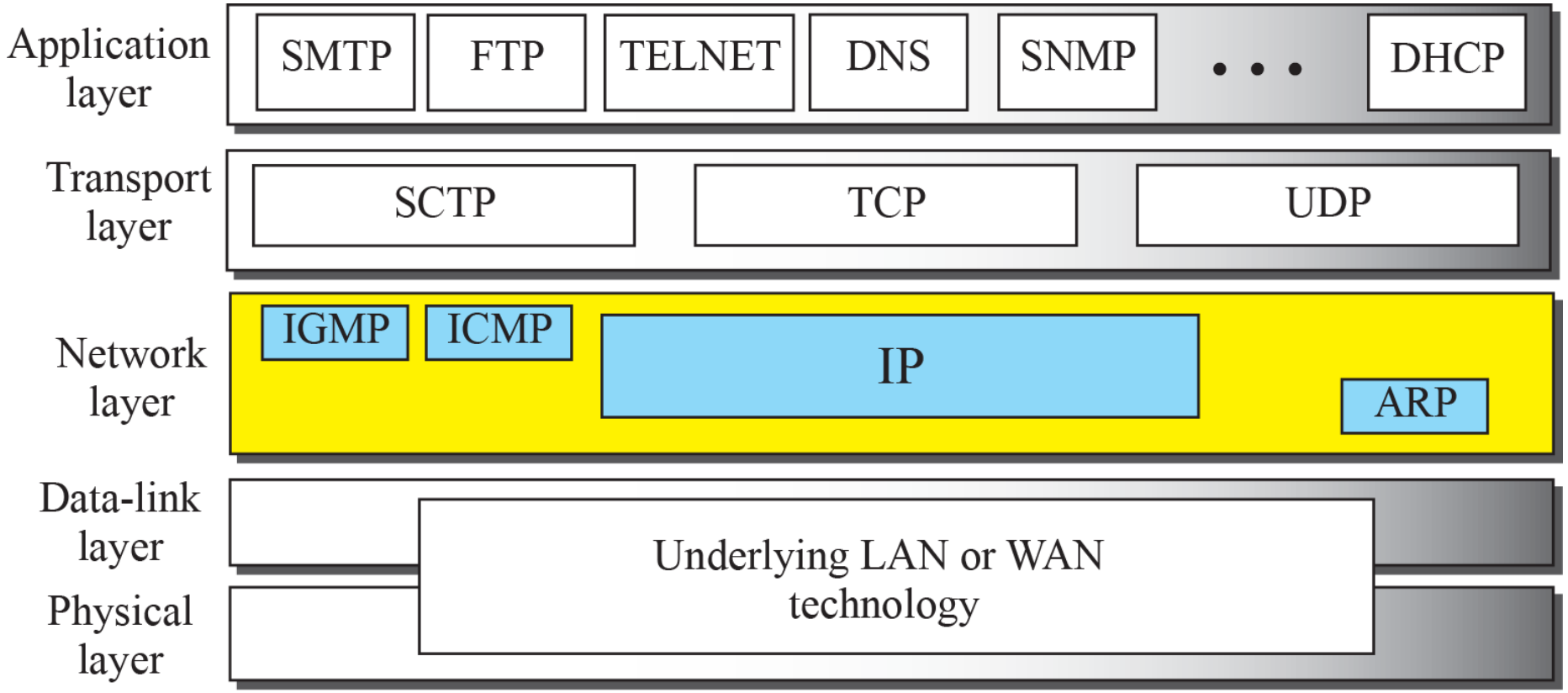
4.4.1 IPv4

4.4.2 ICMPv4

NETWORK-LAYER PROTOCOLS

- The network layer in TCP/IP protocol is responsible for host-to-host delivery of data grams.
- It provides services to Transport Layer and receives services from DLL.
- The main protocol, IPv4, is responsible for packetizing, forwarding, and delivery of a packet.
- The ICMPv4 is auxiliary protocol helps IPv4 to handle some errors that may occur in delivery.
- The IGMP is used to help IPv4 in multicasting.
- ARP is used in address mapping.

Position of IP and other network-layer protocols in TCP/IP protocol suite



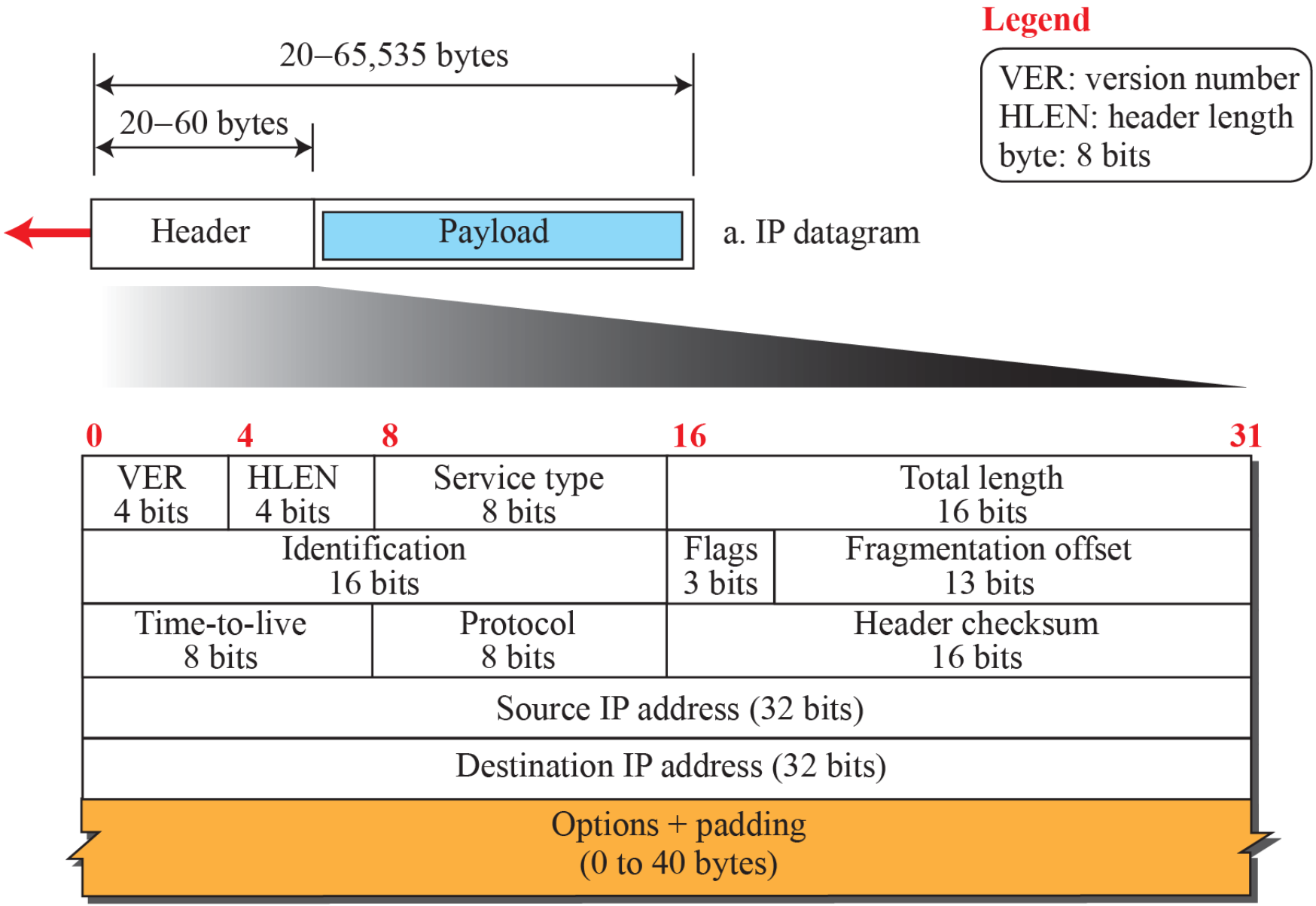
Datagram Format

- Packets used by the IP are called data grams.
- Figure shows the IPv4 datagram format.
- A datagram is a variable-length packet consisting of two parts:

1. Header
2. Payload (data).

- The header is 20 to 60 bytes in length and contains information essential to routing and delivery.
- It is customary in TCP/IP to show the header in 4-byte sections..

IP datagram


Legend
VER: version number
HLEN: header length
byte: 8 bits
20–65,535 bytes
20–60 bytes
Header
Payload
a. IP datagram
0
4
8
16
31
VER
4 bits
HLEN
4 bits
Service type
8 bits
Total length
16 bits
Identification
16 bits
Flags
3 bits
Fragmentation offset
13 bits
Time-to-live
8 bits
Protocol
8 bits
Header checksum
16 bits
Source IP address (32 bits)
Destination IP address (32 bits)
Options + padding
(0 to 40 bytes)
b. Header format

IPV4 data gram format

- A data gram is a variable length packet consisting of two parts: Header and data.
- A header is 20 to 60 bytes in length and contains information essential to routing and delivery.

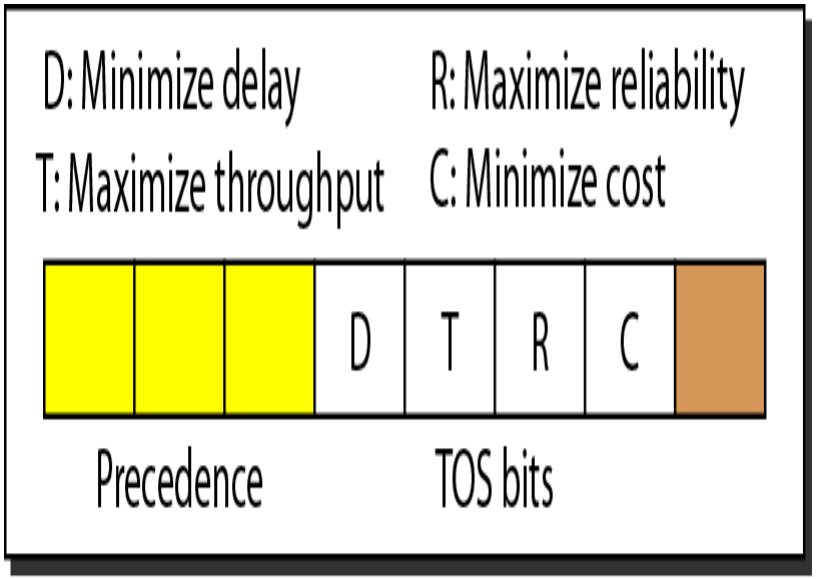
1. Version(VER): 4-bit field defines version of IPV4.This field tells IPV4 s/w running in the processing machine that the data gram has the format of version 4.
2. Header Length(HLEN): Defines the total length of the data gram header in 4 byte words.

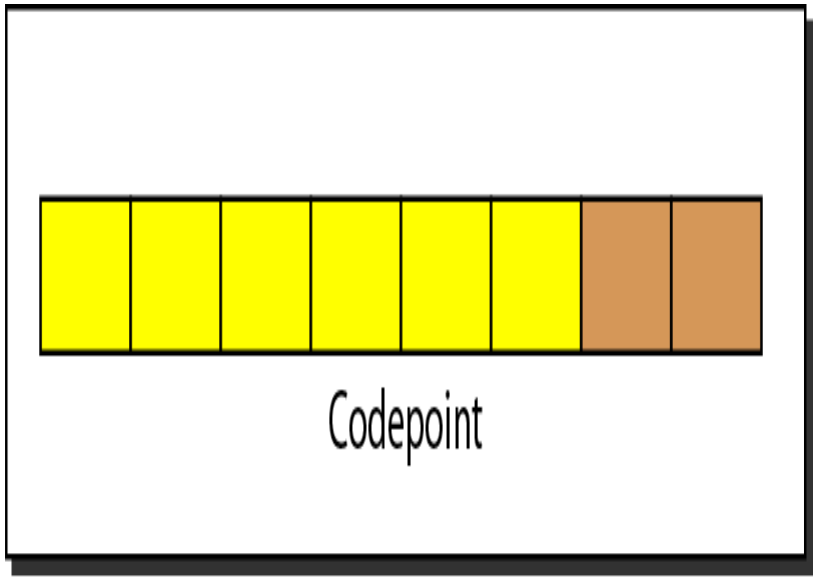
   The length of the header field is variable(20 to 60 bytes).When this data grams is received, receiver should know when header stops & when data encapsulated starts.

   Total length is divided by 4 and the value is inserted in the field. Receiver multiply the value of this field by 4 to find total length.

IPV4 fields

Services: In the original design of IP header, this field was called TOS. In 1990s ,IETF has changed the name of this field to differentiated services.





IPV4 fields

Total Length: 16 bit field defines the total length(header+ data) of IPV4 data grams in bytes.

Length of data= Total Length – (HLEN*4).

- Since field length is 16 bits, total length of IPV4 is limited to 65,535 bytes(2^{16}) bytes, of which 20 to 60 bytes are header and rest is data from upper layer.
- Identification: Used for fragmentation.
- Flags: Used in fragmentation.
- Fragmentation offset: used in fragmentation.
- Time to live: Data gram has limited life time in the Internet. Value in this field defines Total number of hops visited by the data gram. When the source sends the data gram, it stores number in this field.

Datagram header(Continued..)

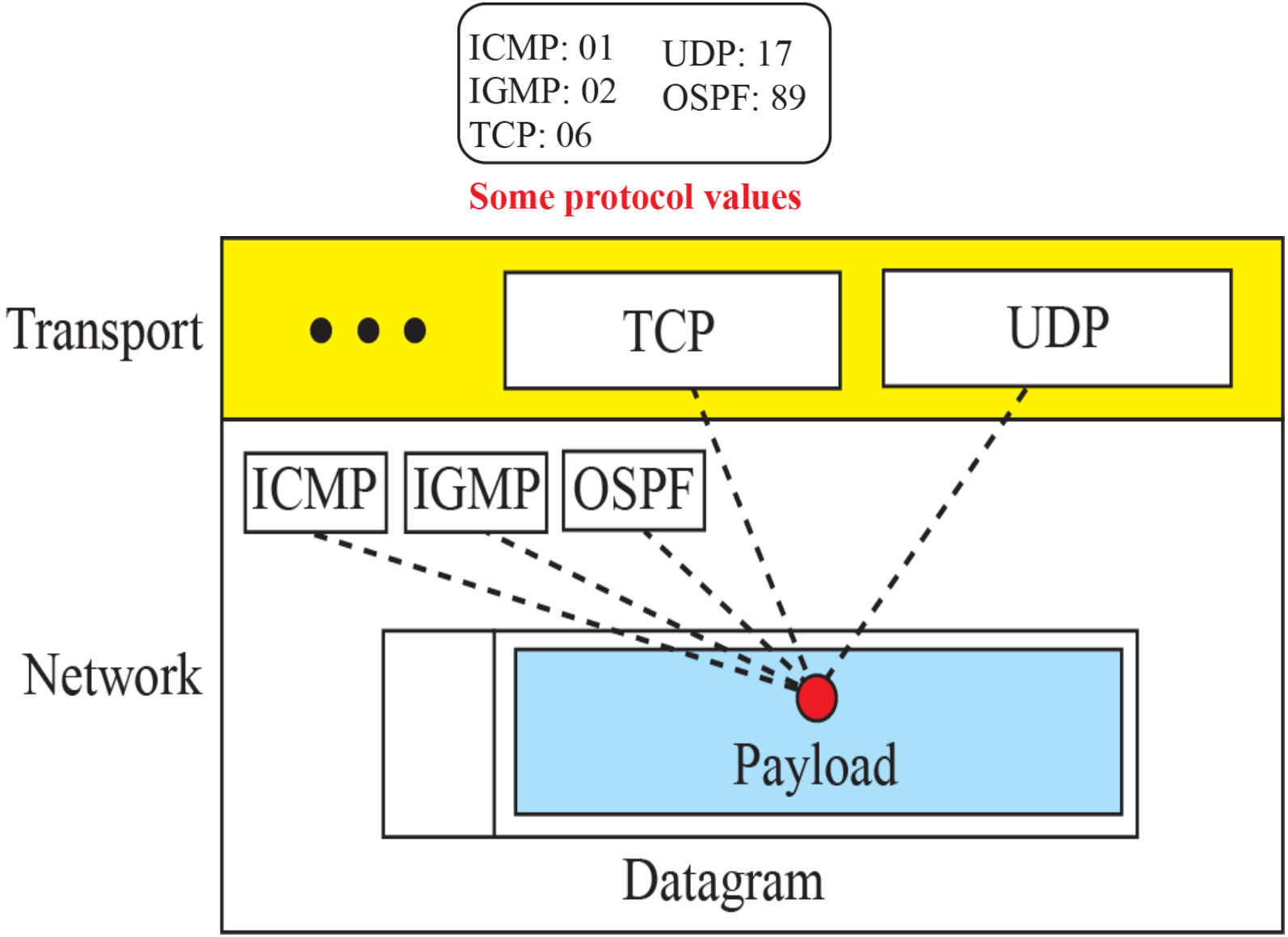
- This value will be two times maximum number of routers between any two hosts. Each time router process the data gram, decrements this value by 1. if its value is 0 , it is discarded.

Protocol field: 8-bit field defines the higher level protocol that uses the services of the IPV4 layer.

- An IPV4 datagram encapsulates data from several higher level protocols such as TCP,UDP,ICMP etc.,.
- This field specifies the final destination protocol to which the IPV4 data gram is delivered.

(i.e) IPV4 protocol carries data from different protocols, the value of this field helps the receiving n/w layer know to which protocol the data belong.

Multiplexing and de multiplexing using the value of the protocol field



IPV4 fields

Header Checksum: IP is not a reliable protocol, it does not check if the data encapsulated in the data gram is corrupted during transmission.

→ IP puts this burden of error checking on the payload to upper layer protocols TCP/UDP.

→ Datagram header is added by IP and its error checking is IP's responsibility.

Eg: 1)if datagram header is corrupted, the packet is delivered to wrong destination.

2) If protocol field is corrupted , it will be delivered to wrong protocol.

3) If fields related to fragmentation is corrupted, the datagram can not be reassembled at the destination.

→ For these reasons, IP adds header checksum field to check the header

IPV4 fields

Source and destination address: 32 bit IPV4 address of the source and destination. The value of these fields must remain unchanged as IP datagram travels from source to destination.

OPTIONS: Datagram can have up to 40 bytes of options. Options can be used for network testing & debugging.

- Existence of options in the header creates some burden on the datagram handling.
- Some options can be changed by the routers, which forces router to recalculate the header checksum.

PAYLOAD:Is the data coming from the upper layer protocols that use the services of IP.

→Pay load is the content of a package.

Options

- The header of the IPv4 datagram is made of two parts:
  a fixed part and a variable part.
- The fixed part is 20 bytes long.
- The variable part comprises the options that can be a maximum of 40 bytes (in multiples of 4-bytes) to preserve the boundary of the header.
- Not required for a datagram, but can be used for network testing and debugging.
- Option processing is required for IPV4 software.
- Options are

i)Single-byte

ii)Multiple byte options

Options(Continued..)

➢Single byte options:-

i) No operation:1byte option used as a filter between options.

ii) End of option: 1 byte option used for padding at the end of the option field.

➢Multiple –byte option:-

i. Record Route: Records the internet routers that handle the datagram. Used for management and debugging purpose.

ii. Strict source Route: Used by source to predetermine a route for datagram as it travels through the Internet.

Eg: 1.Sender can choose a route with specific type of service (minimum delay/Maximum throughput).

2. Choose a safe/reliable route(Datagram not to pass through competitor's network.)

Options(Continued..)

iii. Loose source route: Route is not rigid. Data grams can visit other routers also.

iv. Time stamp: To record the time taken by a router to process the datagram.(expressed in ms(Universal time))

→ Help users & managers to track the behavior of the routers in the Internet.

→ Possible to estimate the time it takes for a datagram to go from one router to another.

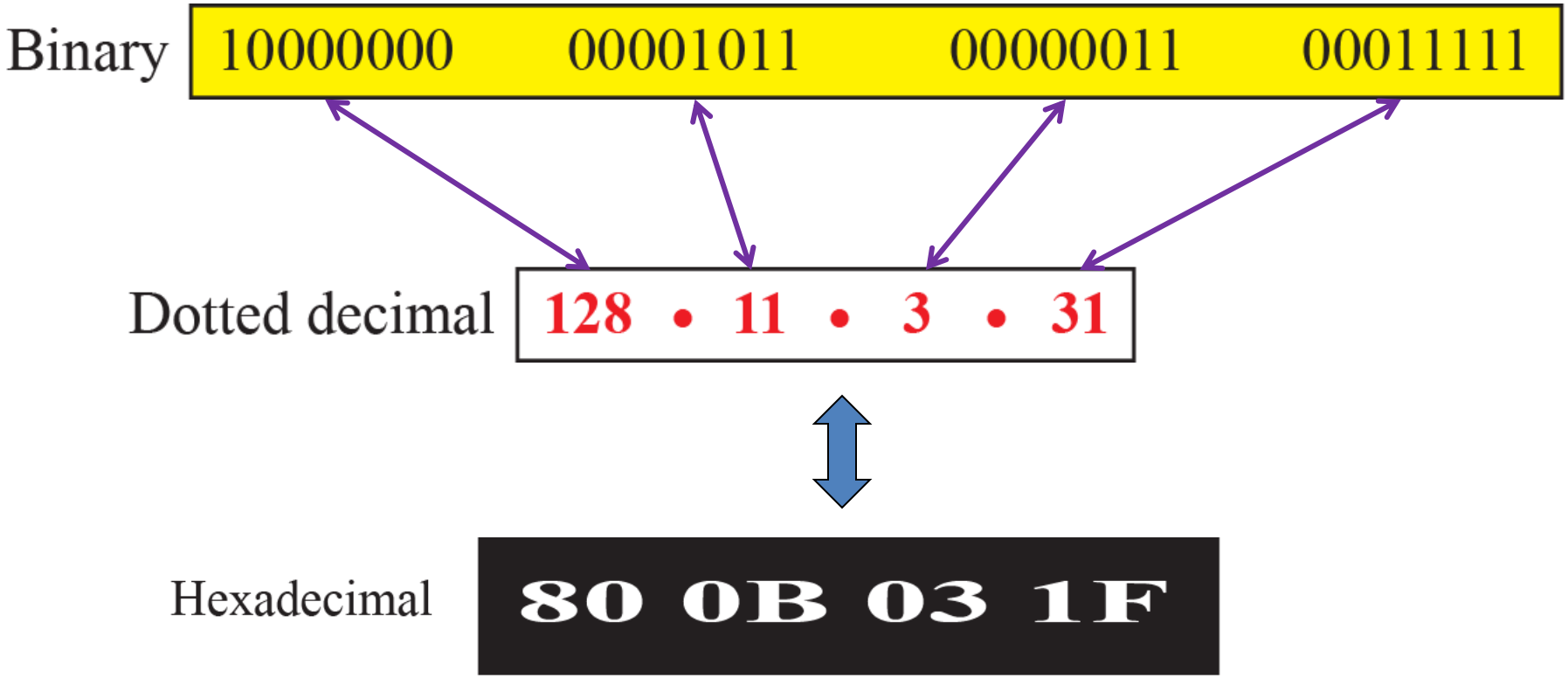
IPv4 ADDRESSES

- The identifier used in the IP layer of the TCP/IP protocol suite to identify the connection of each device to the Internet is called the Internet address or IP address.
- An IPv4 address is a 32-bit address that uniquely and universally defines the connection of a host or a router to the Internet.
- The IP address is the address of the connection, not the host or the router.

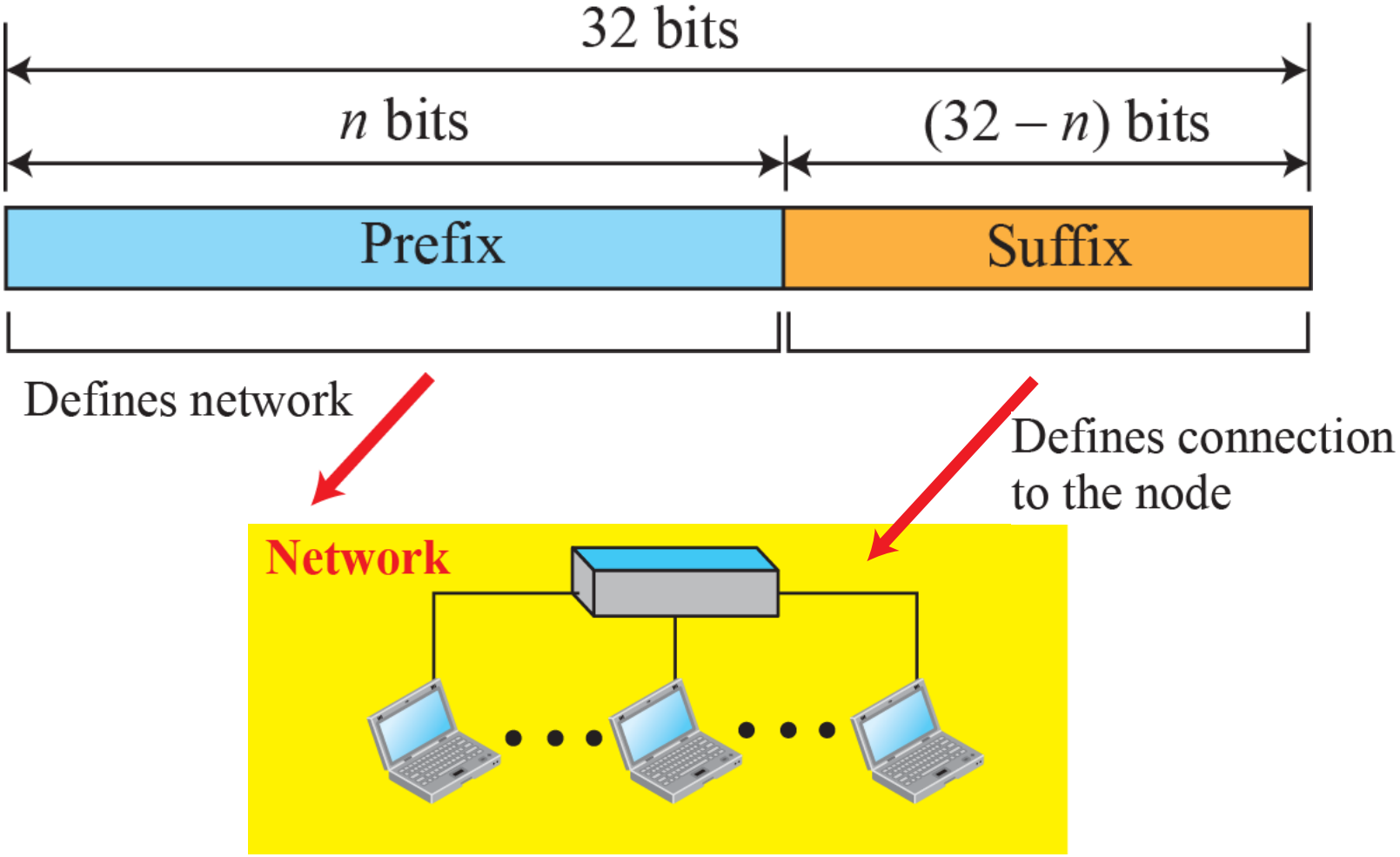
Address Space

- IPV4 protocol defines addresses that has an address space.
- An address space is the total number of addresses used by the protocol.
- IPv4 uses 32-bit addresses, which means that the address space is 2^{32} or 4,294,967,296 (more than four billion).
- If there were no restrictions, more than 4 billion devices could be connected to the Internet.

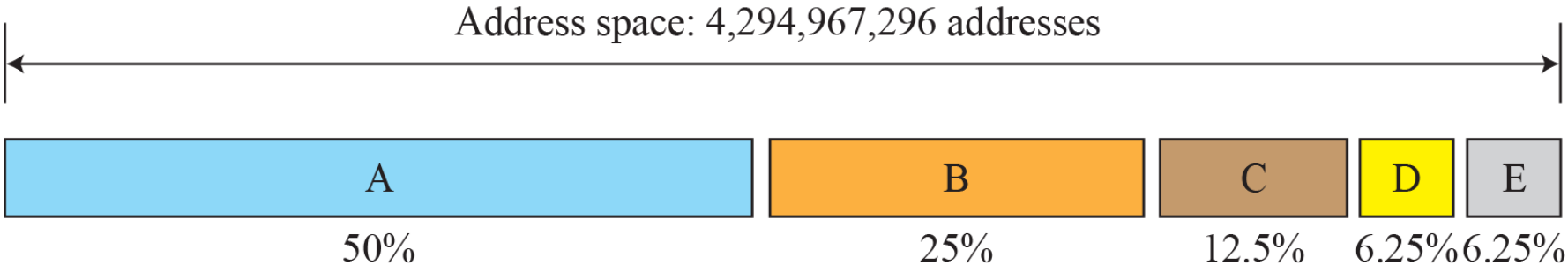
Three different notations in IPv4 addressing

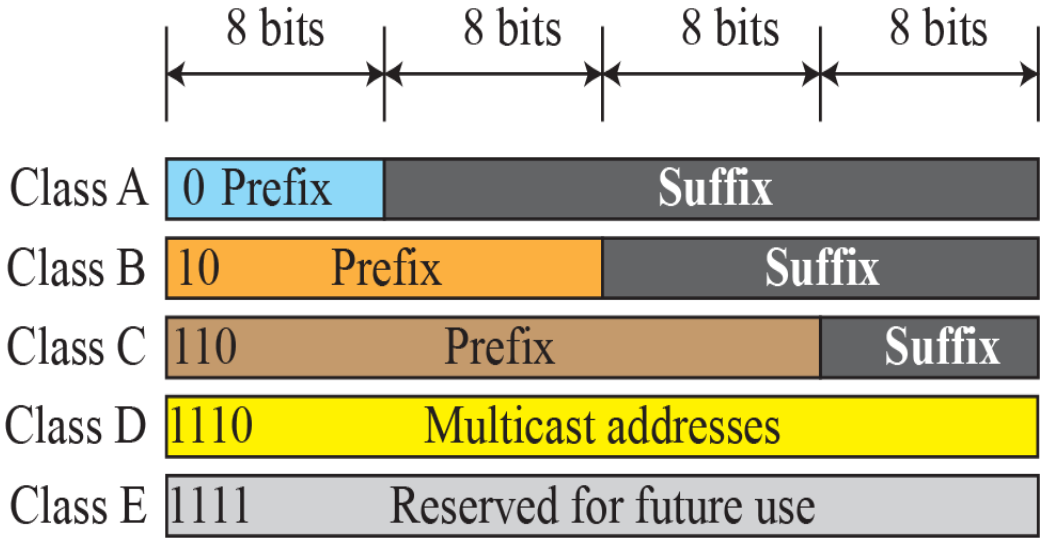


Hierarchy in addressing



Occupation of the address space in class full addressing





Class	Prefixes	First byte
A	n = 8 bits	0 to 127
B	n = 16 bits	128 to 191
C	n = 24 bits	192 to 223
D	Not applicable	224 to 239
E	Not applicable	240 to 255

Example 19.1

An IPv4 packet has arrived with the first 8 bits as $(01000010)_2$ The receiver discards the packet. Why?.

Solution

There is an error in this packet.

The 4 leftmost bits $(0100)_2$ show the version, which is correct.

The next 4 bits $(0010)_2$ show an invalid header length $(2 \times 4 = 8)$.

The minimum number of bytes in the header must be 20.

The packet has been corrupted in transmission.

Example 19.2

In an IPv4 packet, the value of HLEN is $(1000)_2$. How many bytes of options are being carried by this packet?

Solution

→The HLEN value is 8, which means the total number of bytes in the header is

$$8 \times 4 = 32 \text{bytes}.$$

→The first 20 bytes are the base header, the next 12 bytes are the options.

Example 19.3

In an IPv4 packet, the value of HLEN is 5, and the value of the total length field is $(0028)_{16}$. How many bytes of data are being carried by this packet?

Solution

→The HLEN value is 5, which means the total number of bytes in the header is 5×4, or 20 bytes (no options).

→The total length is $(0028)_{16}$ or 40 bytes, which means the packet is carrying 20 bytes of data $(40 - 20)$.

Example 19.4

An IPv4 packet has arrived with the first few hexadecimal digits as shown.

$$(45000028000100000102 \ldots)_{16}$$

How many hops can this packet travel before being dropped? The data belong to what upper-layer protocol?

Solution

→To find the time-to-live field, we skip 8 bytes (16 hexadecimal digits).

→The time-to-live field is the ninth byte, which is $(01)_{16}$. This means the packet can travel only one hop.

→ The protocol field is the next byte $(02)_{16}$, which means that the upper-layer protocol is IGMP.

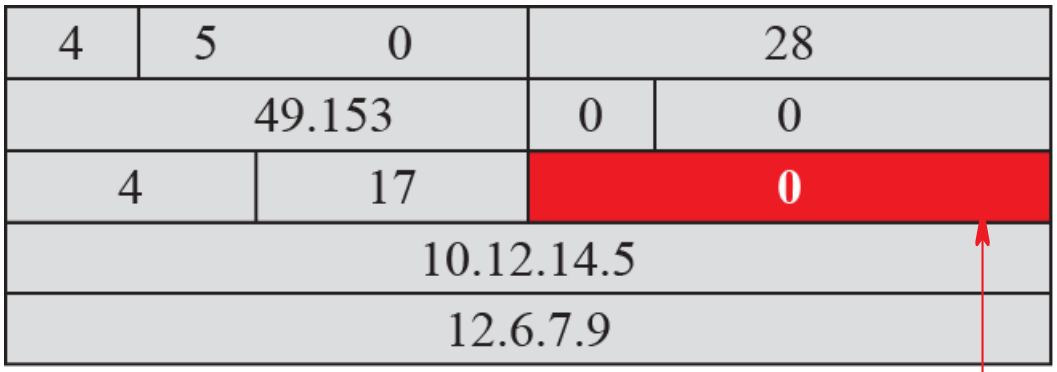
Example

Figure below shows an example of a checksum calculation for an IPv4 header without options. The header is divided into 16-bit sections. All the sections are added and the sum is complemented after wrapping the leftmost digit. The result is inserted in the checksum field.

→Note that the calculation of wrapped sum and checksum can also be done as follows in hexadecimal:

$$\text{Wrapped Sum} = \text{Sum mod FFFF}$$

$$\text{Checksum} = \text{FFFF} - \text{Wrapped Sum}$$



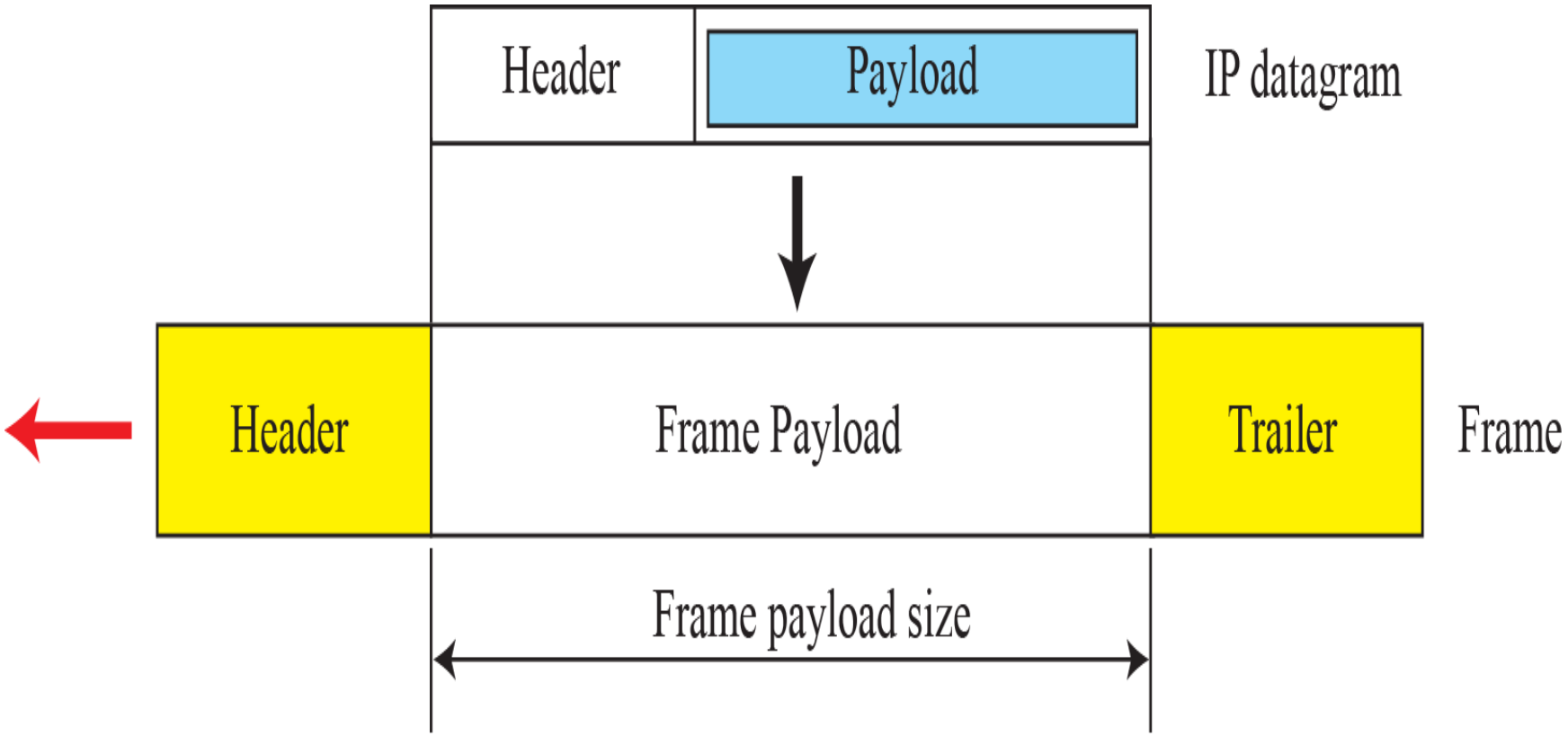
4, 5, and 0	→	4	5	0	0
28	→	0	0	1	C
1	→	C	0	0	1
0 and 0	→	0	0	0	0
4 and 17	→	0	4	1	1
0	→	0	0	0	0
10.12	→	0	A	0	C
14.5	→	0	E	0	5
12.6	→	0	C	0	6
7.9	→	0	7	0	9
Sum	→	1 3	4	4	E
Wrapped sum	→	3	4	4	F
Checksum	→	**C**	**B**	**B**	**0**

Fragmentation

- A datagram can travel through different networks. Each router de capsulates the IP datagram from the frame it receives, processes it, and then encapsulates it in another frame.
- The format and size of the received frame depend on the protocol used by the physical network through which the frame has just traveled.
- The format and size of the sent frame depend on the protocol used by the physical network through which the frame is going to travel.

Eg: if a router connects a LAN to a WAN, it receives a frame in the LAN format and sends a frame in the WAN format.

Maximum transfer unit (MTU)



MTU: Maximum size of frame payload

Maximum Transfer Unit(MTU)

- Each link layer has its own frame format.
- When the datagram is encapsulated in the frame, the total size of the data gram must be less than this maximum size.
- This restriction is imposed by the hardware and soft ware used in the network.
- Value of MTU differs from one physical network to another.
- Eg: The value for a LAN is 1500 bytes.
- In order to make IP protocol independent of the physical network, the maximum length of IP data gram= 65,535 bytes, which makes transmission more efficient.

Maximum transfer unit (MTU)

- If data grams pass through Physical n/w, whose MTU is less than the size of the data gram, the data grams is divided to make it possible to pass through these networks. This is called Fragmentation.
- Source does not fragment the data. It is Transport Layer which segment the data into the size that can be accommodated by the IPV4.
- When the data gram is fragmented, each fragment has its own header with most of the fields repeated.
- If fragmented data gram encounters a n/w with smaller MTU, fragmented data gram itself will be fragmented.
- The host or router that fragments a data gram must changes the value of three fields.
- Flags, fragmentation offset and total length.
- The rest of the fields must be copied.

Maximum transfer unit (MTU)

Fields related to fragmentation are

Identification: 16-bit field identifies a data gram originating from the source host.

→ A combination of identification and the source IPV4 address must uniquely defines a data gram as it leaves the source host.

→ When the data gram is fragmented, value of identification filed is copied to all the fragments.

i.e all fragments have same identification number, which helps the destination in reassembling the data grams.

→ It knows that all fragments with same identification number must be assembled in to one data gram.

Fields related to fragmentation

- Flags : 3-bit field. →first bit is reserved.

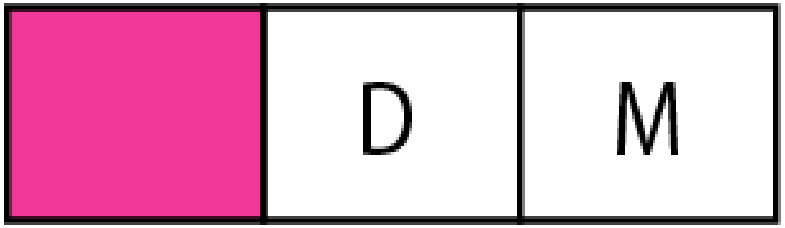
  →second bit is called don't fragment bit. If D=1, the machine must not fragment the data gram.

  →If D=0, the data gram can be fragmented if necessary.

  → Third bit is called more fragment bit.

→If M=1, the data gram is not the last fragment, there are more fragments after this one.

→ If M=0, this is the last/only fragment

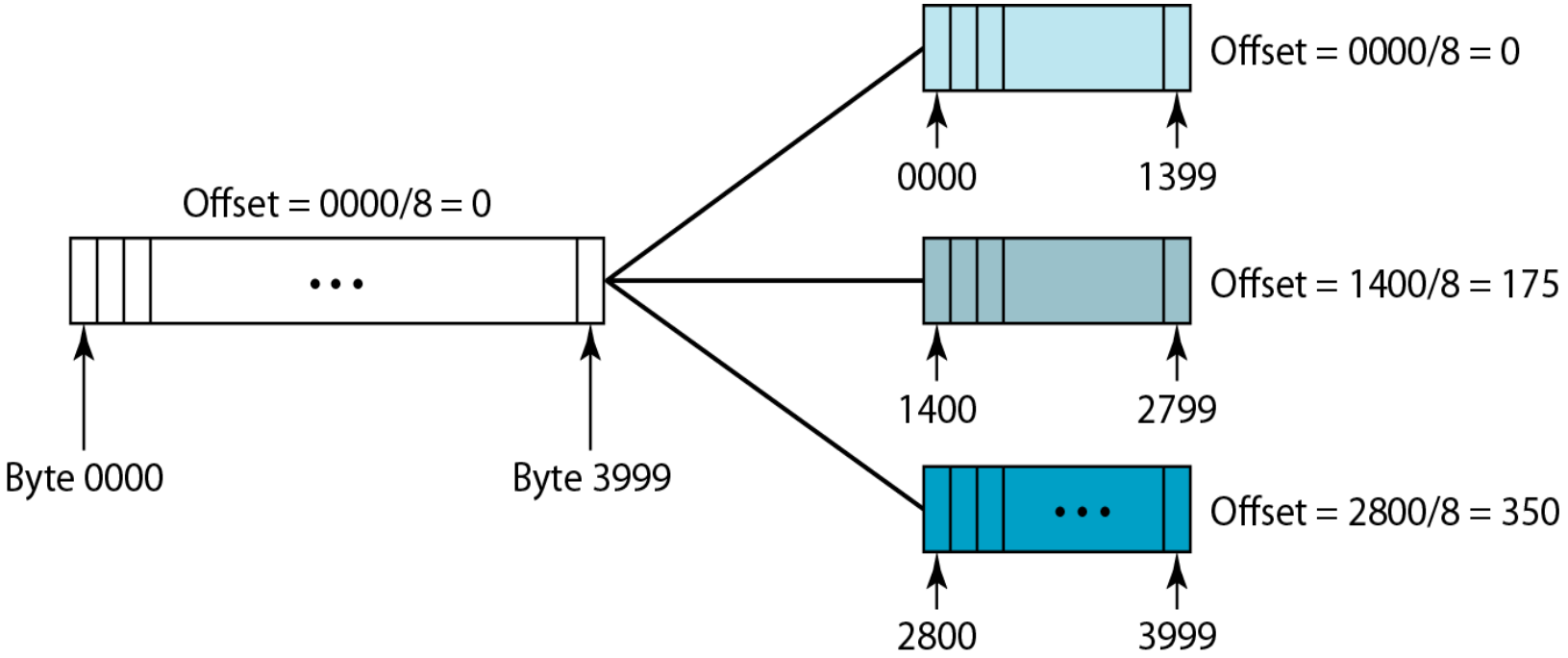


D: Do not fragment

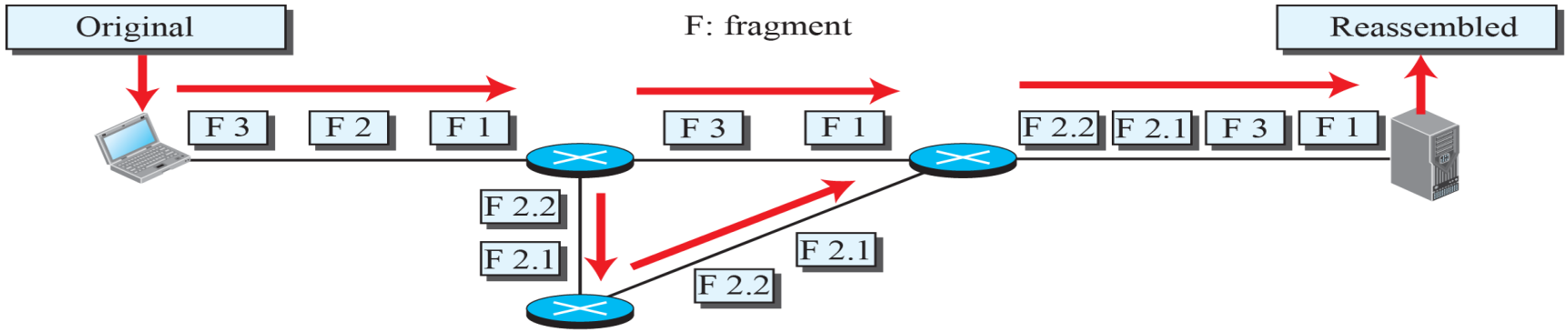
M: More fragments

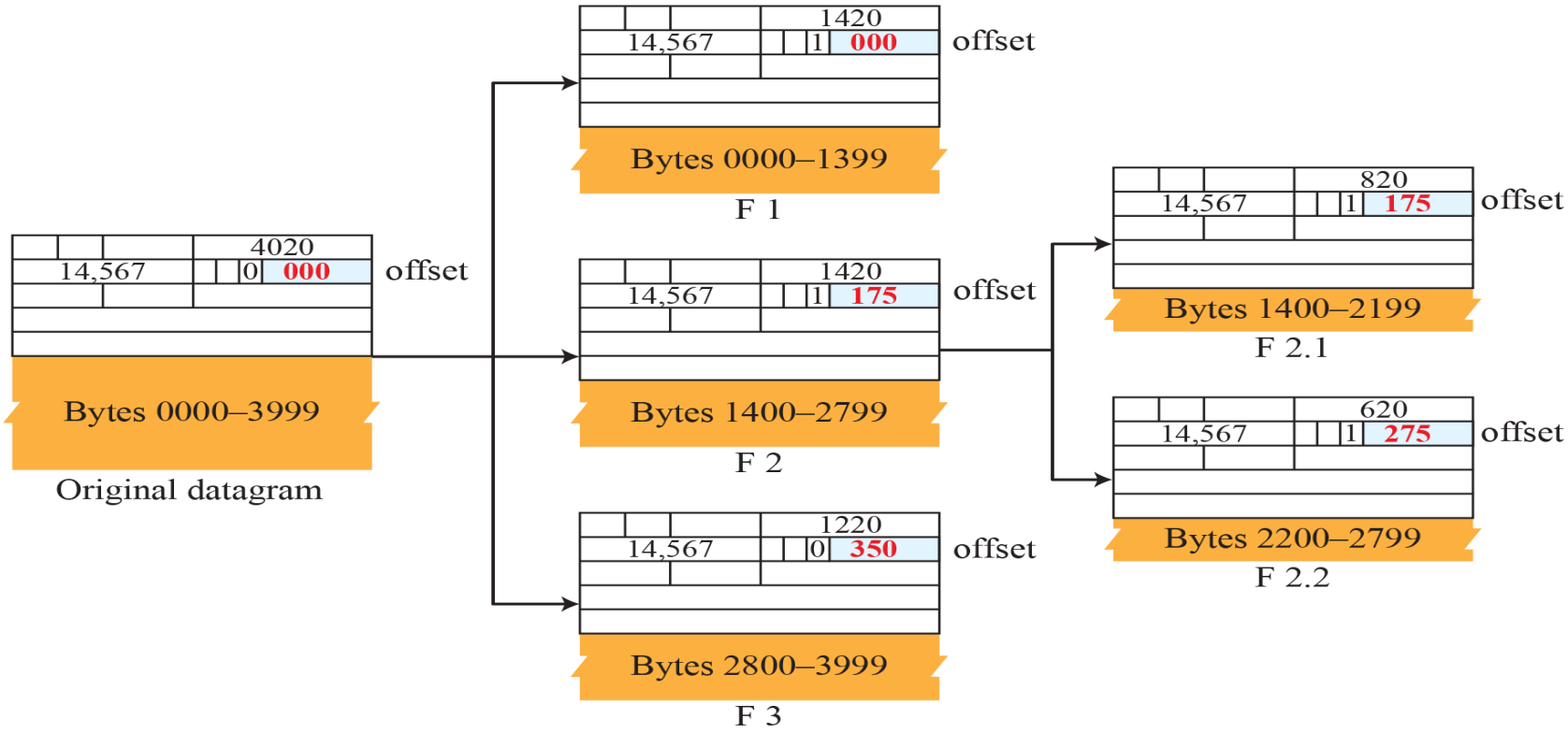
Fragmentation offset.

- 13 bit field shows the relative position of this fragment w.r.t the whole data gram.
- It is the offset of the data in the original data gram measured in units of 8 bytes.



Detailed fragmentation example





Example 19.6

A packet has arrived with an M bit value of 0. Is this the first fragment, the last fragment, or a middle fragment? Do we know if the packet was fragmented?

Solution

→If the M bit=0, it means that there are no more fragments; the fragment is the last one.

However, we cannot say if the original packet was fragmented or not. A non fragmented packet is considered the last fragment.

Example 19.7

A packet has arrived with an M bit value of 1. Is this the first fragment, the last fragment, or a middle fragment? Do we know if the packet was fragmented?

Solution

→If the M=1, it means that there is at least one more fragment.

→This fragment can be the first one or a middle one, but not the last one.

→cannot identify if it is the first one or a middle one which need more information (the value of the fragmentation offset).

Example 19.8

A packet has arrived with an M bit value of 1 and a fragmentation offset value of 0. Is this the first fragment, the last fragment, or a middle fragment?

Solution

→Because the M =1, it is either the first fragment or a middle one.

→Because the offset value =0, it is the first fragment.

Example 19.9

A packet has arrived in which the offset value is 100. What is the number of the first byte? Do we know the number of the last byte?

Solution

→To find the number of the first byte,

Multiply the offset value by 8.

This means that the first byte number is 800.

→Cannot determine the number of the last byte unless length of the data is known.

Example 19.10

A packet has arrived in which the offset value is 100, the value of HLEN is 5, and the value of the total length field is 100. What are the numbers of the first byte and the last byte?

Solution

→The first byte number is $100 \times 8 = 800$.

→The total length is 100 bytes, and the header length is 20 bytes (5×4), which means that there are 80 bytes in this datagram.

→If the first byte number is 800, the last byte number must be 879.

Debugging Tools

- There are several tools that can be used in the Internet for debugging.
- Determine the viability of a host or router.
- For tracing the route of a packet, two tools that use ICMP for debugging: ping and trace route.
- Ping program is to find if a host alive and responding.
- Source host sends ICMP echo request messages to destination, if alive ,destination responds with ICMP echo reply message.
- Ping program sets identifier in the echo request & reply messages and starts with sequence number 0.
- This number is incremented by 1 when new message is sent.
- Ping calculated RTT(Inserts sending time in data section of the message.
- When packet arrives, it subtracts

  (arrival time-departure time=RTT)

Example 19.11

The following shows how we send a ping message to the auniversity.edu site. We set the identifier field in the echo request and reply message and start the sequence number from 0; this number is incremented by one each time a new message is sent. Note that ping can calculate the round-trip time. It inserts the sending time in the data section of the message. When the packet arrives, it subtracts the arrival time from the departure time to get the round-trip time (*rtt*).

```
$ ping auniversity.edu
PING auniversity.edu (152.181.8.3)   56 (84)  bytes of data.
64 bytes from auniversity.edu (152.181.8.3): icmp_seq=0     ttl=62     time=1.91 ms
```

Example 19.11(continued)

```
64 bytes from auniversity.edu (152.181.8.3): icmp_seq=1    ttl=62    time=2.04 ms
64 bytes from auniversity.edu (152.181.8.3): icmp_seq=2    ttl=62    time=1.90 ms
64 bytes from auniversity.edu (152.181.8.3): icmp_seq=3    ttl=62    time=1.97 ms
64 bytes from auniversity.edu (152.181.8.3): icmp_seq=4    ttl=62    time=1.93 ms
64 bytes from auniversity.edu (152.181.8.3): icmp_seq=5    ttl=62    time=2.00 ms
--- auniversity.edu statistics ---
6 packets transmitted, 6 received, 0% packet loss
rtt min/avg/max = 1.90/1.95/2.04 ms
```

Example

- Microsoft Windows [Version 10.0.16299.371]
- (c) 2017 Microsoft Corporation. All rights reserved.
- **C:\Users\Ravi>ping vtu.ac.in**
- Pinging vtu.ac.in [210.212.207.171] with 32 bytes of data:
- Reply from 210.212.207.171: bytes=32 time=45ms TTL=57
- Reply from 210.212.207.171: bytes=32 time=47ms TTL=57
- Reply from 210.212.207.171: bytes=32 time=45ms TTL=57
- Reply from 210.212.207.171: bytes=32 time=44ms TTL=57

Ping statistics for 210.212.207.171:

- Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
- Approximate round trip times in milli-seconds:
- Minimum = 44ms, Maximum = 47ms, Average = 45ms

Trace route

- Trace route program is used to trace the file path of a packet from a source to the destination.
- It can find the IP addresses of all the routers that are visited along the path.
- Trace route program gets help from two error reporting message

1. Time-exceeded 2.Destination unreachable

- It is Application layer program , requires only the client program
- Client program never reaches the application layer in the destination host.
- Trace route application program is encapsulated in a UDP user datagram.
- It intentionally use the port number that is not available at the destination.

Chapter 22: Outline

22.1 IPv6 ADDRESSING

22.2 THE IPv6 PROTOCOL

Chapter 22: Objective

- *The first section discusses the addressing mechanism in the new generation of the Internet. The section first describes the representation and address space. It then shows the allocation in the address space. The section finally explains auto-configuration and renumbering, which makes it easy for a host to move from one network to another.*

- *The second section discusses IPv6 protocol. First the new packet format is described. The section then shows how use of extension headers can replace the options.*

22-1 IPv6 Addressing

The main reason for migration from IPv4 to IPv6 is the small size of the address space in IPv4. In this section, we show how the huge address space of IPv6 prevents address depletion in the future. We also discuss how the new addressing responds to some problems in the IPv4 addressing mechanism. An IPv6 address is 128 bits or 16 bytes (octets) long, four times the address length in IPv4.

22.22.1 Representation

A computer normally stores the address in binary, but it is clear that 128 bits cannot easily be handled by humans. Several notations have been proposed to represent IPv6 addresses when they are handled by humans. The following shows two of these notations: binary and colon hexadecimal.

Binary (128 bits)	1111111011110110	...	1111111100000000
Colon Hexadecimal	FEF6:BA98:7654:3210:ADEF:BBFF:2922:FF00		

22.22.2 Address Space

The address space of IPv6 contains 2^{128} addresses. This address space is 2^{96} times the IPv4 address—definitely no address depletion—as shown, the size of the space is

340, 282, 366, 920, 938, 463, 374, 607, 431, 768, 211, 456.

22.22.3 Address Space Allocation

Like the address space of IPv4, the address space of IPv6 is divided into several blocks of varying size and each block is allocated for a special purpose. Most of the blocks are still unassigned and have been set aside for future use. Table 22.1 shows only the assigned blocks. In this table, the last column shows the fraction each block occupies in the whole address space..

Table 22.1: Prefixes for assigned IPv6 addresses

Block prefix	*CIDR*	*Block assignment*	*Fraction*
0000 0000	0000::/8	Special addresses	1/256
001	**2000::/3**	**Global unicast**	**1/8**
1111 110	FC00::/7	Unique local unicast	1/128
1111 1110 10	FE80::/10	Link local addresses	1/1024
1111 1111	FF00::/8	Multicast addresses	1/256

Figure 22.1: *Global unicast address*

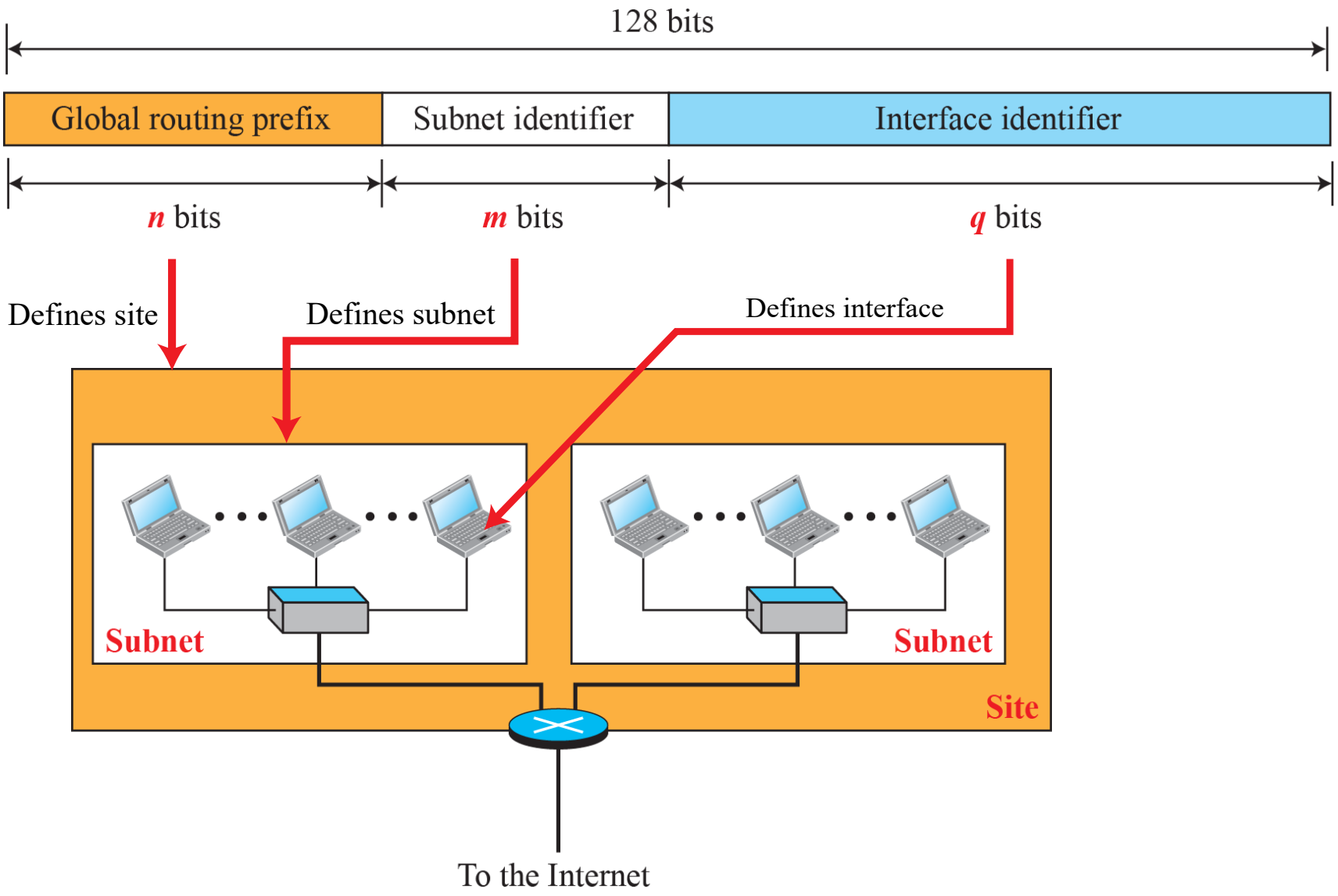


Figure 22.2: Mapping for EUI-64

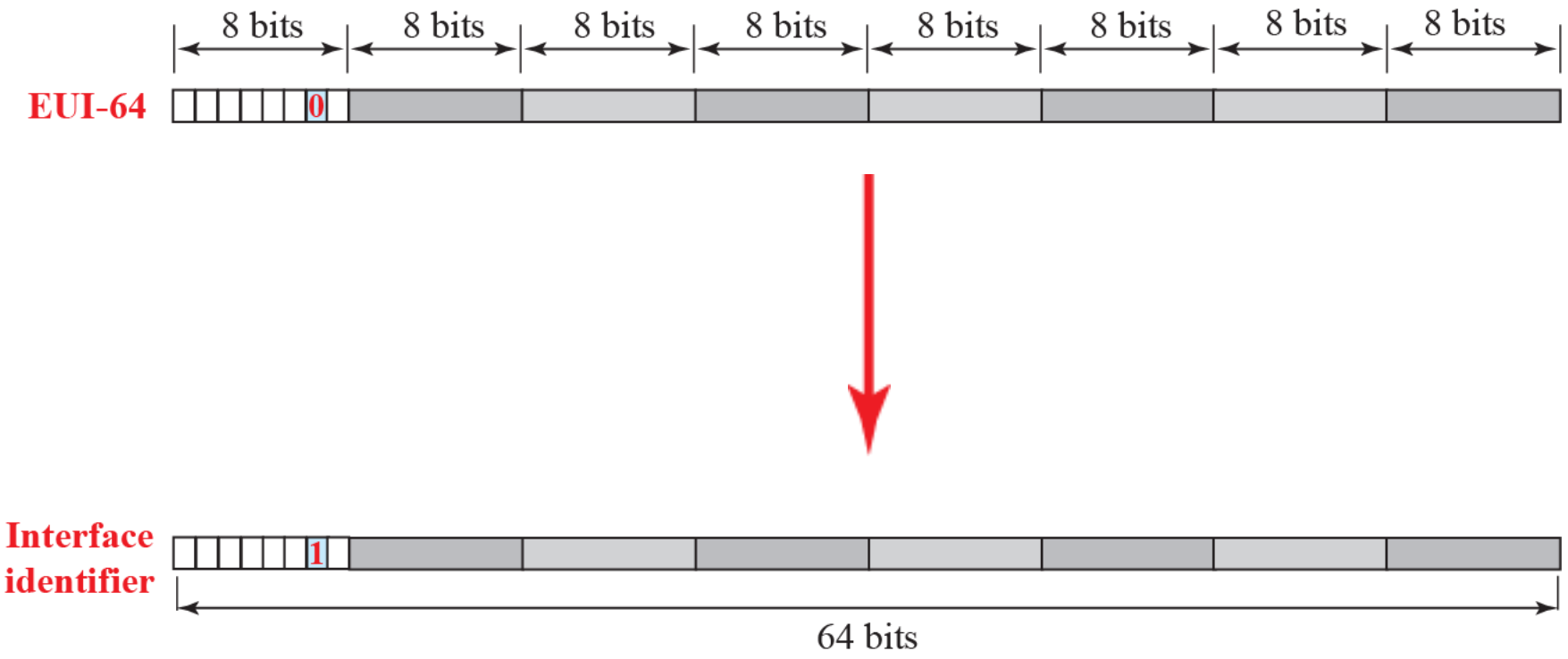
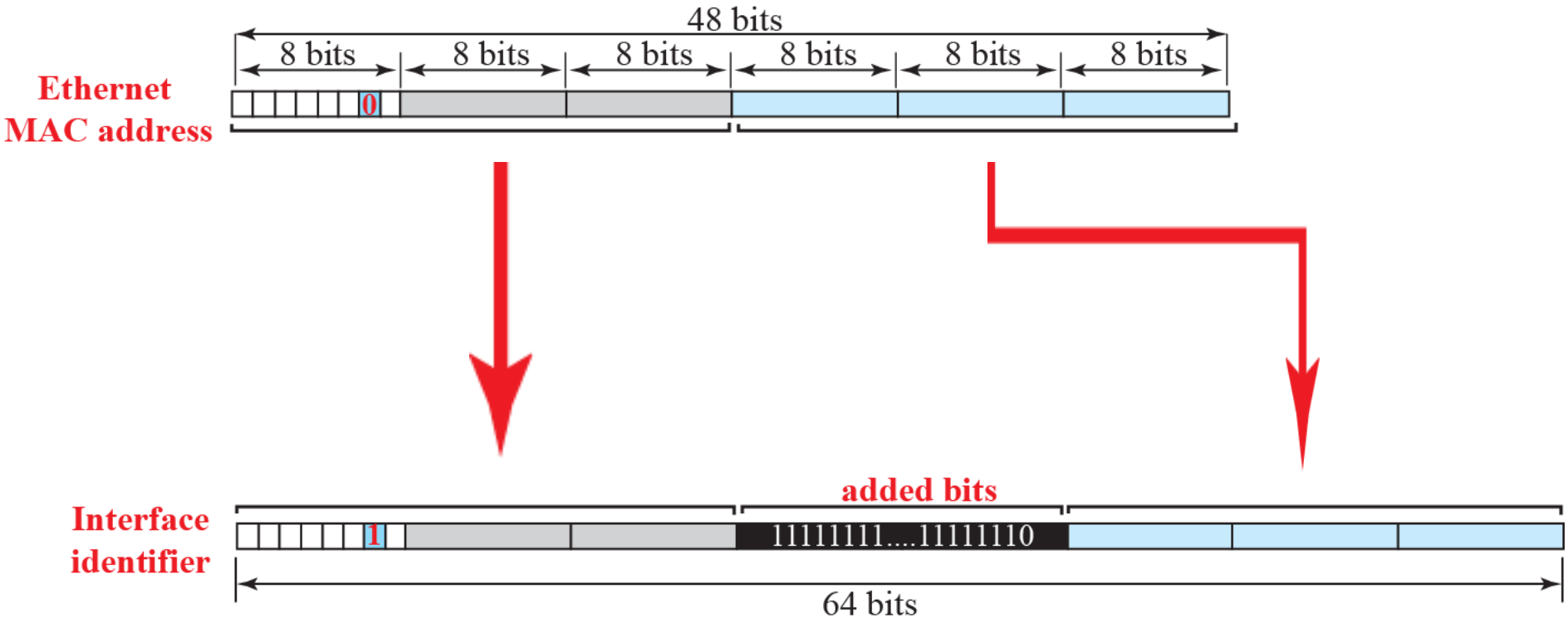


Figure 22.3: Mapping for Ethernet MAC



Example 22.1

An organization is assigned the block 2000:1456:2474/48. What is the CIDR notation for the blocks in the first and second subnets in this organization?

Solution

Theoretically, the first and second subnets should use the blocks with subnet identifier 0001_{16} and 0002_{16}. This means that the blocks are 2000:1456:2474:0000/64 and 2000:1456:2474:0001/64.

Example 22.2

Using the format we defined for Ethernet addresses, find the interface identifier if the physical address in the EUI is $(F5\text{-}A9\text{-}23\text{-}EF\text{-}07\text{-}14\text{-}7A\text{-}D2)_{16}$.

Solution

We only need to change the seventh bit of the first octet from 0 to 1 and change the format to colon hex notation. The result is F7A9:23EF:0714:7AD2.

Example 22.3

Using the format we defined for Ethernet addresses, find the interface identifier if the Ethernet physical address is (F5-A9-23-14-7A-D2)16.

Solution

We only need to change the seventh bit of the first octet from 0 to 1, insert two octets $FFFE_{16}$ and change the format to colon hex notation. The result is F7A9:23FF:FE14:7AD2 in colon hex.

Example 22.4

An organization is assigned the block 2000:1456:2474/48. What is the IPv6 address of an interface in the third subnet if the IEEE physical address of the computer is (F5-A9-23-14-7A-D2)16?

Solution

The interface identifier for this interface is F7A9:23FF:FE14:7AD2 (see Example 22.3). If we append this identifier to the global prefix and the subnet identifier, we get:

2000:1456:2474:0003:F7A9:23FF:FE14:7AD2/128

Figure 22.4: Special addresses

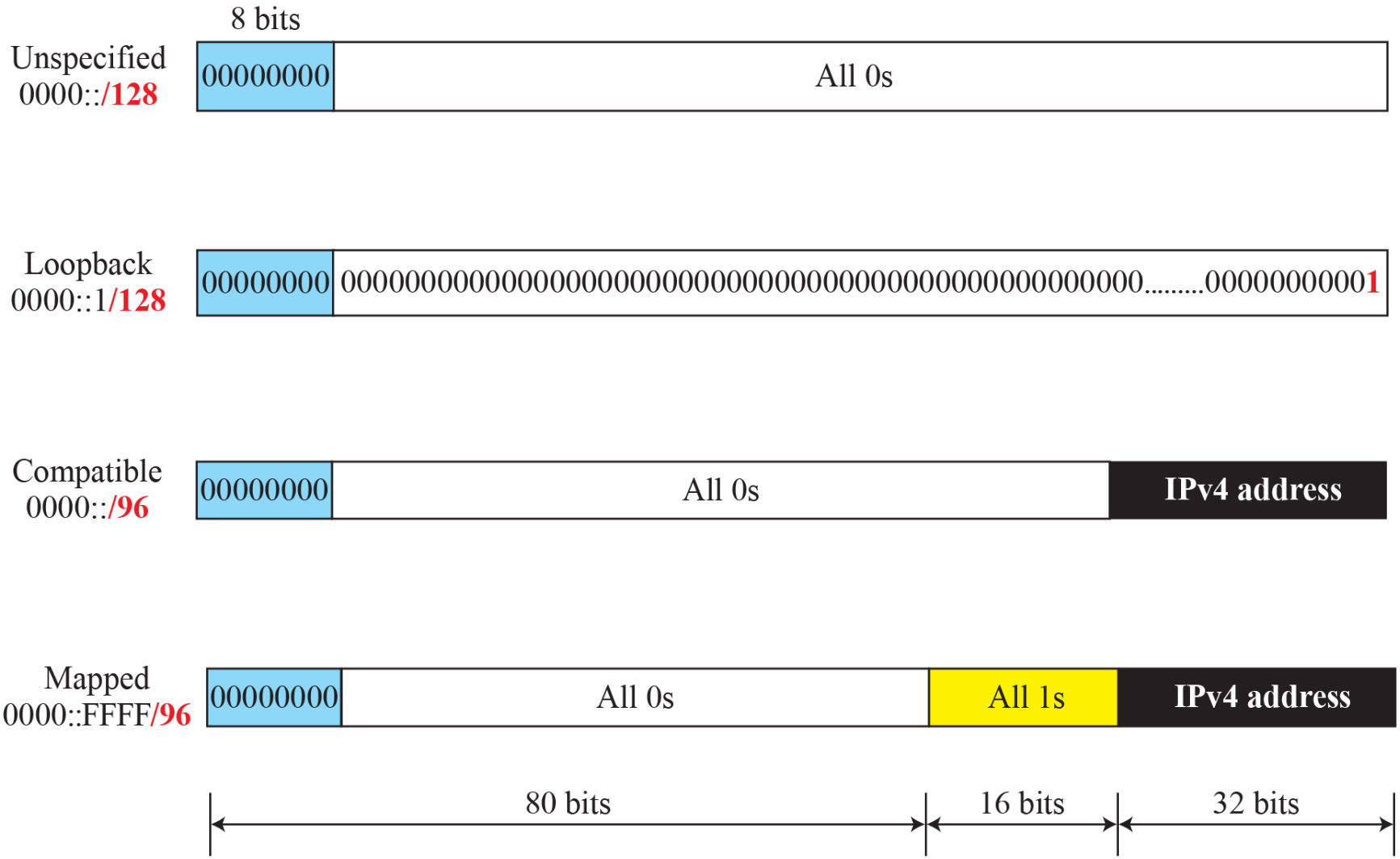
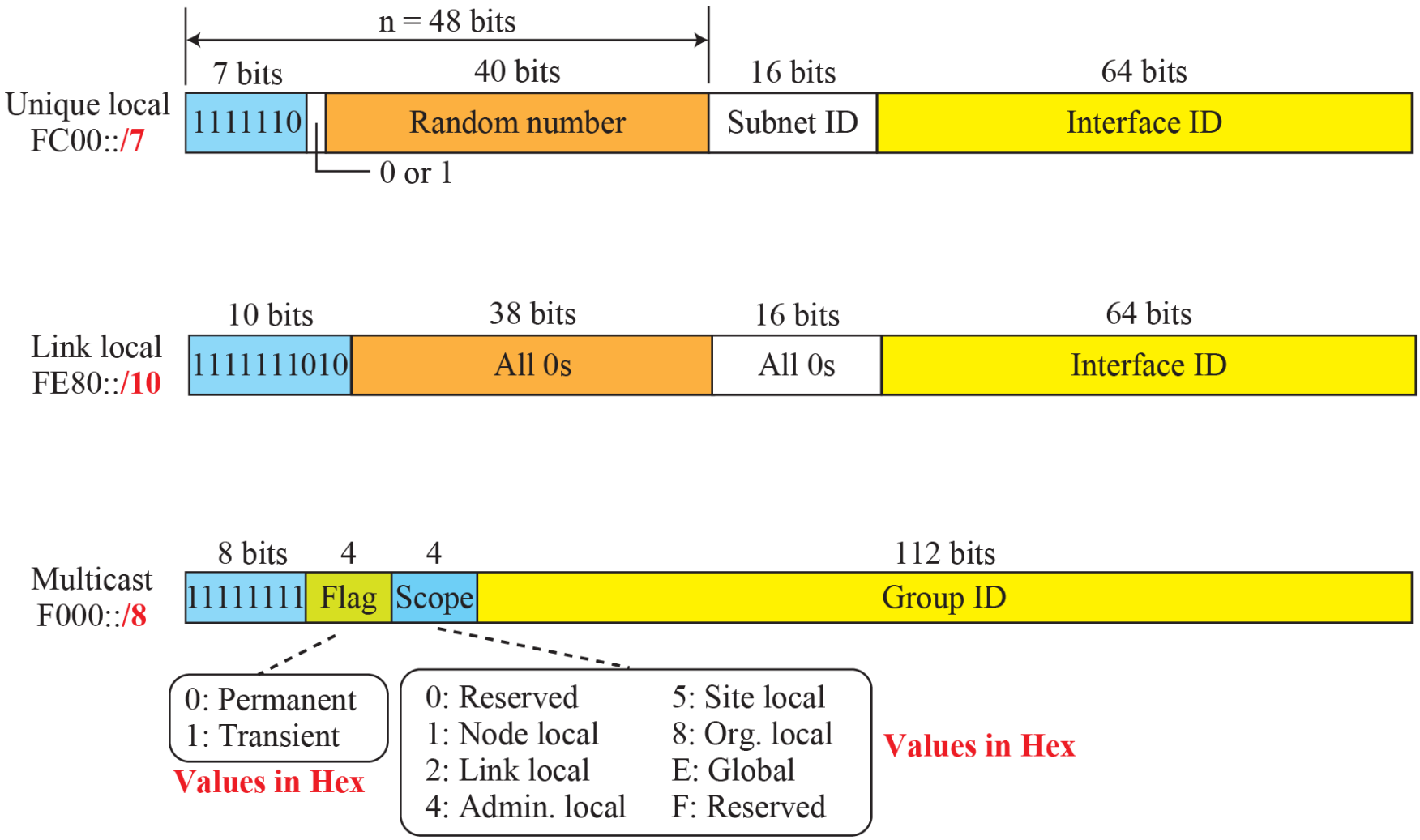


Figure 22.5: Unique local unicast block



22-2 THE IPv6 PROTOCOL

The change of the IPv6 address size requires the change in the IPv4 packet format. The designer of IPv6 decided to implement remedies for other shortcomings now that a change is inevitable. The following shows other changes implemented in the protocol in addition to changing address size and format.

22.2.1 Packet Format

The IPv6 packet is shown in Figure 22.6. Each packet is composed of a base header followed by the payload. The base header occupies 40 bytes, whereas payload can be up to 65,535 bytes of information. The description of fields follows.

Figure 22.6: IPv6 datagram

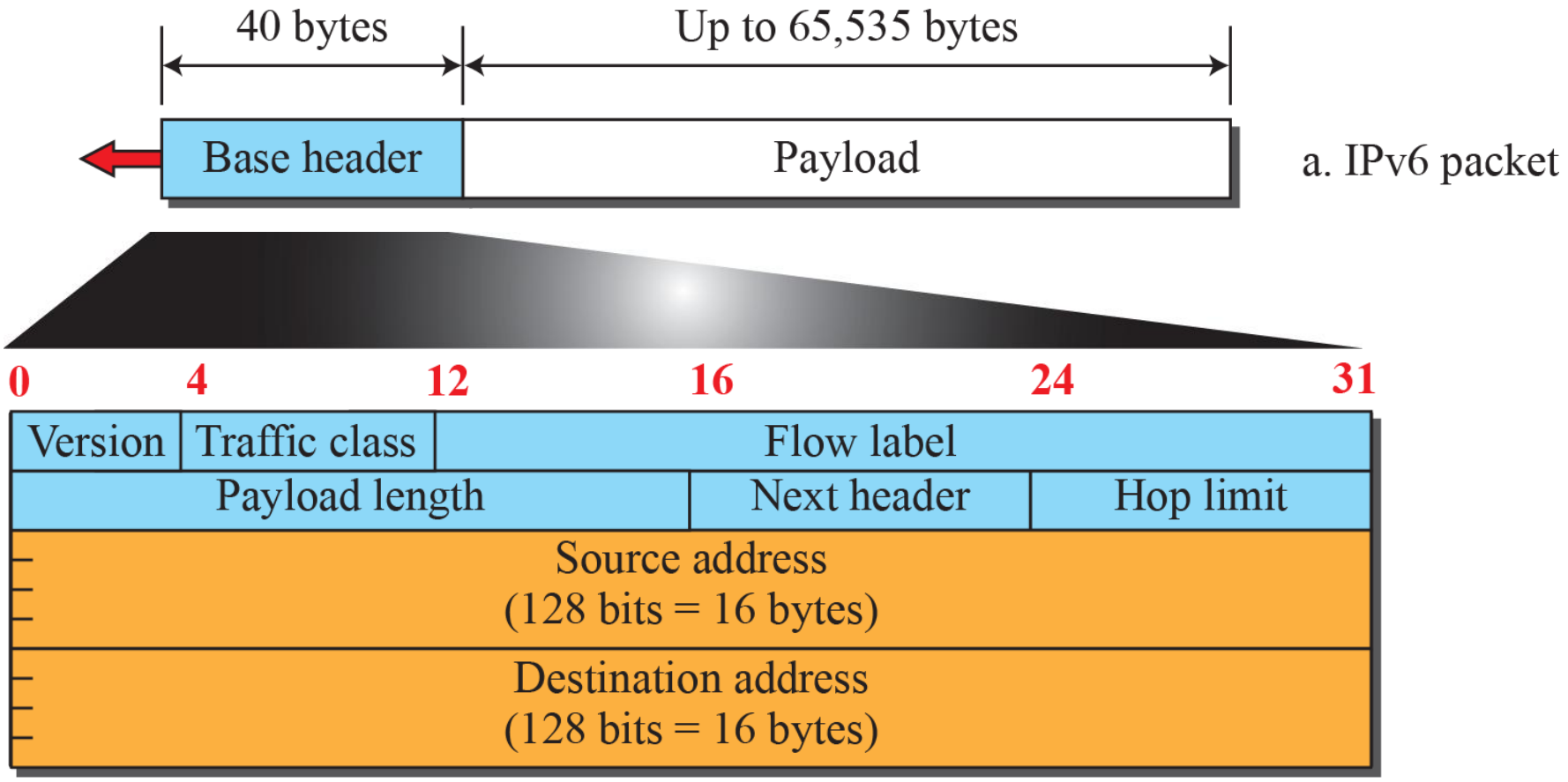
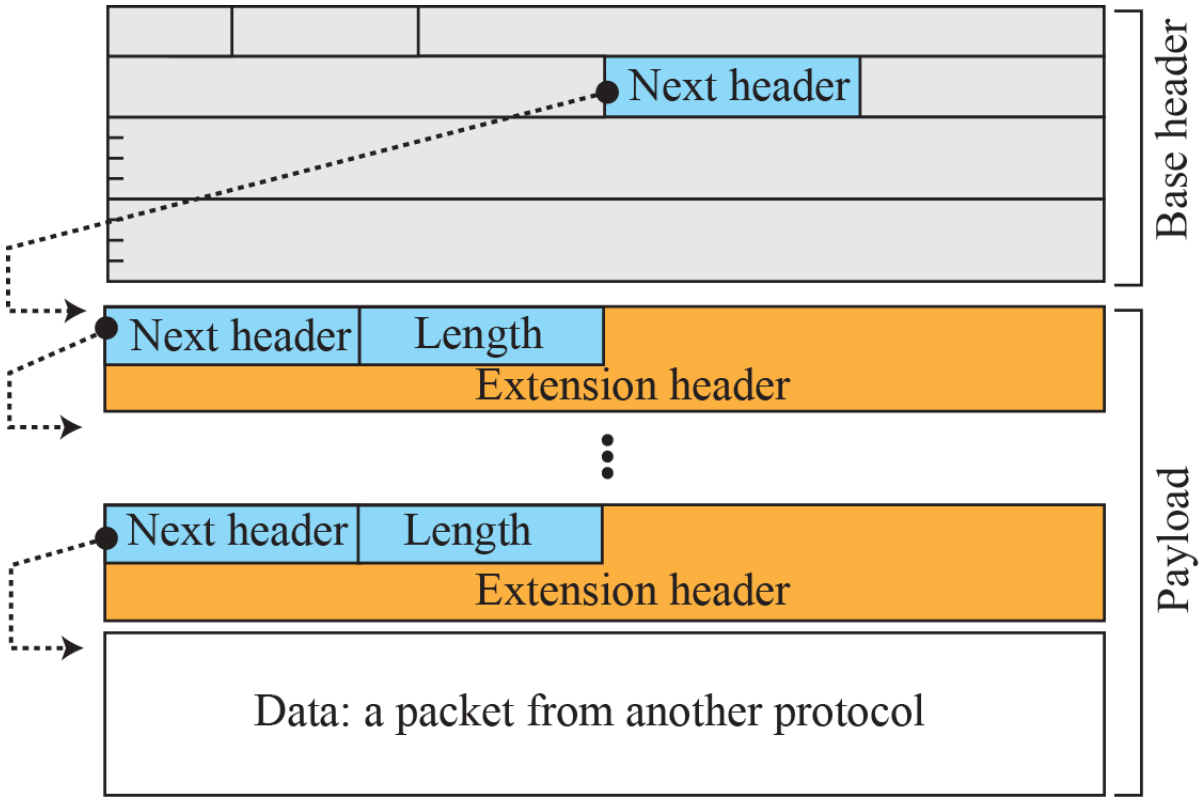


Figure 22.7: Payload in an IPv6 datagram



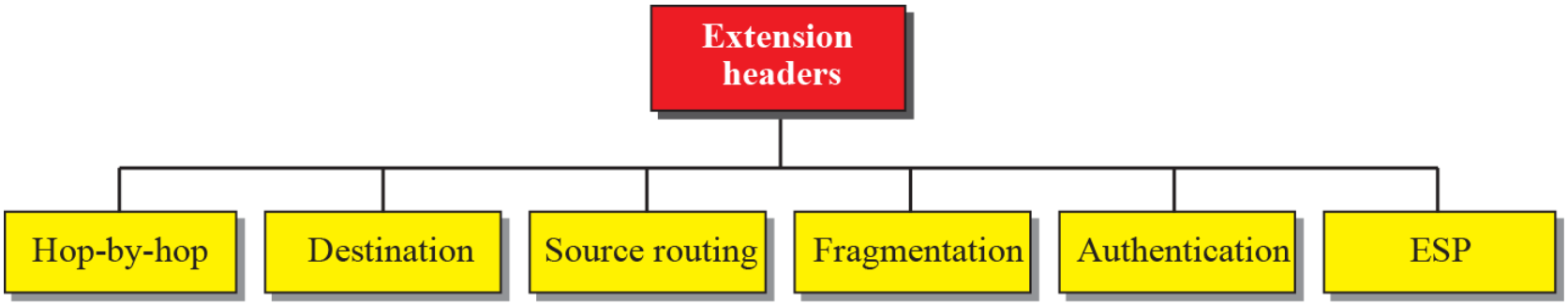
Some next-header codes

00: Hop-by-hop option
02: ICMPv6
06: TCP
17: UDP
43: Source-routing option
44: Fragmentation option
50: Encrypted security payload
51:Authentication header
59: Null (no next header)
60: Destination option

22.2.2 Extension Header

An IPv6 packet is made of a base header and some extension headers. The length of the base header is fixed at 40 bytes. However, to give more functionality to the IP datagram, the base header can be followed by up to six extension headers. Many of these headers are options in IPv4. Six types of extension headers have been defined. These are hop-by-hop option, source routing, fragmentation, authentication, encrypted security payload, and destination option (see Figure 22.8).

Figure 22.8: Extension Header Types



22-2 THE IPv6 PROTOCOL

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Figure 22.6: IPv6 datagram

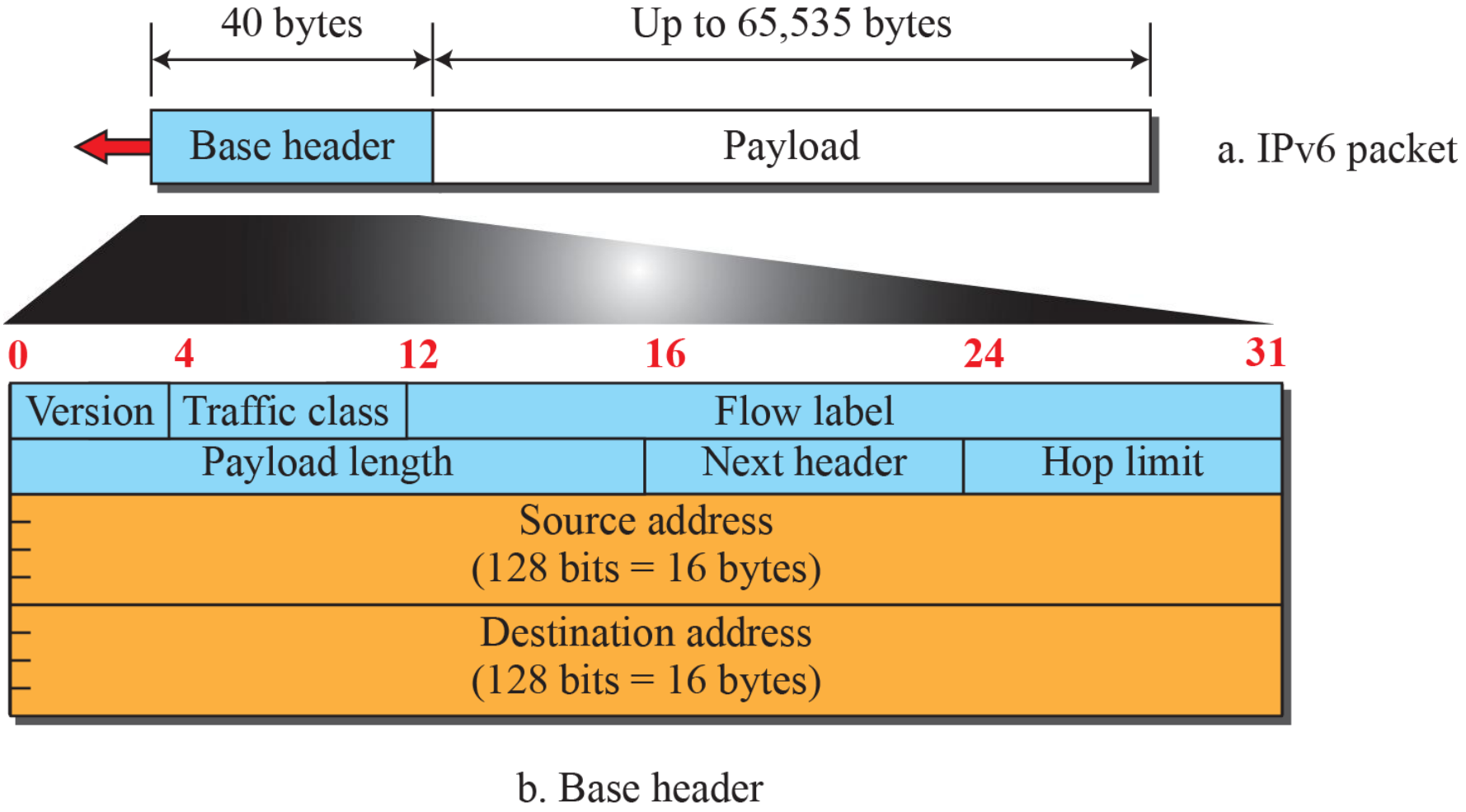
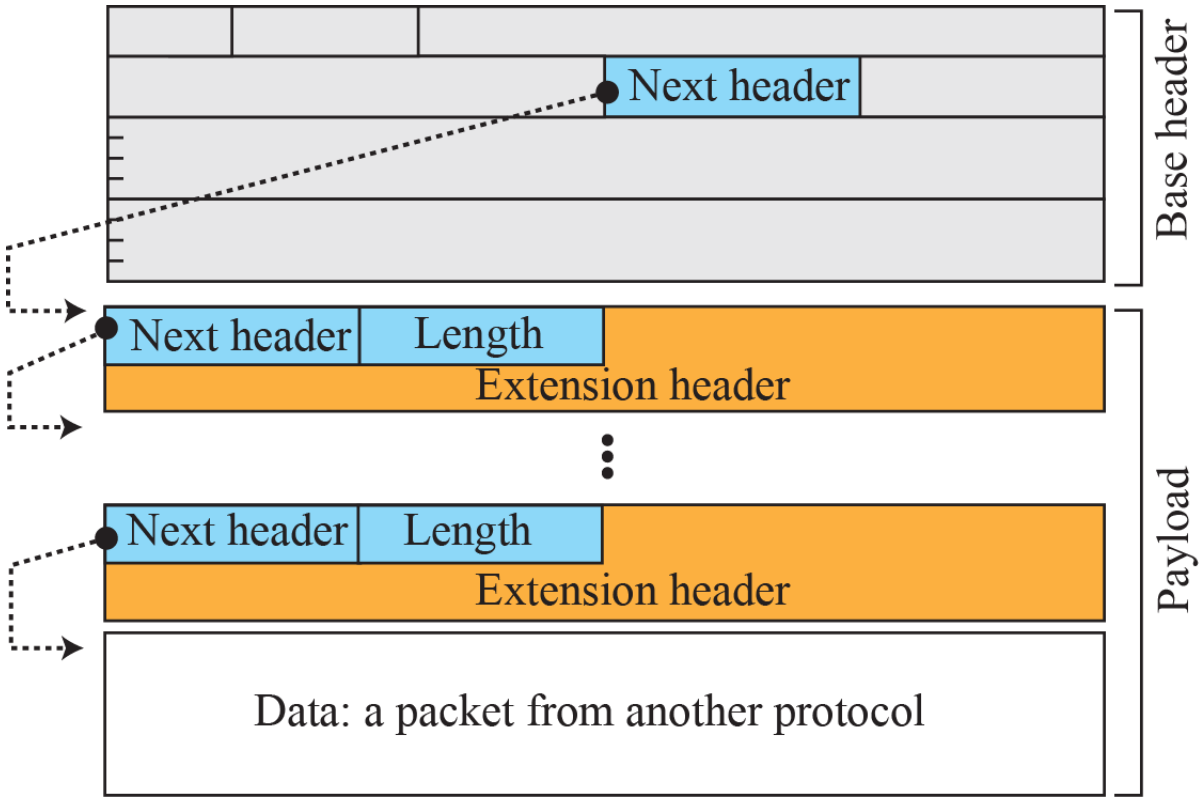


Figure 22.7: Payload in an IPv6 datagram



Some next-header codes

00: Hop-by-hop option
02: ICMPv6
06: TCP
17: UDP
43: Source-routing option
44: Fragmentation option
50: Encrypted security payload
51:Authentication header
59: Null (no next header)
60: Destination option

22.2.2 Extension Header

An IPv6 packet is made of a base header and some extension headers. The length of the base header is fixed at 40 bytes. However, to give more functionality to the IP datagram, the base header can be followed by up to six extension headers. Many of these headers are options in IPv4. Six types of extension headers have been defined. These are hop-by-hop option, source routing, fragmentation, authentication, encrypted security payload, and destination option (see Figure 22.8).

Figure 22.8: Extension Header Types

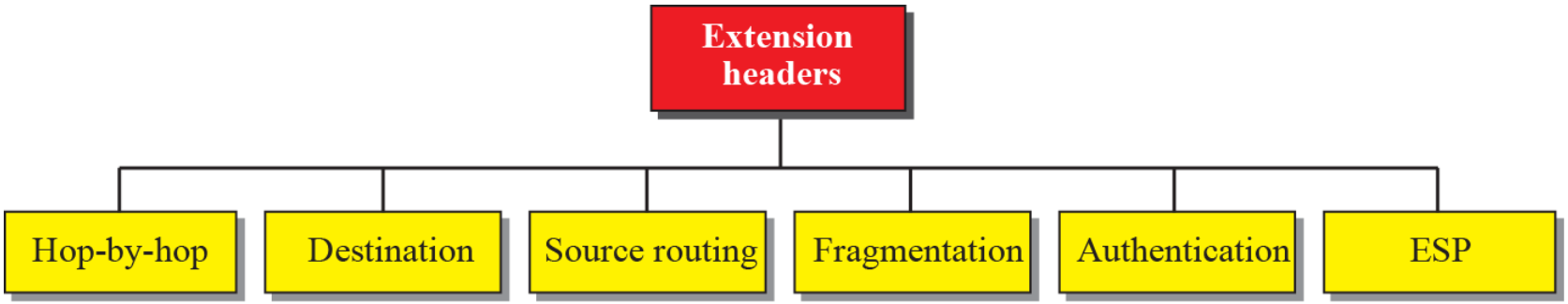
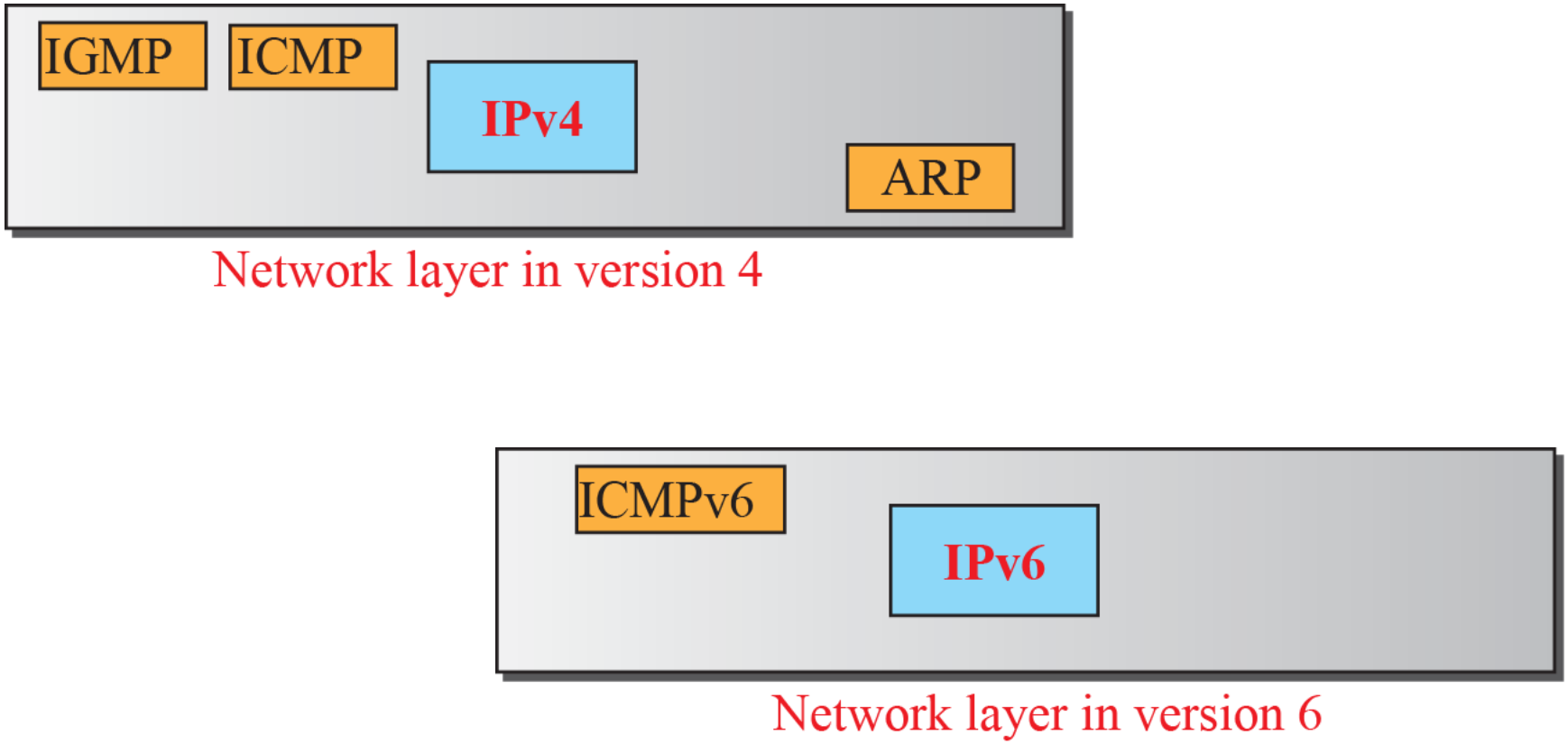


Figure 22.9: ***Comparison of network layer in version 4 and version 6***



18.4.4 Dynamic Host Configuration Protocol (DHCP).

- After a block of addresses are assigned to an organization, the network administration can manually assign addresses to the individual hosts or routers.

- However, address assignment in an organization can be done automatically using the DHCP.

- DHCP is an application-layer program, using the client-server paradigm, that actually helps TCP/IP at the network layer.

DHCP Features

- Automatic allocation: DHCP assigns a permanent IP address to a client.
- Manual allocation :The IP address for the client is assigned by the administrator. DHCP conveys the address to the client.
- Dynamic allocation: DHCP assigns, or leases, an IP address to the client for a limited period of time.

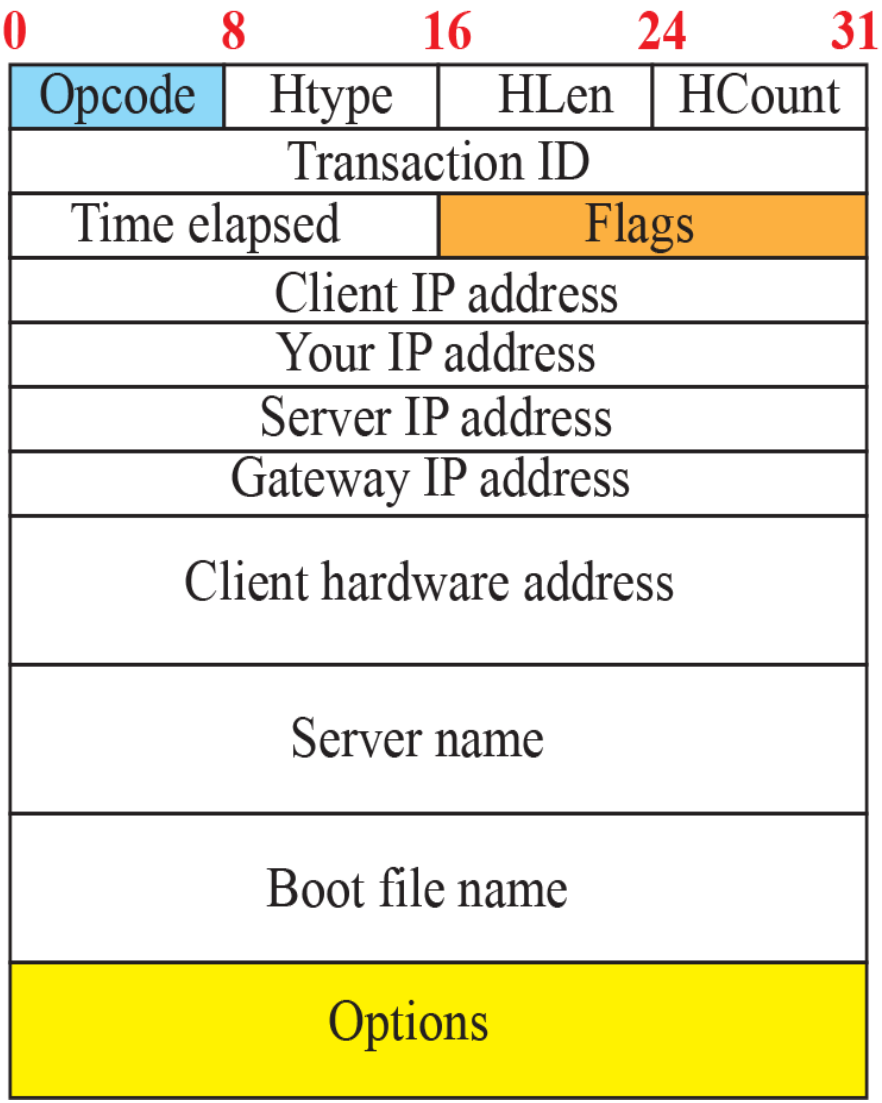
DHCP Address Pooling

- DHCP enables DHCP clients on an IP network to obtain
- Works in Client/Server mode
- Host (DHCP Client) on a network obtain configurations from a Server (DHCP Server
- The Server manages the configurations by answering requests from Clients
- Cannot be used for devices that need STATIC addresses i.e. routers, switches & servers.
- The Server issues LEASES
- Client requests use UDP port 67, server responses use UDP port 68.

DHCP Operation

- DHCP client broadcasts a Discover packet on the local subnet.
- DHCP servers send an Offer packet with lease information.
- DHCP client selects lease and broadcasts a Request packet. Selected DHCP server sends ACK packet.

Figure : DHCP message format



Fields:

Opcode: Operation code, request (1) or reply (2)

Htype: Hardware type (Ethernet, ...)

HLen: Lengh of hardware address

HCount: Maximum number of hops the packet can travel

Transaction ID: An integer set by client and repeated by the server

Time elapsed: The number of seconds since the client started to boot

Flags: First bit defines unicast (0) or multicast (1); other 15 bits not used

Client IP address: Set to 0 if the client does not know it

Your IP address: The client IP address sent by the server

Server IP address: A broadcast IP address if client does not know it

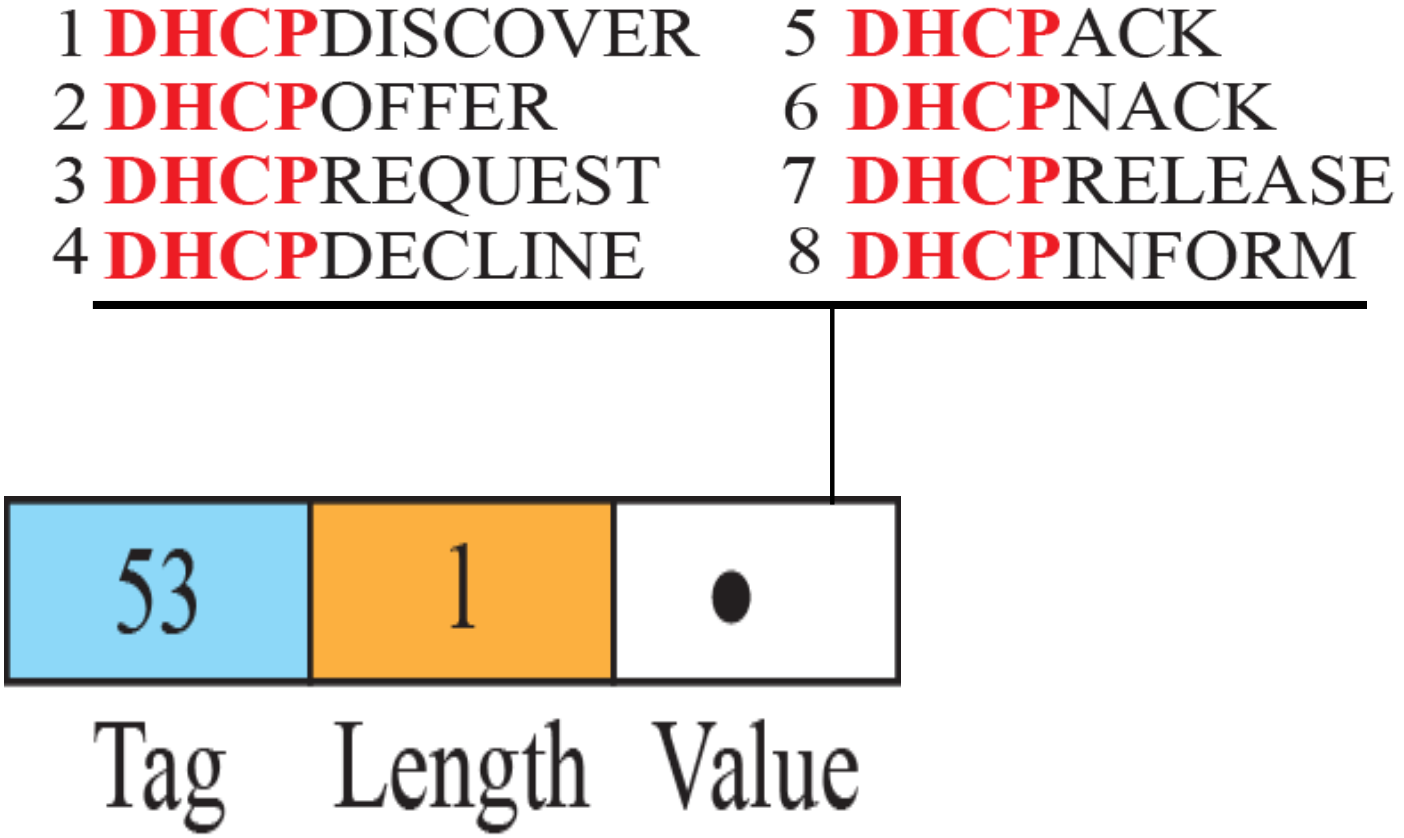
Gateway IP address: The address of default router

Server name: A 64-byte domain name of the server

Boot file name: A 128-byte file name holding extra information

Options: A 64-byte field with dual purpose described in text

Figure :*DHCP Address Discovery*



Discover packet :Client broadcasts Discover packet that identifies client's hardware address Header contains Source IP address Preferred address Typically the last address the client used.

Offer Packet : Send by DHCP server to offer IP address to DHCP client Includes IP address that is offered to the client Sometimes, answers to the requested options in the DHCP Discover packet.

Request Packet: Client can either accept offer by issuing a DHCP Request packet, orReject offer by sending a DHCP Decline packet.

Acknowledgment Packet: Sent from server to client to indicate the completion of the four-packet DHCP Discovery processResponse

Figure :Operation of DHCP

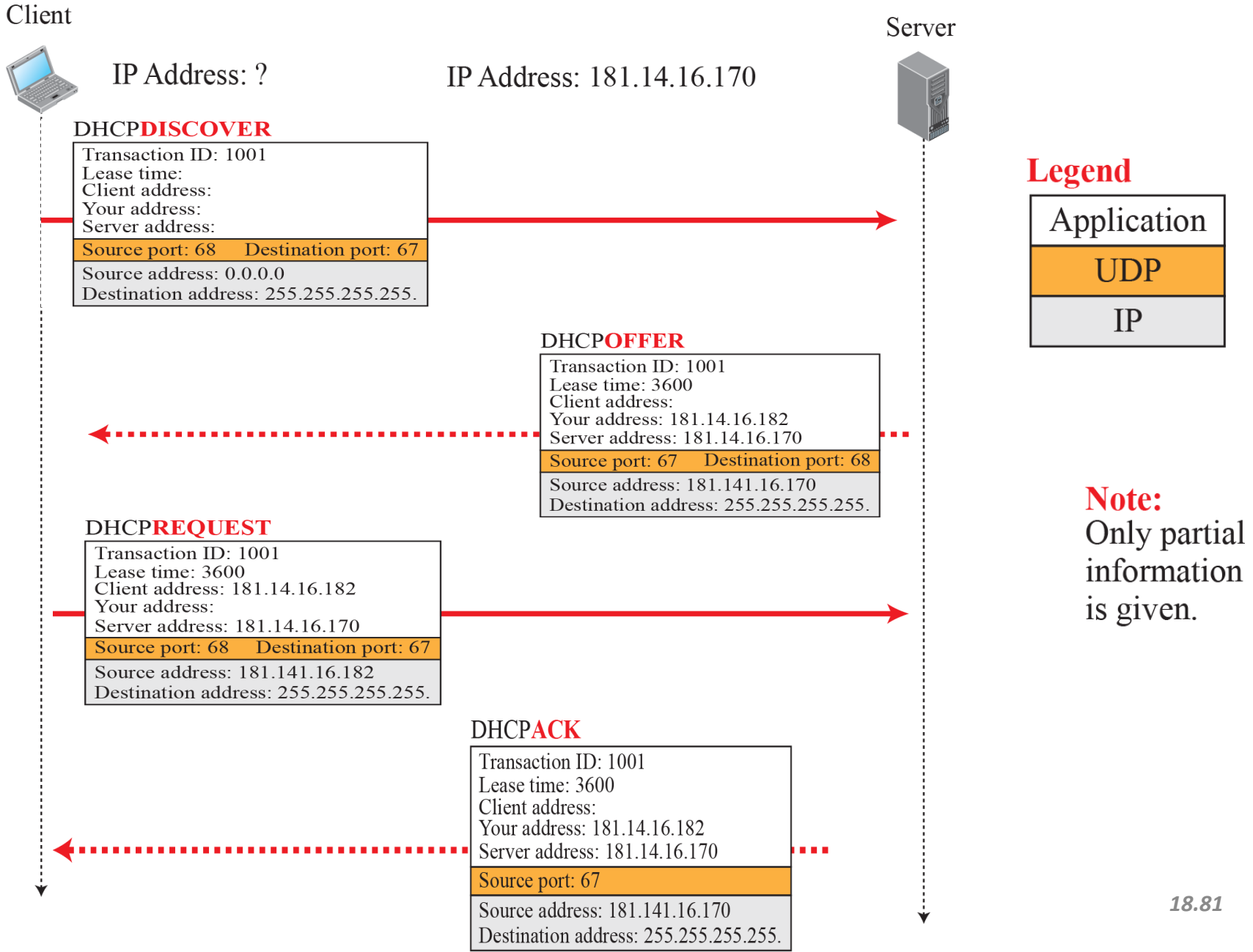
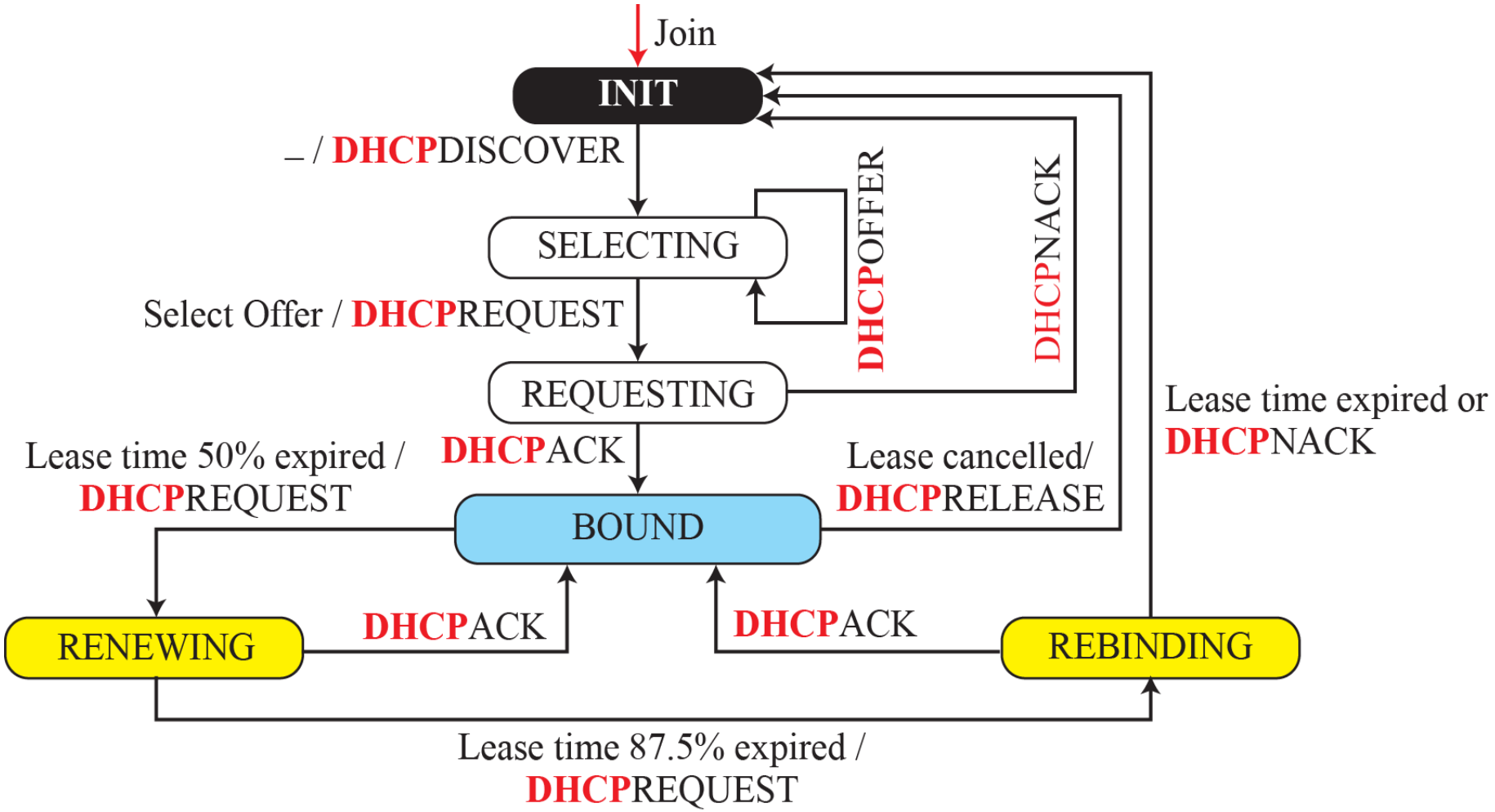


Figure :FSM for the DHCP client

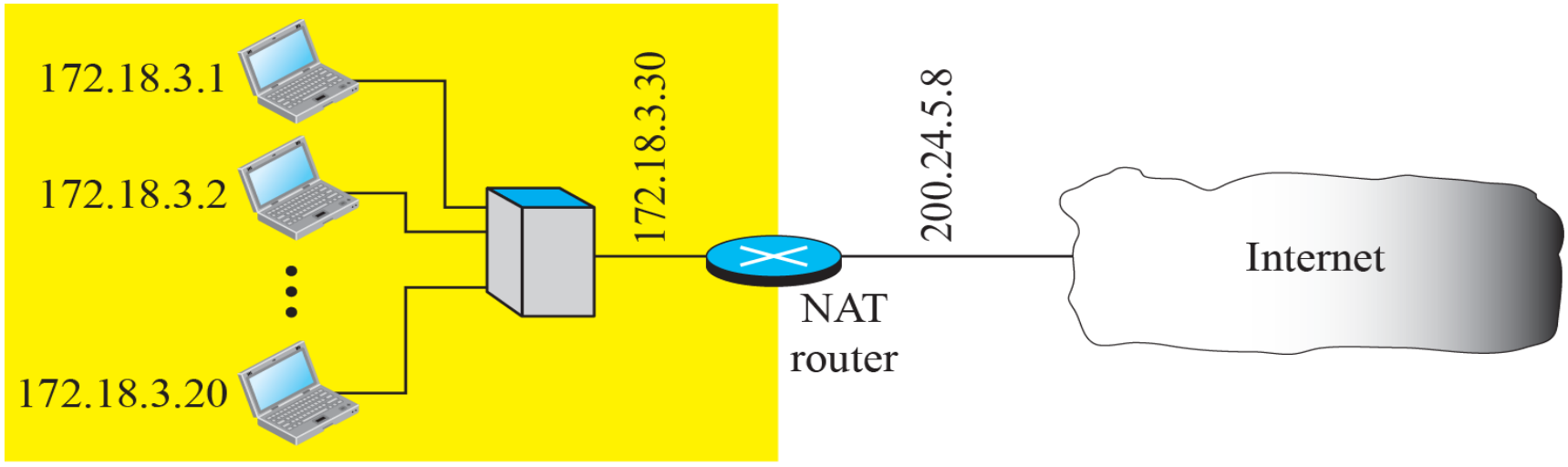


Network Address Translation(NAT)

- NAT is a method that enables hosts on private networks to communicate with hosts on the Internet
- A technology that can provide the mapping between the private and universal addresses, and at the same time support virtual private Networks, NAT.
- NAT is run on routers that connect private networks to the public Internet, to replace the IP address-port pair of an IP packet with another IP address-port pair.

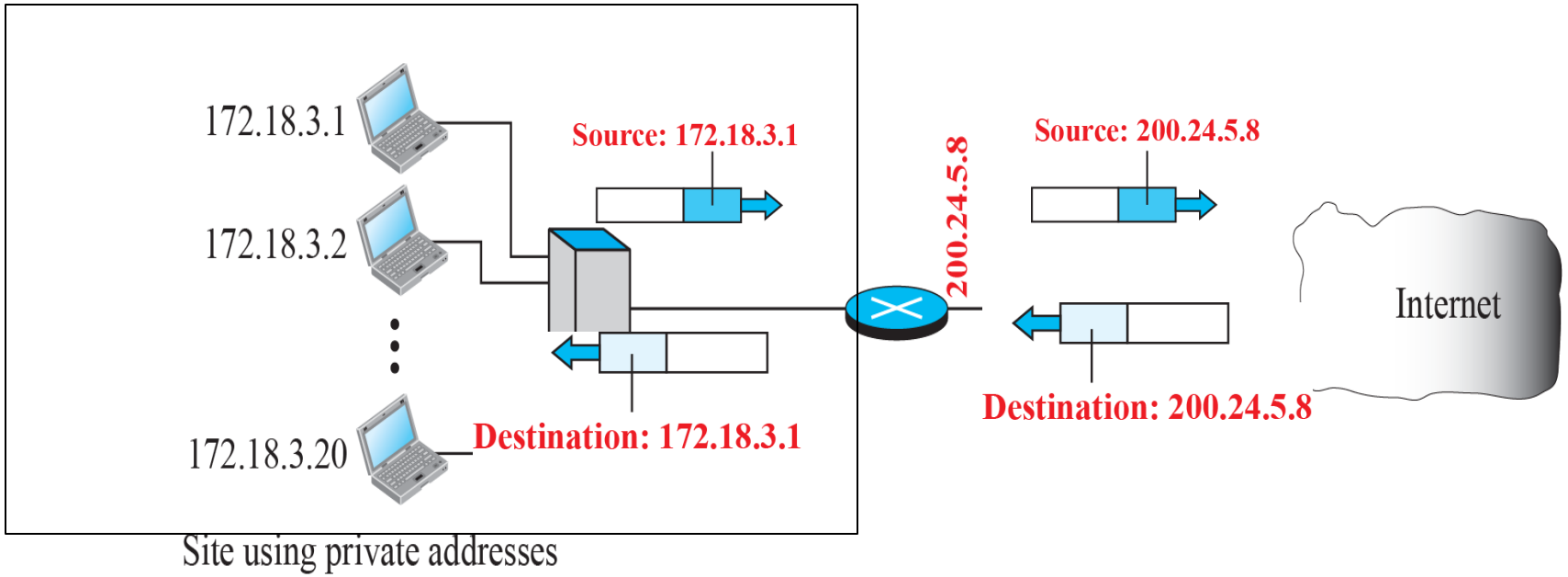
NAT

All the outgoing packets go through the NAT router, which replaces the source address in the packet with the global NAT address. All incoming packets also pass through the NAT router, which replaces the destination address in the packet (the NAT router global address) with the appropriate private address

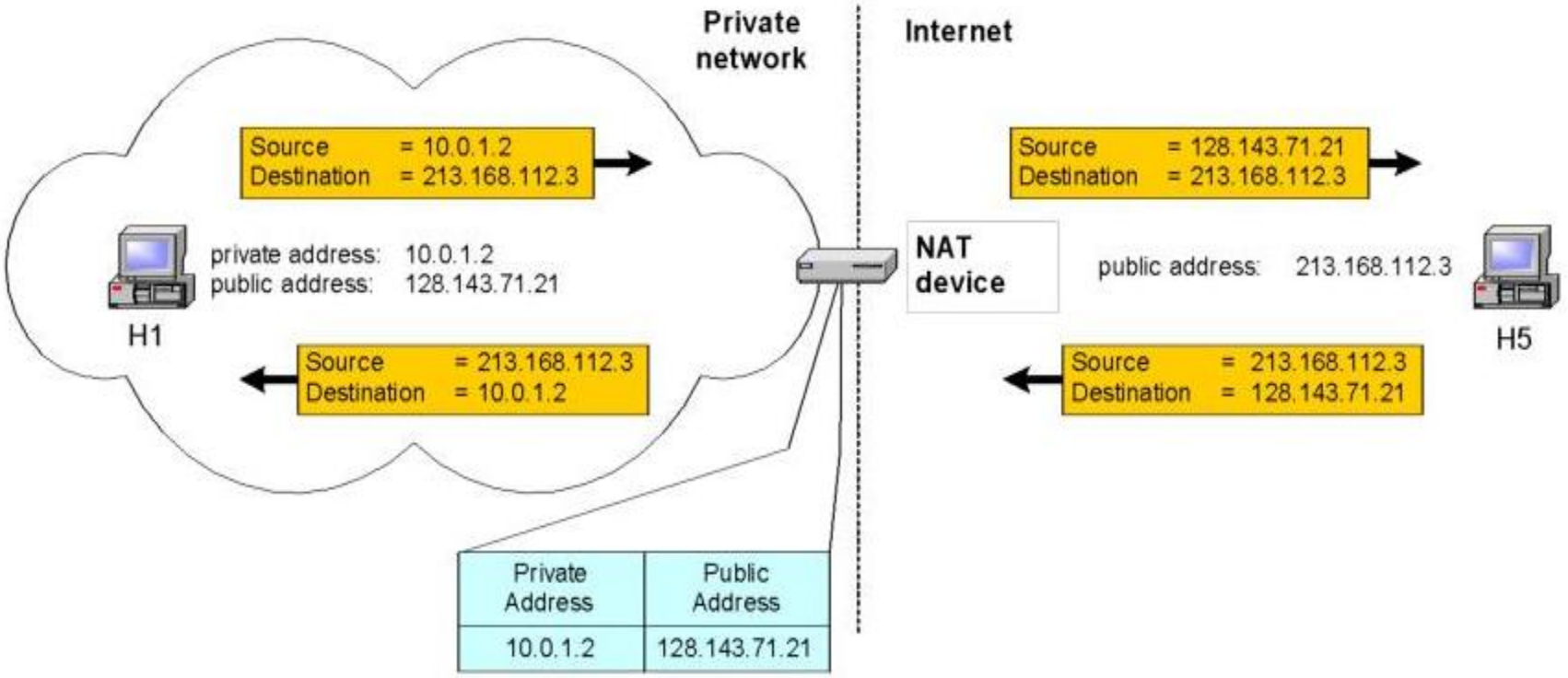


Site using private addresses

Figure : Address translation

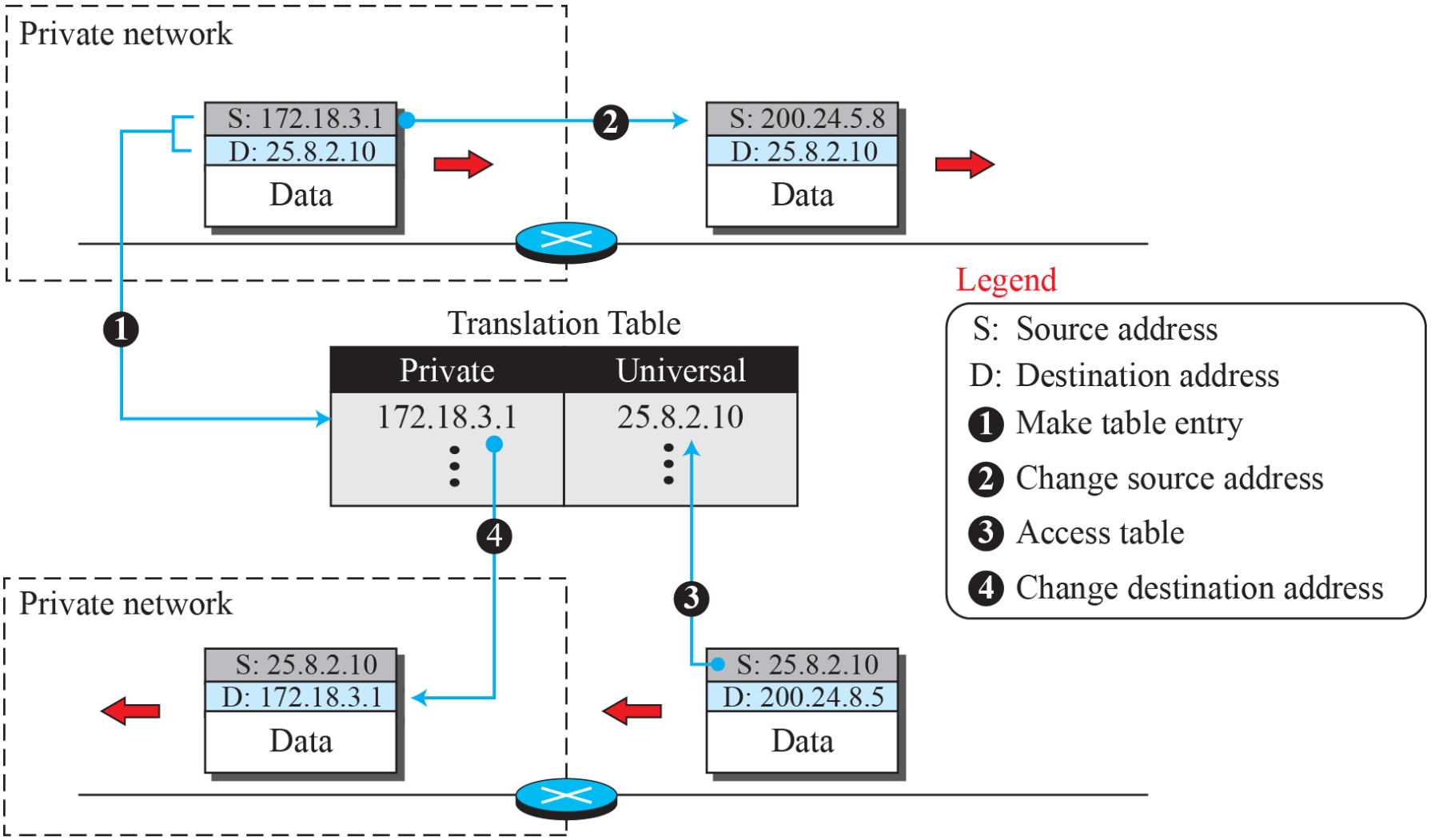


Basic operation of NAT



- NAT device has address translation table

Figure : Translation(using one IP Address)



- The communication must always be initiated by the private network.
- NAT is used mostly by ISPs which assign one single address to a customer.
- The customer, however, may be a member of a private network that has many private addresses
- In this case, communication with the Internet is always initiated from the customer site, using a client program such as HTTP, TELNET, or FTP to access the corresponding server program.

- private network cannot run a server program for clients outside of its network if it is using NAT technology

- Using a Pool of IP Addresses Since the NAT router has only one global address, only one private network host can access the same external host.

For example, instead of using only one global address (200.24.5.8), the NAT router can use four addresses (200.24.5.8, 200.24.5.9, 200.24.5.10, and 200.24.5.11).

- Four private network hosts can communicate with the same external host at the same time because each pair of addresses defines a connection.
- No private-network host can access two external server programs (e.g., HTTP and FfP) at the same time.

Table : Five-column translation table

Private address	*Private port*	*External address*	*External port*	*Transport protocol*
172.18.3.1	1400	25.8.3.2	80	TCP
172.18.3.2	1401	25.8.3.2	80	TCP
⋮	⋮	⋮	⋮	⋮

Note that when the response from HTTP comes back, the combination of source address (25.8.3.2) and destination port number (1400) defines the-private network host to which the response should be directed. Note also that for this translation to work, the temporary port numbers (1400 and 1401) must be unique.

Security of IPv4 Data grams

- The IPv4 protocol and the Internet was started when the Internet users trusted each other.
- No security was provided for the IPv4 protocol.
- Today the situation is different; the Internet is the not secured one.
- The three security issues in IP protocol and the solutions that are particularly applicable to the IP protocol:

1. Packet sniffing
2. Packet modification
3. IP spoofing.

Fragmentation

- A datagram can travel through different networks. Each router de capsulates the IP datagram from the frame it receives, processes it, and then encapsulates it in another frame.
- The format and size of the received frame depend on the protocol used by the physical network through which the frame has just traveled.
- The format and size of the sent frame depend on the protocol used by the physical network through which the frame is going to travel.

Eg: if a router connects a LAN to a WAN, it receives a frame in the LAN format and sends a frame in the WAN format.

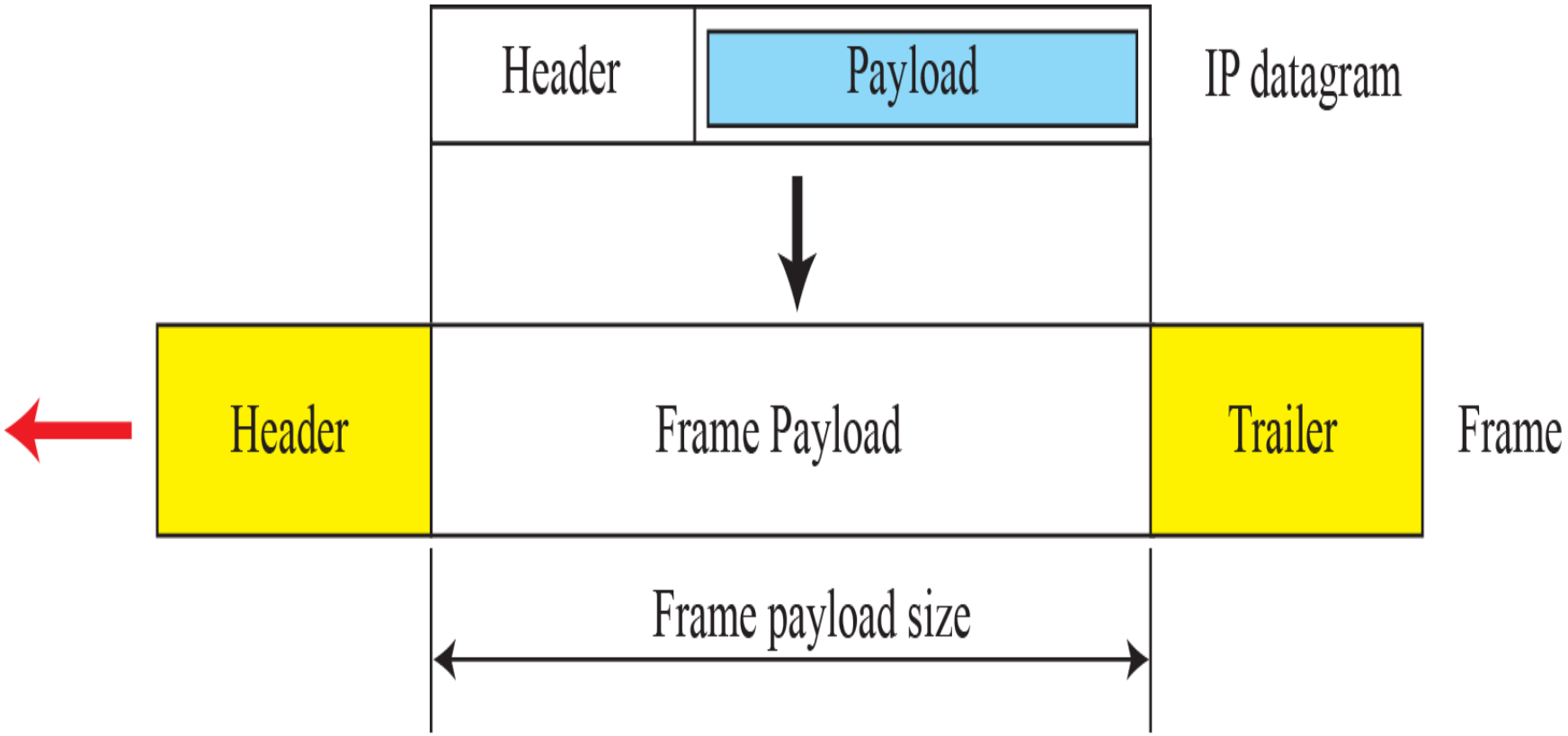
Fragmentation at Network Layer

Fragmentation is done by the network layer when the maximum size of datagram is greater than maximum size of data that can be held a frame

i.e., its Maximum Transmission Unit (MTU). The network layer divides the datagram received from transport layer into fragments so that data flow is not disrupted.

Since there are 16 bits for total length in IP header so, maximum size of IP datagram = $2^{16} - 1 = 65,535$ bytes.

Maximum transfer unit (MTU)



MTU: Maximum size of frame payload

Maximum Transfer Unit(MTU)

- Each link layer has its own frame format.
- When the datagram is encapsulated in the frame, the total size of the data gram must be less than this maximum size.
- This restriction is imposed by the hardware and soft ware used in the network.
- Value of MTU differs from one physical network to another.
- Eg: The value for a LAN is 1500 bytes.
- In order to make IP protocol independent of the physical network, the maximum length of IP data gram= 65,535 bytes, which makes transmission more efficient.

Maximum transfer unit (MTU)

- If data grams pass through Physical n/w, whose MTU is less than the size of the data gram, the data grams is divided to make it possible to pass through these networks. This is called Fragmentation.
- Source does not fragment the data. It is Transport Layer which segment the data into the size that can be accommodated by the IPV4.
- When the data gram is fragmented, each fragment has its own header with most of the fields repeated.
- If fragmented data gram encounters a n/w with smaller MTU, fragmented data gram itself will be fragmented.

- Flags, fragmentation offset and total length.

Maximum transfer unit (MTU)

Fields related to fragmentation are

Identification: 16-bit field identifies a data gram originating from the source host.

→ A combination of identification and the source IPV4 address must uniquely defines a data gram as it leaves the source host.

→ When the data gram is fragmented, value of identification filed is copied to all the fragments.

i.e all fragments have same identification number, which helps the destination in reassembling the data grams.

→ It knows that all fragments with same identification number must be assembled in to one data gram.

Fields related to fragmentation

- Flags : 3-bit field. →first bit is reserved.

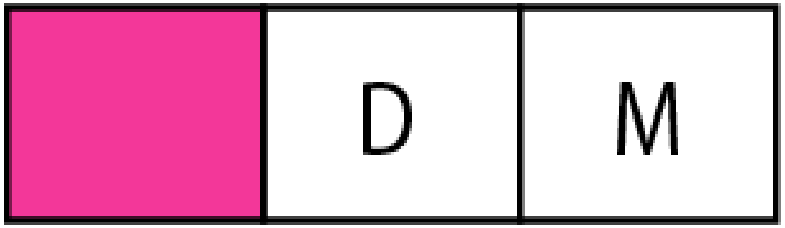
  →second bit is called don't fragment bit. If D=1, the machine must not fragment the data gram.

  →If D=0, the data gram can be fragmented if necessary.

  → Third bit is called more fragment bit.

→If M=1, the data gram is not the last fragment, there are more fragments after this one.

→ If M=0, this is the last/only fragment

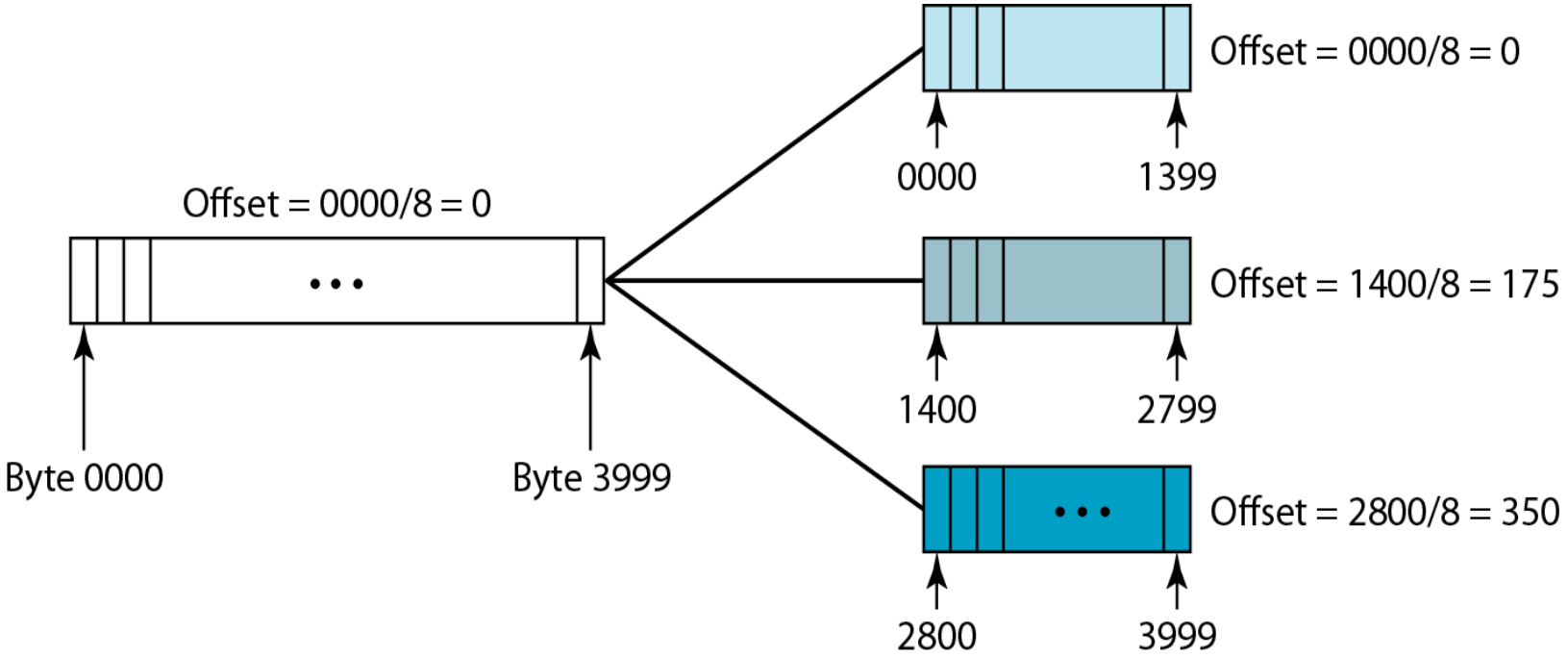


D: Do not fragment

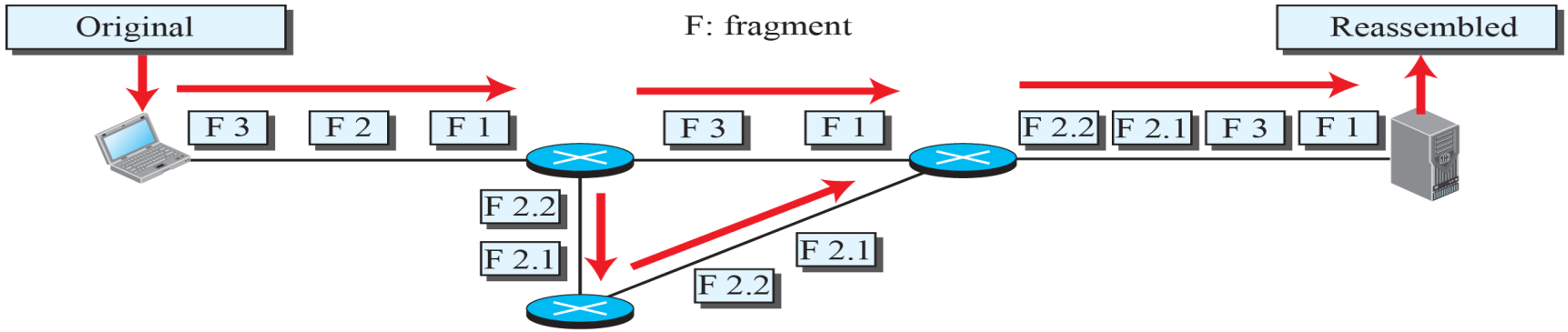
M: More fragments

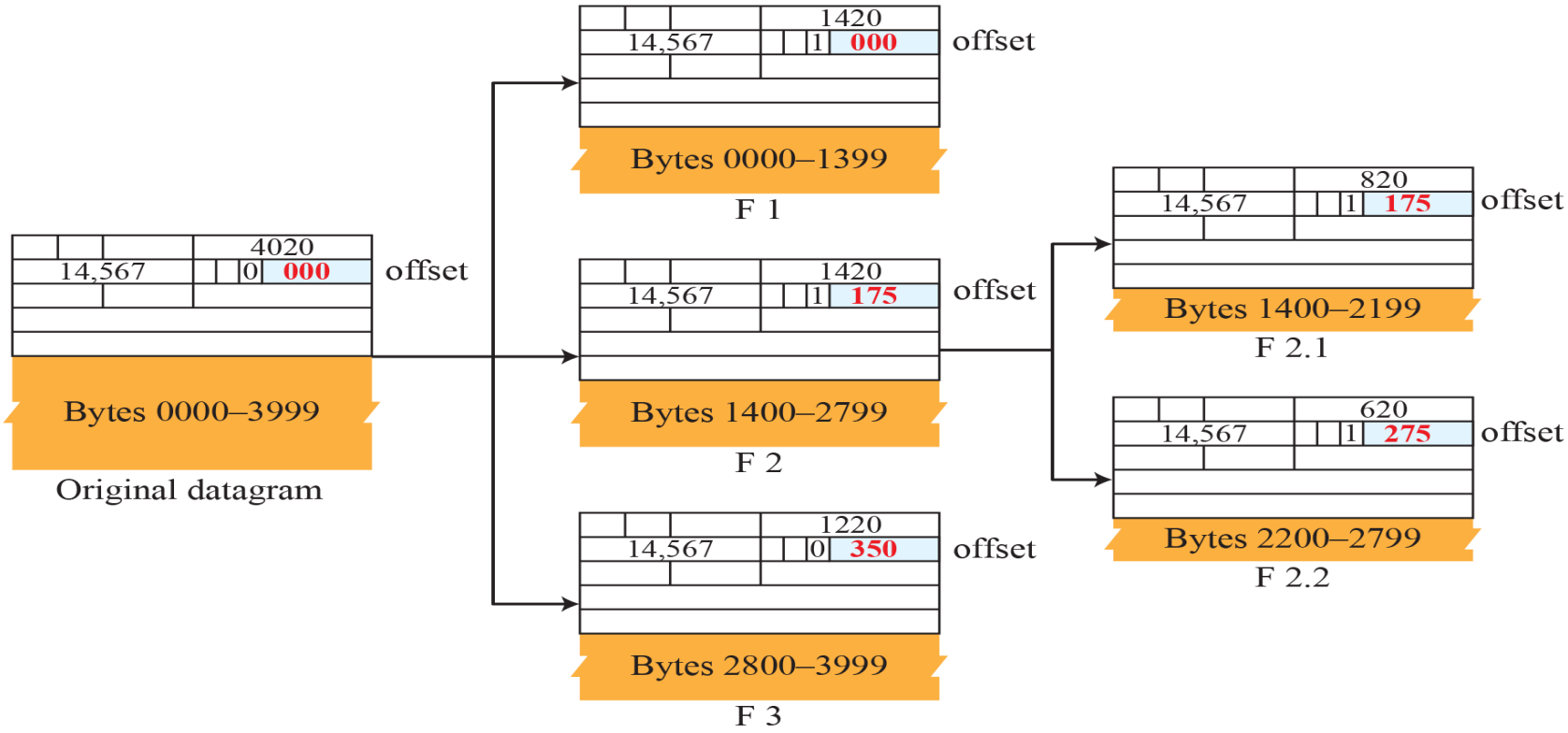
Fragmentation offset.

- 13 bit field shows the relative position of this fragment w.r.t the whole data gram.
- It is the offset of the data in the original data gram measured in units of 8 bytes.



Detailed fragmentation example

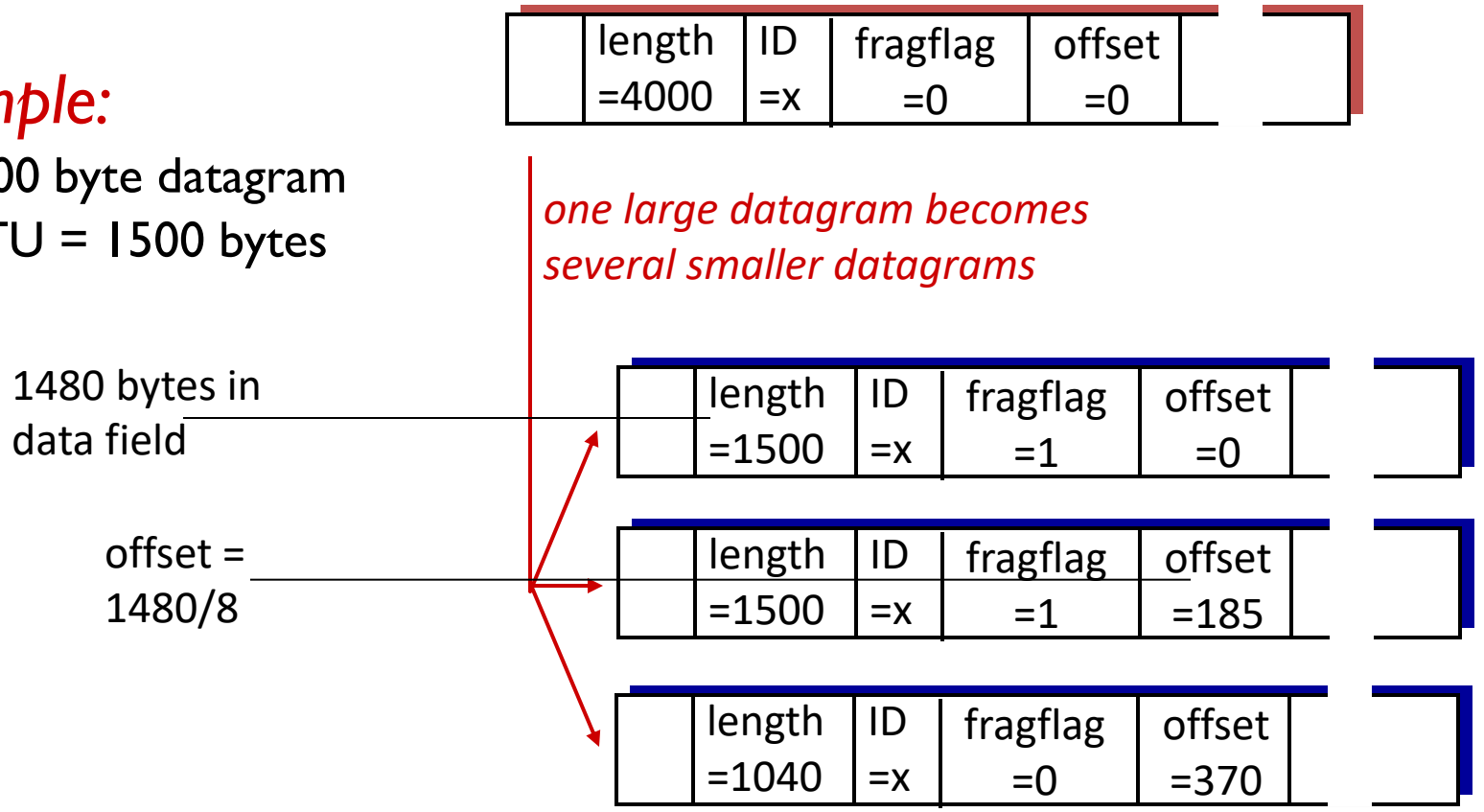




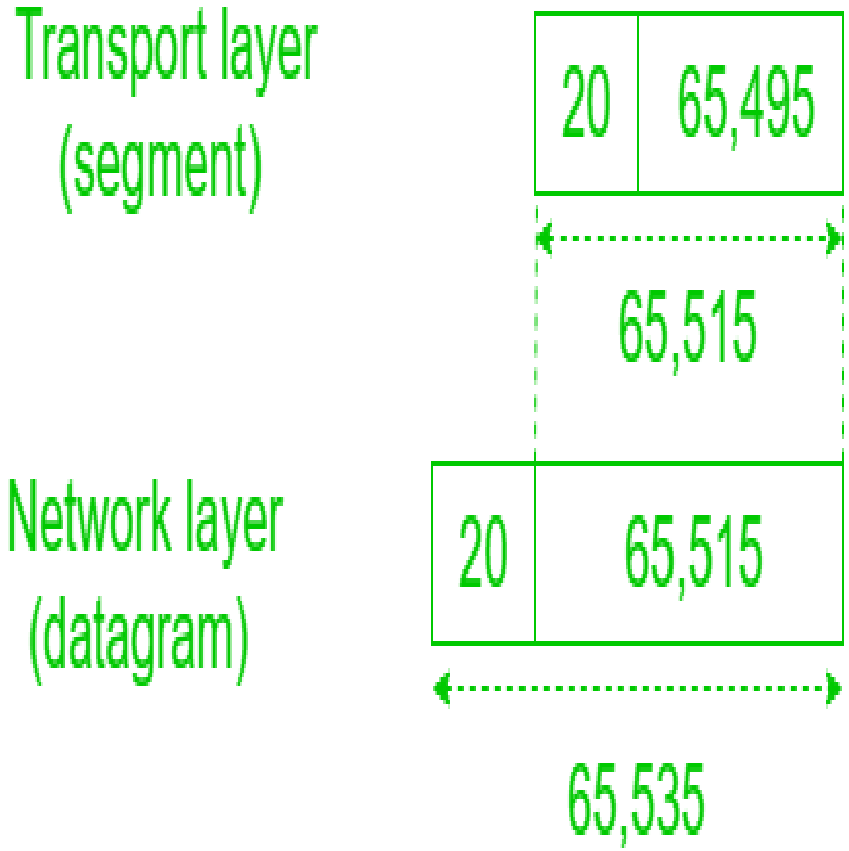
IP fragmentation, reassembly

example:

- 4000 byte datagram
- MTU = 1500 bytes



Fragmentation at Network Layer



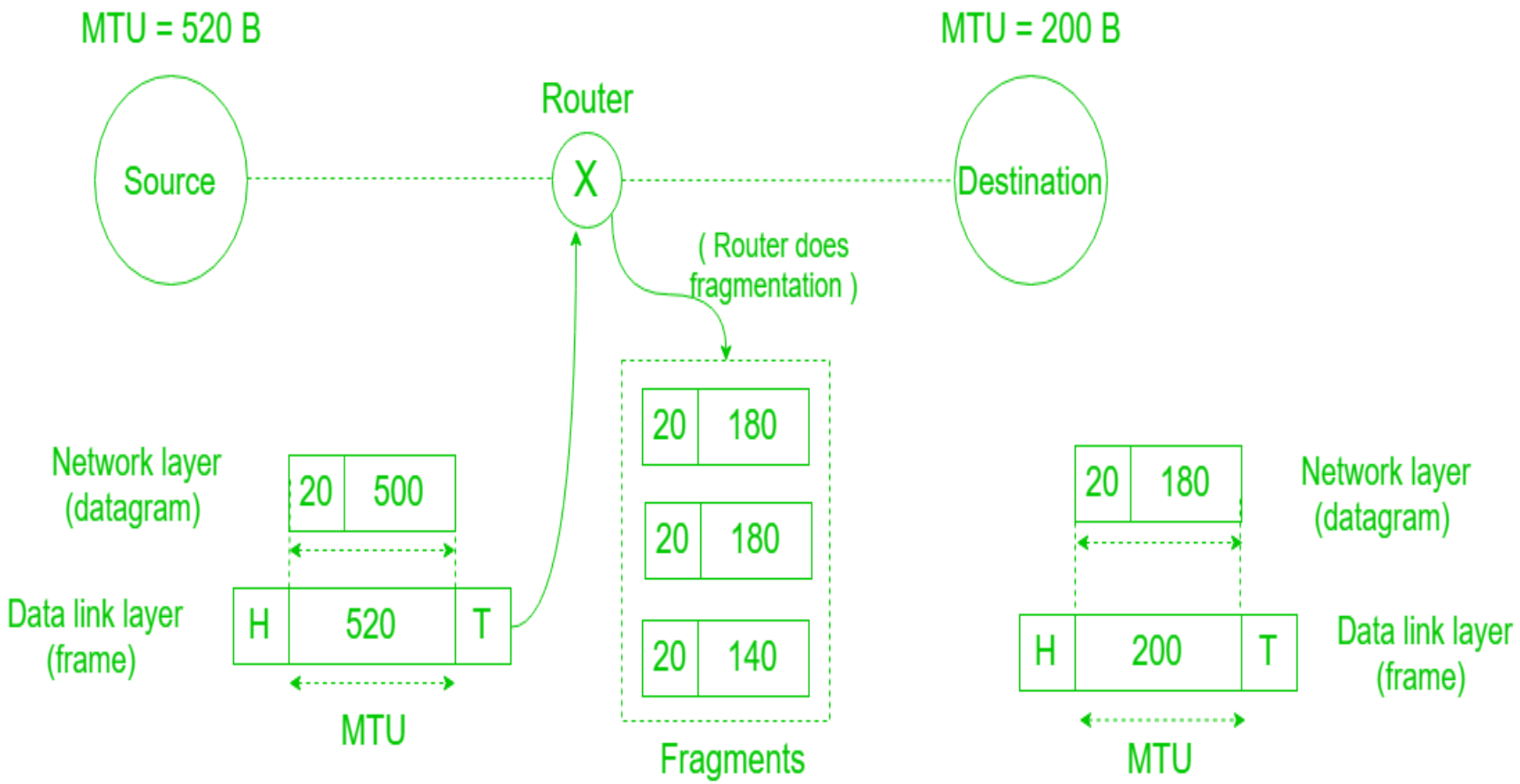
Max size of data in segment
= 65515 - 20 = 65495 B

Max size of data in datagram
= 65535 - 20 = 65515 B

Contd…

- It is done by network layer at the destination side and is usually done at routers.
- Source side does not require fragmentation due to wise (good) segmentation by transport layer i.e. instead of doing segmentation at transport layer and fragmentation at network layer,
- The transport layer looks at datagram data limit and frame data limit and does segmentation in such a way that resulting data can easily fit in a frame without the need of fragmentation

Fragmentation



- Receiver identifies the frame with the **identification (16 bits)** field in IP header. Each fragment of a frame has same identification number.
- Receiver identifies sequence of frames using the **fragment offset(13 bits)** field in IP header
- **Fragment offset (13 bits) –** use to identify sequence of fragments in the frame. It generally indicates number of data bytes preceeding or ahead of the fragment. Maximum fragment offset possible = (65535 – 20) – 1 = 65514

- **More fragments (MF = 1 bit) –** tells if more fragments ahead of this fragment i.e. if MF = 1, more fragments are ahead of this fragment and if MF = 0, it is the last fragment.
- **Don't fragment (DF = 1 bit) –** if we don't want the packet to be fragmented then DF is set i.e. DF = 1.
- **Reassembly of Fragments –**

It takes place only at destination and not at routers since packets take independent path(datagram packet switching), so all may not meet at a router and hence a need of fragmentation may arise again. The fragments may arrive out of order also.

- An overhead at network layer is present due to extra header introduced due to fragmentation.

Reserverd MSB bit	DF	MF
	0	0
	0	1
	1	0
	1	1

Algorithm –

- Destination should identify that datagram is fragmented from MF, Fragment offset field.
- Destination should identify all fragments belonging to same datagram from Identification field.
- Identify the 1st fragment(offset = 0).
- Identify subsequent fragment using header length, fragment offset.
- Repeat until MF = 0.

Example 19.6

A packet has arrived with an M bit value of 0. Is this the first fragment, the last fragment, or a middle fragment? Do we know if the packet was fragmented?

Solution

→If the M bit=0, it means that there are no more fragments; the fragment is the last one.

However, we cannot say if the original packet was fragmented or not. A non fragmented packet is considered the last fragment.

Example 19.7

A packet has arrived with an M bit value of 1. Is this the first fragment, the last fragment, or a middle fragment? Do we know if the packet was fragmented?

Solution

→If the M=1, it means that there is at least one more fragment.

→This fragment can be the first one or a middle one, but not the last one.

→cannot identify if it is the first one or a middle one which need more information (the value of the fragmentation offset).

Example 19.8

A packet has arrived with an M bit value of 1 and a fragmentation offset value of 0. Is this the first fragment, the last fragment, or a middle fragment?

Solution

→Because the M =1, it is either the first fragment or a middle one.

→Because the offset value =0, it is the first fragment.

Example 19.9

A packet has arrived in which the offset value is 100. What is the number of the first byte? Do we know the number of the last byte?

Solution

→To find the number of the first byte,

Multiply the offset value by 8.

This means that the first byte number is 800.

→Cannot determine the number of the last byte unless length of the data is known.

Example 19.10

A packet has arrived in which the offset value is 100, the value of HLEN is 5, and the value of the total length field is 100. What are the numbers of the first byte and the last byte?

Solution

→The first byte number is $100 \times 8 = 800$.

→The total length is 100 bytes, and the header length is 20 bytes (5×4), which means that there are 80 bytes in this datagram.

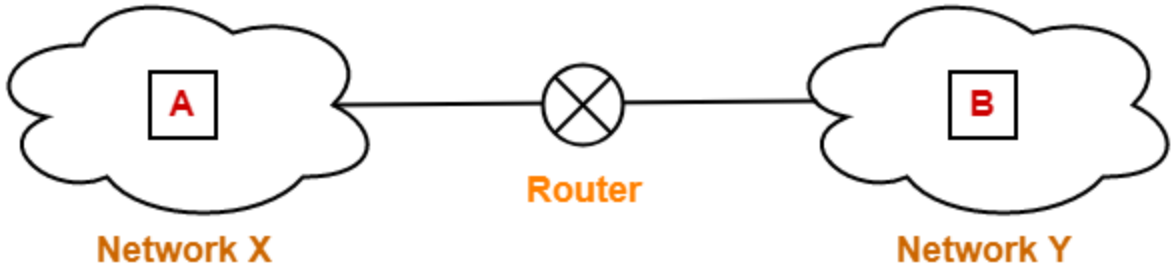
→If the first byte number is 800, the last byte number must be 879.

Example – An IP router with a Maximum Transmission Unit (MTU) of 200 bytes has received an IP packet of size 520 bytes with an IP header of length 20 bytes. The values of the relevant fields in the IP header.

Soln.

Since MTU is 200 bytes and 20 bytes is header size so, the maximum length of data = 180 bytes but it can't be represented in fragment offset since it is not divisible by 8

so, the maximum length of data feasible = 176 bytes. Number of fragments = (520/200) = 3.



➢Consider router receives a datagram from host A having-

•Header length = 20 bytes

•Payload length = 500 bytes

•Total length = 520 bytes

•DF bit set to 0

➢Router examines the datagram and finds-

•Size of the datagram = 520 bytes

•Destination is network Y having MTU = 200 bytes

•DF bit is set to 0

Router knows-

MTU of the destination network = 200 bytes.

So, maximum total length of any fragment can be only 200 bytes.

Out of 200 bytes, 20 bytes will be taken by the header.

So, maximum amount of data that can be sent in any fragment = 180 bytes

	20 | 176	20 | 176	20 | 148
Fragment Offset	0	22	44
MF	1	1	0
Header length	5	5	5
Total length	196	196	168

➢Router knows

•MTU of the destination network = 200 bytes.

•So, maximum total length of any fragment can be only 200 bytes.

•Out of 200 bytes, 20 bytes will be taken by the header.

•So, maximum amount of data that can be sent in any fragment = 180 bytes.

•Router uses the following rule to choose the amount of data that will be transmitted in one fragment-

RULE

The amount of data sent in one fragment is chosen such that-

•It is as large as possible but less than or equal to MTU.

•It is a multiple of 8 so that pure decimal value can be obtained for the fragment offset field.

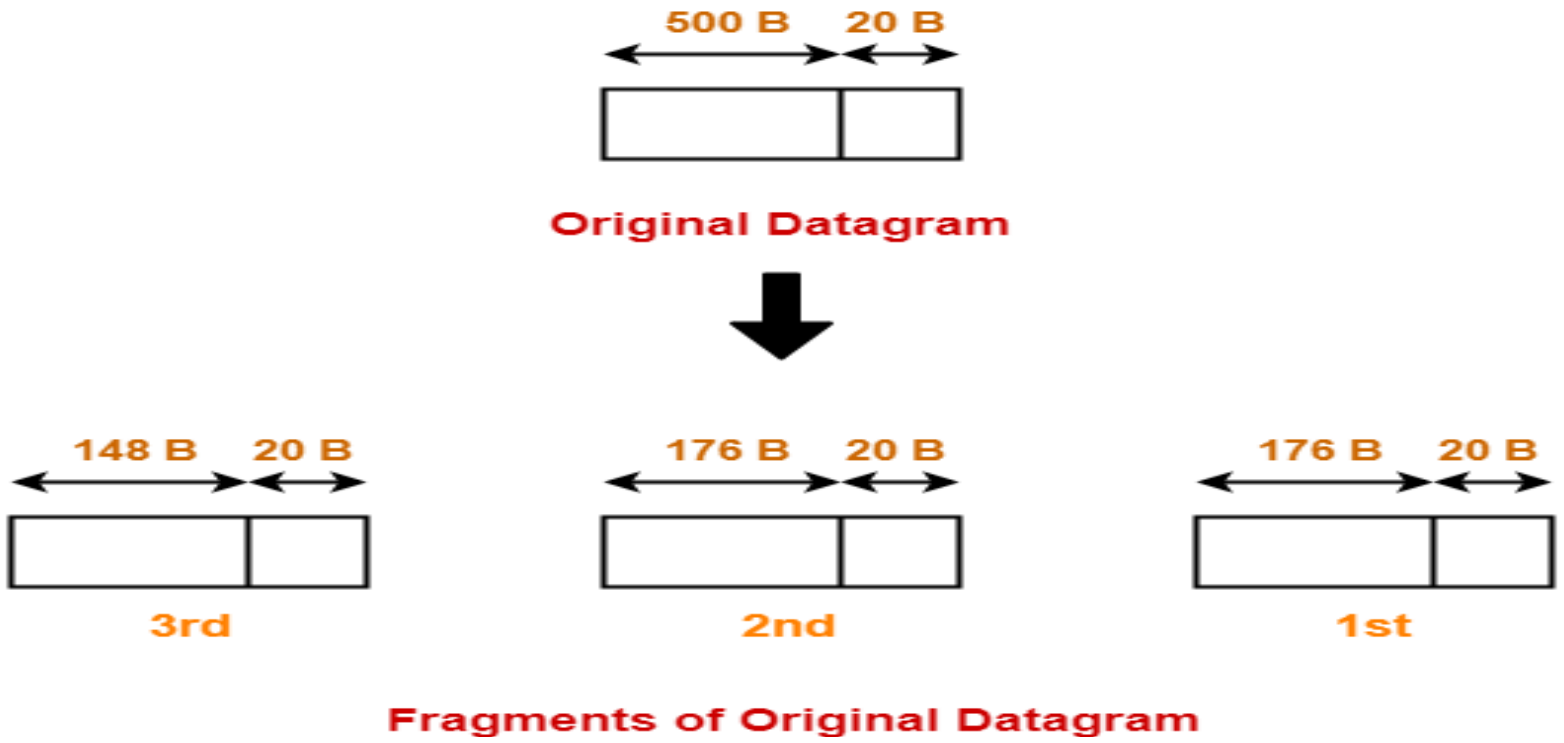
NOTE

• It is not compulsory for the last fragment to contain the amount of data that is a multiple of 8.

•This is because it does not have to decide the fragment offset value for any other fragment.

Router creates three fragments of the original datagram where-

- First fragment contains the data = 176 bytes
- Second fragment contains the data = 176 byes
- Third fragment contains the data = 148 bytes



➢ Header Information Of 1st Fragment-

- Header length field value = 20 / 4 = 5
- Total length field value = 176 + 20 = 196
- MF bit = 1
- Fragment offset field value = 0
- Header checksum is recalculated.
- Identification number is same as that of original datagram.

➢ Header Information Of 2nd Fragment-

- Header length field value = 20 / 4 = 5
- Total length field value = 176 + 20 = 196
- MF bit = 1
- Fragment offset field value = 176 / 8 = 22
- Header checksum is recalculated.
- Identification number is same as that of original datagram.

➢ Header Information Of 3rd Fragment-

- Header length field value = 20 / 4 = 5
- Total length field value = 148 + 20 = 168
- MF bit = 0
- Fragment offset field value = (176 + 176) / 8 = 44
- Header checksum is recalculated.
- Identification number is same as that of original datagram.

Example-2:Data gram of 3000 Bytes(20B of header+2980B IP payload)reached at router and must be forwarded to link with MTU of 500Bytes .How many fragments will be generated and also write ?MF,offset,total length value for all.

➢Router knows

-MTU of the destination network = 500bytes.

-The maximum total length of any fragment can be only 500 bytes.

-Out of 500 bytes, 20 bytes will be taken by the header.

-So, maximum amount of data that can be sent in any fragment = 480 bytes.

-No of Fragments=[2980/480]=7fragments

1st Fragment-

Total length field value = 480 + 20 = 500

MF bit = 1

Fragment offset field value = 0

2nd Fragment

•Total length field value = 480 + 20 = 500

•MF bit = 1

•Fragment offset field value = 480 / 8 = 60

3rd Fragment

•Total length field value = 480 + 20 = 500

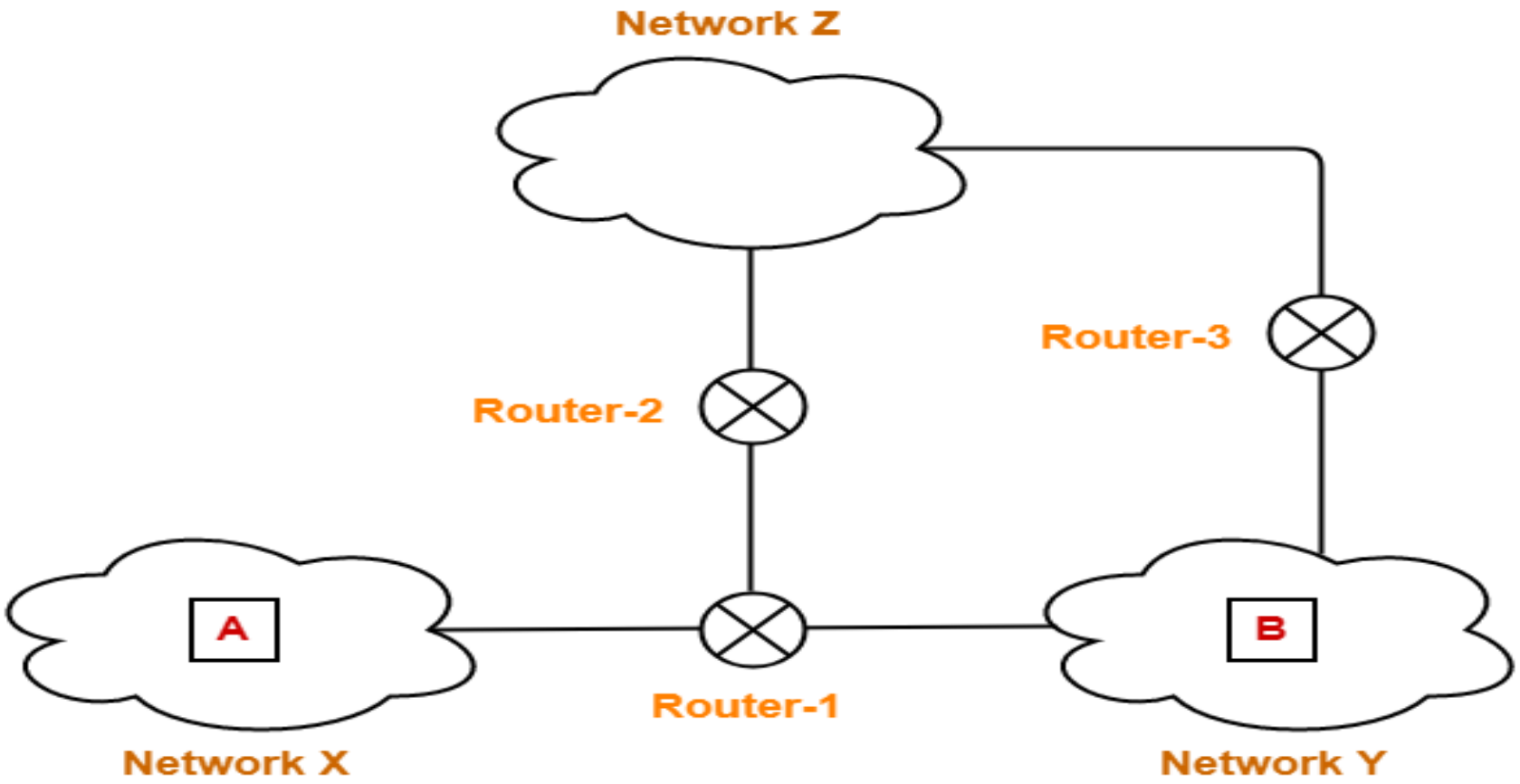
•MF bit = 1

•Fragment offset field value = (480+480) / 8 = 120

Example-02:

Consider-

- There is a host A present in network X having MTU = 520 bytes.
- There is a host B present in network Y having MTU = 200 bytes.
- There exists a network Z having MTU = 110 bytes.
- Host A wants to send a message to host B.



Consider Router-1 divides the datagram into 3 fragments as discussed in Example-01.

Then,

First fragment contains the data = 176 bytes

Second fragment contains the data = 176 byes

Third fragment contains the data = 148 bytes

Now, consider-

First and third fragment reaches the destination directly.

However, second fragment takes its way through network Z and reach the destination through Router-3.

Journey Of Second Fragment-

Router-2 receives a datagram (second fragment of original datagram) where-
Header length = 20 bytes
Payload length = 176 bytes
Total length = 196 bytes
DF bit set to 0

Now, Router-2 works in the following steps-

Step-01:

Router-2 examines the datagram and finds-
Size of the datagram = 196 bytes
Destination is network Z having MTU = 110 bytes
DF bit is set to 0

Router-2 concludes-
Size of the datagram is greater than MTU.
So, it will have to divide the datagram into fragments.
DF bit is set to 0.

Step-02:

Router-2 decides the amount of data that it should transmit in each fragment.

- Router-2 knows-

MTU of the destination network = 110 bytes.

So, maximum total length of any fragment can be only 110 bytes.

Out of 110 bytes, 20 bytes will be taken by the header.

So, maximum amount of data that can be sent in any fragment = 90 bytes.

Following the rule,

Router-2 decides to send maximum 88 bytes of data in one fragment.

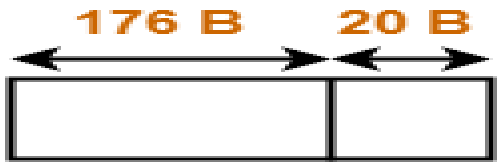
This is because it is the greatest value that is a multiple of 8 and less than MTU.

Step-03:

Router-2 creates two fragments of the received datagram where-

First fragment contains the data = 88 bytes

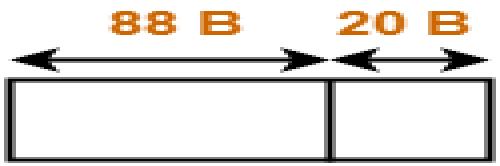
Second fragment contains the data = 88 byes



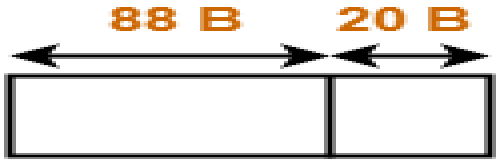
Received Datagram

(Second Fragment From Router-1)





2nd



1st

Fragments of Received Datagram

The information contained in the IP header of each fragment is-

Header Information Of 1st Fragment-

- Header length field value = 20 / 4 = 5
- Total length field value = 88 + 20 = 108
- MF bit = 1
- Fragment offset field value = 176 / 8 = 22

NOTE-

- This fragment is **NOT** the first fragment of the original datagram.
- It is the first fragment of the datagram received by Router-2.
- The datagram received by Router-2 is the second fragment of the original datagram.
- This datagram will serve as the second fragment of the original datagram.
- Therefore, fragment offset field is set according to the first fragment of the original datagram.

Header Information Of 2nd Fragment-

- Header length field value = 20 / 4 = 5
- Total length field value = 88 + 20 = 108
- MF bit = 1
- Fragment offset field value = (176 + 88) / 8 = 33

NOTE-

• This fragment is **NOT** the last fragment of the original datagram.

•It is the last fragment of the datagram received by Router-2.

•The datagram received by Router-2 is the second fragment of the original datagram.

•This datagram will serve as the third fragment of the original datagram.

•There is another fragment of the original datagram that follows it.

•That is why, here MF bit is not set to 0.

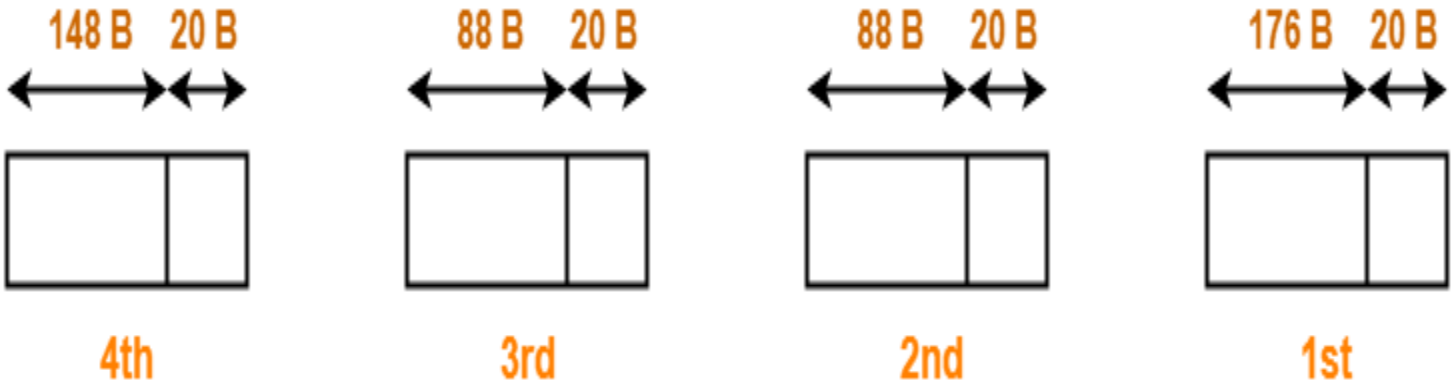
• Router-2 transmits both the fragments which reaches the destination through Router-3.

•Router-3 performs no fragmentation.

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Step-04:

At destination side,
Receiver receives 4 fragments of the datagram.
Reassembly algorithm is applied to combine all the fragments to obtain the original datagram.



Fragments Received By Receiver

Fragmentation Overhead-

Fragmentation of datagram increases the overhead.
This is because after fragmentation, IP header has to be attached with each fragment.

- Total Overhead= (Total number of fragmented datagrams – 1) x size of IP header
- Efficiency = Useful bytes transferred / Total bytes transferred

OR

- Efficiency = Data without header / Data with header
- Bandwidth Utilization or Throughput = Efficiency x Bandwidth